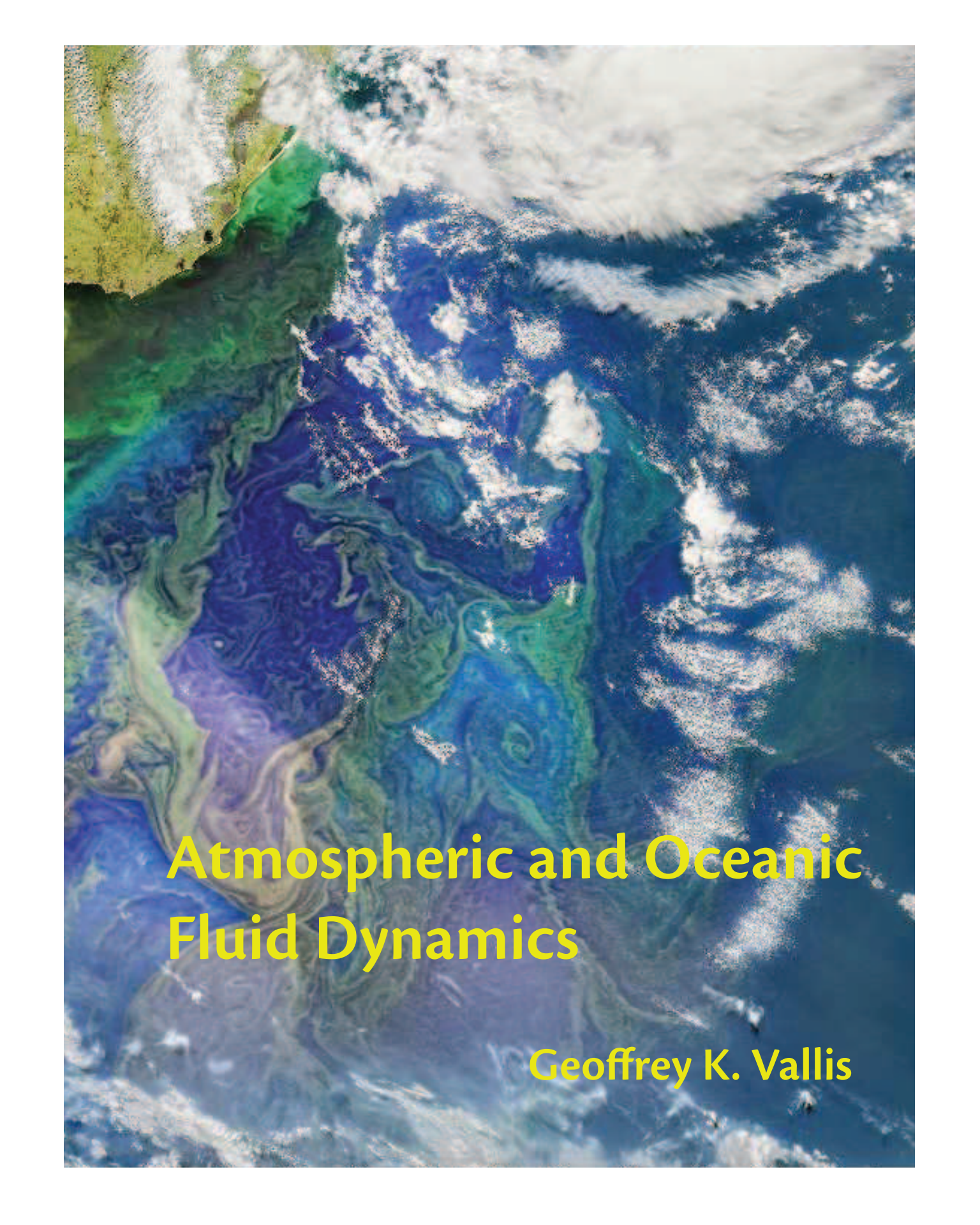


A satellite image of Earth showing swirling ocean currents and clouds. The image is a composite of various colors, including green, blue, and purple, representing different oceanic features. The text "Atmospheric and Oceanic Fluid Dynamics" is overlaid in yellow. The author's name "Geoffrey K. Vallis" is also overlaid in yellow at the bottom right.

Atmospheric and Oceanic Fluid Dynamics

Geoffrey K. Vallis

Placeholder

Contents

Preface

ix

I FUNDAMENTALS OF GEOPHYSICAL FLUID DYNAMICS

1	Equations of Motion	3
1.1	Time Derivatives for Fluids	3
1.2	The Mass Continuity Equation	7
1.3	The Momentum Equation	11
1.4	The Equation of State	13
1.5	Thermodynamic Relations	14
1.6	Thermodynamic Equations for Fluids	21
1.7	Thermodynamics of Seawater	30
1.8	Sound Waves.	39
1.9	Compressible and Incompressible Flow	40
1.10	The Energy Budget.	42
1.11	An Introduction to Non-Dimensionalization and Scaling.	46
1.A	Appendix A: Thermodynamics of an Ideal gas from the Gibbs function . . .	47
1.B	Appendix B: The First Law of Thermodynamics for Fluids.	49
2	Effects of Rotation and Stratification	55
2.1	Equations in a Rotating Frame	55
2.2	Equations of Motion in Spherical Coordinates	59
2.3	Cartesian Approximations: The Tangent Plane.	69
2.4	The Boussinesq Approximation	70
2.5	The Anelastic Approximation	75
2.6	Changing Vertical Coordinate	79
2.7	Scaling for Hydrostatic Balance	83
2.8	Geostrophic and Thermal Wind Balance	87
2.9	Gradient Wind Balance	94
2.10	Static Instability and the Parcel Method	97
2.A	Appendix: Asymptotic Derivation of the Boussinesq Equations	101

3	Shallow Water Systems	105
3.1	Dynamics of a Single Shallow Layer of Fluid	105
3.2	Reduced Gravity Equations.	110
3.3	Multi-Layer Shallow Water Equations	112
3.4	From Continuous Stratification to Shallow Water	114
3.5	Geostrophic Balance and Thermal wind	118
3.6	Form Stress.	119
3.7	Conservation Properties of Shallow Water Systems	120
3.8	Shallow Water Waves	123
3.9	Geostrophic Adjustment	127
3.10	Isentropic Coordinates	134
3.11	Available Potential Energy	137
4	Vorticity and Potential Vorticity	143
4.1	Vorticity and Circulation	143
4.2	The Vorticity Equation	145
4.3	Vorticity and Circulation Theorems	147
4.4	Vorticity Equation in a Rotating Frame	153
4.5	Potential Vorticity Conservation.	156
4.6	Potential Vorticity in the Shallow Water System	161
4.7	Potential Vorticity in Approximate, Stratified Models.	162
4.8	The Impermeability of Isentropes to Potential Vorticity	165
5	Geostrophic Theory	171
5.1	Geostrophic Scaling	171
5.2	The Planetary-Geostrophic Equations.	176
5.3	The Shallow Water Quasi-Geostrophic Equations	180
5.4	The Continuously Stratified Quasi-Geostrophic System	187
5.5	Quasi-geostrophy and Ertel Potential Vorticity	195
5.6	Energetics of Quasi-Geostrophy	198
5.7	The Ekman Layer.	201
II	WAVES, INSTABILITIES AND TURBULENCE	
6	Wave Fundamentals	215
6.1	Fundamentals and Formalities	215
6.2	Group Velocity.	220
6.3	Ray Theory	224
6.4	Rossby Waves	226
6.5	Rossby Waves in Stratified Quasi-Geostrophic Flow.	231
6.6	Energy Propagation and Reflection of Rossby Waves	234
6.7	Group Velocity, Revisited	240
6.8	Energy Propagation of Poincaré Waves	244
6.A	Appendix: The wkb Approximation for Linear Waves.	247
7	Gravity Waves	251
7.1	Surface Gravity Waves	251
7.2	Shallow Water Waves on Fluid Interfaces	257
7.3	Internal Waves in a Continuously Stratified Fluid	259
7.4	Internal Wave Reflection	268

7.5	Internal Waves in a Fluid with Varying Stratification	271
7.6	Internal Waves in a Rotating Frame of Reference.	277
7.7	Topographic Generation of Internal Waves.	283
7.8	Acoustic-Gravity Waves in an Ideal Gas.	293
8	Linear Dynamics at Low Latitudes	297
8.1	Co-existence of Rossby and Gravity Waves	298
8.2	Waves on the Equatorial Beta Plane	303
8.3	Ray Tracing and Equatorial Trapping	314
8.4	Forced-dissipative Wavelike Flow	316
8.5	Forced, Steady Flow: the Matsuno–Gill Problem	321
8.A	Appendix: Nondimensionalization and Parabolic Cylinder Functions.	330
8.B	Appendix: Mathematical Relations in the Matsuno–Gill Problem.	333
9	Barotropic and Baroclinic Instability	335
9.1	Kelvin–Helmholtz Instability.	335
9.2	Instability of Parallel Shear Flow	337
9.3	Necessary Conditions for Instability.	345
9.4	Baroclinic Instability.	347
9.5	The Eady Problem	351
9.6	Two-Layer Baroclinic Instability	356
9.7	A Kinematic View of Baroclinic Instability	363
9.8	The Energetics of Linear Baroclinic Instability	367
9.9	Beta, Shear and Stratification in a Continuous Model.	369
10	Waves, Mean-Flows, and their Interaction	379
10.1	Quasi-geostrophic Wave–Mean-Flow Interaction	380
10.2	The Eliassen–Palm Flux	383
10.3	The Transformed Eulerian Mean	387
10.4	The Non-acceleration Result	394
10.5	Influence of Eddies on the Mean Flow in the Eady Problem	399
10.6	Necessary Conditions for Instability.	403
10.7	Necessary Conditions for Instability: Use of Pseudoenergy.	406
11	Basics of Incompressible Turbulence	413
11.1	The Fundamental Problem of Turbulence	413
11.2	The Kolmogorov Theory	416
11.3	Two-Dimensional Turbulence	423
11.4	Predictability of Turbulence	433
11.5	Spectra of Passive Tracers.	437
12	Geostrophic Turbulence and Baroclinic Eddies	445
12.1	Effects of Differential Rotation	445
12.2	Stratified Geostrophic Turbulence.	454
12.3	A Scaling Theory for Geostrophic Turbulence	460
12.4	Phenomenology of Baroclinic Eddies in the Atmosphere and Ocean	464

13	Turbulent Diffusion and Eddy Transport	473
13.1	Diffusive Transport	473
13.2	Turbulent Diffusion	475
13.3	Two-Particle Diffusivity	480
13.4	Mixing Length Theory.	484
13.5	Homogenization of a Scalar that is Advected and Diffused.	487
13.6	Diffusive Fluxes and Skew Fluxes.	490
13.7	Eddy Diffusion in the Atmosphere and Ocean	493
13.8	Thickness and Potential Vorticity Diffusion.	502
III	LARGE-SCALE ATMOSPHERIC CIRCULATION	
14	The Overturning Circulation: Hadley and Ferrel Cells	511
14.1	Basic Features of the Atmosphere	511
14.2	A Steady Model of the Hadley Cell	516
14.3	A Shallow Water Model of the Hadley Cell	524
14.4	Asymmetry Around the Equator.	526
14.5	Eddy effects on the Hadley Cell	528
14.6	The mechanism of eddy driving	532
14.7	The Ferrel Cell	533
15	Zonally Averaged Mid-Latitude Atmospheric Circulation	539
15.1	Surface Westerlies and the Maintenance of a Barotropic Jet	540
15.2	Layered Models of the Mid-latitude Circulation	549
15.3	Eddy Fluxes and an Example of a Closed Model	562
15.4	A Stratified Model and the Real Atmosphere	566
15.5	The Tropopause and the Stratification of the Atmosphere	572
15.6	A Model for both Stratification and Tropopause Height	578
15.A	Appendix: TEM for the Primitive Equations in Spherical Coordinates	581
16	Planetary Waves and Zonal Asymmetries	585
16.1	Rossby Wave Propagation in a Slowly Varying Medium.	585
16.2	Horizontal Propagation of Rossby Waves	588
16.3	Waves Near a Critical Line	594
16.4	A WKB wave–mean-flow problem for Rossby waves	598
16.5	Vertical Propagation of Rossby waves	599
16.6	Vertical Propagation of Rossby Waves in Shear.	606
16.7	Forced and Stationary Rossby Waves	609
16.8	Effects of Thermal Forcing	615
16.9	Wave Propagation using Ray Theory	621
17	The Stratosphere	627
17.1	A Descriptive Overview.	627
17.2	Waves in the Stratosphere	634
17.3	Wave Momentum Transport and Deposition	638
17.4	Phenomenology of the Residual Overturning Circulation	642
17.5	Dynamics of the Residual Overturning Circulation	644
17.6	The Quasi-Biennial Oscillation	651
17.7	Variability and Extra-Tropical Wave–Mean-Flow Interaction.	663

18	Water Vapour and the Tropical Atmosphere	673
18.1	A Moist Ideal Gas	673
18.2	The Distribution of Relative Humidity.	680
18.3	Atmospheric Convection.	690
18.4	Convection in a Moist Atmosphere	695
18.5	Radiative Equilibrium	701
18.6	Radiative-Convective Equilibrium	704
18.7	Vertically-Constrained Equations of Motion for Large Scales.	708
18.8	Scaling and Balanced Dynamics for Large-scale Flow in the Tropics	711
18.9	Scaling and Balance for Large-Scale Flow with Diabatic Sources	715
18.10	Convectively Coupled Gravity Waves and the MJO	718
18.A	Appendix A: Moist Thermodynamics and the Gibbs Function.	720
18.B	Appendix B: Equations of Radiative Transfer	724
18.C	Appendix C: Analytic Approximation of Tropopause Height.	725
IV	LARGE-SCALE OCEANIC CIRCULATION	
19	Wind-Driven Gyres	731
19.1	The Depth Integrated Wind-Driven Circulation	733
19.2	Using Viscosity Instead of Drag	740
19.3	Zonal Boundary Layers	744
19.4	The Nonlinear Problem	745
19.5	Inertial Solutions.	747
19.6	Topographic Effects on Western Boundary Currents	753
20	Structure of the Upper Ocean	761
20.1	Vertical Structure of the Wind-Driven Circulation.	761
20.2	A Model with Continuous Stratification	767
20.3	The Main Thermocline	774
20.4	Scaling and Simple Dynamics of the Main Thermocline	776
20.5	The Internal Thermocline.	780
20.6	The Ventilated Thermocline	786
20.A	Appendix: Miscellaneous Relationships in a Layered Model	796
21	The Meridional Overturning Circulation and the ACC	801
21.1	Sideways Convection	802
21.2	The Maintenance of Sideways Convection	808
21.3	Simple Box Models	813
21.4	A Laboratory Model of the Abyssal Circulation	818
21.5	A Model for Oceanic Abyssal Flow	821
21.6	A Model of Deep Wind-Driven Overturning.	829
21.7	The Antarctic Circumpolar Current	836
21.8	A Dynamic Model of the Meridional Overturning Circulation.	846
21.9	An Interhemispheric Circulation.	853
22	Equatorial Circulation and El Niño	861
22.1	Observational Preliminaries	861
22.2	Dynamical Preliminaries	862
22.3	A Local Model of the Equatorial Undercurrent.	865
22.4	An Ideal Fluid Model of the Equatorial Undercurrent.	876

22.5	An Introduction to El Niño and the Southern Oscillation.	886
22.6	The Walker Circulation	892
22.7	The Oceanic Response	893
22.8	Coupled Models and Unstable Interactions	895
22.9	Simple Conceptual and Numerical Models of ENSO	898
22.10	Numerical Solutions of the Shallow Water Equations	901
22.A	Appendix: Derivation of a Delayed-Oscillator Model	903

References	909
------------	-----

Index	936
-------	-----

In the main text, sections that are more advanced or that contain material that is peripheral to the main narrative are marked with a black diamond, ♦. Sections that contain material that is still not settled or are at the edge of knowledge are marked with a dagger, †.

Look again at that dot. That's here. That's home. That's us... Our planet is a lonely speck in the great enveloping cosmic dark. In our obscurity, in all this vastness, there is no hint that help will come from elsewhere to save us from ourselves... Like it or not, for the moment the Earth is where we make our stand.
Carl Sagan, *Pale Blue Dot*, 1994.

Preface

December 3, 2016.

The preface is yet to be written,

Placeholder

Part I

FUNDAMENTALS OF GEOPHYSICAL FLUID DYNAMICS

*To begin at the beginning:
 'It is spring, moonless night in the small town,
 starless and bible-black, the cobblestreets silent
 and the hunched, courters'-and-rabbits' wood limping invisible down
 to the sloeback, slow, black, crowblack, fishingboat-bobbing sea.*
 Dylan Thomas, *Under Milk Wood*, 1954.

CHAPTER 1

Equations of Motion

HAVING NOTHING BUT A BLANK SLATE, we begin by establishing the governing equations of motion for a fluid, with particular attention to the fluids of Earth's atmosphere and ocean. These equations determine how a fluid flows and evolves when forces are applied to it, or when it is heated or cooled, and so involve both dynamics and thermodynamics. And because the equations of motion are nonlinear the two become intertwined and at times inseparable.

1.1 TIME DERIVATIVES FOR FLUIDS

The equations of motion of fluid mechanics differ from those of rigid-body mechanics because fluids form a continuum, and because fluids flow and deform. Thus, even though the same relatively simple physical laws (Newton's laws and the laws of thermodynamics) govern both solid and fluid media, the expression of these laws differs between the two. To determine the equations of motion for fluids we must clearly establish what the time derivative of some property of a fluid actually means, and that is the subject of this section.

1.1.1 Field and material viewpoints

In solid-body mechanics one is normally concerned with the position and momentum of identifiable objects — the angular velocity of a spinning top or the motions of the planets around the Sun are two well-worn examples. The position and velocity of a particular object is then computed as a function of time by formulating equations of the form

$$\frac{dx_i}{dt} = F(\{x_i\}, t), \quad (1.1)$$

where $\{x_i\}$ is the set of positions and velocities of all the interacting objects and the operator F on the right-hand side is formulated using Newton's laws of motion. For example, two massive point objects interacting via their gravitational field obey

$$\frac{d\mathbf{r}_i}{dt} = \mathbf{v}_i, \quad \frac{d\mathbf{v}_i}{dt} = \frac{Gm_j}{(\mathbf{r}_i - \mathbf{r}_j)^2} \hat{\mathbf{r}}_{i,j}, \quad i = 1, 2; \quad j = 3 - i. \quad (1.2)$$

We thereby predict the positions, $\mathbf{r}_i$, and velocities, $\mathbf{v}_i$, of the objects given their masses, m_i , and the gravitational constant G , and where $\hat{\mathbf{r}}_{i,j}$ is a unit vector directed from $\mathbf{r}_i$ to $\mathbf{r}_j$.

In fluid dynamics such a procedure would lead to an analysis of fluid motions in terms of the positions and momenta of different fluid parcels, each identified by some label, which might simply be their position at an initial time. We call this a *material* point of view, because we are concerned with identifiable pieces of material; it is also sometimes called a *Lagrangian* view, after J.-L. Lagrange. The procedure is perfectly acceptable in principle, and if followed would provide a complete description of the fluid dynamical system. However, from a practical point of view it is much more than we need, and it would be extremely complicated to implement. Instead, for most problems we would like to know what the values of velocity, density and so on are at *fixed points* in space as time passes. (A weather forecast we might care about tells us how warm it will be where we live, and if we are given that we do not particularly care where a fluid parcel comes from, or where it subsequently goes.) Since the fluid is a continuum, this knowledge is equivalent to knowing how the fields of the dynamical variables evolve in space and time, and this is often known as the *field* or *Eulerian* viewpoint, after L. Euler.¹ Thus, whereas in the material view we consider the time evolution of identifiable fluid elements, in the field view we consider the time evolution of the fluid field from a particular frame of reference. That is, we seek evolution equations of the general form

$$\frac{\partial}{\partial t}\varphi(x, y, z, t) = \mathcal{G}(\varphi, x, y, z, t), \quad (1.3)$$

where the field $\varphi(x, y, z, t)$ represents all the dynamical variables (velocity, density, temperature, etc.) and $\mathcal{G}$ is some operator to be determined from Newton's laws of motion and appropriate thermodynamic laws.

Although the field viewpoint will often turn out to be the most practically useful, the material description is invaluable both in deriving the equations and in the subsequent insight it frequently provides. This is because the important quantities from a fundamental point of view are often those which are associated with a given fluid element: it is these which directly enter Newton's laws of motion and the thermodynamic equations. It is thus important to have a relationship between the rate of change of quantities associated with a given fluid element and the local rate of change of a field. The material or advective derivative provides this relationship.

1.1.2 The material derivative of a fluid property

A *fluid element* is an infinitesimal, indivisible, piece of fluid — effectively a very small fluid parcel of fixed mass. The *material derivative* is the rate of change of a property (such as temperature or momentum) of a particular fluid element or finite mass of fluid; that is, it is the total time derivative of a property of a piece of fluid. It is also known as the 'substantive derivative' (the derivative associated with a parcel of fluid substance), the 'advective derivative' (because the fluid property is being advected), the 'convective derivative' (convection is a slightly old-fashioned name for advection, still used in some fields), or the 'Lagrangian derivative' (after Lagrange).

Let us suppose that a fluid is characterized by a given velocity field $\mathbf{v}(\mathbf{x}, t)$, which determines its velocity throughout. Let us also suppose that the fluid has another property φ , and let us seek an expression for the rate of change of φ of a fluid element. Since φ is changing in time and in space we use the chain rule,

$$\delta\varphi = \frac{\partial\varphi}{\partial t}\delta t + \frac{\partial\varphi}{\partial x}\delta x + \frac{\partial\varphi}{\partial y}\delta y + \frac{\partial\varphi}{\partial z}\delta z = \frac{\partial\varphi}{\partial t}\delta t + \delta\mathbf{x} \cdot \nabla\varphi. \quad (1.4)$$

This is true in general for any δt , δx , etc. The total time derivative is then

$$\frac{d\varphi}{dt} = \frac{\partial\varphi}{\partial t} + \frac{d\mathbf{x}}{dt} \cdot \nabla\varphi. \quad (1.5)$$

If this equation is to represent a material derivative we must identify the time derivative in the second term on the right-hand side with the rate of change of position of a fluid element, namely

its velocity. Hence, the material derivative of the property φ is

$$\frac{d\varphi}{dt} = \frac{\partial\varphi}{\partial t} + \mathbf{v} \cdot \nabla\varphi. \quad (1.6)$$

The right-hand side expresses the material derivative in terms of the local rate of change of φ plus a contribution arising from the spatial variation of φ , experienced only as the fluid parcel moves. Because the material derivative is so common, and to distinguish it from other derivatives, we denote it by the operator D/Dt . Thus, the material derivative of the field φ is

$$\frac{D\varphi}{Dt} = \frac{\partial\varphi}{\partial t} + (\mathbf{v} \cdot \nabla)\varphi. \quad (1.7)$$

The brackets in the last term of this equation are helpful in reminding us that $(\mathbf{v} \cdot \nabla)$ is an operator acting on φ . The operator $\partial/\partial t + (\mathbf{v} \cdot \nabla)$ is the *Eulerian representation of the Lagrangian derivative as applied to a field*. We use the notation D/Dt rather generally for Lagrangian derivatives, but the operator may take a different form when applied to other objects, such as a fluid volume.

Material derivative of vector field

The material derivative may act on a vector field $\mathbf{b}$, in which case

$$\frac{D\mathbf{b}}{Dt} = \frac{\partial\mathbf{b}}{\partial t} + (\mathbf{v} \cdot \nabla)\mathbf{b}. \quad (1.8)$$

In Cartesian coordinates this is

$$\frac{D\mathbf{b}}{Dt} = \frac{\partial\mathbf{b}}{\partial t} + u\frac{\partial\mathbf{b}}{\partial x} + v\frac{\partial\mathbf{b}}{\partial y} + w\frac{\partial\mathbf{b}}{\partial z}, \quad (1.9)$$

and for a particular component of $\mathbf{b}$, b^x say,

$$\frac{Db^x}{Dt} = \frac{\partial b^x}{\partial t} + u\frac{\partial b^x}{\partial x} + v\frac{\partial b^x}{\partial y} + w\frac{\partial b^x}{\partial z}, \quad (1.10)$$

and similarly for b^y and b^z . In Cartesian tensor notation the expression becomes

$$\frac{Db_i}{Dt} = \frac{\partial b_i}{\partial t} + v_j \frac{\partial b_i}{\partial x_j} = \frac{\partial b_i}{\partial t} + v_j \partial_j b_i, \quad (1.11)$$

where the subscripts denote the Cartesian components, repeated indices are summed, and $\partial_j b_i \equiv \partial b_i / \partial x_j$. In coordinate systems other than Cartesian the advective derivative of a vector is not simply the sum of the advective derivative of its components, because the coordinate vectors themselves change direction with position; this will be important when we deal with spherical coordinates. Finally, we remark that the advective derivative of the position of a fluid element, $\mathbf{r}$ say, is its velocity, and this may easily be checked by explicitly evaluating $D\mathbf{r}/Dt$.

1.1.3 Material derivative of a volume

The volume that a given, unchanging, mass of fluid occupies is deformed and advected by the fluid motion, and there is no reason why it should remain constant. Rather, the volume will change as a result of the movement of each element of its bounding material surface, and in particular will change if there is a non-zero normal component of the velocity at the fluid surface. That is, if the volume of some fluid is $\int dV$, then

$$\frac{D}{Dt} \int_V dV = \int_S \mathbf{v} \cdot d\mathbf{S}, \quad (1.12)$$

where the subscript V indicates that the integral is a definite integral over some finite volume V , although the limits of the integral will be functions of time if the volume is changing. The integral on the right-hand side is over the closed surface, S , bounding the volume. Although intuitively apparent (to some), this expression may be derived more formally using Leibniz's formula for the rate of change of an integral whose limits are changing. Using the divergence theorem on the right-hand side, (1.12) becomes

$$\frac{D}{Dt} \int_V dV = \int_V \nabla \cdot \mathbf{v} dV. \quad (1.13)$$

The rate of change of the volume of an infinitesimal fluid element of volume ΔV is obtained by taking the limit of this expression as the volume tends to zero, giving

$$\lim_{\Delta V \rightarrow 0} \frac{1}{\Delta V} \frac{D\Delta V}{Dt} = \nabla \cdot \mathbf{v}. \quad (1.14)$$

We will often write such expressions informally as

$$\frac{D\Delta V}{Dt} = \Delta V \nabla \cdot \mathbf{v}, \quad (1.15)$$

with the limit implied.

Consider now the material derivative of some fluid property, ξ say, multiplied by the volume of a fluid element, ΔV . Such a derivative arises when ξ is the amount per unit volume of ξ -substance — the mass density or the amount of a dye per unit volume, for example. Then we have

$$\frac{D}{Dt}(\xi \Delta V) = \xi \frac{D\Delta V}{Dt} + \Delta V \frac{D\xi}{Dt}. \quad (1.16)$$

Using (1.15) this becomes

$$\frac{D}{Dt}(\xi \Delta V) = \Delta V \left(\xi \nabla \cdot \mathbf{v} + \frac{D\xi}{Dt} \right), \quad (1.17)$$

and the analogous result for a finite fluid volume is just

$$\frac{D}{Dt} \int_V \xi dV = \int_V \left(\xi \nabla \cdot \mathbf{v} + \frac{D\xi}{Dt} \right) dV. \quad (1.18)$$

This expression is to be contrasted with the Eulerian derivative for which the volume, and so the limits of integration, are fixed and we have

$$\frac{d}{dt} \int_V \xi dV = \int_V \frac{\partial \xi}{\partial t} dV. \quad (1.19)$$

Now consider the material derivative of a fluid property φ multiplied by the mass of a fluid element, $\rho \Delta V$, where ρ is the fluid density. Such a derivative arises when φ is the amount of φ -substance per unit mass (note, for example, that the momentum of a fluid element is $\rho \mathbf{v} \Delta V$). The material derivative of $\varphi \rho \Delta V$ is given by

$$\frac{D}{Dt}(\varphi \rho \Delta V) = \rho \Delta V \frac{D\varphi}{Dt} + \varphi \frac{D}{Dt}(\rho \Delta V). \quad (1.20)$$

But $\rho \Delta V$ is just the mass of the fluid element, and that is constant — that is how a fluid element is defined. Thus the second term on the right-hand side vanishes and

$$\frac{D}{Dt}(\varphi \rho \Delta V) = \rho \Delta V \frac{D\varphi}{Dt} \quad \text{and} \quad \frac{D}{Dt} \int_V \varphi \rho dV = \int_V \rho \frac{D\varphi}{Dt} dV, \quad (1.21a,b)$$

Material and Eulerian Derivatives

The material derivative of a scalar (φ) and a vector ($\mathbf{b}$) field are given by:

$$\frac{D\varphi}{Dt} = \frac{\partial\varphi}{\partial t} + \mathbf{v} \cdot \nabla\varphi, \quad \frac{D\mathbf{b}}{Dt} = \frac{\partial\mathbf{b}}{\partial t} + (\mathbf{v} \cdot \nabla)\mathbf{b}. \quad (\text{D.1})$$

Various material derivatives of integrals are:

$$\frac{D}{Dt} \int_V \varphi dV = \int_V \left(\frac{D\varphi}{Dt} + \varphi \nabla \cdot \mathbf{v} \right) dV = \int_V \left(\frac{\partial\varphi}{\partial t} + \nabla \cdot (\varphi \mathbf{v}) \right) dV, \quad (\text{D.2})$$

$$\frac{D}{Dt} \int_V dV = \int_V \nabla \cdot \mathbf{v} dV, \quad (\text{D.3})$$

$$\frac{D}{Dt} \int_V \rho \varphi dV = \int_V \rho \frac{D\varphi}{Dt} dV. \quad (\text{D.4})$$

These formulae also hold if φ is a vector. The Eulerian derivative of an integral is:

$$\frac{d}{dt} \int_V \varphi dV = \int_V \frac{\partial\varphi}{\partial t} dV, \quad (\text{D.5})$$

so that

$$\frac{d}{dt} \int_V dV = 0 \quad \text{and} \quad \frac{d}{dt} \int_V \rho \varphi dV = \int_V \frac{\partial \rho \varphi}{\partial t} dV. \quad (\text{D.6})$$

where (1.21b) applies to a finite volume. That expression may also be derived more formally using Leibniz's formula for the material derivative of an integral, and the result also holds when φ is a vector. The result is quite different from the corresponding Eulerian derivative, in which the volume is kept fixed; in that case we have:

$$\frac{d}{dt} \int_V \varphi \rho dV = \int_V \frac{\partial}{\partial t} (\varphi \rho) dV. \quad (1.22)$$

Various material and Eulerian derivatives are summarized in the shaded box above.

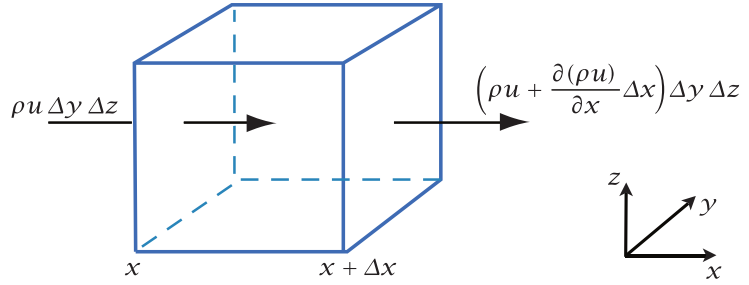
1.2 THE MASS CONTINUITY EQUATION

In classical mechanics mass is absolutely conserved and in solid-body mechanics we normally do not need an explicit equation of mass conservation. However, in fluid mechanics fluid flows into and away from regions, and fluid density may change, and an equation that explicitly accounts for the flow of mass is one of the equations of motion of the fluid.

1.2.1 An Eulerian derivation

We will first derive the mass conservation equation from an Eulerian point of view; that is to say, our reference frame is fixed in space and the fluid flows through it.

Fig. 1.1 Mass conservation in an Eulerian cuboid control volume. The mass convergence, $-\partial(\rho u)/\partial x$ (plus contributions from the y and z directions), must be balanced by a density increase, $\partial\rho/\partial t$.



Cartesian derivation

Consider an infinitesimal, rectangular cuboid, control volume, $\Delta V = \Delta x \Delta y \Delta z$ that is fixed in space, as in Fig. 1.1. Fluid moves into or out of the volume through its surface, including through its faces in the y - z plane of area $\Delta A = \Delta y \Delta z$ at coordinates x and $x + \Delta x$. The accumulation of fluid within the control volume due to motion in the x -direction is evidently

$$\Delta y \Delta z [(\rho u)(x, y, z) - (\rho u)(x + \Delta x, y, z)] = - \left. \frac{\partial(\rho u)}{\partial x} \right|_{x,y,z} \Delta x \Delta y \Delta z. \quad (1.23)$$

To this must be added the effects of motion in the y - and z -directions, namely

$$- \left[\frac{\partial(\rho v)}{\partial y} + \frac{\partial(\rho w)}{\partial z} \right] \Delta x \Delta y \Delta z. \quad (1.24)$$

This net accumulation of fluid must be accompanied by a corresponding increase of fluid mass within the control volume. This is

$$\frac{\partial}{\partial t} (\text{density} \times \text{volume}) = \Delta x \Delta y \Delta z \frac{\partial \rho}{\partial t}, \quad (1.25)$$

because the volume is constant. Thus, because mass is conserved, (1.23), (1.24) and (1.25) give

$$\Delta x \Delta y \Delta z \left[\frac{\partial \rho}{\partial t} + \frac{\partial(\rho u)}{\partial x} + \frac{\partial(\rho v)}{\partial y} + \frac{\partial(\rho w)}{\partial z} \right] = 0. \quad (1.26)$$

The quantity in square brackets must be zero and we therefore have

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0. \quad (1.27)$$

This is called the *mass continuity equation* for it recognizes the continuous nature of the mass field in a fluid. There is no diffusion term in (1.27), no term like $\kappa \nabla^2 \rho$. This is because mass is transported by the macroscopic movement of molecules; even if this motion appears diffusion-like any net macroscopic molecular motion constitutes, by definition, a velocity field.

Vector derivation

Consider an arbitrary control volume V bounded by a surface S , fixed in space, with by convention the direction of $\mathbf{S}$ being toward the outside of V , as in Fig. 1.2. The rate of fluid loss due to flow through the closed surface S is then given by

$$\text{fluid loss} = \int_S \rho \mathbf{v} \cdot d\mathbf{S} = \int_V \nabla \cdot (\rho \mathbf{v}) dV, \quad (1.28)$$

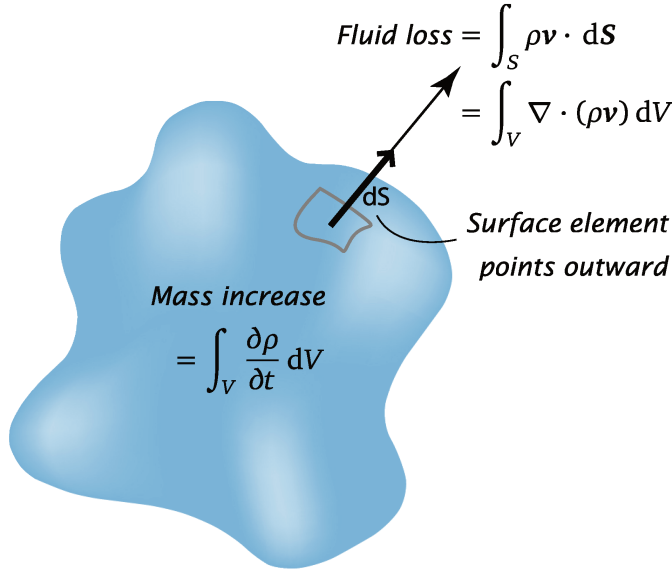


Fig. 1.2 Mass conservation in an arbitrary Eulerian control volume V bounded by a surface S . The mass increase, $\int_V (\partial \rho / \partial t) dV$ is equal to the mass flowing into the volume, $-\int_S (\rho \mathbf{v}) \cdot d\mathbf{S} = -\int_V \nabla \cdot (\rho \mathbf{v}) dV$.

using the divergence theorem.

This must be balanced by a change in the mass M of the fluid within the control volume, which, since its volume is fixed, implies a density change. That is

$$\text{fluid loss} = -\frac{dM}{dt} = -\frac{d}{dt} \int_V \rho dV = -\int_V \frac{\partial \rho}{\partial t} dV. \quad (1.29)$$

Equating (1.28) and (1.29) yields

$$\int_V \left[\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) \right] dV = 0. \quad (1.30)$$

Because the volume is arbitrary, the integrand must vanish and we recover (1.27).

1.2.2 Mass continuity via the material derivative

We now derive the mass continuity equation (1.27) from a material perspective. This is the most fundamental approach of all since the principle of mass conservation states simply that the mass of a given element of fluid is, by definition of the element, constant. Thus, consider a small mass of fluid of density ρ and volume ΔV . Then conservation of mass may be represented by

$$\frac{D}{Dt}(\rho \Delta V) = 0. \quad (1.31)$$

Both the density and the volume of the parcel may change, so

$$\Delta V \frac{D\rho}{Dt} + \rho \frac{D\Delta V}{Dt} = \Delta V \left(\frac{D\rho}{Dt} + \rho \nabla \cdot \mathbf{v} \right) = 0, \quad (1.32)$$

where the second expression follows using (1.15). Since the volume element is arbitrary, the term in brackets must vanish and

$$\frac{D\rho}{Dt} + \rho \nabla \cdot \mathbf{v} = 0. \quad (1.33)$$

After expansion of the first term this becomes identical to (1.27). This result may be derived more formally by rewriting (1.31) as the integral expression

$$\frac{D}{Dt} \int_V \rho dV = 0. \quad (1.34)$$

Expanding the derivative using (1.18) gives

$$\frac{D}{Dt} \int_V \rho dV = \int_V \left(\frac{D\rho}{Dt} + \rho \nabla \cdot \mathbf{v} \right) dV = 0. \quad (1.35)$$

Because the volume over which the integral is taken is arbitrary the integrand itself must vanish and we recover (1.33). Summarizing, equivalent partial differential equation representing conservation of mass are:

$$\frac{D\rho}{Dt} + \rho \nabla \cdot \mathbf{v} = 0, \quad \frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0. \quad (1.36a,b)$$

1.2.3 A general continuity equation

The derivation of continuity equation for a general scalar property of a fluid is similar to that for density, except that there may be an external source or sink, and potentially a means of transferring the property from one location to another than by fluid motion, for example by diffusion. If ξ is the amount of some property of the fluid per unit volume (the volume concentration, sometimes simply called the concentration), and if the net effect per unit volume of all non-conservative processes is denoted by $Q_{[v,\xi]}$, then the continuity equation for concentration may be written:

$$\frac{D}{Dt} (\xi \Delta V) = Q_{[v,\xi]} \Delta V. \quad (1.37)$$

Expanding the left-hand side and using (1.15) we obtain

$$\frac{D\xi}{Dt} + \xi \nabla \cdot \mathbf{v} = Q_{[v,\xi]}, \quad \text{or} \quad \frac{\partial \xi}{\partial t} + \nabla \cdot (\xi \mathbf{v}) = Q_{[v,\xi]}. \quad (1.38)$$

If we are interested in a tracer that is normally measured per unit mass of fluid (which is typical when considering thermodynamic quantities) then the conservation equation would be written

$$\frac{D}{Dt} (\varphi \rho \Delta V) = Q_{[m,\varphi]} \rho \Delta V, \quad (1.39)$$

where φ is the tracer mixing ratio or mass concentration — that is, the amount of tracer per unit fluid mass — and $Q_{[m,\varphi]}$ represents non-conservative sources of φ per unit mass. Then, since $\rho \Delta V$ is constant we obtain

$$\frac{D\varphi}{Dt} = Q_{[m,\varphi]} \quad \text{or} \quad \frac{\partial(\rho\varphi)}{\partial t} + \nabla \cdot (\rho\varphi \mathbf{v}) = \rho Q_{[m,\varphi]}, \quad (1.40)$$

using the mass continuity equation, (1.36), to obtain the equation on the right. The source term $Q_{[m,\varphi]}$ is evidently equal to the rate of change of φ of a fluid element. When this is so, we often write it simply as $\dot{\varphi}$, so that

$$\frac{D\varphi}{Dt} = \dot{\varphi}. \quad (1.41)$$

A tracer obeying (1.41) with $\dot{\varphi} = 0$ is said to be *materially conserved*. If a tracer is materially conserved except for the effects of sources or sinks, or diffusion terms, then it is sometimes said to be an ‘adiabatically conserved’ variable (although adiabatic properly means at constant entropy or with no heat sources). If those sources and sinks are in the form of the divergence of a flux with φ satisfying $\rho D\varphi/Dt = \nabla \cdot \mathbf{F}_\varphi$, then φ is said to be a ‘conservative’ variable because, with no flux boundary conditions, $\int \rho \varphi dV = \text{constant}$. Although momentum as a whole is conserved, momentum is not a materially conserved variable, as we are about to see.

1.3 THE MOMENTUM EQUATION

The momentum equation is a partial differential equation that describes how the velocity or momentum of a fluid responds to internal and imposed forces. We will derive it using material methods, with a very heuristic treatment of the terms representing pressure and viscous forces.

1.3.1 Advection

Let $\mathbf{m}(x, y, z, t)$ be the momentum-density field (momentum per unit volume) of the fluid. Thus, $\mathbf{m} = \rho \mathbf{v}$ and the total momentum of a volume of fluid is given by the volume integral $\int_V \mathbf{m} dV$. Now, for a fluid the rate of change of a momentum of an identifiable fluid mass is given by the material derivative, and by Newton's second law this is equal to the force acting on it. Thus,

$$\frac{D}{Dt} \int_V \rho \mathbf{v} dV = \int_V \mathbf{F} dV, \quad (1.42)$$

where $\mathbf{F}$ is the force per unit volume. Now, using (1.21b) (with φ replaced by $\mathbf{v}$) to transform the left-hand side of (1.42), we obtain

$$\int_V \left(\rho \frac{D\mathbf{v}}{Dt} - \mathbf{F} \right) dV = 0. \quad (1.43)$$

Because the volume is arbitrary the integrand itself must vanish and we obtain

$$\rho \frac{D\mathbf{v}}{Dt} = \mathbf{F} \quad \text{or} \quad \frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v} \cdot \nabla) \mathbf{v} = \frac{\mathbf{F}}{\rho}, \quad (1.44a,b)$$

having used (1.8) to expand the material derivative.

We have thus obtained an expression for how a fluid accelerates if subject to known forces. These forces are however not all external to the fluid itself: a stress arises from the direct contact between one fluid parcel and another, giving rise to pressure and viscous forces, sometimes referred to as *contact* forces. Because a complete treatment of these forces would be very lengthy, and is available elsewhere, we treat them informally and intuitively.

1.3.2 Pressure and viscous forces

Pressure

Within or at the boundary of a fluid the pressure is the normal force per unit area due to the collective action of molecular motion. Thus

$$d\hat{\mathbf{F}}_p = -p d\mathbf{S}, \quad (1.45)$$

where p is the pressure, $\hat{\mathbf{F}}_p$ is the pressure force and $d\mathbf{S}$ an infinitesimal surface element. If we grant ourselves this intuitive notion, it is a simple matter to assess the influence of pressure on a fluid, for the pressure force on a volume of fluid is the integral of the pressure over the its boundary and so

$$\hat{\mathbf{F}}_p = - \int_S p d\mathbf{S}. \quad (1.46)$$

The minus sign arises because the pressure force is directed inwards, whereas $\mathbf{S}$ is a vector normal to the surface and directed outwards. Applying a form of the divergence theorem to the right-hand side gives

$$\hat{\mathbf{F}}_p = - \int_V \nabla p dV, \quad (1.47)$$

where the volume V is bounded by the surface S . The pressure force per unit volume, $\mathbf{F}_p$, is therefore just $-\nabla p$, and the force per unit mass is $\nabla p/\rho$. The force is evidently non-zero only if the pressure varies in space and for this reason it is more properly known as the *pressure-gradient* force.

Table 1.1 Experimental values of viscosity for air, water and mercury at room temperature and pressure.

	μ (kg m ⁻¹ s ⁻¹)	ν (m ² s ⁻¹)
Air	1.8×10^{-5}	1.5×10^{-5}
Water	1.1×10^{-3}	1.1×10^{-6}
Mercury	1.6×10^{-3}	1.2×10^{-7}

Viscosity

Viscosity, like pressure, is a force due to the internal motion of molecules. The effects of viscosity are apparent in many situations — the flow of treacle or volcanic lava are obvious examples. In other situations, for example large-scale flow in the atmosphere, viscous effects are negligible. However, for a constant density fluid viscosity is the *only* way that energy may be removed from the fluid, so that if energy is being added in some way viscosity must ultimately become important if the fluid is to reach an equilibrium in which energy input equals energy dissipation. When tea is stirred in a cup, it is viscous effects that cause the fluid to eventually stop spinning after we have removed our spoon.

A number of textbooks show that, for most Newtonian fluids, the viscous force per unit volume is approximately equal to $\mu \nabla^2 \mathbf{v}$, where μ is the viscosity. Obtaining this expression involves making an incompressibility assumption and is not exact, but it is in fact a good approximation for most liquids and gases. With this term and the pressure-gradient force the momentum equation becomes,

$$\frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v} \cdot \nabla) \mathbf{v} = -\frac{1}{\rho} \nabla p + \nu \nabla^2 \mathbf{v} + \mathbf{F}_b, \quad (1.48)$$

where $\nu \equiv \mu/\rho$ is the *kinematic viscosity* and $\mathbf{F}_b$ represents external body forces (per unit mass) such as gravity, $\mathbf{g}$. Equation (1.48) is sometimes called the Navier–Stokes equation. For gases, dimensional arguments suggest that the magnitude of ν should be given by

$$\nu \sim \text{mean free path} \times \text{mean molecular velocity}, \quad (1.49)$$

which for a typical molecular velocity of 300 m s⁻¹ and a mean free path of 7×10^{-8} m gives the not unreasonable estimate of 2.1×10^{-5} m² s⁻¹, within a factor of two of the experimental value (Table 1.1). Interestingly, the kinematic viscosity is smaller for water and mercury than it is for air.

1.3.3 Hydrostatic balance

The vertical component — the component parallel to the gravitational force, $\mathbf{g}$ — of the momentum equation is

$$\frac{Dw}{Dt} = -\frac{1}{\rho} \frac{\partial p}{\partial z} - g, \quad (1.50)$$

where w is the vertical component of the velocity and $\mathbf{g} = -g\mathbf{k}$. If the fluid is static the gravitational term is balanced by the pressure term and we have

$$\frac{\partial p}{\partial z} = -\rho g \quad (1.51)$$

and this relation is known as *hydrostatic balance*, or hydrostasy. It is clear in this case that the pressure at a point is given by the weight of the fluid above it, provided $p = 0$ at the top of the fluid. It might also appear that (1.51) would be a good *approximation* to (1.50) provided vertical accelerations, Dw/Dt , are small compared to gravity, which is nearly always the case in the atmosphere and ocean. While this statement is true if we need only a reasonable approximate value of

the pressure at a point or in a column, the satisfaction of this condition is *not* sufficient to ensure that (1.51) provides an accurate enough pressure to determine the horizontal pressure gradients responsible for producing motion. We return to this point in section 2.7.

1.4 THE EQUATION OF STATE

In three dimensions the momentum and continuity equations provide four equations, but contain five unknowns — three components of velocity, density and pressure. Obviously other equations are needed, and an *equation of state* is an expression that diagnostically relates the various thermodynamic variables to each other. The *conventional* equation of state, or the *thermal* equation of state, is an expression that relates temperature, pressure, composition (the mass fraction of the various constituents) and density, and we may write it, rather generally, as

$$p = p(\rho, T, \varphi_n), \quad (1.52)$$

where φ_n is mass fraction of the n th constituent. An equation of this form is not the most fundamental equation of state from a thermodynamic perspective, an issue we visit later, but it connects readily measurable quantities.

For an ideal gas (and the air in the Earth's atmosphere is very close to ideal) the thermal equation of state is

$$p = \rho RT, \quad (1.53)$$

where R is the gas constant for the gas in question and T is temperature. R is a specific constant, and is related to the universal gas constant R^* by $R = R^*/\bar{\mu}$, where $\bar{\mu}$ is the mean molar mass (molecular weight in kg/mol) of the constituents of the gas. Equivalently, $R = n_m k_B$, where k_B is Boltzmann's constant and n_m is the number of molecules per unit mass, so that R is proportional to the number of molecules contained in a unit mass. Since $R^* = 8.314 \text{ J mol}^{-1} \text{ K}^{-1}$ and, for dry air, $\bar{\mu} = 29.0 \times 10^{-3} \text{ kg mol}^{-1}$ we obtain $R = 287 \text{ J kg}^{-1} \text{ K}^{-1}$. Air has virtually constant composition except for variations in water vapour; these variations makes the gas constant, R , in the equation of state for air a weak function of the water vapour content but for now we regard R as a constant.

For a liquid such as seawater no simple expression akin to (1.53) is easily derivable, and semi-empirical equations are usually resorted to. For water in a laboratory setting a reasonable approximation of the equation of state is $\rho = \rho_0[1 - \beta_T(T - T_0)]$, where β_T is a thermal expansion coefficient and ρ_0 and T_0 are constants. In the ocean the density is also significantly affected by pressure and dissolved salts: seawater is a solution of many ions in water — chloride ($\approx 1.9\%$ by weight) sodium (1%), sulfate (0.26%), magnesium (0.13%) and so on, with a total average concentration of about 35‰ (ppt, or parts per thousand). The ratios of the fractions of these salts are almost constant throughout the ocean, and their total concentration may be parameterized by a single measure, the *salinity*, S .² Given this, the density of seawater is a function of three variables — pressure, temperature, and salinity — and we may write the conventional equation of state as

$$\rho = \rho(T, S, p) \quad \text{or} \quad \alpha = \alpha(T, S, p), \quad (1.54)$$

where $\alpha = 1/\rho$ is the specific volume, or inverse density. For small variations around a reference value we have

$$d\alpha = \left(\frac{\partial \alpha}{\partial T}\right)_{S,p} dT + \left(\frac{\partial \alpha}{\partial S}\right)_{T,p} dS + \left(\frac{\partial \alpha}{\partial p}\right)_{T,S} dp = \alpha(\beta_T dT - \beta_S dS - \beta_p dp), \quad (1.55)$$

where the rightmost expression serves to define the thermal expansion coefficient β_T , the saline contraction coefficient β_S , and the compressibility coefficient (or inverse bulk modulus) β_p . In general these quantities are not constants, but for small variations around a reference state they may be treated as such and we have

$$\alpha = \alpha_0 [1 + \beta_T(T - T_0) - \beta_S(S - S_0) - \beta_p(p - p_0)]. \quad (1.56)$$

Typical values of these parameters, with variations typically encountered through the ocean, are: $\beta_T \approx 2(\pm 1.5) \times 10^{-4} \text{ K}^{-1}$, $\beta_S \approx 7.6(\pm 0.2) \times 10^{-4} \text{ ppt}^{-1}$, $\beta_p \approx 4.4(\pm 0.5) \times 10^{-10} \text{ Pa}^{-1}$. The value of β_p is also related to the speed of sound, c_s , by $\beta_p = \alpha_0/c_s^2$. Since the variations around the mean density are small (1.56) implies that

$$\rho = \rho_0 \left[1 - \beta_T(T - T_0) + \beta_S(S - S_0) + \beta_p(p - p_0) \right]. \quad (1.57)$$

The pressure terms lead to larger density changes than either the salinity or temperature terms but they are not normally as important for the dynamics, because they are dominated by the hydrostatic pressure. They thus give rise to density gradients in vertical, whereas it is the lateral variations that are usually more important. (In fact, the pressure terms may be combined with the density to form the potential density, which varies only with T and S , as we come to later.)

A linear equation of state for seawater is emphatically *not* accurate enough for quantitative oceanography; as mentioned the β parameters in (1.56) themselves vary with pressure, temperature and (more weakly) salinity so introducing nonlinearities to the equation. The most important of these are captured by an equation of state of the form

$$\alpha = \alpha_0 \left[1 + \beta_T(1 + \gamma^* p)(T - T_0) + \frac{\beta_T^*}{2}(T - T_0)^2 - \beta_S(S - S_0) - \beta_p(p - p_0) \right]. \quad (1.58)$$

The starred constants β_T^* and γ^* capture the leading nonlinearities: γ^* is the *thermobaric parameter*, which determines the extent to which the thermal expansion depends on pressure, and β_T^* is the second thermal expansion coefficient. Even this equation of state has some quantitative deficiencies and more complicated semi-empirical formulae are often used if high accuracy is needed.³ An illustration of the variation of density of seawater with temperature, salinity and pressure is illustrated in Fig. 1.3, with more discussion in section 1.7.2.

Clearly, the equation of state introduces, in general, a sixth unknown, temperature, and we will have to introduce another physical principle — one coming from thermodynamics — to obtain a complete set of equations. However, if the equation of state were such that it linked only density and pressure, without introducing another variable, then the equations would be complete; the simplest case of all is a constant density fluid for which the equation of state is just $\rho = \text{constant}$. A fluid for which the density is a function of pressure alone is called a *barotropic fluid*; otherwise, it is a *baroclinic fluid*. (In this context, ‘barotropic’ is a shortening of the original phrase ‘auto-barotropic’.) Barotropic equations of state of the form $p = C\rho^\gamma$, where γ is a constant, are called ‘polytropic’.

1.5 THERMODYNAMIC RELATIONS

In this section we review a few aspects of thermodynamics. We provide neither a complete nor an a priori development of the subject; rather, we focus on aspects that are particularly relevant to fluid dynamics, and that are needed to derive a ‘thermodynamic equation’ for fluids. Readers whose interest is solely in an ideal gas or a simple Boussinesq fluid may skim this section and then refer back to it as needed.

1.5.1 A few fundamentals

A fundamental postulate of thermodynamics is that the internal energy of a system in equilibrium is a function of its extensive properties volume, entropy, and the mass of its various constituents. Extensive means that the property value is proportional to the amount of material present, in contrast to an intensive property such as temperature. For our purposes it is more convenient to divide all of these quantities by the mass of fluid present, so expressing the internal energy per unit mass (or the specific internal energy) I , as a function of the specific volume $\alpha = \rho^{-1}$, the specific entropy

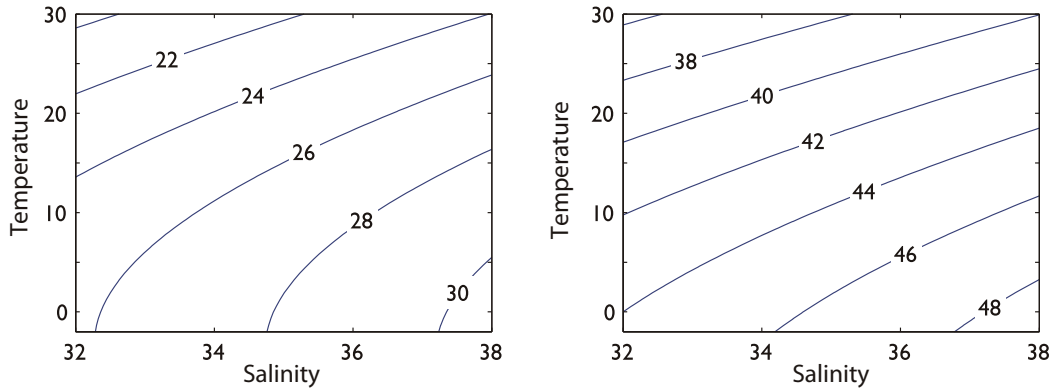


Fig. 1.3 Contours of density as a function of temperature and salinity for seawater, calculated using an accurate equation of state. Contour labels are $(\text{density} - 1000) \text{ kg m}^{-3}$. Left panel: at sea-level ($p = 10^5 \text{ Pa}$, or 1000 mb). Right panel: at $p = 4 \times 10^7 \text{ Pa}$ (about 4 km depth). In both cases the contours are slightly convex, so that if two parcels at the same density but different temperatures and salinities are mixed, the resulting parcel is of higher density.

η , and the mass fractions of its various components. (However, dividing by the mass does not turn an extensive variable into an intensive one!) Our interest is in two-component fluids (dry air and water vapour, or water and salinity) so that we may parameterize the composition by a single parameter, S . (We follow conventional thermodynamical notation as much as possible, except that we use I instead of u for internal energy, since u is a fluid velocity, and η instead of S for entropy, since S is salinity.) We can also write entropy in terms of internal energy, density and salinity, and thus we have

$$I = I(\alpha, \eta, S) \quad \text{or} \quad \eta = \eta(I, \alpha, S). \quad (1.59\text{a,b})$$

Given the functional forms on the right-hand sides, either of these expressions constitutes a complete description of the macroscopic state of a system in equilibrium, and we call either of them the *fundamental equation of state*. The thermal equation of state can be derived from (1.59), but not vice versa. The first differential of (1.59a) gives, formally,

$$dI = \left(\frac{\partial I}{\partial \eta} \right)_{\alpha, S} d\eta + \left(\frac{\partial I}{\partial \alpha} \right)_{\eta, S} d\alpha + \left(\frac{\partial I}{\partial S} \right)_{\alpha, \eta} dS. \quad (1.60)$$

We will now ascribe physical meaning to these differentials.

Conservation of energy states that the internal energy of a body may change because of work done by or on it, or because of a heat input, or because of a change in its chemical composition. We write this as

$$dI = dQ + dW + dC, \quad (1.61)$$

where dW is the work done on the body, dQ is the heat input to the body, and dC is the change in internal energy caused by a change in its chemical composition (e.g., its salinity, or water vapour content), sometimes called the ‘chemical work’. The quantities on the right-hand side (denoted with a d) are imperfect differentials or infinitesimals: Q , W and C are not functions of the state of a body, and the internal energy cannot be regarded as the sum of a ‘heat’ and a ‘work’. We should think of heat and work as having meaning only as fluxes of energy, or rates of energy input, and not as amounts of energy; their sum changes the internal energy of a body, which is a function of its state. Equation (1.61) is sometimes called ‘the first law of thermodynamics’ (discussed more in the

appendix on page 49). We first consider the causes of variations of the quantities on the right-hand side, and then make a connection to (1.60).

Heat input: In an infinitesimal quasi-static or reversible process, if an amount of heat dQ (per unit mass) is externally supplied then the specific entropy η will change according to

$$T d\eta = dQ. \quad (1.62)$$

The entropy is a function of the state of a body and is, by definition, an adiabatic invariant. As we are dealing with the amount of a quantity per unit mass, η is the specific entropy, although we will often refer to it just as the entropy. We may regard (1.61) as *defining* the heat input, dQ , by way of a statement of conservation of energy, and (1.62) then says that there is a function of state, the entropy, that changes by an amount equal to the heat input divided by the temperature.

Work done: The work done on a body during a reversible process is equal to the pressure times its change in volume, and the work is positive if the volume change is negative. Thus if an infinitesimal amount of work dW (per unit mass) is applied to a body then its thermodynamic state will change according to

$$-p d\alpha = dW, \quad (1.63)$$

where $\alpha = 1/\rho$ is the specific volume of the fluid and p is the pressure.

Composition: The change in internal energy due to a small change in composition, dS , is given by

$$dC = \mu dS, \quad (1.64)$$

where μ is the *chemical potential* of the solution. In the ocean, salinity is the compositional variable and changes arise through precipitation and evaporation at the surface and molecular diffusion. When salinity changes the internal energy of a fluid parcel changes by (1.64), but this change is usually small compared to other changes in internal energy. In practice, the most important effect of salinity is that it changes the density of seawater. In the atmosphere the composition of a parcel of air varies mainly according to the amount of water vapour present. Since water vapour and dry air have different chemical potentials these variations cause changes in internal energy, but in the absence of phase-changes the changes are small. An important compositional effect does arise when condensation or evaporation occurs, for then energy is released (or required), as discussed in chapter 18.

Collecting equations (1.61)–(1.64) together we have

$$dI = T d\eta - p d\alpha + \mu dS. \quad (1.65)$$

We refer to this as *the fundamental thermodynamic relation*. The fundamental equation of state, (1.59), describes the properties of a particular fluid, and the fundamental relation, (1.65), is associated with conservation of energy. Much of classical thermodynamics follows from these two expressions.

1.5.2 Thermodynamic potentials and Maxwell relations

Given the fundamental thermodynamic relation, various other ‘thermodynamic potentials’ and relations between variables can be derived that prove extremely useful. The thermodynamic potentials are, like internal energy and entropy, functions of the state but they have different natural variables by which they are expressed.

If we begin with the internal energy itself, then from (1.60) and (1.65) it follows that

$$T = \left(\frac{\partial I}{\partial \eta} \right)_{\alpha, S}, \quad p = - \left(\frac{\partial I}{\partial \alpha} \right)_{\eta, S}, \quad \mu = \left(\frac{\partial I}{\partial S} \right)_{\eta, \alpha}. \quad (1.66a,b,c)$$

These may be regarded as the defining relations for these variables; because of the connection between (1.61) and (1.65) these are not just formal definitions, and the pressure and temperature so defined are indeed related to our intuitive concepts of these variables and to the motion of the fluid molecules. If we write

$$d\eta = \frac{1}{T} dI + \frac{p}{T} d\alpha - \frac{\mu}{T} dS, \quad (1.67)$$

it is also clear that

$$p = T \left(\frac{\partial \eta}{\partial \alpha} \right)_{I, S}, \quad T^{-1} = \left(\frac{\partial \eta}{\partial I} \right)_{\alpha, S}, \quad \mu = -T \left(\frac{\partial \eta}{\partial S} \right)_{I, \alpha}. \quad (1.68a,b,c)$$

We also see that I and α (and S) are the natural variables for entropy.

Because the right-hand side of (1.65) is equal to an exact differential, the second derivatives are independent of the order of differentiation. That is,

$$\frac{\partial^2 I}{\partial \eta \partial \alpha} = \frac{\partial^2 I}{\partial \alpha \partial \eta}, \quad (1.69)$$

and therefore, using (1.66)

$$\left(\frac{\partial T}{\partial \alpha} \right)_{\eta} = - \left(\frac{\partial p}{\partial \eta} \right)_{\alpha}. \quad (1.70)$$

This is one of the *Maxwell relations*, which are a collection of four similar relations that follow directly from the fundamental thermodynamic relation (1.65) and simple relations between second derivatives. (Additional Maxwell-like relations exist if we consider chemical effects.) To derive the other Maxwell relations we will introduce thermodynamic potentials enthalpy, h , the Gibbs function, g , and the free energy, f . These are all closely related to the internal energy and they are all extensive functions (and then denoted with an uppercase letter), but for fluid-dynamical purposes it is convenient to divide them by the mass and use their specific forms, denoted with a lowercase letter (with the exception of I itself, the specific internal energy).

Define the *enthalpy* of a fluid by

$$h \equiv I + p\alpha \quad (1.71)$$

and then (1.65) becomes

$$dh = T d\eta + \alpha dp + \mu dS. \quad (1.72)$$

Evidently, the natural variables for enthalpy are entropy and pressure so that, in general,

$$dh = \left(\frac{\partial h}{\partial \eta} \right)_{p, S} d\eta + \left(\frac{\partial h}{\partial p} \right)_{\eta, S} dp + \left(\frac{\partial h}{\partial S} \right)_{\eta, p} dS. \quad (1.73)$$

Comparing the last two equations we have

$$T = \left(\frac{\partial h}{\partial \eta} \right)_{p, S}, \quad \alpha = \left(\frac{\partial h}{\partial p} \right)_{\eta, S}, \quad \mu = \left(\frac{\partial h}{\partial S} \right)_{\eta, p}. \quad (1.74)$$

Noting that

$$\frac{\partial^2 h}{\partial \eta \partial p} = \frac{\partial^2 h}{\partial p \partial \eta} \quad (1.75)$$

we evidently must have

$$\left(\frac{\partial T}{\partial p}\right)_\eta = \left(\frac{\partial \alpha}{\partial \eta}\right)_p, \quad (1.76)$$

and this is our second Maxwell relation.

To obtain the third, we write

$$dI = T d\eta - p d\alpha + \mu dS = d(T\eta) - \eta dT - d(p\alpha) + \alpha dp + \mu dS, \quad (1.77)$$

or

$$dg = -\eta dT + \alpha dp + \mu dS \quad \text{where} \quad g \equiv I - T\eta + p\alpha = h - TS. \quad (1.78a,b)$$

The quantity g is the *Gibbs function*, also known as the ‘Gibbs free energy’ or ‘Gibbs potential’. (We use g for gravity and g for the specific Gibbs function.) Now, formally, we have,

$$dg = \left(\frac{\partial g}{\partial T}\right)_{p,S} dT + \left(\frac{\partial g}{\partial p}\right)_{T,S} dp + \left(\frac{\partial g}{\partial S}\right)_{T,p} dS. \quad (1.79)$$

Comparing the last two equations we see that

$$\eta = -\left(\frac{\partial g}{\partial T}\right)_{p,S}, \quad \alpha = \left(\frac{\partial g}{\partial p}\right)_{T,S}, \quad \mu = \left(\frac{\partial g}{\partial S}\right)_{T,p}. \quad (1.80)$$

Furthermore, because

$$\frac{\partial^2 g}{\partial p \partial T} = \frac{\partial^2 g}{\partial T \partial p} \quad (1.81)$$

we have our third Maxwell equation,

$$\left(\frac{\partial \eta}{\partial p}\right)_T = -\left(\frac{\partial \alpha}{\partial T}\right)_p. \quad (1.82)$$

The Gibbs function is unique among the thermodynamic potentials in that its natural variables, T and p , are intensive quantities.

The fourth Maxwell equation makes use of the specific free energy or Helmholtz function, f , where

$$f \equiv I - T\eta, \quad \text{and} \quad df = -\eta dT - p d\alpha + \mu dS, \quad (1.83)$$

giving

$$\left(\frac{\partial \eta}{\partial \alpha}\right)_T = \left(\frac{\partial p}{\partial T}\right)_\alpha, \quad (1.84)$$

and all four of these Maxwell equations are summarized in the box on the next page. All of them follow from the fundamental thermodynamic relation, (1.65), which is the real silver hammer of thermodynamics. The fundamental relation can also be written in the following form connecting internal energy, enthalpy and entropy,

$$dI + p d\alpha = dh - \alpha dp = T d\eta + \mu dS, \quad (1.85)$$

which turns out to be useful for fluid dynamical applications.

Thermodynamic Functions and Maxwell Relations

The four canonical Maxwell relations, with their associated potentials, are

Internal Energy:

$$I = I,$$

$$dI = T d\eta - p d\alpha + \mu dS,$$

$$\left(\frac{\partial T}{\partial \alpha}\right)_{\eta,S} = -\left(\frac{\partial p}{\partial \eta}\right)_{\alpha,S}.$$

Enthalpy:

$$h = I + p\alpha,$$

$$dh = T d\eta + \alpha dp + \mu dS,$$

$$\left(\frac{\partial T}{\partial p}\right)_{\eta,S} = \left(\frac{\partial \alpha}{\partial \eta}\right)_{p,S}.$$

Gibbs function:

$$g = I - T\eta + p\alpha,$$

$$dg = -\eta dT + \alpha dp + \mu dS,$$

$$\left(\frac{\partial \eta}{\partial p}\right)_{T,S} = -\left(\frac{\partial \alpha}{\partial T}\right)_{p,S}.$$

Helmholtz free energy:

$$f = I - T\eta,$$

$$df = -\eta dT - p d\alpha + \mu dS,$$

$$\left(\frac{\partial \eta}{\partial \alpha}\right)_{T,S} = \left(\frac{\partial p}{\partial T}\right)_{\alpha,S}.$$

Meaning of the state functions

Internal energy, enthalpy, the Gibbs function, the free energy and the entropy are all state functions from which other thermodynamic quantities can be derived, but with different meanings and uses. The utility of these quantities will become apparent as we proceed, but here is a brief summary.

The internal energy of a body is the total energy within a body, excluding the kinetic energy and the potential energy due to external fields like gravity. It is an invariant if the volume is fixed and there is no heating or chemical change to the body (this is the first law). The other state functions are related to internal energy by Legendre transformations, but they are not direct measures of the energy that a body contains. The enthalpy is the Legendre transformation of the internal energy (a function of entropy, density and composition) to a function of entropy, pressure and composition, and it is important in fluids because the energy transfer between a fluid parcel and its environment is associated with a flux of enthalpy, not internal energy. If a parcel is adiabatically displaced its change in potential energy will be balanced by a change in enthalpy, for it is enthalpy that accounts for the work done by pressure forces. When two adjacent parcels at the same pressure mix their total enthalpy is conserved.

The Gibbs function is useful because it is constant for systems at constant temperature, pressure and composition. Its natural variables, temperature, pressure and composition are all measurable and for that reason it finds use as the fundamental state function from which all other thermodynamic variables may be derived. The Helmholtz free energy is also sometimes used as a fundamental state variable, and is useful for systems at constant temperature, density and composition for then it is constant. For small isothermal and isohaline changes, the increase of free energy is equal to the work done on the system. Free energy is not in fact commonly used in the atmospheric or oceanic sciences. Finally, whole books have been written on entropy. Suffice it to say here that it is the state function that responds directly to heating, and that is a measure of disorder of a system.⁴

Fundamental equation of state

The fundamental equation of state (1.59) gives complete information about a fluid in thermodynamic equilibrium, and given this we can obtain expressions for the temperature, pressure and

chemical potential using (1.66). These expressions are also equations of state; however, each of them, taken individually, contains less information than the fundamental equation because a derivative has been taken. Equivalent to the fundamental equation of state are, using (1.72), an expression for the enthalpy as a function of its natural variables pressure, entropy and composition, or, using (1.78) the Gibbs function as a function of pressure, temperature and composition. Of these, the Gibbs function is particularly useful in practice because the pressure, temperature and composition may all be measured in the laboratory. Given the fundamental equation of state, the thermodynamic state of a body is fully specified by a knowledge of any two of p, ρ, T, η and I , plus its composition. The thermal equation of state, (1.52), is obtained by using (1.59a) to eliminate entropy from (1.66a) and (1.66b).

One simple fundamental equation of state is to take the internal energy to be a function of density and not entropy; that is, $I = I(\alpha)$. Bodies with such a property are called *homentropic*. Using (1.66), temperature and chemical potential have no role in the fluid dynamics and the density is a function of pressure alone — the defining property of a barotropic fluid.⁵ Neither water nor air are, in general, homentropic but under some circumstances the flow may be adiabatic and $p = p(\rho)$.

In an ideal gas the molecules do not interact except by elastic collisions, and the volume of the molecules is presumed to be negligible compared to the total volume they occupy. The internal energy of the gas then depends only on temperature, and not on density. A *simple* ideal gas, also called a *perfect* gas (although nomenclature in the literature varies), is an ideal gas for which the heat capacity is constant, so that

$$I = cT, \quad (1.86)$$

where c is a constant. Using this and the conventional ideal gas equation, $p = \rho RT$ (where R is also constant), along with the fundamental thermodynamic relation (1.65), we can infer the fundamental equation of state; however, we will defer that until we discuss potential temperature in section 1.6.1 — if curious, look ahead to page 25. A general ideal gas also obeys $p = \rho RT$, but has heat capacities that may be a function of temperature (but only of temperature, if composition is fixed), but in this book we only deal with simple ideal gases.

Internal energy and specific heats

We can obtain some useful relations between the internal energy and specific heat capacities, and some useful estimates of their values, by some simple manipulations of the fundamental thermodynamic relation. Assuming that the composition of the fluid is constant (1.65) is

$$T d\eta = dI + p d\alpha, \quad (1.87)$$

so that, taking I to be a function of α and T ,

$$T d\eta = \left(\frac{\partial I}{\partial T} \right)_{\alpha,S} dT + \left[\left(\frac{\partial I}{\partial \alpha} \right)_{T,S} + p \right] d\alpha. \quad (1.88)$$

From this, we see that the heat capacity at constant volume (i.e., constant α) c_v is given by

$$c_v \equiv T \left(\frac{\partial \eta}{\partial T} \right)_{\alpha,S} = \left(\frac{\partial I}{\partial T} \right)_{\alpha,S}. \quad (1.89)$$

Thus, c in (1.86) is equal to c_v .

Similarly, using (1.72) we have

$$T d\eta = dh - \alpha dp = \left(\frac{\partial h}{\partial T} \right)_{p,S} dT + \left[\left(\frac{\partial h}{\partial p} \right)_{T,S} - \alpha \right] dp. \quad (1.90)$$

The heat capacity at constant pressure, c_p , is then given by

$$c_p \equiv T \left(\frac{\partial \eta}{\partial T} \right)_{p,S} = \left(\frac{\partial h}{\partial T} \right)_{p,S}. \quad (1.91)$$

For an ideal gas $h = I + RT = T(c_v + R)$. But $c_p = (\partial h / \partial T)_p$, and hence $c_p = c_v + R$. For future use we define $\gamma \equiv c_p / c_v$ and $\kappa \equiv R / c_p$, and $(\gamma - 1) / \gamma = \kappa$. Statistical mechanics tells us that for a simple ideal gas the internal energy is equal to $kT/2$ per molecule, or $RT/2$ per unit mass, for each excited degree of freedom, where k is the Boltzmann constant and R the gas constant. The diatomic molecules N_2 and O_2 that comprise most of our atmosphere have two rotational and three translational degrees of freedom, so that $I \approx 5RT/2$, and so $c_v \approx 5R/2$ and $c_p \approx 7R/2$, both being constants. These are in fact very good approximations to the measured values for the Earth's atmosphere, and give $c_v \approx 714 \text{ J kg}^{-1} \text{ K}^{-1}$ and $c_p \approx 1000 \text{ J kg}^{-1} \text{ K}^{-1}$ (whereas c_p is measured to be $1003 \text{ J kg}^{-1} \text{ K}^{-1}$). The internal energy is simply $c_v T$ and the enthalpy is $c_p T$. For a liquid, especially one like seawater that contains dissolved salts, no such simple relations are possible: the heat capacities are functions of the state of the fluid, and the internal energy is a function of pressure (or density) as well as temperature and composition.

1.6 THERMODYNAMIC EQUATIONS FOR FLUIDS

The thermodynamic relations — for example (1.65) — apply to identifiable bodies or systems; thus, the heat input affects the fluid parcel to which it is applied, and we can apply the material derivative to the above thermodynamic relations to obtain equations of motion for a moving fluid. In doing so we make two assumptions:

- (i) That locally the fluid is in thermodynamic equilibrium. This means that, although the thermodynamic quantities like temperature, pressure and density vary in space and time, locally they are related by the thermodynamic relations such as the equation of state and Maxwell's relations.
- (ii) That macroscopic fluid motions are reversible and so not entropy producing. Thus, such effects as the viscous dissipation of energy, radiation, and conduction may produce entropy whereas the macroscopic fluid motion itself does not.

The first point requires that the temperature variation on the macroscopic scales must be slow enough that there can exist a volume that is small compared to the scale of macroscopic variations, so that temperature is effectively constant within it, but that is also sufficiently large to contain enough molecules so that macroscopic variables such as temperature have a proper meaning.

Now from (1.61) conservation of energy for a infinitesimal fluid parcel may be written as

$$dI = -p d\alpha + dQ_E \quad (1.92)$$

where $p d\alpha$ is the work done by the parcel and dQ_E is the total energy input to the parcel with contributions from heating and changes in composition. Given the first assumption above, we may form the material derivative of (1.92) to obtain

$$\frac{DI}{Dt} + p \frac{D\alpha}{Dt} = \dot{Q}_E \quad (1.93)$$

where $\dot{Q}_E$ is the rate of total energy input, per unit mass, with (as for dQ_E) possible contributions from thermal fluxes (including radiative heating, thermal diffusion and heat generated by viscous damping) and fluxes of composition, and note that $\dot{Q}_E$ does not include any mechanical effects. (In general, both the diffusion of heat and composition depend on the gradients of both temperature and composition, although the thermal flux is largely determined by the temperature gradient, and the compositional flux by the gradient of composition.)

Using the mass continuity equation in the form $D\alpha/Dt = \alpha \nabla \cdot \mathbf{v}$, we write (1.93) as

$$\frac{DI}{Dt} + p\alpha \nabla \cdot \mathbf{v} = \dot{Q}_E. \quad (1.94)$$

This is the *internal energy equation* for a fluid. Internal energy is not a conservative variable because of the compression term involving $\nabla \cdot \mathbf{v}$. The internal energy equation may also be written in terms of enthalpy, and using (1.71) and (1.93) we obtain the equivalent equation,

$$\frac{Dh}{Dt} - \alpha \frac{Dp}{Dt} = \dot{Q}_E. \quad (1.95)$$

Because $\dot{Q}_E$ contains, in general, energy fluxes due to changes in composition, we need to know that composition. The composition of a fluid parcel is carried with it as it moves, and changes only if there are non-conservative sources and sinks, such as diffusive fluxes. Thus, and analogously to (1.41), the evolution of composition is determined by

$$\frac{DS}{Dt} = \dot{S}, \quad (1.96)$$

where $\dot{S}$ represents all the non-conservative terms.

Rather than use an internal energy equation, we may use the fundamental thermodynamic relation to infer an evolution equation for entropy. Thus, forming the material derivative from (1.65) and using (1.94) and (1.96), we obtain the *entropy equation*

$$\frac{D\eta}{Dt} = \frac{1}{T} \dot{Q}_E - \frac{\mu}{T} \dot{S} \equiv \frac{1}{T} \dot{Q}, \quad (1.97)$$

where $\dot{Q}$ is the heating rate per unit mass and which is defined by the right-hand side of the above relationship. This equation is simply the material derivative of (1.62), along with assumption (ii) above. The heating of a fluid parcel generally needs to be *derived*, since it involves the viscous dissipation of energy as well as radiative and diffusive fluxes. The derivation is the topic of the appendix on page 49, and from now on we assume the heating and energy input are known quantities.

The entropy equation is not independent of the internal energy equation, but is connected via the thermodynamic relations of sections 1.5.1 and 1.5.2, and by the equation of state. If we use the internal energy equation we can in principle then calculate the entropy using the equation of state in the form $\eta = \eta(I, \alpha, S)$, or if we use the entropy equation we can then calculate the internal energy using $I = I(\eta, \alpha, S)$. Indeed, the internal energy equation and the entropy equation are both commonly referred to as ‘the thermodynamic equation’. The heating term, $\dot{Q}$, is, however, not always easy to accurately determine in practice — it is affected by gradients of composition as well viscosity and radiative fluxes — and the use of the internal energy equation may be more straightforward than the use of an entropy equation, especially in multi-component fluids. On the other hand, internal energy is affected by the $\nabla \cdot \mathbf{v}$ term in (1.94) and in a liquid this is small but non zero and that may cause difficulties. See section 1.7.3 for more discussion.

In any case, given evolution equations for composition and internal energy or entropy, and given the fundamental equation of state, we have, in principle, a complete set of equations for a fluid, as summarized in the shaded box on the facing page. Let us now look at the special, but very important, case of a dry ideal gas.

Fundamental Equations of Motion of a Fluid

The following equations constitute, in principle, a complete set of equations for a fluid heated at a rate $\dot{Q}$ and whose composition, S , changes at a rate $\dot{S}$.

Evolution equations for velocity, density and composition

$$\frac{D\mathbf{v}}{Dt} = -\frac{1}{\rho}\nabla p + \nu\nabla^2\mathbf{v} + \mathbf{F}, \quad \frac{D\rho}{Dt} + \rho\nabla\cdot\mathbf{v} = 0, \quad \frac{DS}{Dt} = \dot{S}. \quad (\text{F.1})$$

where $\mathbf{F}$ is some body force, such as gravity.

Internal energy equation or entropy equation (the 'thermodynamic' equation)

$$\frac{DI}{Dt} + \frac{p}{\rho}\nabla\cdot\mathbf{v} = \dot{Q}_E \quad \text{or} \quad \frac{D\eta}{Dt} = \frac{1}{T}\dot{Q}, \quad (\text{F.2})$$

where $\dot{Q}$ is the heating and $\dot{Q}_E = \dot{Q} + \mu\dot{S}$ is the total rate of energy input.

Fundamental equation of state for internal energy, I , entropy, η , or Gibbs function, g

$$I = I(\rho, \eta, S), \quad \eta = \eta(\rho, I, S) \quad \text{or} \quad g = g(T, p, S). \quad (\text{F.3})$$

Diagnostic equations for temperature and pressure

$$T = \left(\frac{\partial I}{\partial \eta}\right)_{\alpha, S} \quad \text{or} \quad T^{-1} = \left(\frac{\partial \eta}{\partial I}\right)_{\alpha, S}. \quad (\text{F.4})$$

$$p = -\left(\frac{\partial I}{\partial \alpha}\right)_{\eta, S} = \rho^2 \left(\frac{\partial I}{\partial \rho}\right)_{\eta, S} \quad \text{or} \quad p = T \left(\frac{\partial \eta}{\partial \alpha}\right)_{I, S}. \quad (\text{F.5})$$

The actual method of solution of these equations will depend on the equation of state. For example, for an ideal gas $T = I/c_v$ and the thermal equation of state, $p = \rho RT$, may be used to infer pressure.

The equations describing fluid motion are called the *Euler equations* if the viscous term is omitted, and the *Navier–Stokes equations* if viscosity is included.⁶ These appellations are often taken to mean only the momentum and mass conservation equations, and the Euler equations are sometimes taken to be the equations for a fluid of constant density.

1.6.1 Thermodynamic equation for an ideal gas

For a dry ideal gas the internal energy is a function of temperature only and $dI = c_v dT$. The first law of thermodynamics becomes

$$dQ = c_v dT + p d\alpha, \quad \text{or} \quad dQ = c_p dT - \alpha dp, \quad (1.98a,b)$$

where the second expression is derived using $\alpha = RT/p$ and $c_p - c_v = R$. Forming the material derivative of the above gives two forms of the internal energy equation:

$$c_v \frac{DT}{Dt} + p \frac{D\alpha}{Dt} = \dot{Q} \quad \text{or} \quad c_p \frac{DT}{Dt} - \frac{RT}{p} \frac{Dp}{Dt} = \dot{Q}. \quad (1.99a,b)$$

Using the mass continuity equation, (1.99a) may be written as

$$c_v \frac{DT}{Dt} + p\alpha \nabla \cdot \mathbf{v} = \dot{Q}. \quad (1.100)$$

This is one of the most common and useful forms of the thermodynamic equation for the atmosphere. A less common but equivalent form arises if we use the ideal gas equation to eliminate T in favour of p , giving

$$\frac{Dp}{Dt} + \gamma p \nabla \cdot \mathbf{v} = \dot{Q} \frac{\rho R}{c_v}. \quad (1.101)$$

The Earth's atmosphere also contains water vapour with mixing ratio q (commonly referred to as specific humidity), and an evolution equation for it takes the form

$$\frac{Dq}{Dt} = \dot{q}, \quad (1.102)$$

where $\dot{q}$ represents the effects of condensation and evaporation. The main thermodynamic effects of water vapour occurs when it condenses and latent heat is released; this heating appears in thermodynamic equation, with $\dot{Q} = -L\dot{q}$, where L is the latent heat of condensation, as discussed in chapter 18.

Potential temperature, potential density and entropy

We can use entropy instead of temperature for our thermodynamic equation, and this corresponds to using (1.97) instead of (1.94). However, it is common in meteorology to express entropy in terms of a temperature-like quantity, potential temperature, which has a more intuitive appeal to some.⁷ Seawater also has a potential temperature variable, but with a different form.

We begin with the observation that when a fluid parcel changes pressure adiabatically, it will expand or contract and, using (1.98b) with $dQ = 0$, its temperature change is determined by $c_p dT = \alpha dp$. This temperature change is plainly not caused by heating, but we may construct a temperature-like quantity that changes *only* if diabatic effects are present; specifically *potential temperature*, θ , is defined to be the temperature that a fluid would have if moved adiabatically and at constant composition to some reference pressure (usually taken to be 1000 hPa, which is close to the pressure at the Earth's surface). Thus, in adiabatic flow the potential temperature obeys $D\theta/Dt = 0$.

In order to relate θ to the other thermodynamic variables we use (1.98b) and the equation of state for an ideal gas to write the fundamental thermodynamic relation as

$$d\eta = c_p d \ln T - R d \ln p. \quad (1.103)$$

If we move a parcel adiabatically ($d\eta = 0$) from p to p_R the temperature changes, by definition, from T to θ , and (1.103) gives

$$\int_T^\theta c_p d \ln T - \int_p^{p_R} R d \ln p = 0. \quad (1.104)$$

For constant c_p and R this equation may be solved to give

$$\theta = T \left(\frac{p_R}{p} \right)^\kappa, \quad (1.105)$$

where p_R is the reference pressure and $\kappa \equiv R/c_p$. (Another derivation of this result is given in the appendix on page 49.) It follows from (1.103) and (1.105) that potential temperature is related to entropy by

$$d\eta = c_p d \ln \theta \quad (1.106)$$

and, if c_p is constant, which it nearly is for Earth's atmosphere,

$$\eta = c_p \ln \theta. \quad (1.107)$$

Forming the material derivative of (1.107) and using (1.97) we obtain

$$c_p \frac{D\theta}{Dt} = \frac{\theta}{T} \dot{Q}, \quad (1.108)$$

with θ given by (1.105). Equations (1.100), (1.101) and (1.108) are all equivalent forms of the thermodynamic equation for an ideal gas.

The *potential density*, ρ_θ , is the density that a fluid parcel would have if moved adiabatically and at constant composition to a reference pressure, p_R . If the equation of state is written as $\rho = f(p, T)$ then the potential density is just $\rho_\theta = f(p_R, \theta)$ and for an ideal gas we therefore have

$$\rho_\theta = \frac{p_R}{R\theta} = \rho \left(\frac{p_R}{p} \right)^{1/\gamma}. \quad (1.109)$$

Finally, for later use, a little manipulation of the equation of state for an ideal gas, for small variations around a reference state, reveals that

$$\frac{\delta\theta}{\theta} = \frac{\delta T}{T} - \kappa \frac{\delta p}{p} = \frac{1}{\gamma} \frac{\delta p}{p} - \frac{\delta\rho}{\rho}. \quad (1.110)$$

The fundamental equation of state for an ideal gas

Equations (1.107) and (1.105) are closely related to the fundamental equation of state, and using $I = c_v T$ and the equation of state $p = \rho RT$, we can express the entropy explicitly in terms of the density and the internal energy. Similarly, we may derive an expression for Gibbs function as a function of pressure and temperature, and we find

$$\eta = c_v \ln I - R \ln \rho + A, \quad g = c_p T(1 - \ln T) + RT \ln p + BT + C. \quad (1.111a,b)$$

where A, B and C are constants. Either of these two expressions may be regarded as *the fundamental equation of state for a simple ideal gas*. They could in fact be used to define a simple ideal gas (although we have not motivated that approach) and if we were to *begin* with either of them we could derive all the other thermodynamic quantities of interest. For example, using (1.68a) and (1.111a) we immediately recover $p = \rho RT$, and using (1.68b) we obtain $T = c_v/I$. Similarly, from the Gibbs function we obtain $\alpha = (\partial g/\partial p)_T = RT/p$, and the entropy satisfies $\eta = -(\partial g/\partial T)_p$. We provide a more complete set of derivations from the Gibbs function in the appendix on page 47 and, for moist air, on page 720.

1.6.2 ♦ Other forms of the thermodynamic equation

For a liquid such as seawater no simple exact equation of state exists and writing down a useful thermodynamic equation is not easy. Thus, although (1.97) holds in general, we need to be able to evaluate the heating and we need an expression relating entropy to the other thermodynamic variables — that is, an equation of state. For quantitative modelling and observational work such an equation of state must be quite accurate and we come back to this in section 1.7. However, we can gain understanding — for both liquids and gases — by beginning with the entropy equation and making simplifications to it, as follows.

Forms of the Thermodynamic Equation

General form

$$\frac{DI}{Dt} + p \frac{D\alpha}{Dt} = \dot{Q}_E \quad \text{or} \quad \frac{DI}{Dt} + p\alpha \nabla \cdot \mathbf{v} = \dot{Q}_E, \quad (\text{T.1a,b})$$

where I is the internal energy and $\dot{Q}_E$ is the rate of energy input, per unit mass. This may be written in terms of enthalpy, h , or entropy, η

$$\frac{Dh}{Dt} - \alpha \frac{Dp}{Dt} = \dot{Q}_E, \quad T \frac{D\eta}{Dt} = \dot{Q}_E - \mu \dot{S} = \dot{Q} \quad (\text{T.2a,b})$$

where $\dot{Q}$ is the heating rate and $\dot{S}$ the rate of change of composition. For a fluid parcel of constant composition, $c_p d \ln \theta = d\eta$ and (T.2b) may be written as a potential temperature equation.

Ideal gas

For an ideal gas $dI = c_v dT$, $dh = c_p dT$, $d\eta = c_p d \ln \theta$ and the adiabatic thermodynamic equation may be written in the following equivalent, exact, forms:

$$\begin{aligned} c_p \frac{DT}{Dt} - \alpha \frac{Dp}{Dt} &= 0, & \frac{Dp}{Dt} + \gamma p \nabla \cdot \mathbf{v} &= 0, \\ c_v \frac{DT}{Dt} + p\alpha \nabla \cdot \mathbf{v} &= 0, & \frac{D\theta}{Dt} &= 0, \end{aligned} \quad (\text{T.3})$$

where $\theta = T(p_R/p)^\kappa$, and energy or heating terms in various forms appear on the right-hand sides as needed. The two expressions on the second line are usually the most useful in modelling and theoretical work in meteorology.

1. Thermodynamic equation using pressure and density

If we regard η as a function of pressure and density, and salinity S where appropriate, we obtain

$$\begin{aligned} T d\eta &= T \left(\frac{\partial \eta}{\partial \rho} \right)_{p,S} d\rho + T \left(\frac{\partial \eta}{\partial p} \right)_{\rho,S} dp + T \left(\frac{\partial \eta}{\partial S} \right)_{\rho,p} dS \\ &= T \left(\frac{\partial \eta}{\partial \rho} \right)_{p,S} d\rho - T \left(\frac{\partial \eta}{\partial \rho} \right)_{p,S} \left(\frac{\partial \rho}{\partial p} \right)_{\eta,S} dp + T \left(\frac{\partial \eta}{\partial S} \right)_{\rho,p} dS. \end{aligned} \quad (\text{1.112})$$

Forming the material derivative, and using (1.97) and (1.96), we obtain for a moving fluid

$$T \left(\frac{\partial \eta}{\partial \rho} \right)_{p,S} \frac{D\rho}{Dt} - T \left(\frac{\partial \eta}{\partial \rho} \right)_{p,S} \left(\frac{\partial \rho}{\partial p} \right)_{\eta,S} \frac{Dp}{Dt} = \dot{Q} - T \left(\frac{\partial \eta}{\partial S} \right)_{\rho,p} \dot{S}. \quad (\text{1.113})$$

But $(\partial p / \partial \rho)_{\eta,S} = c_s^2$ where c_s is the speed of sound (see section 1.8). This is a measurable quantity in a fluid, and often nearly constant, and so useful to keep in an equation. The thermodynamic equation may then be written in the form

$$\frac{D\rho}{Dt} - \frac{1}{c_s^2} \frac{Dp}{Dt} = Q_{[\rho,p]}, \quad (\text{1.114})$$

where $Q_{[\rho,p]} \equiv (\partial\rho/\partial\eta)_{p,S}\dot{Q}/T + (\partial\rho/\partial S)_{\eta,p}\dot{S}$ represents the effects of entropy and salinity source terms. This form of the thermodynamic equation is valid for both liquids and gases.

Approximations using pressure and density

The speed of sound in a fluid is related to its compressibility — the less compressible the fluid, the greater the sound speed. In a liquid, sound speed is often sufficiently high that the second term in (1.114) can be neglected, and the thermodynamic equation takes the simple form:

$$\frac{D\rho}{Dt} = Q_{[\rho,p]}. \quad (1.115)$$

The above equation is a very good approximation for many laboratory fluids. It is a *thermodynamic* equation, arising from the principle of conservation of energy for a liquid; it is a very different equation from the mass conservation equation, which for compressible fluids is also an evolution equation for density.

In the ocean the enormous pressures resulting from columns of seawater kilometres deep mean that although the second term in (1.114) may be small, it is not negligible, and a better approximation results if we suppose that the pressure is given by the weight of the fluid above it — the hydrostatic approximation. In this case $dp = -\rho g dz$ and (1.114) becomes

$$\frac{D\rho}{Dt} + \frac{\rho g}{c_s^2} \frac{Dz}{Dt} = Q_{[\rho,p]}. \quad (1.116)$$

In the second term the height field varies much more than the density field, so a good approximation is to replace ρ by a constant, ρ_0 , in this term only. Taking the speed of sound also to be constant gives

$$\frac{D}{Dt} \left(\rho + \frac{\rho_0 z}{H_\rho} \right) = Q_{[\rho,p]}, \quad \text{where} \quad H_\rho = c_s^2/g. \quad (1.117a,b)$$

H_ρ is the *density scale height* of the ocean. In water, $c_s \approx 1500 \text{ m s}^{-1}$ so that $H_\rho \approx 200 \text{ km}$. The quantity in brackets on the left-hand side of (1.117a) is (in this approximation) the *potential density*, this being the density that a parcel would have if moved adiabatically and with constant composition to the reference height $z = 0$. The adiabatic lapse rate of density is the rate at which the density of a parcel changes when undergoing an adiabatic displacement. From (1.117) it is approximately

$$-\left(\frac{\partial\rho}{\partial z}\right)_\eta \approx \frac{\rho_0 g}{c_s^2} \approx 5 \text{ (kg m}^{-3}\text{)/km}, \quad (1.118)$$

so that if a parcel is moved adiabatically from the surface to the deep ocean (5 km depth, say) its density will increase by about 25 kg m^{-3} , a fractional change of about 1/40 or 2.5%.

II. Thermodynamic equation using pressure and temperature

Taking entropy to be a function of pressure and temperature (and salinity if appropriate) we have

$$\begin{aligned} T d\eta &= T \left(\frac{\partial\eta}{\partial T} \right)_{p,S} dT + T \left(\frac{\partial\eta}{\partial p} \right)_{T,S} dp + T \left(\frac{\partial\eta}{\partial S} \right)_{T,p} dS \\ &= c_p dT + T \left(\frac{\partial\eta}{\partial p} \right)_{T,S} dp + T \left(\frac{\partial\eta}{\partial S} \right)_{T,p} dS. \end{aligned} \quad (1.119)$$

For a moving fluid, and using (1.97) and (1.96), this implies,

$$\frac{DT}{Dt} + \frac{T}{c_p} \left(\frac{\partial\eta}{\partial p} \right)_{T,S} \frac{Dp}{Dt} = Q_{[T,p]}, \quad (1.120)$$

where $Q_{[T,p]} \equiv \dot{Q}/c_p - Tc_p^{-1}\dot{S}(\partial\eta/\partial S)$. Now substitute the Maxwell relation (1.82) in the form

$$\left(\frac{\partial\eta}{\partial p}\right)_T = \frac{1}{\rho^2} \left(\frac{\partial\rho}{\partial T}\right)_p \quad (1.121)$$

to give

$$\frac{DT}{Dt} + \frac{T}{c_p\rho^2} \left(\frac{\partial\rho}{\partial T}\right)_p \frac{Dp}{Dt} = Q_{[T,p]}, \quad \text{or} \quad \frac{DT}{Dt} - \frac{T}{c_p} \left(\frac{\partial\alpha}{\partial T}\right)_p \frac{Dp}{Dt} = Q_{[T,p]}. \quad (1.122a,b)$$

The density and temperature are related through a coefficient of thermal expansion β_T where

$$\left(\frac{\partial\rho}{\partial T}\right)_p = -\beta_T\rho. \quad (1.123)$$

Equation (1.122) then becomes

$$\frac{DT}{Dt} - \frac{\beta_T T}{c_p\rho} \frac{Dp}{Dt} = Q_{[T,p]}. \quad (1.124)$$

This form of the thermodynamic equation is valid for both liquids and gases. In an ideal gas we have $\beta_T = 1/T$, whereas in a liquid β_T is usually quite small.

Approximations using pressure and temperature

In the hydrostatic approximation we suppose that the pressure in (1.124) varies according only to the weight of the fluid above it. Then $dp = -\rho g dz$ and (1.124) becomes

$$\frac{1}{T} \frac{DT}{Dt} + \frac{\beta_T g}{c_p} \frac{Dz}{Dt} = \frac{Q_{[T,p]}}{T}. \quad (1.125)$$

For an ideal gas we have $\beta_T = 1/T$, whence, if c_p is constant,

$$\frac{D}{Dt} (c_p T + gz) = c_p Q_{[T,p]}. \quad (1.126)$$

The quantity $c_p T + gz$ is known as the dry static energy and we will encounter it again throughout the book. The above equation is closely related to the potential temperature form of the thermodynamic equation, since in hydrostatic balance a little manipulation reveals that

$$\frac{T}{\theta} \frac{\partial\theta}{\partial z} = \frac{\partial T}{\partial z} + \frac{g}{c_p}. \quad (1.127)$$

The quantity $T + gz/c_p$ is a height form of potential temperature for an ideal gas in hydrostatic balance, being the temperature that a fluid at a level z and temperature T would have if moved adiabatically to a reference level of $z = 0$.

If β_T is constant, which is a fair approximation for many liquids, then for small variations of temperature around the value T_0 , (1.125) simplifies to

$$\frac{D}{Dt} \left(T + \frac{T_0 z}{H_T} \right) = Q_{[T,p]}, \quad \text{where} \quad H_T = \frac{c_p}{\beta_T g}. \quad (1.128a,b)$$

The quantity H_T is the *temperature scale height* of the fluid, and with the oceanic values $\beta_T \approx 2 \times 10^{-4} \text{ K}^{-1}$ and $c_p \approx 4 \times 10^3 \text{ J kg}^{-1} \text{ K}^{-1}$ we obtain $H_T \approx 2000 \text{ km}$. The field $T + T_0 z/H_T$ is a height form of potential temperature for liquids; that is,

$$\theta \approx T + \frac{\beta_T g T_0}{c_p} z. \quad (1.129)$$

The temperature changes because of the work done by or on the fluid parcel as it expands or is compressed. In seawater the expansion coefficient β_T and c_p are functions of pressure and (1.129) is not good enough if high accuracy is required, whereas in a laboratory setting we can often simply neglect the term involving β_T .

The adiabatic lapse rate of temperature is the rate at which the temperature of a parcel changes in the vertical when undergoing an adiabatic displacement. From (1.125) it is

$$\Gamma_z = - \left(\frac{\partial T}{\partial z} \right)_\eta = \frac{T g \beta_T}{c_p}. \quad (1.130)$$

In general Γ_z is a function of temperature, salinity and pressure, but it is a calculable quantity if β_T is known and, with the oceanic values above, it is approximately 0.15 K km^{-1} . Equation (1.130) is not accurate enough for quantitative oceanography because the expansion coefficient is a function of pressure; nor is it a good measure of stability, because of the effects of salt. In a dry atmosphere the ideal gas relationship gives $\beta_T = 1/T$ and so

$$\Gamma_z = \frac{g}{c_p}, \quad (1.131)$$

which is approximately 10 K km^{-1} . The only approximation involved in deriving this is the use of the hydrostatic relationship.

It is noteworthy that the scale heights given by (1.117b) and (1.128b) differ so much. The first is due to the pressure compressibility of seawater [and so related to c_s^2 , or β_p in (1.57)] whereas the second is due to the change of density with temperature [β_T in (1.57)], and is the distance over which the difference between temperature and potential temperature changes by an amount equal to the temperature itself (i.e., by about 273 K). The two heights differ so much because the value of the thermal expansion coefficient is not directly related to the pressure compressibility — for example, fresh water at 4°C has zero thermal expansivity, and so would have an infinite temperature scale height, but its pressure compressibility differs little from water at 20°C .

III. Thermodynamic equation using density and temperature

Taking entropy to be a function of density and temperature (and salinity if appropriate) we have

$$\begin{aligned} T d\eta &= T \left(\frac{\partial \eta}{\partial T} \right)_{\alpha, S} dT + T \left(\frac{\partial \eta}{\partial \alpha} \right)_{T, S} d\alpha + T \left(\frac{\partial \eta}{\partial S} \right)_{T, \alpha} dS \\ &= c_v dT + T \left(\frac{\partial \eta}{\partial \alpha} \right)_{T, S} d\alpha + T \left(\frac{\partial \eta}{\partial S} \right)_{T, \alpha} dS. \end{aligned} \quad (1.132)$$

For a moving fluid this implies,

$$\frac{DT}{Dt} + \frac{T}{c_v} \left(\frac{\partial \eta}{\partial \alpha} \right)_{T, S} \frac{D\alpha}{Dt} = Q_{[T, \alpha]}, \quad (1.133)$$

where $Q_{[T, \alpha]} \equiv \dot{Q}/c_v - T c_v^{-1} \dot{S}(\partial \eta / \partial S)$.

For an ideal gas, and using (1.84), which is one of Maxwell's relations, (1.133) may be written as

$$c_v \frac{DT}{Dt} + p \frac{D\alpha}{Dt} = \dot{Q}, \quad (1.134)$$

so recovering the internal energy equation (1.99a). On the other hand, for a liquid of nearly constant density, the second term on the left-hand side of (1.133) is small, and $c_p \approx c_v$, and we have to a first approximation $DT/Dt = Q_{[T,\alpha]}$.

Various forms of the thermodynamic equation are summarized on page 26. For an ideal gas, (1.114) and (1.124) are exactly equivalent to (1.100) or (1.101), and numerical models of an ideal gas usually use either a prognostic equation for internal energy $c_v T$ or potential temperature. A discussion of an accurate thermodynamic equation for seawater is given below.

1.7 ♦ THERMODYNAMICS OF SEAWATER

We now discuss the thermodynamics of liquids such as seawater in a little more detail. Readers whose interest is mainly in an ideal gas may skim this section. We begin with a discussion of potential temperature, since this is commonly used in oceanography, but the more general discussion begins with section 1.7.2.

1.7.1 Potential temperature, potential density and entropy

Potential temperature

The potential temperature is defined as the temperature that a parcel would have if moved adiabatically and at constant composition to a given reference pressure p_R , often taken as 10^5 Pa (or 1000 hPa, or 1000 mb, approximately the pressure at the sea-surface). Thus it may be calculated, at least in principle, by the integral

$$\theta(S, T, p, p_R) = T + \int_p^{p_R} \Gamma'_{ad}(S, T, p') dp', \quad (1.135)$$

where $\Gamma'_{ad} = (\partial T / \partial p)_{\eta, S}$. Such integrals may be hard to calculate, and if we know the equation of state in the form $\eta = \eta(S, T, p)$ then we can calculate potential temperature more directly because potential temperature must satisfy

$$\eta(S, T, p) = \eta(S, \theta, p_R). \quad (1.136)$$

Solving this equation for θ gives, in principle, $\theta = \theta(\eta, S, p_R) = \theta(T, S, p)$, and examples will be given in (1.150) and in the appendix on page 47. Potential temperature is not a materially conserved variable in the presence of salinity changes.

For a parcel of constant composition, changes in entropy are directly related to changes in potential temperature because, from the right-hand side of (1.136),

$$d\eta = \left(\frac{\partial \eta(S, \theta, p_R)}{\partial \theta} \right)_S d\theta. \quad (1.137)$$

Thus, if a fluid parcel moves adiabatically and at constant composition then $d\eta = 0$ and $d\theta = 0$. Furthermore, if we express entropy as a function of temperature and pressure then

$$T d\eta = T \left(\frac{\partial \eta}{\partial T} \right)_{p,S} dT + T \left(\frac{\partial \eta}{\partial p} \right)_{T,S} dp + T \left(\frac{\partial \eta}{\partial S} \right)_{p,T} dS = c_p dT + T \left(\frac{\partial \eta}{\partial p} \right)_{T,S} dp + T \left(\frac{\partial \eta}{\partial S} \right)_{p,T} dS, \quad (1.138)$$

Thermodynamics of Liquids

Liquids, unlike ideal gases, do not have a simple equation of state and this has ramifications for the thermodynamic equation. For seawater, a very accurate equation of state is given by IOC *et al.* (2010) and known as TEOS-10. The simpler expression, (1.144), is a good approximation for the Gibbs function and using it we can derive the thermal equation of state and some useful forms of the thermodynamic equation.

Thermal equation of state

A very useful expression for many purposes is given by

$$\alpha = \alpha_0 \left[1 + \beta_T(1 + \gamma^* p)(T - T_0) + \frac{\beta_T^*}{2}(T - T_0)^2 - \beta_S(S - S_0) - \beta_p(p - p_0) \right]. \quad (\text{TL.1})$$

For laboratory situations where pressure variations are not large a useful approximation is

$$\alpha = \alpha_0 [1 + \beta_T(T - T_0) - \beta_S(S - S_0)]. \quad (\text{TL.2})$$

Thermodynamic Equation

The near incompressibility of liquids has consequences for the thermodynamic equation. Using entropy or potential enthalpy, as discussed in section 1.7.3, as a primary thermodynamic variable is the best way to proceed if high accuracy is required. For idealized or laboratory work we can make further approximations, as follows.

Entropy evolution:

Using the hydrostatic approximation and simplifying (1.146) gives

$$\frac{D\eta}{Dt} = 0, \quad \eta = c_{p0} \ln \frac{T}{T_0} \left[1 + \beta_S^*(S - S_0) \right] + gz\beta_T. \quad (\text{TL.3})$$

Given entropy and salinity we can infer temperature and, using (TL.1), density.

Potential temperature evolution:

An alternative is to use potential temperature as the primary evolution variable, giving

$$\frac{D\theta}{Dt} = 0, \quad \theta = T + \frac{\beta_T g T_0 z}{c_{p0}}. \quad (\text{TL.4})$$

Given salinity, we can infer density directly from θ using (1.153) or a simplification thereof. The terms involving β_T in the two equations above are very small in most laboratory settings. Temperature itself is not needed, since we may directly derive the density from entropy or potential temperature, and salinity.

using the definition of c_p . If we evaluate this expression at the reference pressure, where $T = \theta$ and $dp = 0$, and consider isohaline changes with $dS = 0$, then we have $\theta d\eta = c_p(p_R, \theta, S) d\theta$, and therefore

$$d\eta = c_p(p_R, \theta, S) \frac{d\theta}{\theta}, \quad (1.139)$$

and $d\eta/d\theta = c_p(p_R, \theta, S)/\theta$. In the special case of constant c_p integration yields

$$\eta = c_p \ln \theta + \text{constant}, \quad (1.140)$$

as for a simple ideal gas — see (1.107). Given (1.139), the thermodynamic equation can be written

$$c_p \frac{D\theta}{Dt} = \frac{\theta}{T} \dot{Q}, \quad (1.141)$$

where the right-hand side represents heating. We can use (1.141) as a thermodynamic equation instead of (1.97), although we must now relate θ instead of η to the other state variables via an equation of state.

The notion of potential temperature is useful because it is connected to the actual temperature, with which we are familiar; roughly speaking, potential temperature is temperature plus a correction for the effects of compressibility. Entropy, on the other hand, may seem alien and unnecessarily exotic. However, the use of potential temperature brings no simplification to the equations of motion beyond that already afforded by the use of entropy as a thermodynamic variable. We discuss thermodynamic evolution equations for seawater further in section 1.7.3.

Potential density

Potential density, ρ_θ , is the density that a parcel would have if it were moved adiabatically and with fixed composition to a given reference pressure, p_R , that is often, but not always, taken as 10^5 Pa, or 1 bar. If the equation of state is of the form $\rho = \rho(S, T, p)$ then by definition we have

$$\rho_\theta = \rho(S, \theta, p_R). \quad (1.142)$$

For a parcel moving adiabatically and at fixed salinity its potential density is therefore conserved. For an ideal gas (1.142) gives $\rho_\theta = p_R/(R\theta)$ [as in (1.109)] and potential density provides no more information than potential temperature, but in general it is the vertical gradient of potential density that provides the appropriate measure of stability (as we find in section 2.10.1). An approximate expression for ρ_θ is given by (1.117).

In oceanography, and because the density of seawater is nearly constant, it is common to subtract the amount 1000 kg m^{-3} before quoting its value, and depending on whether we are referring to *in situ* density or the potential density the results are called σ_T ('sigma-tee') or σ_θ ('sigma-theta') respectively. Thus,

$$\sigma_T = \rho(p, T, S) - 1000, \quad \sigma_\theta = \rho(p_R, \theta, S) - 1000. \quad (1.143a,b)$$

Instead of the subscript θ , a number can be used to denote the level to which potential density is referenced. Thus, σ_0 is the potential density referenced to the surface and σ_2 is the potential density referenced to 200 bars of pressure, or about 2 kilometres depth. It is the 'locally referenced' potential density that is most relevant for stability of a fluid. Thus, for a parcel near 2 km, σ_2 is more relevant than σ_0 . The 'neutral density' (really the approximate-neutral or quasi-neutral density) is a semi-empirical way to avoid this reference-level difficulty. Neutral density is, by construction, a quantity such that the buoyancy force is locally perpendicular to its iso-surfaces, so that a parcel that is displaced adiabatically along a neutral density iso-surface will remain neutrally buoyant.⁸ We do not discuss neutral density further, and wherever it appears in figures in this book it can be assumed that potential density would look similar.

1.7.2 Equation of state and thermodynamic properties of seawater

Oceanographers go to great lengths to obtain an accurate equation of state and other physical properties of seawater, and we noted in section 1.4 that seawater has some nonlinear properties that, although small, are nevertheless important. We need to be able to calculate these properties, and in

Parameter	Description	Value
ρ_0	Reference density	$1.027 \times 10^3 \text{ kg m}^{-3}$
α_0	Reference specific volume	$9.738 \times 10^{-4} \text{ m}^3 \text{ kg}^{-1}$
T_0	Reference temperature	283 K
S_0	Reference salinity	35 ppt = 35 g kg ⁻¹
c_{s0}	Reference sound speed	1490 m s ⁻¹
β_T	First thermal expansion coefficient	$1.67 \times 10^{-4} \text{ K}^{-1}$
β_T^*	Second thermal expansion coefficient	$1.00 \times 10^{-5} \text{ K}^{-2}$
β_S	Haline contraction coefficient	$0.78 \times 10^{-3} \text{ ppt}^{-1}$
β_p	Compressibility coefficient ($= \alpha_0/c_{s0}^2$)	$4.39 \times 10^{-10} \text{ m s}^2 \text{ kg}^{-1}$
γ^*	Thermobaric parameter ($\approx \gamma'^*$)	$1.1 \times 10^{-8} \text{ Pa}^{-1}$
c_{p0}	Specific heat capacity at constant pressure	$3986 \text{ J kg}^{-1} \text{ K}^{-1}$
β_S^*	Haline heat capacity coefficient	$1.5 \times 10^{-3} \text{ ppt}^{-1}$

Table 1.2 Various thermodynamic and equation-of-state parameters for appropriate for the seawater equations of state (1.58) and (1.144). The unit ppt (or ‰) is parts per thousand by weight, or g/kg.

this section we illustrate how the thermodynamic variables for seawater — the conventional equation of state, an expression for potential temperature, and so on — can be obtained directly from the fundamental equation of state. Writing the fundamental equation in the form $I = I(\eta, S, \alpha)$ is not practically useful, because the variables are not easily measured in the laboratory. However, if we cast the fundamental equation in terms of the Gibbs function, $g = I - T\eta + p\alpha$, then $dg = -\eta dT + \alpha dp + \mu dS$ and the independent variables are the familiar and measurable (T, S, p). A similar, but simpler, procedure is carried out for an ideal gas in the appendix on page 47.

A Gibbs function that reproduces the properties of seawater with high accuracy is very complicated, but we can write down a Gibbs function that, although slightly less accurate, captures the most important properties with a certain degree of economy and transparency. The expression is⁹

$$\begin{aligned}
 g = g_0 - \eta_0(T - T_0) + \mu_0(S - S_0) - c_{p0}T \left[\ln(T/T_0) - 1 \right] \left[1 + \beta_S^*(S - S_0) \right] \\
 + \alpha_0(p - p_0) \left[1 + \beta_T(T - T_0) - \beta_S(S - S_0) - \frac{\beta_p}{2}(p - p_0) \right. \\
 \left. + \frac{\beta_T \gamma^*}{2}(p - p_0)(T - T_0) + \frac{\beta_T^*}{2}(T - T_0)^2 \right].
 \end{aligned} \tag{1.144}$$

In this equation the variables are g, T, S and p . The parameters [which all have subscripts or stars, with the starred parameters giving rise to nonlinear effects; refer also to (1.58)] are all constants that could in principle be determined in the laboratory with the help of the derived quantities like heat capacity, and their approximate values are given in Table 1.2. We will take $p_0 = 0$ and $\beta_p = \alpha_0/c_{s0}^2$, where c_{s0} is a reference sound speed. Equation (1.144) is quite accurate for many oceanographic situations, and from it we may derive the following quantities of interest.

- The conventional or thermal equation of state, $\alpha = (\partial g / \partial p)_{T,S}$:

$$\alpha = \alpha_0 \left[1 + \beta_T(1 + \gamma^* p)(T - T_0) + \frac{\beta_T^*}{2}(T - T_0)^2 - \beta_S(S - S_0) - \beta_p p \right]. \tag{1.145}$$

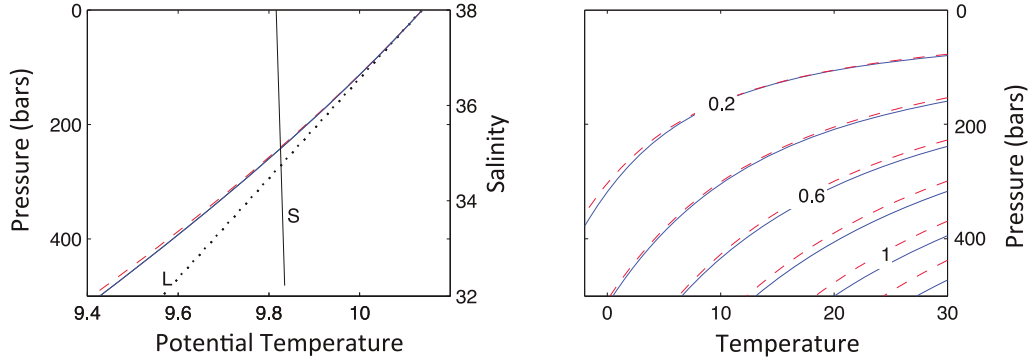


Fig. 1.4 Examples of the variation of potential temperature of seawater with pressure, temperature and salinity. Left panel: the sloping lines show potential temperature as a function of pressure at fixed salinity ($S = 35$ psu) and temperature (10°C). The solid line is computed using an accurate, empirical equation of state, the almost-coincident dashed line uses the simpler expression (1.150) and the dotted line (labelled L) uses the linear expression (1.151c). The near vertical solid line, labelled S, shows the variation of potential temperature with salinity at fixed temperature and pressure. Right panel: Contours of the difference between temperature and potential temperature, $(T - \theta)$ in the pressure–temperature plane, for $S = 35$ psu. The dashed lines use (1.150) and the solid lines use an accurate empirical formula. (Note: 100 bars of pressure (10^7 Pa or 10 MPa) is approximately 1 km depth.)

- The entropy, $\eta = -(\partial g / \partial T)_{p,S}$:

$$\eta = \eta_0 + c_{p0} \ln \frac{T}{T_0} \left[1 + \beta_S^* (S - S_0) \right] - \alpha_0 p \left[\beta_T + \beta_T \gamma^* \frac{p}{2} + \beta_T^* (T - T_0) \right]. \quad (1.146)$$

For temperatures in the range 0° – 30° Celsius, entropy increases linearly with temperature to within a few percent.

- The heat capacity, $c_p = T(\partial \eta / \partial T)_{p,S}$:

$$c_p = c_{p0} \left[1 + \beta_S^* (S - S_0) \right] - \alpha_0 p \beta_T^* T. \quad (1.147)$$

This is to a first approximation constant, varying mildly with salinity and more weakly with temperature and pressure.

- The thermal expansion coefficient, $\hat{\beta}_T = \alpha^{-1}(\partial \alpha / \partial T)_{S,p}$:

$$\hat{\beta}_T = (\alpha_0 / \alpha) \left[\beta_T + \beta_T \gamma^* p + \beta_T^* (T - T_0) \right], \quad (1.148)$$

where α is given by (1.145).

- The adiabatic lapse rate, $\Gamma = (\partial T / \partial p)_{\eta,S}$. Using (1.138) gives

$$\Gamma = \left(\frac{\partial T}{\partial p} \right)_{\eta,S} = -\frac{T}{c_p} \left(\frac{\partial \eta}{\partial p} \right)_{T,S} = \frac{T}{c_p} \alpha_0 [\beta_T (1 + \gamma^* p) + \beta_T^* (T - T_0)]. \quad (1.149)$$

where c_p is given by (1.147).

Potential temperature and other approximations

An expression for the potential temperature, θ , may be obtained by solving (1.136) for θ . In general, such an equation must be solved numerically, but for our equation of state, using (1.146) and taking $p_R = 0$, we find

$$\theta = T \exp \left\{ -\frac{\alpha_0 \beta_T p}{c_p'} \left[1 + \frac{1}{2} \gamma^* p + \frac{\beta_T^*}{\beta_T} (T - T_0) \right] \right\}. \quad (1.150)$$

where $c_p' = c_{p0} [1 + \beta_S^* (S - S_0)]$. Equation (1.150) is a relationship between T , θ and p analogous to (1.105) for an ideal gas. The exponent itself is small, the second and third terms in square brackets are small compared to unity, the deviations of both T and θ from T_0 are also presumed to be small, and c_p' is nearly constant. Taking advantage of all of this enables (1.150) to be rewritten, with increasing levels of approximation, as

$$T' \approx \frac{T_0 \alpha_0 \beta_T}{c_{p0}} p \left(1 + \frac{1}{2} \gamma^* p + T_0 \frac{\alpha_0 \beta_T^*}{c_{p0}} p \right) + \theta' \left(1 + T_0 \frac{\alpha_0 \beta_T^*}{c_{p0}} p \right) \quad (1.151a)$$

$$\approx \frac{T_0 \alpha_0 \beta_T}{c_{p0}} p \left(1 + \frac{1}{2} \gamma^* p \right) + \theta' \left(1 + T_0 \frac{\alpha_0 \beta_T^*}{c_{p0}} p \right) \quad (1.151b)$$

$$\approx \frac{T_0 \alpha_0 \beta_T}{c_{p0}} p + \theta', \quad (1.151c)$$

where $T' = T - T_0$ and $\theta' = \theta - T_0$. The last of the three, (1.151c), holds for a linear equation of state, and is useful for calculating approximate differences between temperature and potential temperature; making use of the hydrostatic approximation reveals that it is essentially the same as (1.129). Plots of the difference between temperature and potential temperature, using both a highly accurate empirical equation of state and using the simplified equation, (1.150), are given in Fig. 1.4, and some examples of the density variation of seawater are given in Fig. 1.5.

It is sometimes useful to have equation of state that gives density in terms of potential temperature, pressure, and salinity. Using (1.151b) in the equation of state (1.145) we find that

$$\alpha \approx \alpha_0 \left[1 - \frac{\alpha_0 p}{c_{s0}'} + \beta_T (1 + \gamma'^* p) \theta' + \frac{1}{2} \beta_T^* \theta'^2 - \beta_S (S - S_0) \right], \quad (1.152)$$

where $\gamma'^* = \gamma^* + T_0 \beta_T^* \alpha_0 / c_{p0} \approx \gamma^*$ and $c_{s0}'^{-2} = c_{s0}^{-2} - \beta_T^2 T_0 / c_p \approx c_{s0}^{-2}$ (γ^* and γ'^* differ by a few percent, and c_s^2 and $c_s'^2$ differ by only a few parts in a thousand). We may approximate (1.152) further by using the hydrostatic pressure instead of the actual pressure; thus, letting $p = -g(z - z_0) / \alpha_0$ where z_0 is the nominal value of z at which $p = 0$, we obtain

$$\alpha \approx \alpha_0 \left[1 + \frac{g(z - z_0)}{c_{s0}^2} + \beta_T \left(1 - \gamma'^* \frac{g(z - z_0)}{\alpha_0} \right) \theta' + \frac{\beta_T^*}{2} \theta'^2 - \beta_S (S - S_0) \right]. \quad (1.153)$$

Using z instead of p in the equation of state entails a slight loss of accuracy, but is necessary to ensure that the Boussinesq equations maintain good conservation properties, as will be discussed in sections 2.4 and 4.7.1.

1.7.3 Potential enthalpy as a thermodynamic variable

As discussed in section 1.6 there are various forms of thermodynamic equation that are, in principle, equivalent. However, they are not equivalent in practice, even if the fundamental equation of state of is known. Thus, for example, for a nearly incompressible fluid like seawater the velocity

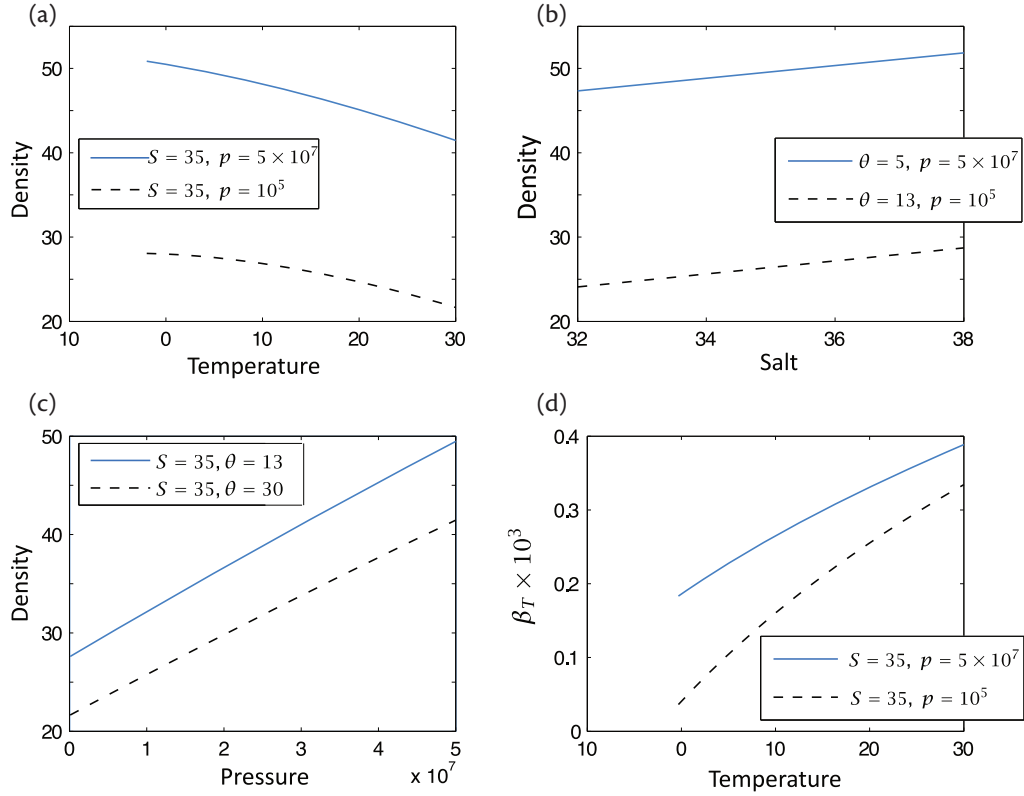


Fig. 1.5 Examples of the variation of density of seawater, ($\rho - 1000$) kg m⁻³. (a) With potential temperature (°C); (b) with salt (g/kg); and (c) with pressure (Pa), for seawater. Panel (d), shows the thermal expansion coefficient, $\beta_T = -\rho^{-1}(\partial\rho/\partial T)_{p,S}$ K⁻¹, for each of the two curves in panel (a).

divergence is small but non-zero and numerically integrating the internal energy equation, (1.94), may be awkward and inaccurate. An ideal choice might be some conservative variable that obeys

$$\rho \frac{D\chi}{Dt} = \nabla \cdot \mathbf{F}_\chi \quad \text{or equivalently} \quad \frac{\partial}{\partial t}(\rho\chi) + \nabla \cdot (\rho\mathbf{v}\chi) = \nabla \cdot \mathbf{F}_\chi, \quad (1.154)$$

where χ is a thermodynamical variable from which all the other variables can be inferred using the fundamental equation of state, and $\mathbf{F}_\chi$ represents molecular and radiative fluxes. Unfortunately, there is no variable that exactly has an advective left-hand side and a right-hand side that is the divergence of a flux — but some are nearly so, as we discuss.

Entropy

An obvious choice for a primary thermodynamic variable is entropy, an advected variable satisfying

$$\frac{D\eta}{Dt} = \frac{\dot{Q}}{T}, \quad (1.155)$$

where $\dot{Q}$ is the heating. The main difficulties with an entropic approach are the determination of the heating and the accurate treatment of that heating once determined. Consider even the case in which the heating is due solely to diffusive molecular fluxes and radiation. The right-hand side still does not have a conservative form, because the division by temperature means that the right-hand side is not a diffusive flux of entropy, nor of temperature plus salinity. A further, albeit

minor, complication is that heating is affected by irreversible molecular fluxes of salt in the interior. Further difficulties arise in using (1.155) in a model of a turbulent ocean. First, heating itself is affected not only by thermal effects but also by freshwater fluxes at the ocean surface and unresolved fluxes of salinity in the ocean interior. Care must also be taken to ensure that any parameterized fluxes of temperature and salinity do produce (and not reduce) entropy. Finally, the heating, even once determined, is still not in a conservative form and this can lead to numerical inaccuracies.

Potential temperature, θ , provides no fundamental advantage over entropy, because θ is a function of both entropy and salinity and still requires a computation of heating. Indeed, because $d\eta \approx c_p d\ln\theta$ the evolution equation for potential temperature properly involves the heat capacity, c_p , which is a function (albeit a weak one) of both salinity and pressure. Having said all this, the above problems are not in practice large ones, and entropy and (more commonly) potential temperature have been used very successfully in quantitative ocean models.

Potential enthalpy

An alternative to entropy is to construct a near-conservative variable related to enthalpy, as follows. Consider first the fundamental thermodynamic relation, (1.85), in the form

$$\frac{DI}{Dt} + p \frac{D\alpha}{Dt} = T \frac{D\eta}{Dt} + \mu \frac{DS}{Dt} = \frac{Dh}{Dt} - \alpha \frac{Dp}{Dt} = \dot{Q}_E. \quad (1.156)$$

where $\dot{Q}_E = \dot{Q} + \mu\dot{S}$ is the total rate of non-mechanical energy input per unit mass, $\dot{Q}$ is the heating and $\dot{S}$ represents saline sources and sinks. (With $\dot{Q}_E$, (1.156) constitutes the first law.) Now, it is nearly always easier and more accurate to determine energy input, $\dot{Q}_E$, than the heating, $\dot{Q}$. The reason for this is that $\dot{Q}_E$ is very nearly equal to the divergence of an energy flux; that is $\rho\dot{Q}_E \approx \nabla \cdot \mathbf{F}_E$ where $\mathbf{F}_E$ is an energy flux, with the error mainly arising from the small heating due to viscous dissipation of kinetic energy — see appendix on page 49 for details. This property suggests the use of internal energy or enthalpy as a primary thermodynamical variable, but clearly neither of these are conservative quantities. However we can form a quantity, *potential enthalpy*, that nearly is.¹⁰

The potential enthalpy, h^0 , is defined to be the enthalpy that a fluid parcel has if taken at constant composition and entropy from its current location to a fixed reference pressure, p_R . Thus,

$$h^0(S, \eta, p_R) \equiv h(S, \eta, p_R) + \int_p^{p_R} \left(\frac{\partial h}{\partial p} \right)_{S, \eta} dp' = h(S, \eta, p_R) + \int_p^{p_R} \alpha dp'. \quad (1.157)$$

using (1.74). That is, $h = h^0 + h^d$ where

$$h^d(\eta, p, S) = - \int_p^{p_R} \left(\frac{\partial h}{\partial p} \right)_{S, \eta} dp' = - \int_p^{p_R} \alpha dp' \quad (1.158)$$

is the ‘dynamic enthalpy’. The material derivative of the dynamic enthalpy is given by

$$\frac{Dh^d}{Dt} = \frac{D\eta}{Dt} \frac{\partial h^d}{\partial \eta} + \frac{DS}{Dt} \frac{\partial h^d}{\partial S} + \alpha \frac{Dp}{Dt}. \quad (1.159)$$

To obtain this expression we have used $(\partial h^d / \partial p)_{\eta, S} = (\partial h / \partial p)_{\eta, S} = \alpha$, because h^0 is not a function of pressure. Using (1.159) and (1.157) we obtain

$$\frac{Dh^0}{Dt} = \frac{Dh}{Dt} - \frac{Dh^d}{Dt} = \frac{Dh}{Dt} - \left[\alpha \frac{Dp}{Dt} + \frac{D\eta}{Dt} \frac{\partial h^d}{\partial \eta} + \frac{DS}{Dt} \frac{\partial h^d}{\partial S} \right], \quad (1.160)$$

and using the fundamental thermodynamic relation, (1.156), we can write this equation as

$$\frac{Dh^0}{Dt} = \dot{Q}_E - \frac{\partial h^d}{\partial \eta} \dot{\eta} - \frac{\partial h^d}{\partial S} \dot{S}. \quad (1.161)$$

The second and third terms on the right-hand side are much smaller than the other terms and can be neglected in most oceanographic applications. To see this, realize that an approximate size of the first term on the right-hand side is $c_p d\theta/dt$, whereas the approximate size of the second term is $(\partial h^d/\partial \eta) \times (c_p \theta^{-1} d\theta/dt)$. That is, the second term is smaller than the first by the factor

$$\gamma = \frac{1}{\theta} \frac{\partial h^d}{\partial \eta} = \frac{1}{c_p} \frac{\partial h_d}{\partial \theta} \sim \frac{1}{c_p} \frac{\partial \alpha}{\partial \theta} \Delta p \quad (1.162)$$

using (1.158), where $\Delta p = p - p_R \leq 4 \times 10^7$ Pa. Putting in values from the seawater equation of state, (1.152), we find

$$\gamma \sim \frac{1}{c_p} \frac{\partial \alpha}{\partial \theta} \Delta p \approx \frac{\beta_T \Delta p}{\rho_0 c_p} \approx \frac{1.7 \times 10^{-4}}{10^3} \frac{4 \times 10^7}{4 \times 10^3} \approx 1.7 \times 10^{-3}, \quad (1.163)$$

with a similarly small value for the saline term, where the smallness of these terms ultimately stems from the near incompressibility of seawater. These terms may actually be smaller than (1.163) suggests if we choose p_R appropriately. Most of the energy flux into the ocean occurs at the ocean surface and if we choose $p_R = 0$ the fluxes affect enthalpy and potential enthalpy in the same way.

Given the above arguments, an accurate and computable thermodynamic equation for seawater is (1.161) with the last two terms neglected and the source term in flux form, namely

$$\rho \frac{Dh^0}{Dt} = \nabla \cdot \mathbf{F}_E \quad \text{or} \quad \frac{\partial}{\partial t}(\rho h^0) + \nabla \cdot (\rho \mathbf{v} h^0) = \nabla \cdot \mathbf{F}_E. \quad (1.164)$$

Given that the right-hand side is a flux divergence, in steady state the interior fluxes of potential enthalpy are (in this approximation) in exact balance with the energy fluxes at the ocean boundaries. The integral of ρh^0 is thus a sensible measure of the total non-mechanical energy, or ‘heat content’, of the ocean because it responds almost exactly to energy fluxes at the ocean surface.¹¹

Why does internal energy not have these same advantages since, from (1.94), it too is conservative apart from a small compression term? The reason is that, because of the compression term, internal energy can be changed by processes that are entirely internal to the ocean, whereas enthalpy cannot. When two adjacent fluid parcels mix, the total enthalpy is conserved because of the form of the thermodynamic equation, $\rho Dh/Dt - Dp/Dt = \rho Q_E$. If ρQ_E is the divergence of a flux then the total enthalpy ($\int \rho h dV$) is conserved on mixing (because the parcels are at the same pressure), but this conservative property is not shared by internal energy, because of the $pD\alpha/Dt$ term in the internal energy equation. Indeed, when two parcels at different temperatures and salinities are mixed the density of the resulting parcel is higher than the average of the two (see Fig. 1.3); this effect is called cabbeling and the contraction leads to a small increase in internal energy. The differences in the conservation of enthalpy and internal energy arise from the differences in the equation of motion for enthalpy and internal energy and not directly from the equation of state. In principle, internal energy could be used as a primary thermodynamic variable in the ocean, but it is not a measure of ‘heat content’ and in practice the very small compression term would be hard to treat accurately. Internal energy often is used as a primary thermodynamic variable in ideal-gas atmospheric models, where $I = c_v T$.

Thus, in short, the practical advantages of potential enthalpy are three-fold. (i) Potential enthalpy is nearly conservative, as in (1.164). (ii) As a consequence, potential enthalpy is itself a useful measure of the non-kinetic energy of the ocean. (iii) Energy flux is a little more easily and more accurately computed than heating. Property (ii) is not shared by internal energy, since that is not conserved when parcels mix in the interior. In practice, potential enthalpy is almost proportional to potential temperature, as we now see.

Conservative temperature and an expression for potential enthalpy

To make a connection to the perhaps more familiar potential temperature it is convenient to define *conservative temperature*, Θ , by

$$\Theta \equiv \frac{h^0(\eta, p_R, S)}{c_p^0}, \quad (1.165)$$

where $c_p^0 = 3991.87 \text{ J kg}^{-1} \text{ K}^{-1}$ is the average heat capacity at the ocean surface and $p_R = 0$. (Potential enthalpy can have any reference pressure; conservative temperature has, by definition $p_R = 0$.) As we see below, conservative temperature is then very similar to potential temperature.

The enthalpy is given in terms of the Gibbs function by

$$h = g + T\eta = g - T \left(\frac{\partial g}{\partial T} \right)_{p,S}. \quad (1.166)$$

The potential enthalpy is therefore given by

$$h^0(S, T, p) = h(S, \theta, p_R) = g(S, \theta, p_R) - T \frac{\partial}{\partial T} g(S, \theta, p_R), \quad (1.167)$$

where θ is the potential temperature referenced to $p = p_R$ (and $p_R = 0$) and the last term is the derivative of the Gibbs function evaluated at $T = \theta$ and $p = p_R$. We can evaluate the derivative using the seawater equation of state, (1.144), giving, to within a constant factor,

$$\begin{aligned} h(T, p, S) = & \mu_0 S + c_{p0} T [1 + \beta_S^* (S - S_0)] \\ & + \alpha_0 p \left[1 - \beta_T T_0 - \beta_S (S - S_0) - \frac{\beta_P p}{2} - \frac{\beta_T \gamma^*}{2} p T_0 + \frac{\beta_T}{2} (T_0^2 - T^2) \right]. \end{aligned} \quad (1.168)$$

and

$$h^0(\theta, 0, S) = h(\theta, 0, S) = \theta c_{p0} [1 + \beta_S^* (S - S_0)] + \mu_0 S. \quad (1.169)$$

The last term on the right-hand side of the above equation, $\mu_0 S$, is over two orders of magnitude smaller than the first. Also, from (1.147), the factor multiplying θ in (1.169) is just the heat capacity at $p = 0$, namely c_p^0 , which varies only by few percent over the ocean. Thus, using the definition of conservative temperature given in (1.165), we have to a good approximation

$$c_p^0 \Theta \equiv h^0(\theta, 0, S) \approx c_p^0 \theta. \quad (1.170)$$

That is, to this approximation, conservative temperature equals potential temperature. Finally, if we are to use potential enthalpy (or entropy) as a thermodynamic variable we must manipulate the equation of state to obtain variables such as pressure or density from it, analogous to (1.152).

1.8 SOUND WAVES

Full of sound and fury, signifying nothing.

William Shakespeare, *Macbeth*, c. 1606.

We now consider, rather briefly, one of the most common phenomena in fluid dynamics, yet one which is in most circumstances relatively unimportant for geophysical fluid dynamics — sound waves. Their unimportance stems from the fact that the pressure disturbances produced by sound waves are a tiny fraction of the ambient pressure and are too small to affect the circulation. For example, the ambient surface pressure in the atmosphere is about 10^5 Pa and variations due to large-scale weather phenomena are about 10^3 Pa or larger, whereas sound waves of 70 dB (i.e., a loud

conversation) produce pressure variations of about 0.06 Pa. (To convert, $\text{dBs} = 20 \log_{10}(\Delta p/p_r)$ where Δp is the pressure change in Pascals and $p_r = 2 \times 10^{-5}$.)

The smallness of the disturbance produced by sound waves justifies a linearization of the equations of motion and we do so about a spatially uniform basic state that is a time-independent solution to the equations of motion. Thus, we write $\mathbf{v} = \mathbf{v}_0 + \mathbf{v}'$, $\rho = \rho_0 + \rho'$ (where a subscript 0 denotes a basic state and a prime denotes a perturbation) and so on, substitute in the equations of motion, and neglect terms involving products of primed quantities. By choice of our reference frame we will simplify matters further by setting $\mathbf{v}_0 = 0$. The linearized momentum and mass conservation equations are then, respectively,

$$\rho_0 \frac{\partial \mathbf{v}'}{\partial t} = -\nabla p', \quad \frac{\partial \rho'}{\partial t} = -\rho_0 \nabla \cdot \mathbf{v}'. \quad (1.171a,b)$$

(These linear equations do not in themselves determine the magnitude of the disturbance.) Now, sound waves are largely adiabatic. Thus,

$$\frac{dp}{dt} = \left(\frac{\partial p}{\partial \rho} \right)_\eta \frac{d\rho}{dt}, \quad (1.172)$$

where $(\partial p / \partial \rho)_\eta$ is the derivative at constant entropy, whose particular form is given by the equation of state for the fluid at hand. Then, from (1.171) and (1.172) we obtain a single equation for pressure,

$$\frac{\partial^2 p'}{\partial t^2} = c_s^2 \nabla^2 p', \quad (1.173)$$

where $c_s^2 = (\partial p / \partial \rho)_\eta$. Equation (1.173) is the classical wave equation; solutions propagate at a speed c_s , which may be identified as the speed of sound. For adiabatic flow in an ideal gas, manipulation of the equation of state leads to $p = C\rho^\gamma$, where $\gamma = c_p/c_v$, whence $c_s^2 = \gamma p / \rho = \gamma RT$. Values of γ typically range from 5/3 for a monatomic gas to 7/5 for a diatomic gas and so for air, which is almost entirely diatomic, we find $c_s \approx 350 \text{ m s}^{-1}$ at 300 K. In seawater no such theoretical approximation is easily available, but measurements show that $c_s \approx 1500 \text{ m s}^{-1}$.

1.9 COMPRESSIBLE AND INCOMPRESSIBLE FLOW

Although there may be no fluids of truly constant density, in many cases the density of a fluid will vary so little that it is a very good approximation to consider the density effectively constant. The fluid is then said to be *incompressible*. (Some sources take incompressible to mean that density is unaffected by pressure. We take it to mean that density is also unaffected by temperature and composition.) For example, in the Earth's oceans the density varies by less than 5% (usually much less) even though the pressure at the ocean bottom, a few kilometres below the surface, is several hundred times the atmospheric pressure at the surface. We first consider how the mass conservation equation simplifies when density is truly constant, and then consider conditions under which treating density as constant is a good approximation.

1.9.1 Constant density fluids

If a fluid is strictly of constant density then the mass continuity equation, $D\rho/Dt + \rho \nabla \cdot \mathbf{v} = 0$, simplifies easily by neglecting all derivatives of density yielding

$$\nabla \cdot \mathbf{v} = 0. \quad (1.174)$$

The *prognostic* equation (1.36) has become a *diagnostic* equation (1.174), or a constraint to be satisfied by the velocity. A consequence of (1.174) is that the volume of each material fluid element

remains constant, and to see this recall that conservation of mass is $D(\rho\Delta V)/Dt = 0$. If ρ is constant this reduces to an expression for volume conservation, $D\Delta V/Dt = 0$, whence (1.174) is recovered because $D\Delta V/Dt = \Delta V \nabla \cdot \mathbf{v}$.

1.9.2 Incompressible flows

In reality no fluid is truly incompressible and for (1.174) to approximately hold we just require that

$$\left| \frac{D\rho}{Dt} \right| \ll \rho \left(\left| \frac{\partial u}{\partial x} \right| + \left| \frac{\partial v}{\partial y} \right| + \left| \frac{\partial w}{\partial z} \right| \right); \quad (1.175)$$

that is, the material derivative of density is much smaller than the individual terms constituting the divergence. As a working definition we say that *in an incompressible fluid, density changes are so small that they have a negligible effect on the mass balance*. We do not need to assume that the densities of differing fluid elements are similar to each other, but in the ocean (and in most liquids) variations in density, $\delta\rho$, are in fact everywhere small compared to the mean density, ρ_0 . A sufficient condition for incompressibility, then, is that

$$\frac{\delta\rho}{\rho_0} \ll 1. \quad (1.176)$$

The fact that $\nabla \cdot \mathbf{v} = 0$ does *not* imply that we may independently use $D\rho/Dt = 0$. Indeed for a liquid with an equation of state $\rho = \rho_0(1 - \beta_T(T - T_0))$ and a thermodynamic equation $c_p DT/Dt = \dot{Q}$ we have

$$\frac{D\rho}{Dt} = -\frac{\beta_T \rho_0}{c_p} \dot{Q}. \quad (1.177)$$

Furthermore, incompressibility does not necessarily imply the neglect of density variations in the momentum equation — it is only in the mass continuity equation that density variations are neglected.

Conditions for incompressibility

The conditions under which incompressibility is a good approximation to the full mass continuity equation depend not only on the physical nature of the fluid but also on the flow itself. The condition that density is largely unaffected by pressure gives one necessary condition for the legitimate use of (1.174), as follows. First assume adiabatic flow, and omit the gravitational term. Then

$$\frac{dp}{dt} = \left(\frac{\partial p}{\partial \rho} \right)_\eta \frac{d\rho}{dt} = c_s^2 \frac{d\rho}{dt} \quad (1.178)$$

so that the density and pressure variations of a fluid parcel are related by

$$\delta p \sim c_s^2 \delta \rho. \quad (1.179)$$

From the momentum equation we estimate

$$\frac{U^2}{L} \sim \frac{1}{L} \frac{\delta p}{\rho_0}, \quad (1.180)$$

where U and L are typical velocities and lengths and where ρ_0 is a representative value of the density. Using (1.179) and (1.180) gives $U^2 \sim c_s^2 \delta \rho / \rho_0$. The incompressibility condition (1.176) then becomes

$$\frac{U^2}{c_s^2} \ll 1. \quad (1.181)$$

Thus, for a flow to be incompressible the fluid velocities must be less than the speed of sound; that is, the Mach number, $M \equiv U/c_s$, must be small.

In the Earth's atmosphere it is apparent that density changes with height. To estimate how much density does change, let us first assume hydrostatic balance and an ideal gas, so that $\partial p/\partial z = -\rho g$. If we also assume that atmosphere is isothermal then

$$\frac{\partial p}{\partial z} = \left(\frac{\partial p}{\partial \rho} \right)_T \frac{\partial \rho}{\partial z} = RT_0 \frac{\partial \rho}{\partial z}. \quad (1.182)$$

Using hydrostasy and (1.182) gives

$$\rho = \rho_0 \exp(-z/H_\rho), \quad (1.183)$$

where $H_\rho = RT_0/g$ is the (density) *scale height* of the atmosphere. (It is also the pressure scale height here.) It is easy to see that density changes are negligible only if we concern ourselves with motion less than the scale height, so this is another necessary condition for incompressibility.

In the atmosphere, although the Mach number is small for most flows, vertical displacements often exceed the scale height and the flow cannot then be considered incompressible. In the ocean density changes from all causes are small and in most circumstances the ocean may be considered to contain an incompressible fluid. We return to this in the next chapter when we consider the Boussinesq equations.

1.10 THE ENERGY BUDGET

The total energy of a fluid includes the kinetic, potential and internal energies. Both fluid flow and pressure forces will, in general, move energy from place to place, but we nevertheless expect energy to be conserved in an enclosed volume — in fact we demand it. Let us now consider what form energy conservation takes in a fluid.

1.10.1 Constant density fluid

For a constant density fluid the momentum equation and the mass continuity equation $\nabla \cdot \mathbf{v} = 0$, are sufficient to completely determine the evolution of a system. The momentum equation is

$$\frac{D\mathbf{v}}{Dt} = -\nabla(\phi + \Phi) + \nu \nabla^2 \mathbf{v}, \quad (1.184)$$

where $\phi = p/\rho_0$ and Φ is the potential for any conservative force per unit mass (e.g., gz for a uniform gravitational field). We can rewrite the advective term on the left-hand side using the identity,

$$(\mathbf{v} \cdot \nabla)\mathbf{v} = -\mathbf{v} \times \boldsymbol{\omega} + \nabla(v^2/2), \quad (1.185)$$

where $\boldsymbol{\omega} \equiv \nabla \times \mathbf{v}$ is the *vorticity*, discussed more in later chapters. Then, omitting viscosity, we have

$$\frac{\partial \mathbf{v}}{\partial t} + \boldsymbol{\omega} \times \mathbf{v} = -\nabla B, \quad (1.186)$$

where $B = (\phi + \Phi + v^2/2)$ is the *Bernoulli function* for constant density flow. Consider for a moment steady flows ($\partial \mathbf{v}/\partial t = 0$). Streamlines are, by definition, parallel to $\mathbf{v}$ everywhere, and the vector $\mathbf{v} \times \boldsymbol{\omega}$ is everywhere orthogonal to the streamlines, so that taking the dot product of the steady version of (1.186) with $\mathbf{v}$ gives $\mathbf{v} \cdot \nabla B = 0$. That is, for steady flows the Bernoulli function is constant along a streamline, and $DB/Dt = 0$.

Reverting back to the time-varying case, take the dot product of (1.186) with $\mathbf{v}$ and include the density to yield

$$\frac{1}{2} \frac{\partial \rho_0 \mathbf{v}^2}{\partial t} + \rho_0 \mathbf{v} \cdot (\boldsymbol{\omega} \times \mathbf{v}) = -\rho_0 \mathbf{v} \cdot \nabla B \quad (1.187)$$

The second term on the left-hand side vanishes identically. Defining the kinetic energy density K , or energy per unit volume, by $K = \rho_0 \mathbf{v}^2/2$, (1.187) becomes an expression for the rate of change of K ,

$$\frac{\partial K}{\partial t} + \nabla \cdot (\rho_0 \mathbf{v} B) = 0. \quad (1.188)$$

Because Φ is time-independent this may be written

$$\frac{\partial E}{\partial t} + \nabla \cdot (\rho_0 \mathbf{v} B) = 0. \quad (1.189)$$

where $E = K + \rho_0 \Phi$ is the total energy density (i.e, the total energy per unit volume). This has the form of a general conservation equation in which a local change in a quantity is balanced by the divergence of its flux. However, the energy flux, $\rho_0 \mathbf{v} B = \rho_0 \mathbf{v} (\mathbf{v}^2/2 + \Phi + \phi)$, is *not* simply the velocity times the energy density $\rho_0 (\mathbf{v}^2/2 + \Phi)$; there is an additional term, $\mathbf{v} p$, that represents the energy transfer occurring when work is done by the fluid against the pressure force.

Now consider a volume through which there is no mass flux, for example a domain bounded by rigid walls. The rate of change of energy within that volume is then given by the integral of (1.189),

$$\frac{d}{dt} \int_V E dV = - \int_V \nabla \cdot (\rho_0 \mathbf{v} B) dV = - \int_S \rho_0 B \mathbf{v} \cdot d\mathbf{S} = 0, \quad (1.190)$$

using the divergence theorem. Thus, the total energy within the volume is conserved. The total kinetic energy is also conserved, total gravitational potential energy is equal to $\int_V \rho_0 g z dV$, and this is a constant, unaffected by a rearrangement of the fluid. Thus, in a constant density fluid there is no exchange between kinetic energy and potential energy.

1.10.2 Variable density fluids

We start with the inviscid momentum equation with a time-independent potential Φ ,

$$\rho \frac{D\mathbf{v}}{Dt} = -\nabla p - \rho \nabla \Phi, \quad (1.191)$$

and take its dot product with $\mathbf{v}$ to obtain an equation for the evolution of kinetic energy,

$$\frac{1}{2} \rho \frac{D\mathbf{v}^2}{Dt} = -\mathbf{v} \cdot \nabla p - \rho \mathbf{v} \cdot \nabla \Phi = -\nabla \cdot (p\mathbf{v}) + p \nabla \cdot \mathbf{v} - \rho \mathbf{v} \cdot \nabla \Phi. \quad (1.192)$$

The internal energy equation for adiabatic flow is

$$\rho \frac{DI}{Dt} = -p \nabla \cdot \mathbf{v}. \quad (1.193)$$

Finally, and somewhat trivially, the potential energy density obeys

$$\rho \frac{D\Phi}{Dt} = \rho \mathbf{v} \cdot \nabla \Phi. \quad (1.194)$$

Adding (1.192), (1.193) and (1.194) we obtain

$$\rho \frac{D}{Dt} \left(\frac{1}{2} \mathbf{v}^2 + I + \Phi \right) = -\nabla \cdot (p\mathbf{v}), \quad (1.195)$$

which, on expanding the material derivative and using the mass conservation equation, becomes

$$\frac{\partial}{\partial t} \left[\rho \left(\frac{1}{2} \mathbf{v}^2 + I + \Phi \right) \right] + \nabla \cdot \left[\rho \mathbf{v} \left(\frac{1}{2} \mathbf{v}^2 + I + \Phi + p/\rho \right) \right] = 0. \quad (1.196)$$

This may be written

$$\frac{\partial E}{\partial t} + \nabla \cdot [\mathbf{v}(E + p)] = 0, \quad (1.197)$$

where $E = \rho(\mathbf{v}^2/2 + I + \Phi)$ is the total energy per unit volume of the fluid. This is the energy equation for an unforced, inviscid and adiabatic, compressible fluid. The energy flux term vanishes when integrated over a closed domain with rigid boundaries, implying that the total energy is conserved. However, there can be an exchange of energy between kinetic, potential and internal components. It is the divergent term, $\nabla \cdot \mathbf{v}$, that connects the kinetic energy equation, (1.192), and the internal energy equation, (1.193). In an incompressible fluid this term is absent, and the internal energy is divorced from the other components of energy. This consideration will be important when we consider the Boussinesq equations in section 2.4. Note finally that the flux of energy, $\mathbf{F}_E = \mathbf{v}(E + p)$ is not equal to the velocity times the energy; rather, energy is also transferred by pressure. We may write the energy flux as

$$\mathbf{F}_E = \rho \mathbf{v} \left(\frac{\mathbf{v}^2}{2} + \Phi + h \right). \quad (1.198)$$

where $h = I + p/\rho$ is the enthalpy. That is, the local rate of change of energy is effected by the fluxes of kinetic and potential energy and *enthalpy*, not internal energy.

Bernoulli's theorem

The quantity

$$B = \left(E + \frac{p}{\rho} \right) = \left(\frac{1}{2} \mathbf{v}^2 + I + \Phi + p/\rho \right) = \left(\frac{1}{2} \mathbf{v}^2 + h + \Phi \right), \quad (1.199)$$

is the general form of the Bernoulli function, equal to the sum of the kinetic energy, the potential energy and the enthalpy. Equation (1.197) may be written as

$$\frac{\partial E}{\partial t} + \nabla \cdot (\rho \mathbf{v} B) = 0. \quad (1.200)$$

This equation may also be written $\partial(\rho B)/\partial t + \nabla \cdot (\rho \mathbf{v} B) = \partial p/\partial t$, so obviously the Bernoulli function itself is not conserved, even for adiabatic flow. For steady flow $\nabla \cdot (\rho \mathbf{v}) = 0$, and the $\partial/\partial t$ terms vanish so that (1.200) may be written $\mathbf{v} \cdot \nabla B = 0$, or even $DB/Dt = 0$. The Bernoulli function is then a constant along streamline, a result commonly known as Bernoulli's theorem.¹² For adiabatic flow at constant composition we also have $D\theta/Dt = 0$. Thus, steady flow is both along surfaces of constant θ and along surfaces of constant B , and the vector

$$\mathbf{l} = \nabla \theta \times \nabla B \quad (1.201)$$

is parallel to streamlines. A related result for unsteady flow is given in section 4.8.

Viscous effects

We might expect that viscosity will always act to reduce the kinetic energy of a flow, and we will demonstrate this for a constant density fluid. Retaining the viscous term in (1.184), the energy equation becomes

$$\frac{d\hat{E}}{dt} \equiv \frac{d}{dt} \int_V E dV = \mu \int_V \mathbf{v} \cdot \nabla^2 \mathbf{v} dV. \quad (1.202)$$

The right-hand side is negative definite. To see this we use the vector identity

$$\nabla \times (\nabla \times \mathbf{v}) = \nabla(\nabla \cdot \mathbf{v}) - \nabla^2 \mathbf{v}, \quad (1.203)$$

and because $\nabla \cdot \mathbf{v} = 0$ we have $\nabla^2 \mathbf{v} = -\nabla \times \boldsymbol{\omega}$, where $\boldsymbol{\omega} \equiv \nabla \times \mathbf{v}$. Thus,

$$\frac{d\hat{E}}{dt} = -\mu \int_V \mathbf{v} \cdot (\nabla \times \boldsymbol{\omega}) dV = -\mu \int_V \boldsymbol{\omega} \cdot (\nabla \times \mathbf{v}) dV = -\mu \int_V \boldsymbol{\omega}^2 dV, \quad (1.204)$$

after integrating by parts, providing $\mathbf{v} \times \boldsymbol{\omega}$ vanishes at the boundary. Thus, viscosity acts to extract kinetic energy from the flow. The loss of kinetic energy reappears as an irreversible warming of the fluid (called ‘Joule heating’), and the total energy of the fluid is conserved, but this effect plays no role in a constant density fluid. The heating is normally locally small, at least in the Earth’s ocean and atmosphere, but it is needed to preserve total energy.

1.10.3 Enthalpy, static energy and energy flux

We saw above that it is the quantity

$$B = \text{kinetic energy} + \text{potential energy} + \text{enthalpy}, \quad (1.205)$$

that, when multiplied by $\rho \mathbf{v}$, fluxes the energy. However, the conserved energy contains only the kinetic and potential energies plus the internal energy. The difference between B and energy is the fluid dynamical analogue of the thermodynamical principle that it is the flux of *enthalpy* that changes the energy of a system, because this accounts for work done by the pressure.

For an example, consider an ideal gas in a uniform gravitational field for which $I = c_v T$ and $p\alpha = RT$ so that $B = c_v T + RT + KE + gz = c_p T + KE + gz$. The sum of the enthalpy and the potential energy,

$$h_d^* \equiv c_p T + gz, \quad (1.206)$$

is known as the *dry static energy*. Dry static energy is not an integral conserved quantity nor is it a measure of the total energy itself. To see the importance of enthalpy fluxes, consider an adiabatic rearrangement of a fluid with $d\eta = 0$. From (1.87) we then have $dI = -p d\alpha$, meaning that the internal energy changes because of work done. At the same time, from (1.72) with $d\eta = 0$ we also have $dh = \alpha dp$. For a fluid in hydrostatic balance, (1.51), we have that $dp = -g dz/\alpha$ and thus

$$d(h + gz) = 0. \quad (1.207)$$

That is to say, the change in potential energy of a parcel arises from a change in the enthalpy, and not just from changes in the internal energy, because of the work that must be done by the pressure force to move the parcel. Evidently, the static energy $h + gz$ of a parcel is adiabatically conserved if the pressure changes are hydrostatic.

The conservation law (1.207) is of a different form than the total energy conservation previously derived. Equation (1.197) is an *integral* conservation law, whereas (1.207) is a *parcel* conservation law, which we might write in fluid dynamical form as

$$\frac{D}{Dt}(h + gz) = 0. \quad (1.208)$$

Equation (1.208) is in fact a form of the thermodynamic equation for adiabatic changes to an ideal gas in hydrostatic balance, similar to (1.126) or (1.141). The equivalence arises because changes in h_d^* and θ are related by $dh_d^* = c_p dT + g dz = c_p (T/\theta) d\theta$, as in (1.127).

Potential enthalpy (and see section 1.7.3 for a oceanographic discussion) is defined to be the enthalpy that a parcel would have if adiabatically moved to a reference pressure, and since the

enthalpy for an ideal gas is $c_p T$, the potential enthalpy is just $c_p \theta$. The dry static energy, $c_p T + gz$, is the enthalpy that a parcel at a height z and temperature T would have if moved adiabatically and hydrostatically to $z = 0$, and is thus a special form of potential enthalpy. (The dry static energy is also sometimes called ‘generalized enthalpy’.) Energy and enthalpy both play major roles in the chapters ahead, but let’s round off this chapter with an introduction to scaling in a pure fluid-dynamical setting.

1.11 AN INTRODUCTION TO NON-DIMENSIONALIZATION AND SCALING

The units we use to measure length, velocity and so on are irrelevant to the dynamics, and commonly used ones (e.g., SI units) are not necessarily the most appropriate for a given problem. Rather, it is useful to express the equations of motion, so far as is possible, in terms of ‘non-dimensional’ variables, by which we mean expressing every variable (such as velocity) as the ratio of its value to some reference value. For velocity the reference could be the speed of light — but this would not be very helpful for fluid problems in the atmosphere or oceans! Instead, we should choose the reference value as a natural one for a given flow, in order that, so far as possible, the non-dimensional variables are order-unity quantities, and doing this is called *scaling the equations*. Evidently, there is no reference velocity that is universally appropriate, and much of the art of fluid dynamics lies in choosing sensible scaling factors for the problem at hand. Non-dimensionalization plays an important role in fluid dynamics, and we introduce it here with a simple example.

1.11.1 The Reynolds number

Consider the constant-density momentum equation in Cartesian coordinates. If a typical velocity is U , a typical length is L , a typical time scale is T , and a typical value of the pressure deviation is Φ , then the approximate sizes of the various terms in the momentum equation are given by

$$\frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v} \cdot \nabla) \mathbf{v} = -\nabla \phi + \nu \nabla^2 \mathbf{v} \quad (1.209a)$$

$$\frac{U}{T} \quad \frac{U^2}{L} \quad \sim \quad \frac{\Phi}{L} \quad \nu \frac{U}{L^2}. \quad (1.209b)$$

The ratio of the inertial terms to the viscous terms is $(U^2/L)/(\nu U/L^2) = UL/\nu$, and this is the *Reynolds number*.¹³ More formally, we can non-dimensionalize the momentum equation by writing

$$\hat{\mathbf{v}} = \frac{\mathbf{v}}{U}, \quad \hat{\mathbf{x}} = \frac{\mathbf{x}}{L}, \quad \hat{t} = \frac{t}{T}, \quad \hat{\phi} = \frac{\phi}{\Phi}, \quad (1.210)$$

where the terms with hats on are *non-dimensional* values of the variables and the capitalized quantities are known as *scaling values*, and these are the approximate magnitudes of the variables. We now choose the scaling values so that the non-dimensional variables are of order unity, or $\hat{u} = \mathcal{O}(1)$. Thus, for example, we choose U so that $u = \mathcal{O}(U)$, where this notation should be taken to mean that the magnitude of the variable u is of order U , or that $u \sim U$, and we say that ‘ u scales like U ’. Scaling like this is most easily done when there are, in fact, well-defined scales in the problem.

Because there are no external forces in this problem, appropriate scaling values for time and pressure are

$$T = \frac{L}{U}, \quad \Phi = U^2. \quad (1.211)$$

Substituting (1.210) and (1.211) into the momentum equation gives

$$\frac{U^2}{L} \left[\frac{\partial \hat{\mathbf{v}}}{\partial \hat{t}} + (\hat{\mathbf{v}} \cdot \nabla) \hat{\mathbf{v}} \right] = -\frac{U^2}{L} \nabla \hat{\phi} + \frac{\nu U}{L^2} \nabla^2 \hat{\mathbf{v}}, \quad (1.212)$$

where we use the convention that when ∇ operates on a non-dimensional variable it is a non-dimensional operator. Equation (1.212) simplifies to

$$\frac{\partial \hat{\mathbf{v}}}{\partial \hat{t}} + (\hat{\mathbf{v}} \cdot \nabla) \hat{\mathbf{v}} = -\nabla \hat{\phi} + \frac{1}{Re} \nabla^2 \hat{\mathbf{v}}, \quad (1.213)$$

where

$$Re \equiv \frac{UL}{\nu} \quad (1.214)$$

is, again, the Reynolds number. If we have chosen our length and velocity scales sensibly — that is, if we have scaled them properly — each variable in (1.213) is order unity, with the viscous term being multiplied by the parameter $1/Re$. There are two important conclusions:

- (i) The ratio of the importance of the inertial terms to the viscous terms is given by the *Reynolds number*, defined by (1.214). In the absence of other forces, such as those due to gravity and rotation, the Reynolds number is the only non-dimensional parameter explicitly appearing in the momentum equation. Hence its value, along with the boundary conditions and geometry, controls the behaviour of the system.
- (ii) More generally, by scaling the equations of motion appropriately the parameters determining the behaviour of the system become explicit. *Scaling the equations is intelligent non-dimensionalization.*

Non-dimensionalizing the equations does not, however, absolve the investigator from the responsibility of producing dimensionally correct equations. One should regard ‘non-dimensional’ equations as dimensional equations in units appropriate for the problem at hand.

1.A APPENDIX A: THERMODYNAMICS OF AN IDEAL GAS FROM THE GIBBS FUNCTION

All the thermodynamic quantities of interest for a simple ideal gas, sometimes called a perfect gas, may be derived in a straightforward way from the fundamental equation of state. (An analogous treatment for moist air is given in the appendix to chapter 18 on page 720.) To show this we begin with the specific Gibbs function, $g = I - T\eta + p\alpha$, which for an ideal gas is given by

$$g = \mathcal{C}_p T(1 - \ln T) + \mathcal{R} T \ln p + BT + C. \quad (1.215)$$

where $\mathcal{C}_p$, $\mathcal{R}$, B and C are constants (and our notation anticipates what $\mathcal{C}_p$ and $\mathcal{R}$ really are). The procedure below is especially useful when the Gibbs function is more complex than (1.215), with the main difficulty then lying in obtaining the Gibbs function in the first instance.

Density and the thermal equation of state

From the fundamental relation involving the Gibbs function, (1.78a), specific volume is given by

$$\alpha = \left(\frac{\partial g}{\partial p} \right)_T = \frac{\mathcal{R}T}{p}, \quad \text{or} \quad p = \rho \mathcal{R}T. \quad (1.216)$$

Thus, $\mathcal{R}$, the gas constant used elsewhere in this chapter, is equal to $\mathcal{R}$.

Entropy

$$\eta = - \left(\frac{\partial g}{\partial T} \right)_p = \mathcal{C}_p \ln T - \mathcal{R} \ln p - B \quad (1.217)$$

Using (1.216) and (1.219) in (1.217), the entropy may be expressed in terms of I and α , to wit $\eta = (\mathcal{C}_p - \mathcal{R}) \ln I - \mathcal{R} \ln \rho + \text{constant}$, as in (1.111).

The internal energy

Using (1.216) and (1.217) in the definition of the Gibbs function, $g = I - T\eta + p\alpha$, gives

$$I = g + T\eta - p\alpha = g - T \left(\frac{\partial g}{\partial T} \right)_p - p \left(\frac{\partial g}{\partial p} \right)_T, \quad (1.218)$$

and so

$$I = (\mathcal{C}_p - \mathcal{R})T + C = \mathcal{C}_v T + C \quad (1.219)$$

where $\mathcal{C}_v \equiv \mathcal{C}_p - \mathcal{R}$. Equation (1.219) also suggests we may sensibly take $C = 0$, but no physical result in classical mechanics depends on this choice.

Heat capacities

Let c_p be the heat capacity at constant pressure. We have

$$c_p \equiv T \left(\frac{\partial \eta}{\partial T} \right)_p = \mathcal{C}_p. \quad (1.220)$$

To obtain the heat capacity at constant volume, c_v , first rewrite the entropy using the thermal equation of state as

$$\eta = (\mathcal{C}_p \ln T - \mathcal{R} \ln T + \mathcal{R} \ln \alpha) + \text{constant} \quad (1.221)$$

We then have

$$c_v \equiv T \left(\frac{\partial \eta}{\partial T} \right)_\alpha = \mathcal{C}_p - \mathcal{R}. \quad (1.222)$$

Adiabatic lapse rate

The adiabatic lapse rate, Γ_p , is the rate of change of temperature with pressure at constant entropy. Thus

$$\Gamma_p \equiv \left(\frac{\partial T}{\partial p} \right)_\eta \quad (1.223)$$

Now, from (1.138) we have

$$\left(\frac{\partial T}{\partial p} \right)_\eta = -\frac{T}{\mathcal{C}_p} \left(\frac{\partial \eta}{\partial p} \right)_T, \quad (1.224)$$

and using (1.217) to evaluate the right-hand side gives

$$\Gamma_p = -\frac{T}{\mathcal{C}_p} \left(\frac{\partial \eta}{\partial p} \right)_T = \frac{T}{\mathcal{C}_p} \frac{\mathcal{R}}{p} = \frac{1}{\mathcal{C}_p \rho}. \quad (1.225)$$

using the thermal equation of state. It is common to write this expression in terms of a rate of change of temperature with respect to height by using the hydrostatic approximation, $dp = -\rho g dz$, whence

$$\Gamma_z \equiv -\left(\frac{\partial T}{\partial z} \right)_\eta = \frac{g}{\mathcal{C}_p}. \quad (1.226)$$

The physical significance of this quantity is explored in chapter 2.

Potential temperature

As in (1.136), potential temperature, θ , satisfies

$$\eta(T, p) = \eta(\theta, p_R). \quad (1.227)$$

That is to say, θ is the temperature that a parcel will have if moved at constant entropy from a pressure p to a reference pressure p_R . Using (1.217) gives

$$\mathcal{C}_p \ln T - \mathcal{R} \ln p = \mathcal{C}_p \ln \theta - \mathcal{R} \ln p_R, \quad \text{or} \quad \ln(T/\theta)^{\mathcal{C}_p} = \ln(p/p_R)^{\mathcal{R}}. \quad (1.228)$$

Re-arranging gives

$$\theta = T \left(\frac{p_R}{p} \right)^{\mathcal{R}/\mathcal{C}_p}. \quad (1.229)$$

We can derive the same result by noting that, by definition, potential temperature satisfies

$$\theta \equiv T(p_R) = T(p) + \int_p^{p_R} \left(\frac{\partial T}{\partial p'} \right)_\eta dp' = T(p) + \int_p^{p_R} \frac{\mathcal{R}T}{\mathcal{C}_p p'} dp' \quad (1.230)$$

where the rightmost expression uses (1.225). It is easy to verify that the solution to this integral equation is $T = \theta(p/p_R)^{\mathcal{R}/\mathcal{C}_p}$, although solving the equation ab initio is a little more difficult. Finally, in an ideal gas potential temperature can be related to entropy using (1.217) and (1.229), giving $\eta = \mathcal{C}_p \ln \theta + \text{constant}$.

Enthalpy and potential enthalpy

Enthalpy is related to the Gibbs function by

$$h = g + T\eta = g - T \left(\frac{\partial g}{\partial T} \right)_p = \mathcal{C}_p T. \quad (1.231)$$

Potential enthalpy, h^0 , is the enthalpy that a parcel would have if moved adiabatically to a reference pressure. It therefore satisfies an equation similar to (1.227), namely $\eta(h, p) = \eta(h^0, p_R)$. Since $h = \mathcal{C}_p T$ we immediately obtain

$$h^0 = \mathcal{C}_p \theta. \quad (1.232)$$

That is, the potential enthalpy is equal to the potential temperature times the heat capacity at constant pressure.

1.B APPENDIX B: THE FIRST LAW OF THERMODYNAMICS FOR FLUIDS

In its usual form the first law of thermodynamics states that changes in the internal energy of a body are equal to the sum of heat supplied, the work done, and changes due to composition (the chemical work), and in a reversible process the entropy of a body then changes according to the heat supplied. However, the heating is not known a priori and the first law may be regarded as a definition of heating by way of energy conservation; heating should then be considered a derived quantity, as we noted in our discussion of (1.61) and (1.62). The real problem is to determine what the heating actually is, and this is not wholly trivial for fluids that also have mechanical energy and viscosity, as well as thermal and compositional diffusion. The first law is not a statement of total energy conservation, since it does not involve kinetic energy.

To obtain an unambiguous prescription for the heating — and hence a meaningful thermodynamic equation — we *begin* with total energy conservation and work backwards to obtain an prescription for the heating, and since energy conservation is the more fundamental physical law

this procedure is natural.¹⁴ Effectively, we subtract off the evolution of mechanical energy from an equation for total energy conservation, and the remainder is the thermodynamic equation. Equivalently (and this is how we proceed below) we demand consistency between total energy conservation and an energy equation derived in a forward fashion using a thermodynamic equation representing the first law but with the heating unspecified, and thereby deduce an explicit expression for that heating. Underlying this procedure is the fundamental notion that the work done and the generation of mechanical energy are less ambiguous than heating because they can be measured and/or arise through well-defined forces in the momentum equation. We will assume that energy and energy fluxes are knowable or calculable, but we will not discuss the questions of what energy fundamentally is or why it is conserved, or why entropy increases.

For reference, we first write down the fundamental thermodynamic relation, (1.85), and its inputs,

$$\frac{DI}{Dt} + p \frac{D\alpha}{Dt} = T \frac{D\eta}{Dt} + \mu \frac{DS}{Dt} = \frac{Dh}{Dt} - \alpha \frac{Dp}{Dt} = \dot{Q}_E, \quad (1.233)$$

where $\dot{Q}_E = \dot{Q} + \mu\dot{S}$ accounts for the total energy input to a fluid parcel from both heating, $\dot{Q}$, and compositional changes, $\mu\dot{S}$. The inclusion of the term $\dot{Q}_E$ connects the above equation to the first law, but the first law is not useful until we know what the heating is.

1.13.1 Single component fluid

Consider the energetics of a fluid as in section 1.10.2, with two additional effects: the fluid is viscous, and there is an additional energy source in the fluid, for example radiation or thermal conduction, that does not appear in the momentum equation. We write the momentum equation, (1.191), in Cartesian tensor notation (where repeated indices are summed) as

$$\frac{Dv_i}{Dt} = -\frac{1}{\rho} \frac{\partial p}{\partial x_i} - \frac{\partial \Phi}{\partial x_i} + \nu \frac{\partial^2 v_i}{\partial x_j \partial x_j}. \quad (1.234)$$

where Φ is a time-independent external potential field, such as gz . The form of the viscous term is exact only for incompressible fluids with constant viscosity but it is usually an excellent approximation in the atmosphere and ocean, because the Mach number is small and the scale on which dissipation occurs is very much smaller than the density scale height. (In any case those restrictions can be relaxed.) We also write the entropy equation as

$$T \frac{D\eta}{Dt} = \dot{Q}, \quad (1.235)$$

where $\dot{Q}$ is the heating, whose form we do not yet know. In a single component fluid we can use the fundamental thermodynamic relation, (1.233), to write (1.235) as an internal energy equation,

$$\rho \frac{DI}{Dt} + p \nabla \cdot \mathbf{v} = \rho \dot{Q}, \quad (1.236)$$

also having used $D\alpha/Dt = \alpha \nabla \cdot \mathbf{v}$.

We may obtain a kinetic energy equation by multiplying (1.234) by ρv_i to give

$$\frac{1}{2} \rho \frac{Dv_i^2}{Dt} = -\partial_i(pv_i) + p\partial_i v_i - \rho v_i \partial_i \Phi + \nu v_i \partial_j (\partial_j v_i), \quad (1.237)$$

where $\partial_i \equiv \partial/\partial x_i$. The viscous term may be written as

$$\nu v_i \partial_j (\partial_j v_i) = \nu \left[\partial_j (v_i \partial_j v_i) - (\partial_j v_i)^2 \right]. \quad (1.238)$$

The first term on the right hand side is the divergence of a flux, and so is energy conserving, and the second term is negative definite, representing kinetic energy dissipation.

We now proceed, just as in section 1.10.2, to obtain a total energy equation from (1.236) and (1.237) and the result is

$$\frac{\partial}{\partial t} \left[\rho \left(\frac{1}{2} v_i^2 + I + \Phi \right) \right] + \partial_i \left[\rho v_i \left(\frac{1}{2} v_j^2 + I + \Phi + p/\rho \right) \right] = \nu \left[\partial_j (v_i \partial_j v_i) - (\partial_j v_i)^2 \right] + \rho \dot{Q}. \quad (1.239)$$

Now, the general form of the energy conservation law for a fluid must take the form

$$\frac{\partial}{\partial t} \left[\rho \left(\frac{1}{2} v_i^2 + I + \Phi \right) \right] + \partial_i \left[\rho v_i \left(\frac{1}{2} v_j^2 + I + \Phi + p/\rho \right) \right] = \partial_j \left[\nu v_i \partial_j v_i + F_{Ej} \right], \quad (1.240)$$

where F_{Ej} (or F_E , and $\partial_j F_{Ej} = \nabla \cdot \mathbf{F}_E$) is the total energy flux due to radiation, conduction and any other effects — the important point being that the right-hand side of (1.240) must be the divergence of a flux in order to guarantee energy conservation. We have not used the first law to obtain (1.240), just the fundamental thermodynamic relation, the momentum equation, and energy conservation. The above two equations are consistent only if the heating term has the form $\rho \dot{Q} = \partial_j F_{Ej} + \nu (\partial_j v_i)^2$, and the internal energy and entropy equations are then

$$\rho \frac{DI}{Dt} + p \nabla \cdot \mathbf{v} = \nabla \cdot \mathbf{F}_E + \nu (\partial_j v_i)^2, \quad \rho T \frac{D\eta}{Dt} = \nabla \cdot \mathbf{F}_E + \nu (\partial_j v_i)^2. \quad (1.241a,b)$$

Evidently, the ‘heating’ of a fluid is given by the sum of the energy fluxes and a positive definite term due to viscous dissipation. Either of the above equivalent equations may be considered to be statements of the first law with explicit expressions for energy input and heating, and either of them then provides a useful predictive thermodynamic equation for a fluid.

1.13.2 Multi-component fluids

Consider now a two component fluid, such as dry air and water vapour or water and salinity. We refer to the second component as concentration and we assume it obeys

$$\rho \frac{DS}{Dt} = \nabla \cdot \mathbf{F}_S. \quad (1.242)$$

The only difference with the previous derivation is that, using the fundamental relation, the internal energy equation is now

$$\rho \frac{DI}{Dt} + p \nabla \cdot \mathbf{v} = \dot{Q} + \mu \nabla \cdot \mathbf{F}_S, \quad (1.243)$$

where μ is the chemical potential and the second term on the right-hand side accounts for the effects of concentration fluxes on internal energy (i.e., the chemical work). Proceeding to calculate the energy equation as before, we find that (1.239) then has an additional term $\mu \nabla \cdot \mathbf{F}_S$ on the right-hand side. This is consistent with (1.240) only if the internal energy and entropy equations obey

$$\rho \frac{DI}{Dt} + p \nabla \cdot \mathbf{v} = \nabla \cdot \mathbf{F}_E + \nu (\partial_j v_i)^2, \quad \rho T \frac{D\eta}{Dt} = \nabla \cdot \mathbf{F}_E - \mu \nabla \cdot \mathbf{F}_S + \nu (\partial_j v_i)^2, \quad (1.244a,b)$$

where the energy flux, F_E , now includes the effects of any fluxes of composition. Again, either of the above two equations are statements of the first law, and the right-hand side of (1.244b) is the heating. Additional terms may appear on the right-hand sides of the above equations if there are additional source or sink terms, for example a source of concentration. Note too, for example, that

a wind-stress may increase the kinetic energy of the ocean via a boundary condition in the viscous term, and the recipe above provides a way to decide whether this affects in the heating in any way.

Equation (1.244) superficially differs from the single component case only by the addition of a concentration flux. However, in a two component fluid the diffusion of temperature and concentration are affected by gradients of both temperature and concentration, so that the heat flux itself differs from the single component case. Thermodynamics provides constraints on these fluxes, but the reader must look elsewhere to learn about them¹⁵ — although it is easy to show that, in a single component fluid, total entropy increase follows if the heat flux is proportional to a downgradient temperature flux. Finally, in both oceanography and meteorology the viscous heating term is usually small, at least on Earth, but it must be included if energy balance is desired — for example if incoming solar radiation is to balance outgoing infra-red radiation.

Notes

- 1 Joseph-Louis Lagrange (1736–1813) was a Franco-Italian, born and raised in Turin who then lived and worked mainly in Germany and France. He made notable contributions in analysis, number theory and mechanics and was recognized as one of the greatest mathematicians of the eighteenth century. He laid the foundations of the calculus of variations (to wit, the ‘Lagrange multiplier’) and first formulated the principle of least action, and his treatise *Mécanique Analytique* (1788) provides a unified analytic framework (it contains no diagrams, a feature emulated in Whittaker’s *Treatise on Analytical Dynamics*, 1927) for all Newtonian mechanics.

Leonard Euler (1707–1783), a Swiss mathematician who lived and worked for extended periods in Berlin and St. Petersburg, made important contributions in many areas of mathematics and mechanics, including the analytical treatment of algebra, the theory of equations, calculus, number theory and classical mechanics. He was the first to establish the form of the equations of motion of fluid mechanics, writing down both the field description of fluids *and* what we now call the material or advective derivative.

Truesdell (1954) points out that ‘Eulerian’ and ‘Lagrangian’, especially the latter, are inappropriate eponyms. The so-called Eulerian description was introduced by d’Alembert in 1749 and generalized by Euler in 1752, and the so-called Lagrangian description was introduced by Euler in 1759. (It is sometimes said that advances in mathematics are named after the next person to discover them after Euler — the Coriolis effect is another example.) The modern confusion evidently stems from a monograph by Dirichlet in 1860 that credits Euler in 1757 and Lagrange in 1788 for the respective methods.

Clifford Truesdell (1919–2000) was a remarkable figure himself, known both for his own contributions to many areas of continuum mechanics and for his scholarly investigations on the history of mathematics and science. He also had a trenchant and at times pungent writing style. Ball & James (2002) provide a biography.

- 2 Salinity is a mass fraction and thus a nondimensional number, but it is commonly referred to in units of g/kg. For many years the measure of salinity of seawater that was used in oceanography was based on electrical conductivity and referred to as ‘practical salinity’, S_p , since this was (and still is) more easily measured. In thermodynamical calculations practical salinity is now largely dropped in favour of the true salinity, generally referred to as absolute salinity and denoted S_A . Differences between practical and absolute salinity are small but not negligible (Millero *et al.* 2008, IOC *et al.* 2010).
- 3 The form of (1.58) is given in de Szoeke (2004). Nycander & Roquet (2015) and Roquet *et al.* (2015) show that it can, in fact, be used to give a quantitatively accurate simulation of the ocean.
- 4 It has been said that the early students of ideas related to entropy were unusually prone to suicide, Ludwig Boltzmann and Paul Ehrenfest being notable examples. William Thompson (Lord Kelvin) did brilliant work early in his career on fluids and thermodynamics, but his later work — for example his prediction of the age of the Earth to be between 20–40 million years — was less successful. But he lived to a ripe age.

- 5 Because the word barotropic has other meanings — sometimes it is just taken to mean the vertical average — it might be better to always refer to fluids for which density is a function only of pressure as homentropic. Unfortunately the current usage is deeply ingrained and to insist on homentropic would be tilting at windmills.
- 6 Claude-Louis-Marie-Henri Navier (1785–1836) was a French civil engineer, professor at the École Polytechnique and later at the École des Ponts et Chaussée. He was a expert in road and bridge building (he developed the theory of suspension bridges) and, relatedly, made lasting theoretical contributions to the theory of elasticity, being the first to publish a set of general equations for the dynamics of an elastic solid. In fluid mechanics, he laid down the now-called *Navier–Stokes equations*, including the viscous terms, in 1822.
George Gabriel Stokes (1819–1903). Irish born (in Skreen, County Sligo), he was a professor of mathematics at Cambridge from 1849 until his retirement. As well as having a role in the development of fluid mechanics, especially through his considerations of viscous effects, Stokes worked on the dynamics of elasticity, fluorescence, the wave theory of light, and was (perhaps rather ill-advisedly in hindsight) a proponent of the idea of an ether permeating all space.
- 7 Potential temperature was known to William Thomson in 1857.
- 8 Jackett & McDougall (1997), extending McDougall (1987). A similar quantity was described by Eden & Willebrand (1999). Neutral density is not a thermodynamic state variable and because of the form of the seawater equation of state there is no continuous, unique field of neutral density extending through the ocean (de Szoeke 2000). See Nycander (2011) and Tailleux (2016) for more discussion.
- 9 Building from de Szoeke (2004) and with the help of conversations with W. R. Young. See also Fofonoff (1959) and Warren (2006) for some historical background. A very accurate semi-empirical formulae for the Gibbs function is given by Feistel (2008). Using this, seawater equations of state are now available from TEOS-10 standard (IOC *et al.* 2010) and Roquet *et al.* (2015), and which fit laboratory measurements close to the accuracy of the measurements themselves.
- 10 Potential enthalpy was introduced to oceanography by McDougall (2003) and its use is advocated in IOC *et al.* (2010). The advantages and disadvantages of various thermodynamic variables, including entropy and potential enthalpy, are discussed there and in Graham & McDougall (2013). Useful discussion is also to be found in Warren (1999), Young (2010) and Nycander (2011). I am very grateful to T. McDougall for discussions on these matters.
- 11 Referring to the ‘heat content’ of a fluid borders on dangerous language, because heat itself is a type of energy transfer, like work, and not a state variable. On the other hand, heat content has an intuitive appeal and, provided it is properly understood, conveys a useful meaning in oceanography where kinetic energy is relatively small.
- 12 Bernoulli’s theorem was developed mainly by Daniel Bernoulli (1700–1782). It was based on earlier work on the conservation of energy that Daniel had done with his father, Johann Bernoulli (1667–1748), and so perhaps should be known as Bernoullis’ theorem. The two men fell out when Daniel was a young man, reputedly because of Johann’s jealousy of Daniel’s abilities, and subsequently had a very strained relationship. The Bernoulli family produced several (at least eight) talented mathematicians over three generations in the seventeenth and eighteenth centuries, and is often regarded as the most mathematically distinguished family of all time.
- 13 Osborne Reynolds (1842–1912) was an Irish born (Belfast) physicist who was professor of engineering at Manchester University from 1868–1905. His early work was in electricity and magnetism, but he is now most famous for his work in hydrodynamics. The ‘Reynolds number’, which determines the ratio of inertial to viscous forces, and the ‘Reynolds stress’, which is the stress on the mean flow due to the fluctuating components, are both named after him. He was also one of the first scientists to think about the concept of group velocity.
- 14 See also Landau & Lifshitz (1987) and IOC *et al.* (2010). For an interesting and somewhat idiosyncratic view of the first law applied to the ocean, read Warren (2006).
- 15 Onsager (1931), Salmon (1998).

Further reading

General fluid dynamics

There are numerous books on hydrodynamics, an early one being

Lamb, H., 1932. *Hydrodynamics*, 6th edn.

Lamb's book is a classic in the field, although now too dated to make it useful as an introduction.

Two somewhat more modern references, at a fairly advanced level, are

Batchelor, G. K., 1967. *An Introduction to Fluid Dynamics*.

Landau, L. D. & Lifshitz, E. M., 1987. *Fluid Mechanics*.

These two books both contain a detailed derivation of the equations of motion, including viscous and pressure forces.

At a more elementary level we have

Kundu, P., Cohen, I. & Dowling, D., 2015. *Fluid Mechanics*.

This book is written at the advanced undergraduate/beginning graduate level, is easier-going than Batchelor or Landau & Lifshitz, and contains material on geophysical fluid dynamics.

For the connoisseur, a more specialized treatment is

Truesdell, C., 1954. *The Kinematics of Vorticity*.

Written in Truesdell's inimitable style, this book discusses many aspects of vorticity with numerous historical references. Truesdell's books are all gems in their own way.

Thermodynamics

There are many books on thermodynamics, and two that I have found particularly useful are

Reif, F., 1965. *Fundamentals of Statistical and Thermal Physics*,

Callen, H. B., 1985. *Thermodynamics and an Introduction to Thermostatistics*.

Reif's book has become something of a classic, and Callen provides an axiomatic approach that will be an antidote for those who feel that thermodynamic reasoning is mysterious or even circular.

For the subtopic of atmospheric thermodynamics see the further reading section on page 538.

Geophysical fluid dynamics

Gill, A. E., 1982. *Atmosphere–Ocean Dynamics*.

A richly textured book, especially strong on equatorial dynamics and gravity wave motion.

Holton, J. R. & Hakim, G., 2014. *An Introduction to Dynamical Meteorology*.

A straightforward and well-known textbook at the undergraduate/beginning graduate level.

Salmon, R., 1998. *Lectures on Geophysical Fluid Dynamics*.

Lucidly covers the fundamentals as well as Hamiltonian fluid dynamics, geostrophic turbulence and oceanic circulation.

Pedlosky, J., 1987. *Geophysical Fluid Dynamics*.

A primary reference for flow at low Rossby number. Although the book requires some effort, there is a handsome pay-off for those who study it closely.

White (2002) provides a clear and thorough summary of the equations of motion for meteorology, including the non-hydrostatic and primitive equations.

If a body is moving in any direction, there is a force, arising from the Earth's rotation, which always deflects it to the right in the northern hemisphere, and to the left in the southern.

William Ferrel, *The influence of the Earth's rotation upon the relative motion of bodies near its surface*, 1858.

CHAPTER 2

Effects of Rotation and Stratification

THE ATMOSPHERE AND OCEAN are shallow layers of fluid on a sphere, 'shallow' because their thickness is much less than their horizontal extent. Their motion is strongly influenced by two effects: rotation and stratification, the latter meaning that there is a mean vertical gradient of (potential) density that is often large compared with the horizontal gradient. Here we consider how the equations of motion are affected by these effects. First, we consider some elementary effects of rotation on a fluid and derive the Coriolis and centrifugal forces, and write down the equations of motion appropriate for motion on a sphere. Then we discuss some approximations to the equations of motion that are appropriate for large-scale flow in the ocean and atmosphere, in particular the hydrostatic and geostrophic approximations, and finally we look at the possible static instability of stratified flows.

2.1 THE EQUATIONS OF MOTION IN A ROTATING FRAME OF REFERENCE

Newton's second law of motion, that the acceleration on a body is proportional to the imposed force divided by the body's mass, applies in so-called inertial frames of reference; that is, frames that are stationary or moving only with a constant rectilinear velocity relative to the distant galaxies. Now Earth spins round its own axis with a period of almost 24 hours (23h 56m, the difference due to Earth's rotation around the Sun) and so the surface of the Earth manifestly is not an inertial frame. Nevertheless, it is very convenient to describe the flow relative to Earth's surface (which in fact is moving at speeds of up to a few hundreds of metres per second), rather than in some inertial frame. This necessitates recasting the equations into a form that is appropriate for a rotating frame of reference, and that is the subject of this section.

2.1.1 Rate of change of a vector

Consider first a vector $\mathbf{C}$ of constant length rotating relative to an inertial frame at a constant angular velocity $\boldsymbol{\Omega}$. Then, in a frame rotating with that same angular velocity it appears stationary and constant. If in a small interval of time δt the vector $\mathbf{C}$ rotates through a small angle $\delta\lambda$ then the change in $\mathbf{C}$, as perceived in the inertial frame, is given by (see Fig. 2.1)

$$\delta\mathbf{C} = |\mathbf{C}| \cos \vartheta \delta\lambda \mathbf{m}, \quad (2.1)$$

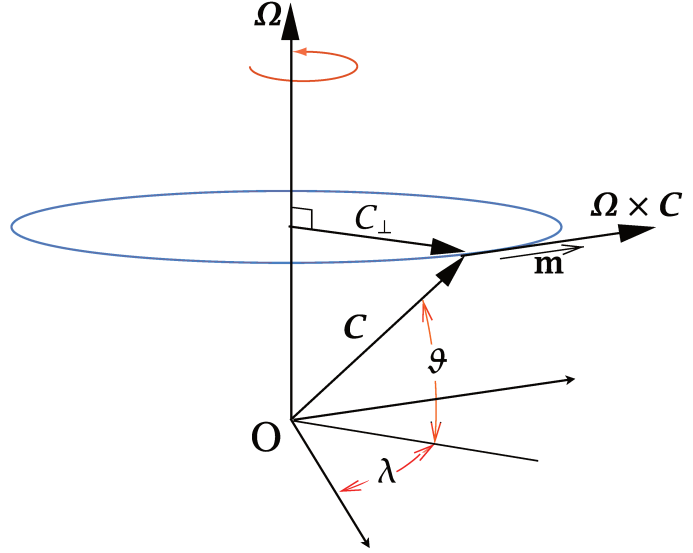


Fig. 2.1 A vector C rotating at an angular velocity Ω . It appears to be a constant vector in the rotating frame, whereas in the inertial frame it evolves according to $(dC/dt)_I = \Omega \times C$.

where the vector $\mathbf{m}$ is the unit vector in the direction of change of $\mathbf{C}$, which is perpendicular to both $\mathbf{C}$ and Ω . But the rate of change of the angle λ is just, by definition, the angular velocity so that $\delta\lambda = |\Omega|\delta t$ and

$$\delta\mathbf{C} = |\mathbf{C}||\Omega| \sin \hat{\vartheta} \mathbf{m} \delta t = \Omega \times \mathbf{C} \delta t. \quad (2.2)$$

using the definition of the vector cross product, where $\hat{\vartheta} = (\pi/2 - \vartheta)$ is the angle between Ω and $\mathbf{C}$. Thus

$$\left(\frac{d\mathbf{C}}{dt} \right)_I = \Omega \times \mathbf{C}, \quad (2.3)$$

where the left-hand side is the rate of change of $\mathbf{C}$ as perceived in the inertial frame.

Now consider a vector $\mathbf{B}$ that changes in the inertial frame. In a small time δt the change in $\mathbf{B}$ as seen in the rotating frame is related to the change seen in the inertial frame by

$$(\delta\mathbf{B})_I = (\delta\mathbf{B})_R + (\delta\mathbf{B})_{rot}, \quad (2.4)$$

where the terms are, respectively, the change seen in the inertial frame, the change due to the vector itself changing as measured in the rotating frame, and the change due to the rotation. Using (2.2) $(\delta\mathbf{B})_{rot} = \Omega \times \mathbf{B} \delta t$, and so the rates of change of the vector $\mathbf{B}$ in the inertial and rotating frames are related by

$$\left(\frac{d\mathbf{B}}{dt} \right)_I = \left(\frac{d\mathbf{B}}{dt} \right)_R + \Omega \times \mathbf{B}. \quad (2.5)$$

This relation applies to a vector $\mathbf{B}$ that, as measured at any one time, is the same in both inertial and rotating frames.

2.1.2 Velocity and acceleration in a rotating frame

The velocity of a body is not measured to be the same in the inertial and rotating frames, so care must be taken when applying (2.5) to velocity. First apply (2.5) to $\mathbf{r}$, the position of a particle to obtain

$$\left(\frac{d\mathbf{r}}{dt} \right)_I = \left(\frac{d\mathbf{r}}{dt} \right)_R + \Omega \times \mathbf{r} \quad (2.6)$$

or

$$\mathbf{v}_I = \mathbf{v}_R + \boldsymbol{\Omega} \times \mathbf{r}. \quad (2.7)$$

We refer to $\mathbf{v}_R$ and $\mathbf{v}_I$ as the relative and inertial velocity, respectively, and (2.7) relates the two. Apply (2.5) again, this time to the velocity $\mathbf{v}_R$ to give

$$\left(\frac{d\mathbf{v}_R}{dt} \right)_I = \left(\frac{d\mathbf{v}_R}{dt} \right)_R + \boldsymbol{\Omega} \times \mathbf{v}_R, \quad (2.8)$$

or, using (2.7)

$$\left(\frac{d}{dt} (\mathbf{v}_I - \boldsymbol{\Omega} \times \mathbf{r}) \right)_I = \left(\frac{d\mathbf{v}_R}{dt} \right)_R + \boldsymbol{\Omega} \times \mathbf{v}_R, \quad (2.9)$$

or

$$\left(\frac{d\mathbf{v}_I}{dt} \right)_I = \left(\frac{d\mathbf{v}_R}{dt} \right)_R + \boldsymbol{\Omega} \times \mathbf{v}_R + \frac{d\boldsymbol{\Omega}}{dt} \times \mathbf{r} + \boldsymbol{\Omega} \times \left(\frac{d\mathbf{r}}{dt} \right)_I. \quad (2.10)$$

Then, noting that

$$\left(\frac{d\mathbf{r}}{dt} \right)_I = \left(\frac{d\mathbf{r}}{dt} \right)_R + \boldsymbol{\Omega} \times \mathbf{r} = (\mathbf{v}_R + \boldsymbol{\Omega} \times \mathbf{r}), \quad (2.11)$$

and assuming that the rate of rotation is constant, (2.10) becomes

$$\left(\frac{d\mathbf{v}_R}{dt} \right)_R = \left(\frac{d\mathbf{v}_I}{dt} \right)_I - 2\boldsymbol{\Omega} \times \mathbf{v}_R - \boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{r}). \quad (2.12)$$

This equation may be interpreted as follows. The term on the left-hand side is the rate of change of the relative velocity as measured in the rotating frame. The first term on the right-hand side is the rate of change of the inertial velocity as measured in the inertial frame (the inertial acceleration, which is, by Newton's second law, equal to the force on a fluid parcel divided by its mass). The second and third terms on the right-hand side (including the minus signs) are the *Coriolis force* and the *centrifugal force* per unit mass. Neither of these are true forces — they may be thought of as quasi-forces (i.e., 'as if' forces); that is, when a body is observed from a rotating frame behaves as if unseen forces are present that affect its motion. If (2.12) is written, as is common, with the terms $+2\boldsymbol{\Omega} \times \mathbf{v}_r$ and $+\boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{r})$ on the left-hand side then these terms should be referred to as the Coriolis and centrifugal *accelerations*.¹

Centrifugal force

If $\mathbf{r}_\perp$ is the perpendicular distance from the axis of rotation (see Fig. 2.1 and substitute $\mathbf{r}$ for $\mathbf{C}$), then, because $\boldsymbol{\Omega}$ is perpendicular to $\mathbf{r}_\perp$, $\boldsymbol{\Omega} \times \mathbf{r} = \boldsymbol{\Omega} \times \mathbf{r}_\perp$. Then, using the vector identity $\boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{r}_\perp) = (\boldsymbol{\Omega} \cdot \mathbf{r}_\perp)\boldsymbol{\Omega} - (\boldsymbol{\Omega} \cdot \boldsymbol{\Omega})\mathbf{r}_\perp$ and noting that the first term is zero, we see that the centrifugal force per unit mass is just given by

$$\mathbf{F}_{ce} = -\boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{r}) = \Omega^2 \mathbf{r}_\perp. \quad (2.13)$$

This may usefully be written as the gradient of a scalar potential,

$$\mathbf{F}_{ce} = -\nabla \Phi_{ce}. \quad (2.14)$$

where $\Phi_{ce} = -(\Omega^2 r_\perp^2)/2 = -(\boldsymbol{\Omega} \times \mathbf{r}_\perp)^2/2$.

Coriolis force

The Coriolis force per unit mass is given by

$$\mathbf{F}_{Co} = -2\boldsymbol{\Omega} \times \mathbf{v}_R. \quad (2.15)$$

It plays a central role in much of geophysical fluid dynamics and will be considered extensively later on. For now, we just note three basic properties.

- (i) There is no Coriolis force on bodies that are stationary in the rotating frame.
- (ii) The Coriolis force acts to deflect moving bodies at right angles to their direction of travel.
- (iii) The Coriolis force does no work on a body because it is perpendicular to the velocity, and so $\mathbf{v}_R \cdot (\boldsymbol{\Omega} \times \mathbf{v}_R) = 0$.

2.1.3 Momentum equation in a rotating frame

Since (2.12) simply relates the accelerations of a particle in the inertial and rotating frames, then in the rotating frame of reference the momentum equation may be written

$$\frac{D\mathbf{v}}{Dt} + 2\boldsymbol{\Omega} \times \mathbf{v} = -\frac{1}{\rho} \nabla p - \nabla \Phi, \quad (2.16)$$

incorporating the centrifugal term into the potential, Φ . We have dropped the subscript R ; henceforth, unless we need to be explicit (as in the next section), all velocities without a subscript will be considered to be relative to the rotating frame.

2.1.4 Mass and tracer conservation in a rotating frame

Let φ be a scalar field that, in the inertial frame, obeys

$$\frac{D\varphi}{Dt} + \varphi \nabla \cdot \mathbf{v}_I = 0. \quad (2.17)$$

Now, observers in both the rotating and inertial frame measure the same value of φ . Further, $D\varphi/Dt$ is simply the rate of change of φ associated with a material parcel, and therefore is reference frame invariant. Thus, without further ado, we write

$$\left(\frac{D\varphi}{Dt} \right)_R = \left(\frac{D\varphi}{Dt} \right)_I, \quad (2.18)$$

where $(D\varphi/Dt)_R = (\partial\varphi/\partial t)_R + \mathbf{v}_R \cdot \nabla\varphi$ and $(D\varphi/Dt)_I = (\partial\varphi/\partial t)_I + \mathbf{v}_I \cdot \nabla\varphi$ and the local temporal derivatives $(\partial\varphi/\partial t)_R$ and $(\partial\varphi/\partial t)_I$ are evaluated at fixed locations in the rotating and inertial frames, respectively.

Further, using (2.7), we have that

$$\nabla \cdot \mathbf{v}_I = \nabla \cdot (\mathbf{v}_R + \boldsymbol{\Omega} \times \mathbf{r}) = \nabla \cdot \mathbf{v}_R \quad (2.19)$$

since $\nabla \cdot (\boldsymbol{\Omega} \times \mathbf{r}) = 0$. Thus, using (2.18) and (2.19), (2.17) is equivalent to

$$\frac{D\varphi}{Dt} + \varphi \nabla \cdot \mathbf{v}_R = 0, \quad (2.20)$$

where all observables are measured in the *rotating* frame. Thus, the equation for the evolution of a scalar whose measured value is the same in rotating and inertial frames is unaltered by the presence of rotation. In particular, the mass conservation equation is unaltered by the presence of rotation.

Although we have taken (2.18) as true a priori, the individual components of the material derivative differ in the rotating and inertial frames. In particular

$$\left(\frac{\partial\varphi}{\partial t}\right)_I = \left(\frac{\partial\varphi}{\partial t}\right)_R - (\boldsymbol{\Omega} \times \mathbf{r}) \cdot \nabla\varphi \quad (2.21)$$

because $\boldsymbol{\Omega} \times \mathbf{r}$ is the velocity, in the inertial frame, of a uniformly rotating body. Similarly,

$$\mathbf{v}_I \cdot \nabla\varphi = (\mathbf{v}_R + \boldsymbol{\Omega} \times \mathbf{r}) \cdot \nabla\varphi. \quad (2.22)$$

Adding the last two equations reprises and confirms (2.18).

2.2 EQUATIONS OF MOTION IN SPHERICAL COORDINATES

The Earth is very nearly spherical and it might appear obvious that we should cast our equations in spherical coordinates. Although this does turn out to be true, the presence of a centrifugal force causes some complications that we should first discuss. The reader who is willing ab initio to treat the Earth as a perfect sphere and to neglect the horizontal component of the centrifugal force may skip the next section.

2.2.1 ♦ The centrifugal force and spherical coordinates

The centrifugal force is a potential force, like gravity, and so we may therefore define an ‘effective gravity’ equal to the sum of the true, or Newtonian, gravity and the centrifugal force. The Newtonian gravitational force is directed approximately toward the centre of the Earth, with small deviations due mainly to the Earth’s oblateness. The line of action of the effective gravity will in general differ slightly from this, and therefore have a component in the ‘horizontal’ plane, that is the plane perpendicular to the radial direction. The magnitude of the centrifugal force is $\Omega^2 r_\perp$, and so the effective gravity is given by

$$\mathbf{g} \equiv \mathbf{g}_{\text{eff}} = \mathbf{g}_{\text{grav}} + \Omega^2 \mathbf{r}_\perp, \quad (2.23)$$

where $\mathbf{g}_{\text{grav}}$ is the Newtonian gravitational force due to the gravitational attraction of the Earth and $\mathbf{r}_\perp$ is normal to the rotation vector (in the direction $\mathbf{C}$ in Fig. 2.2), with $r_\perp = r \cos \vartheta$. Both gravity and centrifugal force are potential forces and therefore we may define the *geopotential*, Φ , such that

$$\mathbf{g} = -\nabla\Phi. \quad (2.24)$$

Surfaces of constant Φ are not quite spherical because $r_\perp$, and hence the centrifugal force, vary with latitude (Fig. 2.2); this has certain ramifications, as we now discuss.

The components of the centrifugal force parallel and perpendicular to the radial direction are $\Omega^2 r \cos^2 \vartheta$ and $\Omega^2 r \cos \vartheta \sin \vartheta$. Newtonian gravity is much larger than either of these, and at the Earth’s surface the ratio of centrifugal to gravitational terms is approximately, and no more than,

$$\alpha \approx \frac{\Omega^2 a}{g} \approx \frac{(7.27 \times 10^{-5})^2 \times 6.4 \times 10^6}{9.8} \approx 3 \times 10^{-3}. \quad (2.25)$$

(At the equator and pole the horizontal component of the centrifugal force is zero and the effective gravity is aligned with Newtonian gravity.) The angle between $\mathbf{g}$ and the line to the centre of the Earth is given by a similar expression and so is also small, typically around 3×10^{-3} radians. However, the horizontal component of the centrifugal force is still large compared to the Coriolis force, the ratio of their magnitudes in mid-latitudes being given by

$$\frac{\text{horizontal centrifugal force}}{\text{Coriolis force}} \approx \frac{\Omega^2 a \cos \vartheta \sin \vartheta}{2\Omega|u|} \approx \frac{\Omega a}{4|u|} \approx 10, \quad (2.26)$$

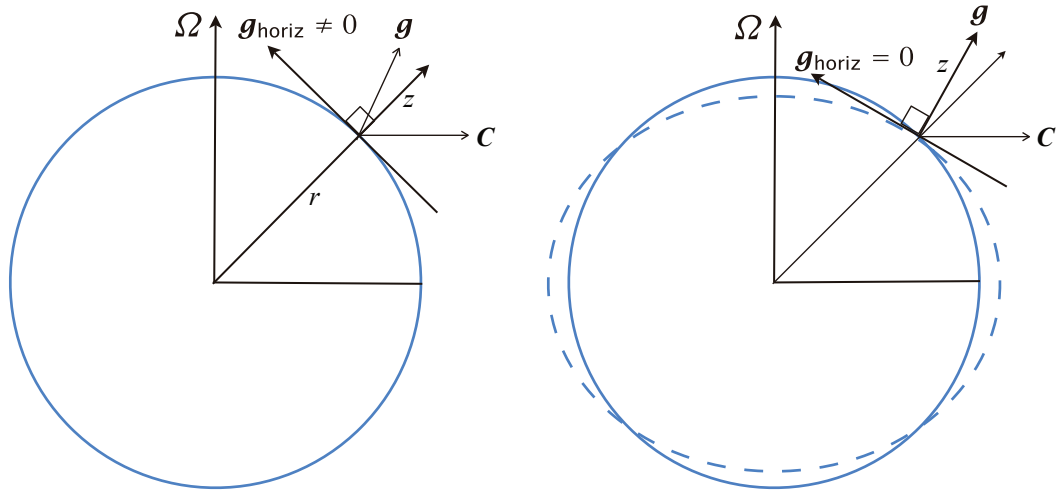


Fig. 2.2 Left: directions of forces and coordinates in true spherical geometry. g is the effective gravity (including the centrifugal force, C) and its horizontal component is evidently non-zero. Right: a modified coordinate system, in which the vertical direction is defined by the direction of g , and so the horizontal component of g is identically zero. The dashed line schematically indicates a surface of constant geopotential. The differences between the direction of g and the direction of the radial coordinate, and between the sphere and the geopotential surface, are much exaggerated and in reality are similar to the thickness of the lines themselves.

using $u = 10 \text{ m s}^{-1}$. The centrifugal term therefore dominates over the Coriolis term, and is largely balanced by a pressure gradient force. Thus, if we adhered to true spherical coordinates, both the horizontal and radial components of the momentum equation would be dominated by a static balance between a pressure gradient and gravity or centrifugal terms. Although in principle there is nothing wrong with writing the equations this way, it obscures the dynamical balances involving the Coriolis force and pressure that determine the large-scale horizontal flow.

A way around this problem is to use the direction of the geopotential force to *define* the vertical direction, and then for all geometric purposes to regard the surfaces of constant Φ as if they were true spheres.² The horizontal component of effective gravity is then identically zero, and we have traded a potentially large dynamical error for a very small geometric error. In fact, over time, the Earth has developed an equatorial bulge to compensate for and neutralize the centrifugal force, so that the effective gravity does act in a direction virtually normal to the Earth's surface; that is, the surface of the Earth is an oblate spheroid of nearly constant geopotential. The geopotential Φ is then a function of the vertical coordinate alone, and for many purposes we can just take $\Phi = gz$; that is, the direction normal to geopotential surfaces, the local vertical, is, in this approximation, taken to be the direction of increasing r in spherical coordinates. It is because the oblateness is very small (the polar diameter is about 12 714 km, whereas the equatorial diameter is about 12 756 km) that using spherical coordinates is a very accurate way to map the spheroid. If the angle between effective gravity and a natural direction of the coordinate system were not small then more heroic measures would be called for.

If the solid Earth did not bulge at the equator, the *behaviour* of the atmosphere and ocean would differ significantly from that of the present system. For example, the surface of the ocean is nearly a geopotential surface, and if the solid Earth were exactly spherical then the ocean would perforce become much deeper at low latitudes and the ocean basins would dry out completely at high latitudes. We could still choose to use the spherical coordinate system discussed above to describe the dynamics, but the shape of the surface of the solid Earth would have to be represented by a

topography, with the topographic height increasing monotonically polewards nearly everywhere.

2.2.2 Some identities in spherical coordinates

The location of a point is given by the coordinates (λ, ϑ, r) where λ is the angular distance eastwards (i.e., longitude), ϑ is angular distance polewards (i.e., latitude) and r is the radial distance from the centre of the Earth — see Fig. 2.3. (In some other fields of study co-latitude is used as a spherical coordinate.) If a is the radius of the Earth, then we also define $z = r - a$. At a given location we may also define the Cartesian increments $(\delta x, \delta y, \delta z) = (r \cos \vartheta \delta \lambda, r \delta \vartheta, \delta r)$.

For a scalar quantity ϕ the material derivative in spherical coordinates is

$$\frac{D\phi}{Dt} = \frac{\partial \phi}{\partial t} + \frac{u}{r \cos \vartheta} \frac{\partial \phi}{\partial \lambda} + \frac{v}{r} \frac{\partial \phi}{\partial \vartheta} + w \frac{\partial \phi}{\partial r}, \quad (2.27)$$

where the velocity components corresponding to the coordinates (λ, ϑ, r) are

$$(u, v, w) \equiv \left(r \cos \vartheta \frac{D\lambda}{Dt}, r \frac{D\vartheta}{Dt}, \frac{Dr}{Dt} \right). \quad (2.28)$$

That is, u is the zonal velocity, v is the meridional velocity and w is the vertical velocity. If we define $(\mathbf{i}, \mathbf{j}, \mathbf{k})$ to be the unit vectors in the direction of increasing (λ, ϑ, r) then

$$\mathbf{v} = \mathbf{i}u + \mathbf{j}v + \mathbf{k}w. \quad (2.29)$$

Note also that $Dr/Dt = Dz/Dt$.

The divergence of a vector $\mathbf{B} = \mathbf{i}B^\lambda + \mathbf{j}B^\vartheta + \mathbf{k}B^r$ is

$$\nabla \cdot \mathbf{B} = \frac{1}{\cos \vartheta} \left[\frac{1}{r} \frac{\partial B^\lambda}{\partial \lambda} + \frac{1}{r} \frac{\partial}{\partial \vartheta} (B^\vartheta \cos \vartheta) + \frac{\cos \vartheta}{r^2} \frac{\partial}{\partial r} (r^2 B^r) \right]. \quad (2.30)$$

The vector gradient of a scalar is:

$$\nabla \phi = \mathbf{i} \frac{1}{r \cos \vartheta} \frac{\partial \phi}{\partial \lambda} + \mathbf{j} \frac{1}{r} \frac{\partial \phi}{\partial \vartheta} + \mathbf{k} \frac{\partial \phi}{\partial r}. \quad (2.31)$$

The Laplacian of a scalar is:

$$\nabla^2 \phi \equiv \nabla \cdot \nabla \phi = \frac{1}{r^2 \cos \vartheta} \left[\frac{1}{\cos \vartheta} \frac{\partial^2 \phi}{\partial \lambda^2} + \frac{\partial}{\partial \vartheta} \left(\cos \vartheta \frac{\partial \phi}{\partial \vartheta} \right) + \cos \vartheta \frac{\partial}{\partial r} \left(r^2 \frac{\partial \phi}{\partial r} \right) \right]. \quad (2.32)$$

The curl of a vector is:

$$\text{curl } \mathbf{B} = \nabla \times \mathbf{B} = \frac{1}{r^2 \cos \vartheta} \begin{vmatrix} \mathbf{i} r \cos \vartheta & \mathbf{j} r & \mathbf{k} \\ \frac{\partial}{\partial \lambda} & \frac{\partial}{\partial \vartheta} & \frac{\partial}{\partial r} \\ B^\lambda r \cos \vartheta & B^\vartheta r & B^r \end{vmatrix}. \quad (2.33)$$

The vector Laplacian $\nabla^2 \mathbf{B}$ (used for example when calculating viscous terms in the momentum equation) may be obtained from the vector identity:

$$\nabla^2 \mathbf{B} = \nabla(\nabla \cdot \mathbf{B}) - \nabla \times (\nabla \times \mathbf{B}). \quad (2.34)$$

Only in Cartesian coordinates does this take the simple form:

$$\nabla^2 \mathbf{B} = \frac{\partial^2 \mathbf{B}}{\partial x^2} + \frac{\partial^2 \mathbf{B}}{\partial y^2} + \frac{\partial^2 \mathbf{B}}{\partial z^2}. \quad (2.35)$$

The expansion in spherical coordinates is of itself, to most eyes, rather uninformative.

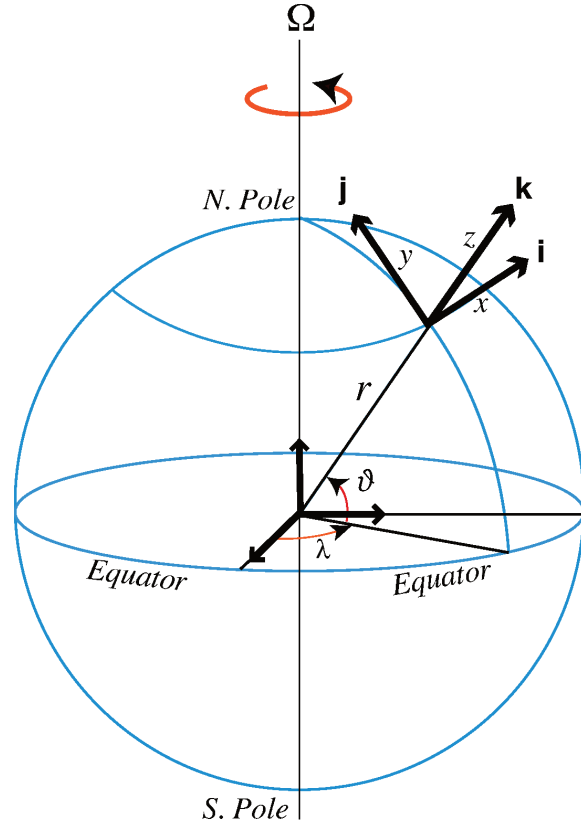


Fig. 2.3 The spherical coordinate system. The orthogonal unit vectors $\mathbf{i}$, $\mathbf{j}$ and $\mathbf{k}$ point in the direction of increasing longitude λ , latitude ϑ , and altitude z . Locally, one may apply a Cartesian system with variables x , y and z measuring distances along $\mathbf{i}$, $\mathbf{j}$ and $\mathbf{k}$.

Rate of change of unit vectors

In spherical coordinates the defining unit vectors are $\mathbf{i}$, the unit vector pointing eastwards, parallel to a line of latitude; $\mathbf{j}$ is the unit vector pointing polewards, parallel to a meridian; and $\mathbf{k}$, the unit vector pointing radially outward. The directions of these vectors change with location, and in fact this is the case in nearly all coordinate systems, with the notable exception of the Cartesian one, and thus their material derivative is not zero. One way to evaluate this is to consider geometrically how the coordinate axes change with position. Another way, and the way that we shall proceed, is to first obtain the effective rotation rate Ω_{flow} , relative to the Earth, of a unit vector as it moves with the flow, and then apply (2.3). Specifically, let the fluid velocity be $\mathbf{v} = (u, v, w)$. The meridional component, v , produces a displacement $r\delta\vartheta = v\delta t$, and this gives rise a local effective vector rotation rate around the local zonal axis of $-(v/r)\mathbf{i}$, the minus sign arising because a displacement in the direction of the north pole is produced by negative rotational displacement around the $\mathbf{i}$ axis. Similarly, the zonal component, u , produces a displacement $\delta\lambda r \cos \vartheta = u\delta t$ and so an effective rotation rate, about the Earth's rotation axis, of $u/(r \cos \vartheta)$. Now, a rotation around the Earth's rotation axis may be written as (see Fig. 2.4)

$$\Omega = \Omega(\mathbf{j} \cos \vartheta + \mathbf{k} \sin \vartheta). \quad (2.36)$$

If the scalar rotation rate is not Ω but is $u/(r \cos \vartheta)$, then the vector rotation rate is

$$\frac{u}{r \cos \vartheta}(\mathbf{j} \cos \vartheta + \mathbf{k} \sin \vartheta) = \mathbf{j} \frac{u}{r} + \mathbf{k} \frac{u \tan \vartheta}{r}. \quad (2.37)$$

Thus, the total rotation rate of a vector that moves with the flow is

$$\boldsymbol{\Omega}_{flow} = -\mathbf{i}\frac{v}{r} + \mathbf{j}\frac{u}{r} + \mathbf{k}\frac{u \tan \vartheta}{r}. \quad (2.38)$$

Applying (2.3) to (2.38), we find

$$\frac{D\mathbf{i}}{Dt} = \boldsymbol{\Omega}_{flow} \times \mathbf{i} = \frac{u}{r \cos \vartheta} (\mathbf{j} \sin \vartheta - \mathbf{k} \cos \vartheta), \quad (2.39a)$$

$$\frac{D\mathbf{j}}{Dt} = \boldsymbol{\Omega}_{flow} \times \mathbf{j} = -\mathbf{i}\frac{u}{r} \tan \vartheta - \mathbf{k}\frac{v}{r}, \quad (2.39b)$$

$$\frac{D\mathbf{k}}{Dt} = \boldsymbol{\Omega}_{flow} \times \mathbf{k} = \mathbf{i}\frac{u}{r} + \mathbf{j}\frac{v}{r}. \quad (2.39c)$$

2.2.3 Equations of motion

Mass Conservation and Thermodynamic Equation

The mass conservation equation, (1.36a), expanded in spherical co-ordinates, is

$$\begin{aligned} \frac{\partial \rho}{\partial t} + \frac{u}{r \cos \vartheta} \frac{\partial \rho}{\partial \lambda} + \frac{v}{r} \frac{\partial \rho}{\partial \vartheta} + w \frac{\partial \rho}{\partial r} \\ + \frac{\rho}{r \cos \vartheta} \left[\frac{\partial u}{\partial \lambda} + \frac{\partial}{\partial \vartheta} (v \cos \vartheta) + \frac{1}{r} \frac{\partial}{\partial r} (wr^2 \cos \vartheta) \right] = 0. \end{aligned} \quad (2.40)$$

Equivalently, using the form (1.36b), this is

$$\frac{\partial \rho}{\partial t} + \frac{1}{r \cos \vartheta} \frac{\partial (u\rho)}{\partial \lambda} + \frac{1}{r \cos \vartheta} \frac{\partial}{\partial \vartheta} (v\rho \cos \vartheta) + \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 w \rho) = 0. \quad (2.41)$$

The thermodynamic equation, (1.108), is a tracer advection equation. Thus, using (2.27), its (adiabatic) spherical coordinate form is

$$\frac{D\theta}{Dt} = \frac{\partial \theta}{\partial t} + \frac{u}{r \cos \vartheta} \frac{\partial \theta}{\partial \lambda} + \frac{v}{r} \frac{\partial \theta}{\partial \vartheta} + w \frac{\partial \theta}{\partial r} = 0, \quad (2.42)$$

and similarly for tracers such as water vapour or salt.

Momentum Equation

Recall that the inviscid momentum equation is:

$$\frac{D\mathbf{v}}{Dt} + 2\boldsymbol{\Omega} \times \mathbf{v} = -\frac{1}{\rho} \nabla p - \nabla \Phi, \quad (2.43)$$

where Φ is the geopotential. In spherical coordinates the directions of the coordinate axes change with position and so the component expansion of (2.43) is

$$\frac{D\mathbf{v}}{Dt} = \frac{Du}{Dt} \mathbf{i} + \frac{Dv}{Dt} \mathbf{j} + \frac{Dw}{Dt} \mathbf{k} + u \frac{D\mathbf{i}}{Dt} + v \frac{D\mathbf{j}}{Dt} + w \frac{D\mathbf{k}}{Dt} \quad (2.44a)$$

$$= \frac{Du}{Dt} \mathbf{i} + \frac{Dv}{Dt} \mathbf{j} + \frac{Dw}{Dt} \mathbf{k} + \boldsymbol{\Omega}_{flow} \times \mathbf{v}, \quad (2.44b)$$

using (2.39). Using either (2.44a) and the expressions for the rates of change of the unit vectors given in (2.39), or (2.44b) and the expression for $\boldsymbol{\Omega}_{flow}$ given in (2.38), (2.44) becomes

$$\frac{D\mathbf{v}}{Dt} = \mathbf{i} \left(\frac{Du}{Dt} - \frac{uv \tan \vartheta}{r} + \frac{uw}{r} \right) + \mathbf{j} \left(\frac{Dv}{Dt} + \frac{u^2 \tan \vartheta}{r} + \frac{vw}{r} \right) + \mathbf{k} \left(\frac{Dw}{Dt} - \frac{u^2 + v^2}{r} \right). \quad (2.45)$$

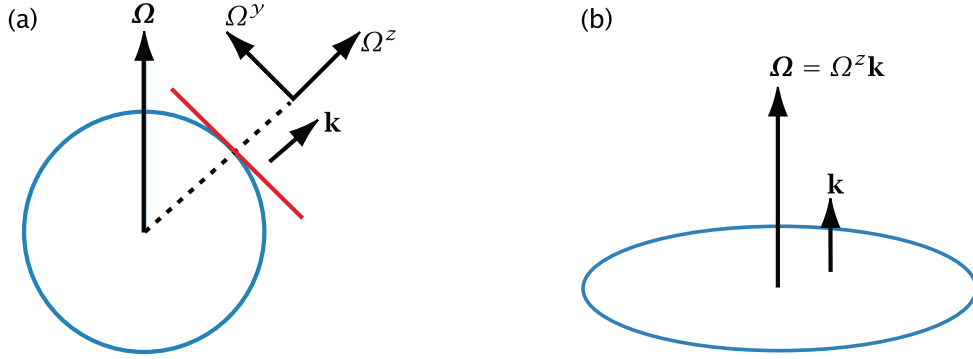


Fig. 2.4 (a) On the sphere the rotation vector Ω can be decomposed into two components, one in the local vertical and one in the local horizontal, pointing toward the pole. That is, $\Omega = \Omega_y \mathbf{j} + \Omega_z \mathbf{k}$ where $\Omega_y = \Omega \cos \vartheta$ and $\Omega_z = \Omega \sin \vartheta$. In geophysical fluid dynamics, the rotation vector in the local vertical is often the more important component in the horizontal momentum equations. On a rotating disk, (b), the rotation vector Ω is parallel to the local vertical $\mathbf{k}$.

Using the definition of a vector cross product the Coriolis term is:

$$\begin{aligned} 2\Omega \times \mathbf{v} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & 2\Omega \cos \vartheta & 2\Omega \sin \vartheta \\ u & v & w \end{vmatrix} \\ &= \mathbf{i} (2\Omega w \cos \vartheta - 2\Omega v \sin \vartheta) + \mathbf{j} 2\Omega u \sin \vartheta - \mathbf{k} 2\Omega u \cos \vartheta. \end{aligned} \quad (2.46)$$

Using (2.45) and (2.46), and the gradient operator given by (2.31), the momentum equation (2.43) becomes:

$$\frac{Du}{Dt} - \left(2\Omega + \frac{u}{r \cos \vartheta} \right) (v \sin \vartheta - w \cos \vartheta) = -\frac{1}{\rho r \cos \vartheta} \frac{\partial p}{\partial \lambda}, \quad (2.47a)$$

$$\frac{Dv}{Dt} + \frac{wv}{r} + \left(2\Omega + \frac{u}{r \cos \vartheta} \right) u \sin \vartheta = -\frac{1}{\rho r} \frac{\partial p}{\partial \vartheta}, \quad (2.47b)$$

$$\frac{Dw}{Dt} - \frac{u^2 + v^2}{r} - 2\Omega u \cos \vartheta = -\frac{1}{\rho} \frac{\partial p}{\partial r} - g. \quad (2.47c)$$

The terms involving Ω are called Coriolis terms, and the quadratic terms on the left-hand sides involving $1/r$ are often called metric terms.

2.2.4 The primitive equations

The so-called *primitive equations* of motion are simplifications of the above equations frequently used in atmospheric and oceanic modelling.³ Three related approximations are involved.

- (i) *The hydrostatic approximation.* In the vertical momentum equation the gravitational term is assumed to be balanced by the pressure gradient term, so that

$$\frac{\partial p}{\partial z} = -\rho g. \quad (2.48)$$

The advection of vertical velocity, the Coriolis terms, and the metric term $(u^2 + v^2)/r$ are all neglected.

- (ii) *The shallow-fluid approximation.* We write $r = a + z$ where the constant a is the radius of the Earth and z increases in the radial direction. The coordinate r is then replaced by a except where it is used as the differentiating argument. Thus, for example,

$$\frac{1}{r^2} \frac{\partial(r^2 w)}{\partial r} \rightarrow \frac{\partial w}{\partial z}. \quad (2.49)$$

- (iii) *The traditional approximation.* Coriolis terms in the horizontal momentum equations involving the vertical velocity, and the still smaller metric terms uw/r and vw/r , are neglected.

The second and third of these approximations should be taken, or not, together, the underlying reason being that they both relate to the presumed small aspect ratio of the motion, so the approximations succeed or fail together. If we make one approximation but not the other then we are being asymptotically inconsistent, and angular momentum and energy conservation are not assured (see section 2.2.7). The hydrostatic approximation also depends on the small aspect ratio of the flow, but in a slightly different way. For large-scale flow in the terrestrial atmosphere and ocean all three approximations are in fact very accurate approximations. We defer a more complete treatment until section 2.7, in part because a treatment of the hydrostatic approximation is done most easily in the context of the Boussinesq equations, derived in section 2.4.

Making these approximations, the momentum equations for a shallow layer are

$$\frac{Du}{Dt} - 2\Omega \sin \vartheta v - \frac{uv}{a} \tan \vartheta = -\frac{1}{a\rho \cos \vartheta} \frac{\partial p}{\partial \lambda}, \quad (2.50a)$$

$$\frac{Dv}{Dt} + 2\Omega \sin \vartheta u + \frac{u^2 \tan \vartheta}{a} = -\frac{1}{\rho a} \frac{\partial p}{\partial \vartheta}, \quad (2.50b)$$

$$0 = -\frac{1}{\rho} \frac{\partial p}{\partial z} - g, \quad (2.50c)$$

where

$$\frac{D}{Dt} = \left(\frac{\partial}{\partial t} + \frac{u}{a \cos \vartheta} \frac{\partial}{\partial \lambda} + \frac{v}{a} \frac{\partial}{\partial \vartheta} + w \frac{\partial}{\partial z} \right). \quad (2.51)$$

We note the ubiquity of the factor $2\Omega \sin \vartheta$, and take the opportunity to define the *Coriolis parameter*, $f \equiv 2\Omega \sin \vartheta$. The associated mass conservation equation for a shallow fluid layer is:

$$\frac{\partial \rho}{\partial t} + \frac{u}{a \cos \vartheta} \frac{\partial \rho}{\partial \lambda} + \frac{v}{a} \frac{\partial \rho}{\partial \vartheta} + w \frac{\partial \rho}{\partial z} + \rho \left[\frac{1}{a \cos \vartheta} \frac{\partial u}{\partial \lambda} + \frac{1}{a \cos \vartheta} \frac{\partial}{\partial \vartheta} (v \cos \vartheta) + \frac{\partial w}{\partial z} \right] = 0, \quad (2.52)$$

or equivalently,

$$\frac{\partial \rho}{\partial t} + \frac{1}{a \cos \vartheta} \frac{\partial (u\rho)}{\partial \lambda} + \frac{1}{a \cos \vartheta} \frac{\partial}{\partial \vartheta} (v\rho \cos \vartheta) + \frac{\partial (w\rho)}{\partial z} = 0. \quad (2.53)$$

2.2.5 Primitive equations in vector form

The primitive equations on a sphere may be written in a compact vector form provided we make a slight reinterpretation of the material derivative of the coordinate axes. Instead of (2.39) we take the material derivative of the unit vectors to be

$$\frac{D\mathbf{i}}{Dt} = \tilde{\Omega}_{flow} \times \mathbf{i} = \mathbf{j} \frac{u \tan \vartheta}{a}, \quad (2.54a)$$

$$\frac{D\mathbf{j}}{Dt} = \tilde{\Omega}_{flow} \times \mathbf{j} = -\mathbf{i} \frac{u \tan \vartheta}{a}, \quad (2.54b)$$

where $\tilde{\Omega}_{flow} = \mathbf{k}u \tan \vartheta/a$, which is the vertical component of (2.38) with r replaced by a . Given (2.54), the primitive equations (2.50a) and (2.50b) may be written as

$$\frac{D\mathbf{u}}{Dt} + \mathbf{f} \times \mathbf{u} = -\frac{1}{\rho} \nabla_z p, \quad (2.55)$$

where $\mathbf{u} = u\mathbf{i} + v\mathbf{j} + 0\mathbf{k}$ is the horizontal velocity, $\nabla_z p = [(a \cos \vartheta)^{-1} \partial p / \partial \lambda, a^{-1} \partial p / \partial \vartheta]$ is the gradient operator at constant z , and $\mathbf{f} = f\mathbf{k} = 2\Omega \sin \vartheta \mathbf{k}$. In (2.55) the material derivative of the horizontal velocity is given by

$$\frac{D\mathbf{u}}{Dt} = \mathbf{i} \frac{Du}{Dt} + \mathbf{j} \frac{Dv}{Dt} + u \frac{D\mathbf{i}}{Dt} + v \frac{D\mathbf{j}}{Dt}, \quad (2.56)$$

The advection of the horizontal wind $\mathbf{u}$ is still by the three-dimensional velocity $\mathbf{v}$.

The vertical momentum equation is the hydrostatic equation, (2.50c), and the mass conservation equation is

$$\frac{D\rho}{Dt} + \rho \nabla \cdot \mathbf{v} = 0 \quad \text{or} \quad \frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0, \quad (2.57)$$

where D/Dt is given by (2.51), and the second expression is written out in full in (2.53).

2.2.6 The vector invariant form of the momentum equation

The ‘vector invariant’ form of the momentum equation is so-called because it appears to take the same form in all coordinate systems — there is no advective derivative of the coordinate system to worry about. With the aid of the identity $(\mathbf{v} \cdot \nabla) \mathbf{v} = -\mathbf{v} \times \boldsymbol{\omega} + \nabla(\mathbf{v}^2/2)$, where $\boldsymbol{\omega} \equiv \nabla \times \mathbf{v}$ is the relative vorticity (which we explore at greater length in chapter 4) the three-dimensional momentum equation, (2.16), may be written:

$$\frac{\partial \mathbf{v}}{\partial t} + (2\boldsymbol{\Omega} + \boldsymbol{\omega}) \times \mathbf{v} = -\frac{1}{\rho} \nabla p - \frac{1}{2} \nabla \mathbf{v}^2 + \mathbf{g}, \quad (2.58)$$

and this is the vector invariant momentum equation. In spherical coordinates the relative vorticity is given by:

$$\begin{aligned} \boldsymbol{\omega} = \nabla \times \mathbf{v} &= \frac{1}{r^2 \cos \vartheta} \begin{vmatrix} \mathbf{i} & r \cos \vartheta & \mathbf{j} & r & \mathbf{k} \\ \partial/\partial \lambda & \partial/\partial \vartheta & \partial/\partial r \\ ur \cos \vartheta & rv & w \end{vmatrix} \\ &= \mathbf{i} \frac{1}{r} \left(\frac{\partial w}{\partial \vartheta} - \frac{\partial(rv)}{\partial r} \right) - \mathbf{j} \frac{1}{r \cos \vartheta} \left(\frac{\partial w}{\partial \lambda} - \frac{\partial}{\partial r}(ur \cos \vartheta) \right) + \mathbf{k} \frac{1}{r \cos \vartheta} \left(\frac{\partial v}{\partial \lambda} - \frac{\partial}{\partial \vartheta}(u \cos \vartheta) \right). \end{aligned} \quad (2.59)$$

We can write the horizontal momentum equations of the primitive equations in a similar way. Making the traditional and shallow fluid approximations, the horizontal components of (2.58) become

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{f} + \mathbf{k} \zeta) \times \mathbf{u} + w \frac{\partial \mathbf{u}}{\partial z} = -\frac{1}{\rho} \nabla_z p - \frac{1}{2} \nabla \mathbf{u}^2, \quad (2.60)$$

where $\mathbf{u} = (u, v, 0)$, $\mathbf{f} = \mathbf{k} 2\Omega \sin \vartheta$ and ∇_z is the horizontal gradient operator (the gradient at a constant value of z). Using (2.59), ζ is given by

$$\zeta = \frac{1}{a \cos \vartheta} \frac{\partial v}{\partial \lambda} - \frac{1}{a \cos \vartheta} \frac{\partial}{\partial \vartheta}(u \cos \vartheta) = \frac{1}{a \cos \vartheta} \frac{\partial v}{\partial \lambda} - \frac{1}{a} \frac{\partial u}{\partial \vartheta} + \frac{u}{a} \tan \vartheta. \quad (2.61)$$

The separate components of the momentum equation are given by:

$$\frac{\partial u}{\partial t} - (f + \zeta)v + w \frac{\partial u}{\partial z} = -\frac{1}{a \cos \vartheta} \left(\frac{1}{\rho} \frac{\partial p}{\partial \lambda} + \frac{1}{2} \frac{\partial \mathbf{u}^2}{\partial \lambda} \right), \quad (2.62)$$

and

$$\frac{\partial v}{\partial t} + (f + \zeta)u + w \frac{\partial v}{\partial z} = -\frac{1}{a} \left(\frac{1}{\rho} \frac{\partial p}{\partial \vartheta} + \frac{1}{2} \frac{\partial u^2}{\partial \vartheta} \right). \quad (2.63)$$

2.2.7 Angular momentum

The zonal momentum equation can be usefully expressed as a statement about axial angular momentum; that is, angular momentum about the rotation axis. The zonal angular momentum per unit mass is the component of angular momentum in the direction of the axis of rotation and it is given by, without making any shallow atmosphere approximation,

$$m = (u + \Omega r \cos \vartheta) r \cos \vartheta. \quad (2.64)$$

The evolution equation for this quantity follows from the zonal momentum equation and has the simple form

$$\frac{Dm}{Dt} = -\frac{1}{\rho} \frac{\partial p}{\partial \lambda}, \quad (2.65)$$

where the material derivative is

$$\frac{D}{Dt} = \frac{\partial}{\partial t} + \frac{u}{r \cos \vartheta} \frac{\partial}{\partial \lambda} + \frac{v}{r} \frac{\partial}{\partial \vartheta} + w \frac{\partial}{\partial r}. \quad (2.66)$$

Using the mass continuity equation, (2.65) can be written as

$$\frac{D\rho m}{Dt} + \rho m \nabla \cdot \mathbf{v} = -\frac{\partial p}{\partial \lambda} \quad (2.67)$$

or

$$\frac{\partial \rho m}{\partial t} + \frac{1}{r \cos \vartheta} \frac{\partial(\rho u m)}{\partial \lambda} + \frac{1}{r \cos \vartheta} \frac{\partial}{\partial \vartheta} (\rho v m \cos \vartheta) + \frac{1}{r^2} \frac{\partial}{\partial r} (\rho m w r^2) = -\frac{\partial p}{\partial \lambda}. \quad (2.68)$$

This is an angular momentum conservation equation.

If the fluid is confined to a shallow layer near the surface of a sphere, then we may replace r , the radial coordinate, by a , the radius of the sphere, in the definition of m , and we define $\bar{m} \equiv (u + \Omega a \cos \vartheta) a \cos \vartheta$. Then (2.65) is replaced by

$$\frac{D\bar{m}}{Dt} = -\frac{1}{\rho} \frac{\partial p}{\partial \lambda}, \quad (2.69)$$

where now

$$\frac{D}{Dt} = \frac{\partial}{\partial t} + \frac{u}{a \cos \vartheta} \frac{\partial}{\partial \lambda} + \frac{v}{a} \frac{\partial}{\partial \vartheta} + w \frac{\partial}{\partial z}. \quad (2.70)$$

In the shallow fluid approximation (2.68) becomes

$$\frac{\partial \rho m}{\partial t} + \frac{1}{a \cos \vartheta} \frac{\partial(\rho u m)}{\partial \lambda} + \frac{1}{a \cos \vartheta} \frac{\partial}{\partial \vartheta} (\rho v m \cos \vartheta) + \frac{\partial}{\partial z} (\rho m w) = -\frac{\partial p}{\partial \lambda}. \quad (2.71)$$

which is an angular momentum conservation equation for a shallow atmosphere.

♦ From angular momentum to the spherical component equations

An alternative way of deriving the three components of the momentum equation in spherical polar coordinates is to *begin* with (2.65) and the principle of conservation of energy. That is, we take the equations for conservation of angular momentum and energy as true a priori and demand that the forms of the momentum equation be constructed to satisfy these. Expanding the material

derivative in (2.65), noting that $Dr/Dt = w$ and $D\cos\vartheta/Dt = -(v/r)\sin\vartheta$, immediately gives (2.47a). Multiplication by u then yields

$$u \frac{Du}{Dt} - 2\Omega uv \sin\vartheta + 2\Omega uw \cos\vartheta - \frac{u^2 v \tan\vartheta}{r} + \frac{u^2 w}{r} = -\frac{u}{\rho r \cos\vartheta} \frac{\partial p}{\partial \lambda}. \quad (2.72)$$

Now suppose that the meridional and vertical momentum equations are of the form

$$\frac{Dv}{Dt} + \text{Coriolis and metric terms} = -\frac{1}{\rho r} \frac{\partial p}{\partial \vartheta} \quad (2.73a)$$

$$\frac{Dw}{Dt} + \text{Coriolis and metric terms} = -\frac{1}{\rho} \frac{\partial p}{\partial r}, \quad (2.73b)$$

but that we do not know what form the Coriolis and metric terms take. To determine that form, construct the kinetic energy equation by multiplying (2.73) by v and w , respectively. Now, the metric terms must vanish when we sum the resulting equations along with (2.72), so that (2.73a) must contain the Coriolis term $2\Omega u \sin\vartheta$ as well as the metric term $u^2 \tan\vartheta/r$, and (2.73b) must contain the term $-2\Omega u \cos\vartheta$ as well as the metric term u^2/r . But if (2.73b) contains the term u^2/r it must also contain the term v^2/r by isotropy, and therefore (2.73a) must also contain the term vw/r . In this way, (2.47) is precisely reproduced, although the sceptic might argue that the uniqueness of the form has not been demonstrated.

A particular advantage of this approach arises in determining the appropriate momentum equations that conserve angular momentum and energy in the shallow-fluid approximation. We begin with (2.69) and expand to obtain (2.50a). Multiplying by u gives

$$u \frac{Du}{Dt} - 2\Omega uv \sin\vartheta - \frac{u^2 v \tan\vartheta}{a} = -\frac{u}{\rho a \cos\vartheta} \frac{\partial p}{\partial \lambda}. \quad (2.74)$$

To ensure energy conservation, the meridional momentum equation must contain the Coriolis term $2\Omega u \sin\vartheta$ and the metric term $u^2 \tan\vartheta/a$, but the vertical momentum equation must have neither of the metric terms appearing in (2.47c). Thus we deduce the following equations:

$$\frac{Du}{Dt} - \left(2\Omega \sin\vartheta + \frac{u \tan\vartheta}{a} \right) v = -\frac{1}{\rho a \cos\vartheta} \frac{\partial p}{\partial \lambda}, \quad (2.75a)$$

$$\frac{Dv}{Dt} + \left(2\Omega \sin\vartheta + \frac{u \tan\vartheta}{a} \right) u = -\frac{1}{\rho a} \frac{\partial p}{\partial \vartheta}, \quad (2.75b)$$

$$\frac{Dw}{Dt} = -\frac{1}{\rho} \frac{\partial p}{\partial r} - g. \quad (2.75c)$$

This equation set, when used in conjunction with the thermodynamic and mass continuity equations, conserves appropriate forms of angular momentum and energy. In the hydrostatic approximation the material derivative of w in (2.75c) is *additionally* neglected. Thus, the hydrostatic approximation is mathematically and physically consistent with the shallow-fluid approximation, but it is an additional approximation with slightly different requirements that one may choose, rather than being required, to make. From an asymptotic perspective, the difference lies in the small parameter necessary for either approximation to hold, namely:

$$\text{shallow fluid and traditional approximations:} \quad \gamma \equiv \frac{H}{a} \ll 1, \quad (2.76a)$$

$$\text{small aspect ratio for hydrostatic approximation:} \quad \alpha \equiv \frac{H}{L} \ll 1, \quad (2.76b)$$

where L is the horizontal scale of the motion and a is the radius of the Earth. For hemispheric or global scale phenomena $L \sim a$ and the two approximations coincide. (Requirement (2.76b) for the hydrostatic approximation will be derived in section 2.7.)

2.3 CARTESIAN APPROXIMATIONS: THE TANGENT PLANE

2.3.1 The f -plane

Although the rotation of the Earth is central for many dynamical phenomena, the sphericity of the Earth is not always so. This is especially true for phenomena on a scale somewhat smaller than global where the use of spherical coordinates becomes awkward, and it is more convenient to use a locally Cartesian representation of the equations. Referring to the red line in Fig. 2.4 we will define a plane tangent to the surface of the Earth at a latitude ϑ_0 , and then use a Cartesian coordinate system (x, y, z) to describe motion on that plane. For small excursions on the plane, $(x, y, z) \approx (a\lambda \cos \vartheta_0, a(\vartheta - \vartheta_0), z)$. Consistently, the velocity is $\mathbf{v} = (u, v, w)$, so that u, v and w are the components of the velocity *in the tangent plane*, in approximately in the east–west, north–south and vertical directions, respectively.

The momentum equations for flow in this plane are then

$$\frac{\partial u}{\partial t} + (\mathbf{v} \cdot \nabla)u + 2(\Omega^y w - \Omega^z v) = -\frac{1}{\rho} \frac{\partial p}{\partial x}, \quad (2.77a)$$

$$\frac{\partial v}{\partial t} + (\mathbf{v} \cdot \nabla)v + 2(\Omega^z u - \Omega^x w) = -\frac{1}{\rho} \frac{\partial p}{\partial y}, \quad (2.77b)$$

$$\frac{\partial w}{\partial t} + (\mathbf{v} \cdot \nabla)w + 2(\Omega^x v - \Omega^y u) = -\frac{1}{\rho} \frac{\partial p}{\partial z} - g, \quad (2.77c)$$

where the rotation vector $\boldsymbol{\Omega} = \Omega^x \mathbf{i} + \Omega^y \mathbf{j} + \Omega^z \mathbf{k}$ and $\Omega^x = 0$, $\Omega^y = \Omega \cos \vartheta_0$ and $\Omega^z = \Omega \sin \vartheta_0$. If we make the traditional approximation, and so ignore the components of $\boldsymbol{\Omega}$ not in the direction of the local vertical, then the above equations become

$$\frac{Du}{Dt} - f_0 v = -\frac{1}{\rho} \frac{\partial p}{\partial x}, \quad \frac{Dv}{Dt} + f_0 u = -\frac{1}{\rho} \frac{\partial p}{\partial y}, \quad \frac{Dw}{Dt} = -\frac{1}{\rho} \frac{\partial p}{\partial z} - g. \quad (2.78a,b,c)$$

where $f_0 = 2\Omega^z = 2\Omega \sin \vartheta_0$. Defining the horizontal velocity vector $\mathbf{u} = (u, v, 0)$, the first two equations may be written as

$$\frac{D\mathbf{u}}{Dt} + \mathbf{f}_0 \times \mathbf{u} = -\frac{1}{\rho} \nabla_z p, \quad (2.79)$$

where $D\mathbf{u}/Dt = \partial \mathbf{u} / \partial t + \mathbf{v} \cdot \nabla \mathbf{u}$, $\mathbf{f}_0 = 2\Omega \sin \vartheta_0 \mathbf{k} = f_0 \mathbf{k}$, and $\mathbf{k}$ is the direction perpendicular to the plane. These equations are, evidently, exactly the same as the momentum equations in a system in which the rotation vector is aligned with the local vertical, as illustrated in the right-hand panel in Fig. 2.4. They will describe flow on the surface of a rotating sphere to a good approximation provided the flow is of limited latitudinal extent so that the effects of sphericity are unimportant; we have made what is known as the f -plane. We may in addition make the hydrostatic approximation, in which case (2.78c) becomes the familiar $\partial p / \partial z = -\rho g$.

2.3.2 The beta-plane approximation

The magnitude of the vertical component of rotation varies with latitude, and this has important dynamical consequences. We can approximate this effect by allowing the effective rotation vector to vary. Thus, noting that, for small variations in latitude,

$$f = 2\Omega \sin \vartheta \approx 2\Omega \sin \vartheta_0 + 2\Omega(\vartheta - \vartheta_0) \cos \vartheta_0, \quad (2.80)$$

then on the tangent plane we may mimic this by allowing the Coriolis parameter to vary as

$$f = f_0 + \beta y, \quad (2.81)$$

where $f_0 = 2\Omega \sin \vartheta_0$ and $\beta = \partial f / \partial y = (2\Omega \cos \vartheta_0)/a$. This important approximation is known as the *beta-plane*, or *β -plane*, approximation; it captures the the most important *dynamical* effects of sphericity, without the complicating *geometric* effects, which are not essential to describe many phenomena. The momentum equations (2.78) are unaltered except that f_0 is replaced by $f_0 + \beta y$ to represent a varying Coriolis parameter. Thus, sphericity combined with rotation is dynamically equivalent to a *differentially rotating* system. For future reference, we write down the β -plane horizontal momentum equations:

$$\frac{D\mathbf{u}}{Dt} + \mathbf{f} \times \mathbf{u} = -\frac{1}{\rho} \nabla_z p, \quad (2.82)$$

where $\mathbf{f} = (f_0 + \beta y)\hat{\mathbf{k}}$. In component form this equation becomes

$$\frac{Du}{Dt} - fv = -\frac{1}{\rho} \frac{\partial p}{\partial x}, \quad \frac{Dv}{Dt} + fu = -\frac{1}{\rho} \frac{\partial p}{\partial y}. \quad (2.83a,b)$$

The mass conservation, thermodynamic and hydrostatic equations in the β -plane approximation are the same as the usual Cartesian, f -plane, forms of those equations.

2.4 EQUATIONS FOR A STRATIFIED OCEAN: THE BOUSSINESQ APPROXIMATION

The density variations in the ocean are quite small compared to the mean density, and we may exploit this to derive somewhat simpler but still quite accurate equations of motion. Let us first examine how much density does vary in the ocean.

2.4.1 Variation of density in the ocean

The variations of density in the ocean are due to three effects: the compression of water by pressure (which we denote as $\Delta_p \rho$), the thermal expansion of water if its temperature changes ($\Delta_T \rho$), and the haline contraction if its salinity changes ($\Delta_S \rho$). How big are these? An appropriate equation of state to approximately evaluate these effects is the linear one

$$\rho = \rho_0 [1 - \beta_T(T - T_0) + \beta_S(S - S_0) + \beta_p p], \quad (2.84)$$

where $\beta_T \approx 2 \times 10^{-4} \text{ K}^{-1}$, $\beta_S \approx 10^{-3} \text{ g/kg}^{-1}$ and $\beta_p = 1/(\rho_0 c_s^2) \approx 4.4 \times 10^{-10} \text{ Pa}^{-1}$ with $c_s \approx 1500 \text{ m s}^{-1}$ (see the table on page 33). The three effects may then be evaluated as follows.

Pressure compressibility. We have $\Delta_p \rho \approx \Delta p / c_s^2 \approx \rho_0 g H / c_s^2$ where H is the depth we evaluate the pressure change (quite accurately) using the hydrostatic approximation. Thus,

$$\frac{|\Delta_p \rho|}{\rho_0} \approx \frac{gH}{c_s^2} \sim 4 \times 10^{-2}, \quad (2.85)$$

with $H = 8 \text{ km}$ and $c_s^2/g \approx 200 \text{ km}$. The latter quantity is the density scale height of the ocean. Thus, the pressure at the bottom of the ocean, enormous as it is, is insufficient to compress the water enough to make a significant change in its density. Changes in density due to dynamical variations of pressure are small if the Mach number is small, and this is also usually the case.

Thermal expansion. We have $\Delta_T \rho \approx -\beta_T \rho_0 \Delta T$ and therefore

$$\frac{|\Delta_T \rho|}{\rho_0} \approx \beta_T \Delta T \sim 4 \times 10^{-3} \quad (2.86)$$

with $\Delta T = 20 \text{ K}$. Evidently we would require temperature differences of order β_T^{-1} , or 5000 K to obtain order one variations in density.

Saline contraction. We have $\Delta_S \rho \approx \beta_S \rho_0 \Delta S$ and therefore

$$\frac{|\Delta_S \rho|}{\rho_0} \approx \beta_S \Delta S \sim 1.5 \times 10^{-3}, \quad (2.87)$$

with $\Delta S = 5 \text{ g kg}^{-1}$. The fractional change in the density of seawater due to salinity variations is thus also very small.

Evidently, fractional density changes in the ocean are very small due to the above effects.

2.4.2 The Boussinesq equations

The *Boussinesq equations* are a set of equations that exploit the smallness of density variations in liquids.⁴ A formal asymptotic derivation is given in the appendix (page 101) but in what follows we are more heuristic. To set notation we write

$$\rho = \rho_0 + \delta\rho(x, y, z, t) \quad (2.88a)$$

$$= \rho_0 + \bar{\rho}(z) + \rho'(x, y, z, t) \quad (2.88b)$$

$$= \bar{\rho}(z) + \rho'(x, y, z, t), \quad (2.88c)$$

where ρ_0 is a constant and we assume that

$$|\bar{\rho}|, |\rho'|, |\delta\rho| \ll \rho_0. \quad (2.89)$$

We need not assume that $|\rho'| \ll |\bar{\rho}|$, but this is often the case in the ocean. The horizontal gradients (i.e., gradients at constant z , ∇_z) satisfy $\nabla_z p = \nabla_z p' = \nabla_z \delta p$. To obtain the Boussinesq equations we will just use (2.88a), but (2.88c) will be useful for the anelastic equations considered later.

Associated with the reference density is a reference pressure that is defined to be in hydrostatic balance with it. That is,

$$p = p_0(z) + \delta p(x, y, z, t), \quad (2.90a,b)$$

where $|\delta p| \ll p_0$, $|p'| \ll \bar{p}$ and

$$\frac{dp_0}{dz} \equiv -g\rho_0, \quad (2.91a,b)$$

Momentum equations

To obtain the Boussinesq equations we use (2.88a), so that $\rho = \rho_0 + \delta\rho$, and we assume that $\delta\rho/\rho_0$ is small. Without approximation, the momentum equation can be written as

$$(\rho_0 + \delta\rho) \left(\frac{D\mathbf{v}}{Dt} + 2\boldsymbol{\Omega} \times \mathbf{v} \right) = -\nabla \delta p - \frac{\partial p_0}{\partial z} \mathbf{k} - g(\rho_0 + \delta\rho) \mathbf{k}, \quad (2.92)$$

and using (2.91a) this becomes, again without approximation,

$$(\rho_0 + \delta\rho) \left(\frac{D\mathbf{v}}{Dt} + 2\boldsymbol{\Omega} \times \mathbf{v} \right) = -\nabla \delta p - g\delta\rho \mathbf{k}. \quad (2.93)$$

If density variations are small this equation becomes

$$\frac{D\mathbf{v}}{Dt} + 2\boldsymbol{\Omega} \times \mathbf{v} = -\nabla \phi + b\mathbf{k}, \quad (2.94)$$

where $\phi = \delta p/\rho_0$ and $b = -g\delta\rho/\rho_0$ is the *buoyancy*. We should not and do not neglect the term $g\delta\rho$, for there is no reason to believe it to be small: $\delta\rho$ may be small, but g is big! Equation (2.94)

is the momentum equation in the Boussinesq approximation, and it is common to say that the Boussinesq approximation ignores all variations of density of a fluid in the momentum equation, except when associated with the gravitational term.

For most large-scale motions in the ocean the *deviation* pressure and density fields are also approximately in hydrostatic balance, and in that case the vertical component of (2.94) becomes

$$\frac{\partial \phi}{\partial z} = b. \quad (2.95)$$

A condition for (2.95) to hold is that vertical accelerations are small *compared to* $g \delta \rho / \rho_0$, and *not compared to the acceleration due to gravity itself*. For more discussion of this point, see section 2.7.

Mass continuity

The unapproximated mass continuity equation is

$$\frac{D\delta\rho}{Dt} + (\rho_0 + \delta\rho)\nabla \cdot \mathbf{v} = 0. \quad (2.96)$$

Provided that time scales advectively — that is to say that D/Dt scales in the same way as $\mathbf{v} \cdot \nabla$ — then we may approximate this equation by

$$\nabla \cdot \mathbf{v} = 0, \quad (2.97)$$

which is the same as that for a constant density fluid. This *absolutely does not* allow one to go back and use (2.96) to say that $D\delta\rho/Dt = 0$; the evolution of density is given by the thermodynamic equation in conjunction with an equation of state, and this should not be confused with the mass conservation equation. Note also that in eliminating the time-derivative of density we eliminate the possibility of sound waves.

Thermodynamic equation and equation of state

The Boussinesq equations are closed by the addition of an equation of state, a thermodynamic equation and, as appropriate, a salinity equation. Neglecting salinity for the moment, a useful starting point is to write the thermodynamic equation, (1.114), as

$$\frac{D\rho}{Dt} - \frac{1}{c_s^2} \frac{Dp}{Dt} = \frac{\dot{Q}}{(\partial\eta/\partial\rho)_p T} \approx -\dot{Q} \left(\frac{\rho_0 \beta_T}{c_p} \right) \quad (2.98)$$

using $(\partial\eta/\partial\rho)_p = (\partial\eta/\partial T)_p (\partial T/\partial\rho)_p \approx -c_p/(T\rho_0\beta_T)$. Given the expansions (2.88a) and (2.90a), (2.98) can be written to a good approximation as

$$\frac{D\delta\rho}{Dt} - \frac{1}{c_s^2} \frac{Dp_0}{Dt} = -\dot{Q} \left(\frac{\rho_0 \beta_T}{c_p} \right), \quad (2.99)$$

or, using (2.91a),

$$\frac{D}{Dt} \left(\delta\rho + \frac{\rho_0 g}{c_s^2} z \right) = -\dot{Q} \left(\frac{\rho_0 \beta_T}{c_p} \right), \quad (2.100)$$

as in (1.117). The severest approximation to this is to neglect the second term in brackets on the left-hand side, and noting that $b = -g\delta\rho/\rho_0$ we obtain

$$\frac{Db}{Dt} = \dot{b}, \quad (2.101)$$

where $\dot{b} = g\beta_T \dot{Q}/c_p$. The momentum equation (2.94), mass continuity equation (2.97) and thermodynamic equation (2.101) then form a closed set, called the *simple Boussinesq equations*.

A somewhat more accurate approach is to include the compressibility of the fluid that is due to the hydrostatic pressure. From (2.100), the potential density is given by $\delta\rho_{\text{pot}} = \delta\rho + \rho_0 gz/c_s^2$, and so the *potential buoyancy*, that is the buoyancy based on potential density, is given by

$$b_\sigma \equiv -g \frac{\delta\rho_{\text{pot}}}{\rho_0} = -\frac{g}{\rho_0} \left(\delta\rho + \frac{\rho_0 gz}{c_s^2} \right) = b - g \frac{z}{H_\rho}, \quad (2.102)$$

where $H_\rho = c_s^2/g$. The thermodynamic equation, (2.100), may then be written

$$\frac{Db_\sigma}{Dt} = \dot{b}_\sigma, \quad (2.103)$$

where $\dot{b}_\sigma = \dot{b}$. Buoyancy itself is obtained from b_σ by the ‘equation of state’, $b = b_\sigma + gz/H_\rho$.

In many applications we may need to use a still more accurate equation of state. In that case (and see section 1.7.3) we replace (2.101) by the thermodynamic equations

$$\frac{D\Theta}{Dt} = \dot{\Theta}, \quad \frac{DS}{Dt} = \dot{S}, \quad (2.104a,b)$$

where Θ is a thermodynamic state variable, such as potential temperature or entropy, S is salinity, and an equation of state then gives the buoyancy. The equation of state has the general form $b = b(\Theta, S, p)$, but to be consistent with the level of approximation in the other Boussinesq equations we replace p by the hydrostatic pressure calculated with the reference density, that is by $-\rho_0 gz$, and the equation of state then takes the general form

$$b = b(\Theta, S, z). \quad (2.105)$$

An example of (2.105) is (1.153), taken with the definition of buoyancy $b = -g\delta\rho/\rho_0$. The closed set of equations (2.94), (2.97), (2.104) and (2.105) are sometimes called the general Boussinesq equations, or, in oceanographic contexts, the seawater Boussinesq equations. Using an accurate equation of state and the Boussinesq approximation is the procedure used in many comprehensive ocean general circulation models. The Boussinesq equations, which with the hydrostatic and traditional approximations are often considered to be the oceanic primitive equations, are summarized in the shaded box on the next page.

♦ Mean stratification and the buoyancy frequency

The processes that cause density to vary in the vertical often differ from those that cause it to vary in the horizontal. For this reason it is sometimes useful to write $\rho = \rho_0 + \tilde{\rho}(z) + \rho'(x, y, z, t)$ and define $\tilde{b}(z) \equiv -g\tilde{\rho}/\rho_0$ and $b' \equiv -g\rho'/\rho_0$. Using the hydrostatic equation to evaluate pressure, the thermodynamic equation (2.98) becomes, to a good approximation,

$$\frac{Db'}{Dt} + N^2 w = 0, \quad (2.106)$$

where D/Dt remains a three-dimensional operator and

$$N^2(z) = \left(\frac{d\tilde{b}}{dz} - \frac{g^2}{c_s^2} \right) = \frac{d\tilde{b}_\sigma}{dz}, \quad (2.107)$$

where $\tilde{b}_\sigma = \tilde{b} - gz/H_\rho$. The quantity N^2 is a measure of the mean stratification of the fluid, and is equal to the vertical gradient of the mean potential buoyancy. N is known as the buoyancy frequency, something we return to in section 2.10. Equations (2.106) and (2.107) also hold in the simple Boussinesq equations, but with $c_s^2 = \infty$.

Summary of Boussinesq Equations

The simple Boussinesq equations are, for an inviscid fluid:

$$\text{momentum equations:} \quad \frac{D\mathbf{v}}{Dt} + \mathbf{f} \times \mathbf{v} = -\nabla\phi + b\mathbf{k}, \quad (\text{B.1})$$

$$\text{mass conservation:} \quad \nabla \cdot \mathbf{v} = 0, \quad (\text{B.2})$$

$$\text{buoyancy equation:} \quad \frac{Db}{Dt} = \dot{b}. \quad (\text{B.3})$$

A more general form replaces the buoyancy equation by:

$$\text{thermodynamic equation:} \quad \frac{D\Theta}{Dt} = \dot{\Theta}, \quad (\text{B.4})$$

$$\text{salinity equation:} \quad \frac{DS}{Dt} = \dot{S}, \quad (\text{B.5})$$

$$\text{equation of state:} \quad b = b(\Theta, S, z). \quad (\text{B.6})$$

If the equation of state is $b = b(\Theta, S, \phi)$ then energy conservation is not assured.

2.4.3 Energetics of the Boussinesq system

In a uniform gravitational field but with no other forcing or dissipation, we write the simple Boussinesq equations as

$$\frac{D\mathbf{v}}{Dt} + 2\boldsymbol{\Omega} \times \mathbf{v} = b\mathbf{k} - \nabla\phi, \quad \nabla \cdot \mathbf{v} = 0, \quad \frac{Db}{Dt} = 0. \quad (2.108a,b,c)$$

From (2.108a) and (2.108b) the kinetic energy density evolution (cf. section 1.10) is given by

$$\frac{1}{2} \frac{Dv^2}{Dt} = bw - \nabla \cdot (\phi\mathbf{v}), \quad (2.109)$$

where the constant reference density ρ_0 is omitted. Let us now define the potential $\Phi \equiv -z$, so that $\nabla\Phi = -\mathbf{k}$ and

$$\frac{D\Phi}{Dt} = \nabla \cdot (\mathbf{v}\Phi) = -w, \quad (2.110)$$

and using this and (2.108c) gives

$$\frac{D}{Dt}(b\Phi) = -wb. \quad (2.111)$$

Adding (2.111) to (2.109) and expanding the material derivative gives

$$\frac{\partial}{\partial t} \left(\frac{1}{2} v^2 + b\Phi \right) + \nabla \cdot \left[\mathbf{v} \left(\frac{1}{2} v^2 + b\Phi + \phi \right) \right] = 0. \quad (2.112)$$

This constitutes an energy equation for the Boussinesq system, and may be compared to (1.196). The energy density (divided by ρ_0) is just $v^2/2 + b\Phi$. What does the term $b\Phi$ represent? Its integral, multiplied by ρ_0 , is the potential energy of the flow minus that of the basic state, or $\int g(\rho - \rho_0)z \, dz$. If there were a heating term on the right-hand side of (2.108c) this would directly provide a source of potential energy, rather than internal energy as in the compressible system. Because the fluid is incompressible, there is no conversion from kinetic and potential energy into internal energy.

♦ *Energetics with a general equation of state*

Now consider the energetics of the general Boussinesq equations. Suppose first that we allow the equation of state to be a function of pressure; the equations of motion are then (2.108) except that (2.108c) is replaced by

$$\frac{D\Theta}{Dt} = 0, \quad \frac{DS}{Dt} = 0, \quad b = b(\Theta, S, \phi). \quad (2.113a,b,c)$$

where Θ is some conservative thermodynamic variable and S is salinity. A little algebraic experimentation will reveal that no energy conservation law of the form (2.112) generally exists for this system! The problem arises because, by requiring the fluid to be incompressible, we eliminate the proper conversion of internal energy to kinetic energy. However, if we use the approximation $b = b(\Theta, S, z)$, the system does conserve an energy, as we now show.⁵

Define the potential, Π , as the integral of b at constant potential temperature and salinity, namely

$$\Pi(\Theta, S, z) \equiv - \int_a^z b \, dz', \quad (2.114)$$

where a is any constant, so that $\partial\Pi/\partial z = -b$. (The quantity Π is related to the dynamic enthalpy of section 1.7.3.) Taking the material derivative of the left-hand side gives

$$\frac{D\Pi}{Dt} = \left(\frac{\partial\Pi}{\partial\Theta} \right)_{S,z} \frac{D\Theta}{Dt} + \left(\frac{\partial\Pi}{\partial S} \right)_{\Theta,z} \frac{DS}{Dt} + \left(\frac{\partial\Pi}{\partial z} \right)_{\Theta,S} \frac{Dz}{Dt} = -bw, \quad (2.115)$$

using (2.113a,b). Combining (2.115) and (2.109) gives

$$\frac{\partial}{\partial t} \left(\frac{1}{2} \mathbf{v}^2 + \Pi \right) + \nabla \cdot \left[\mathbf{v} \left(\frac{1}{2} \mathbf{v}^2 + \Pi + \phi \right) \right] = 0. \quad (2.116)$$

Thus, energetic consistency is maintained with an arbitrary equation of state, provided the pressure is replaced by a function of z . If b is not an explicit function of z in the equation of state, the conservation law is identical to (2.112).

2.5 EQUATIONS FOR A STRATIFIED ATMOSPHERE: THE ANELASTIC APPROXIMATION

2.5.1 Preliminaries

In the atmosphere the density varies significantly, especially in the vertical. However deviations of both ρ and p from a statically balanced state are often quite small, and the relative vertical variation of potential temperature is also small. We can usefully exploit these observations to give a somewhat simplified set of equations, useful both for theoretical and numerical analyses because sound waves are eliminated by way of an ‘anelastic’ approximation.⁶ To begin we set

$$\rho = \bar{\rho}(z) + \delta\rho(x, y, z, t), \quad p = \bar{p}(z) + \delta p(x, y, z, t), \quad (2.117a,b)$$

where we assume that $|\delta\rho| \ll |\bar{\rho}|$ and we define $\tilde{p}$ such that

$$\frac{\partial \tilde{p}}{\partial z} \equiv -g\bar{\rho}(z). \quad (2.118)$$

The notation is similar to that for the Boussinesq case except that, importantly, the density basic state is now a (given) function of vertical coordinate. As with the Boussinesq case, the idea is to ignore dynamic variations of density (i.e., of $\delta\rho$) except where associated with gravity. First recall a couple of ideal gas relationships involving potential temperature, θ . If we define $s = \log \theta$ (so that s is entropy divided by c_p) then

$$s = \log \theta = \log T - \frac{R}{c_p} \log p = \frac{1}{\gamma} \log p - \log \rho, \quad (2.119)$$

where $\gamma = c_p/c_v$, implying

$$\delta s = \frac{\delta \theta}{\theta} = \frac{1}{\gamma} \frac{\delta p}{p} - \frac{\delta \rho}{\rho} \approx \frac{1}{\gamma} \frac{\delta p}{\bar{p}} - \frac{\delta \rho}{\bar{\rho}}. \quad (2.120)$$

Further, if $\tilde{s} \equiv \gamma^{-1} \log \bar{p} - \log \bar{\rho}$ then

$$\frac{d\tilde{s}}{dz} = \frac{1}{\gamma \bar{p}} \frac{d\bar{p}}{dz} - \frac{1}{\bar{\rho}} \frac{d\bar{\rho}}{dz} = -\frac{g\bar{p}}{\gamma \bar{p}} - \frac{1}{\bar{\rho}} \frac{d\bar{\rho}}{dz}. \quad (2.121)$$

In the atmosphere, the left-hand side is, typically, much smaller than either of the two terms on the right-hand side.

2.5.2 The momentum equation

The exact inviscid horizontal momentum equation is

$$(\bar{\rho} + \delta \rho) \left(\frac{D\mathbf{u}}{Dt} + \mathbf{f} \times \mathbf{u} \right) = -\nabla_z \delta p. \quad (2.122)$$

Neglecting $\delta \rho$ where it appears with $\bar{\rho}$ leads to

$$\frac{D\mathbf{u}}{Dt} + \mathbf{f} \times \mathbf{u} = -\nabla_z \phi, \quad (2.123)$$

where $\phi = \delta p / \bar{\rho}$, and this is similar to the corresponding equation in the Boussinesq approximation.

The vertical component of the inviscid momentum equation is, without approximation,

$$(\bar{\rho} + \delta \rho) \frac{Dw}{Dt} = -\frac{\partial \bar{p}}{\partial z} - \frac{\partial \delta p}{\partial z} - g\bar{\rho} - g\delta \rho = -\frac{\partial \delta p}{\partial z} - g\delta \rho. \quad (2.124)$$

using (2.118). Neglecting $\delta \rho$ on the left-hand side we obtain

$$\frac{Dw}{Dt} = -\frac{1}{\bar{\rho}} \frac{\partial \delta p}{\partial z} - g \frac{\delta \rho}{\bar{\rho}} = -\frac{\partial}{\partial z} \left(\frac{\delta p}{\bar{\rho}} \right) - \frac{\delta p}{\bar{\rho}^2} \frac{\partial \bar{\rho}}{\partial z} - g \frac{\delta \rho}{\bar{\rho}}. \quad (2.125)$$

This is not a useful form for a gaseous atmosphere, since the variation of the mean density cannot be ignored. However, we may eliminate $\delta \rho$ in favour of δs using (2.120) to give

$$\frac{Dw}{Dt} = g \delta s - \frac{\partial}{\partial z} \left(\frac{\delta p}{\bar{\rho}} \right) - \frac{g}{\gamma} \frac{\delta p}{\bar{p}} - \frac{\delta p}{\bar{\rho}^2} \frac{\partial \bar{\rho}}{\partial z}, \quad (2.126)$$

and using (2.121) gives

$$\frac{Dw}{Dt} = g \delta s - \frac{\partial}{\partial z} \left(\frac{\delta p}{\bar{\rho}} \right) + \frac{d\tilde{s}}{dz} \frac{\delta p}{\bar{\rho}}. \quad (2.127)$$

What have these manipulations gained us? Two things:

- (i) The gravitational term now involves δs rather than $\delta \rho$ which enables a more direct connection with the thermodynamic equation.
- (ii) The potential temperature scale height (~ 100 km) in the atmosphere is much larger than the density scale height (~ 10 km), and so the last term in (2.127) is small.

The second item thus suggests that we choose our reference state to be one of constant potential temperature. The term $d\bar{s}/dz$ then vanishes and the vertical momentum equation becomes

$$\frac{Dw}{Dt} = g\delta s - \frac{\partial\phi}{\partial z}, \quad (2.128)$$

where $\delta s = \delta\theta/\theta_0$, where θ_0 is a constant. If we define a buoyancy by $b_a \equiv g\delta s = g\delta\theta/\theta_0$, then (2.123) and (2.128) have the same form as the Boussinesq momentum equations, but with a slightly different definition of buoyancy.

2.5.3 Mass conservation

Using (2.117a) the mass conservation equation may be written, without approximation, as

$$\frac{\partial\delta\rho}{\partial t} + \nabla \cdot [(\bar{\rho} + \delta\rho)\mathbf{v}] = 0. \quad (2.129)$$

We neglect $\delta\rho$ where it appears with $\bar{\rho}$ in the divergence term. Further, the local time derivative will be small if time itself is scaled advectively (i.e., $T \sim L/U$ and sound waves do not dominate), giving

$$\nabla \cdot \mathbf{u} + \frac{1}{\bar{\rho}} \frac{\partial}{\partial z}(\bar{\rho}w) = 0. \quad (2.130)$$

It is here that the eponymous anelastic approximation arises: the elastic compressibility of the fluid is neglected, and this serves to eliminate sound waves. For reference, in spherical coordinates the equation is

$$\frac{1}{a \cos \vartheta} \frac{\partial u}{\partial \lambda} + \frac{1}{a \cos \vartheta} \frac{\partial}{\partial \vartheta}(v \cos \vartheta) + \frac{1}{\bar{\rho}} \frac{\partial(w\bar{\rho})}{\partial z} = 0. \quad (2.131)$$

In an ideal gas, the choice of constant potential temperature determines how the reference density $\bar{\rho}$ varies with height. In some circumstances it is convenient to let $\bar{\rho}$ be a constant, ρ_0 (effectively choosing a different equation of state), in which case the anelastic equations become identical to the Boussinesq equations, albeit with the buoyancy interpreted in terms of potential temperature in the former and density in the latter.

2.5.4 Thermodynamic equation

The thermodynamic equation for an ideal gas may be written

$$\frac{D \ln \theta}{Dt} = \frac{\dot{Q}}{Tc_p}. \quad (2.132)$$

In the anelastic equations, $\theta = \theta_0 + \delta\theta$, where θ_0 is constant, and the thermodynamic equation is

$$\frac{D\delta s}{Dt} = \frac{\bar{\theta}}{Tc_p} \dot{Q}. \quad (2.133)$$

Summarizing, the complete set of anelastic equations, with rotation but with no dissipation or diabatic terms, is

$$\begin{aligned} \frac{D\mathbf{v}}{Dt} + 2\boldsymbol{\Omega} \times \mathbf{v} &= \mathbf{k}b_a - \nabla\phi, \\ \frac{Db_a}{Dt} &= 0, \\ \nabla \cdot (\bar{\rho}\mathbf{v}) &= 0, \end{aligned} \quad (2.134a,b,c)$$

where $b_a = g\delta s = g\delta\theta/\theta_0$. The anelastic equations are sometimes called the ‘weak Boussinesq equations’, with the original incompressible set then called the ‘strong Boussinesq equations’.

The main difference between the anelastic and Boussinesq sets is in the mass continuity equation, and when $\bar{\rho} = \rho_0 = \text{constant}$ the two equation sets are identical. However, whereas the Boussinesq approximation is a very good one for ocean dynamics, the anelastic approximation is less so for large-scale atmosphere flow: the constancy of the reference potential temperature state is not a particularly good approximation, and the deviations in density from its reference profile are not especially small, leading to inaccuracies in the momentum equation. Nevertheless, the anelastic equations have been used very productively in limited area ‘large-eddy simulations’ where one does not wish to make the hydrostatic approximation but where sound waves are unimportant.⁷ The equations also provide a good jumping-off point for theoretical studies and for the still simpler models of chapter 5.

2.5.5 ♦ Energetics of the anelastic equations

Conservation of energy follows in much the same way as for the Boussinesq equations, except that $\bar{\rho}$ enters. Take the dot product of (2.134a) with $\bar{\rho}\mathbf{v}$ to obtain

$$\bar{\rho} \frac{D}{Dt} \left(\frac{1}{2} \mathbf{v}^2 \right) = -\nabla \cdot (\phi \bar{\rho} \mathbf{v}) + b_a \bar{\rho} w. \quad (2.135)$$

Now, define a potential $\Phi(z)$ such that $\nabla\Phi = -\mathbf{k}$, and so

$$\bar{\rho} \frac{D\Phi}{Dt} = -w\bar{\rho}. \quad (2.136)$$

Combining this with the thermodynamic equation (2.134b) gives

$$\bar{\rho} \frac{D(b_a \Phi)}{Dt} = -w b_a \bar{\rho}. \quad (2.137)$$

Adding this to (2.135) gives

$$\bar{\rho} \frac{D}{Dt} \left(\frac{1}{2} \mathbf{v}^2 + b_a \Phi \right) = -\nabla \cdot (\phi \bar{\rho} \mathbf{v}), \quad (2.138)$$

or, expanding the material derivative,

$$\frac{\partial}{\partial t} \left[\bar{\rho} \left(\frac{1}{2} \mathbf{v}^2 + b_a \Phi \right) \right] + \nabla \cdot \left[\bar{\rho} \mathbf{v} \left(\frac{1}{2} \mathbf{v}^2 + b_a \Phi + \phi \right) \right] = 0. \quad (2.139)$$

This equation has the form

$$\frac{\partial E}{\partial t} + \nabla \cdot [\mathbf{v}(E + \bar{\rho}\phi)] = 0, \quad (2.140)$$

where $E = \bar{\rho}(\mathbf{v}^2/2 + b_a \Phi)$ is the energy density of the flow. This is a consistent energetic equation for the system, and when integrated over a closed domain the total energy is evidently conserved. The total energy density comprises the kinetic energy and a term $\bar{\rho} b_a \Phi$, which is analogous to the potential energy of a simple Boussinesq system. However, it is not exactly equal to potential energy because b_a is the buoyancy based on potential temperature, not density; rather, the term combines contributions from both the internal energy and the potential energy into an enthalpy-like quantity.

2.6 CHANGING VERTICAL COORDINATE

Although using z as a vertical coordinate is a natural choice given our Cartesian worldview, it is not the only option, nor is it always the most useful one. Any variable that has a one-to-one correspondence with z in the vertical, so any variable that varies monotonically with z , could be used; pressure and, more surprisingly, entropy, are common choices. In the atmosphere pressure almost always falls monotonically with height, and using it instead of z provides a useful simplification of the mass conservation and geostrophic relations, as well as a more direct connection with observations, which are often taken at fixed values of pressure. (In the ocean pressure coordinates are essentially almost the same as height coordinates because density is almost constant.) Entropy seems an exotic vertical coordinate, but it is very useful in adiabatic flow and we consider it in chapter 3.

2.6.1 General relations

First consider a general vertical coordinate, ξ . Any variable Ψ that is a function of the coordinates (x, y, z, t) may be expressed instead in terms of (x, y, ξ, t) by considering ξ to be function of the independent variables (x, y, z, t) . Derivatives with respect to z and ξ are related by

$$\frac{\partial \Psi}{\partial \xi} = \frac{\partial \Psi}{\partial z} \frac{\partial z}{\partial \xi} \quad \text{and} \quad \frac{\partial \Psi}{\partial z} = \frac{\partial \Psi}{\partial \xi} \frac{\partial \xi}{\partial z}. \quad (2.141a,b)$$

Horizontal derivatives in the two coordinate systems are related by the chain rule,

$$\left(\frac{\partial \Psi}{\partial x} \right)_{\xi} = \left(\frac{\partial \Psi}{\partial x} \right)_z + \left(\frac{\partial z}{\partial x} \right)_{\xi} \frac{\partial \Psi}{\partial z}, \quad (2.142)$$

and similarly for time.

The material derivative in ξ coordinates may be derived by transforming the original expression in z coordinates using the chain rule, but because (x, y, ξ, t) are independent coordinates, and noting that the ‘vertical velocity’ in ξ coordinates is just $\dot{\xi}$ (i.e., $D\xi/Dt$, just as the vertical velocity in z coordinates is $w = Dz/Dt$), we can write down

$$\frac{D\Psi}{Dt} = \frac{\partial \Psi}{\partial t} \Big|_{x,y,\xi} + \mathbf{u} \cdot \nabla_{\xi} \Psi + \xi \frac{\partial \Psi}{\partial \xi}, \quad (2.143)$$

where ∇_{ξ} is the gradient operator at constant ξ . The operator D/Dt is the same in z or ξ coordinates because it is the total derivative of some property of a fluid parcel, and this is independent of the coordinate system. However, the individual terms within it will differ between coordinate systems.

2.6.2 Pressure coordinates

In pressure coordinates the analogue of the vertical velocity is $\omega \equiv Dp/Dt$, and the advective derivative itself is given by

$$\frac{D}{Dt} = \frac{\partial}{\partial t} + \mathbf{u} \cdot \nabla_p + \omega \frac{\partial}{\partial p}. \quad (2.144)$$

Note, though, that advective derivative is the same operator as it is in height coordinates, since it is just the total derivative of a given fluid parcel; it is just written with different coordinates.

To obtain an expression for the pressure force, now let $\xi = p$ in (2.142) and apply the relationship to p itself to give

$$0 = \left(\frac{\partial p}{\partial x} \right)_z + \left(\frac{\partial z}{\partial x} \right)_p \frac{\partial p}{\partial z}, \quad (2.145)$$

which, using the hydrostatic relationship, gives

$$\left(\frac{\partial p}{\partial x}\right)_z = \rho \left(\frac{\partial \Phi}{\partial x}\right)_p, \quad (2.146)$$

where $\Phi = gz$ is the *geopotential*. Thus, the horizontal pressure force in the momentum equations is

$$\frac{1}{\rho} \nabla_z p = \nabla_p \Phi, \quad (2.147)$$

where the subscripts on the gradient operator indicate that the horizontal derivatives are taken at constant z or constant p . The horizontal momentum equation thus becomes

$$\frac{D\mathbf{u}}{Dt} + \mathbf{f} \times \mathbf{u} = -\nabla_p \Phi, \quad (2.148)$$

where D/Dt is given by (2.144). The hydrostatic equation in height coordinates is $\partial p / \partial z = -\rho g$ and in pressure coordinates this becomes

$$\frac{\partial \Phi}{\partial p} = -\alpha \quad \text{or} \quad \frac{\partial \Phi}{\partial p} = -\frac{p}{RT}. \quad (2.149)$$

The mass continuity equation simplifies attractively in pressure coordinates, if the hydrostatic approximation is used. Recall that the mass conservation equation can be derived from the material form

$$\frac{D}{Dt}(\rho \delta V) = 0, \quad (2.150)$$

where $\delta V = \delta x \delta y \delta z$ is a volume element. But by the hydrostatic relationship $\rho \delta z = -(1/g) \delta p$ and thus

$$\frac{D}{Dt}(\delta x \delta y \delta p) = 0. \quad (2.151)$$

This is completely analogous to the expression for the material conservation of volume in an incompressible fluid, (1.15). Thus, without further ado, we write the mass conservation in pressure coordinates as

$$\nabla_p \cdot \mathbf{u} + \frac{\partial \omega}{\partial p} = 0, \quad (2.152)$$

where the horizontal derivative is taken at constant pressure.

The (adiabatic) thermodynamic equation is still $D\theta/Dt = 0$, and θ may be related to pressure and temperature using its definition and the ideal gas equation to complete the equation set. However, because the hydrostatic equation is written in terms of temperature and not potential temperature it is convenient to write the thermodynamic equation accordingly. To do this we begin with the thermodynamic equation in the form of (1.99b), namely $c_p DT/Dt - \alpha Dp/Dt = 0$. Since $\omega \equiv Dp/Dt$ this equation is simply

$$c_p \frac{DT}{Dt} - \frac{RT}{p} \omega = 0, \quad (2.153)$$

which is an appropriate thermodynamic equation in pressure coordinates. It is sometimes useful to write this as

$$\frac{\partial T}{\partial t} + u \frac{\partial T}{\partial x} + v \frac{\partial T}{\partial y} - \omega S_p = 0, \quad \text{where} \quad S_p = \frac{\kappa T}{p} - \frac{\partial T}{\partial p} = -\frac{T}{\theta} \frac{\partial \theta}{\partial p}, \quad (2.154a,b)$$

Equations of Motion in Pressure and Log-pressure Coordinates

The adiabatic, inviscid primitive equations in pressure coordinates are:

$$\frac{D\mathbf{u}}{Dt} + \mathbf{f} \times \mathbf{u} = -\nabla_p \Phi, \quad (\text{P.1})$$

$$\frac{\partial \Phi}{\partial p} = \frac{-RT}{p}, \quad (\text{P.2})$$

$$\nabla_p \cdot \mathbf{u} + \frac{\partial \omega}{\partial p} = 0, \quad (\text{P.3})$$

$$c_p \frac{DT}{Dt} - \frac{RT}{p} \omega = 0 \quad \text{or} \quad \frac{\partial T}{\partial t} + u \frac{\partial T}{\partial x} + v \frac{\partial T}{\partial y} - \omega S_p = 0. \quad (\text{P.4})$$

where $S_p = \kappa T/p - \partial T/\partial p$ and $\kappa = R/c_p$. The above equations are, respectively, the horizontal momentum equation, the hydrostatic equation, the mass continuity equation and the thermodynamic equation. Using hydrostasy and the ideal gas relation, the thermodynamic equation may also be written as

$$\frac{\partial T}{\partial t} + u \frac{\partial T}{\partial x} + v \frac{\partial T}{\partial y} + \omega \frac{\partial s}{\partial p} = 0, \quad (\text{P.5})$$

where $s = T + gz/c_p$ is the dry static energy divided by c_p .

The corresponding equations in log-pressure coordinates are

$$\frac{D\mathbf{u}}{Dt} + \mathbf{f} \times \mathbf{u} = -\nabla_Z \Phi, \quad (\text{P.6})$$

$$\frac{\partial \Phi}{\partial Z} = \frac{RT}{H}, \quad (\text{P.7})$$

$$\nabla_Z \cdot \mathbf{u} + \frac{1}{\rho_R} \frac{\partial \rho_R W}{\partial z} = 0, \quad (\text{P.8})$$

$$c_p \frac{DT}{Dt} + W \frac{RT}{H} = 0 \quad \text{or} \quad \frac{\partial T}{\partial t} + u \frac{\partial T}{\partial x} + v \frac{\partial T}{\partial y} + W S_Z = 0. \quad (\text{P.9})$$

where $\rho_R = \rho_0 \exp(-Z/H)$ and $S_Z = \kappa T/H + \partial T/\partial Z$. The thermodynamic equation may also be written as

$$\frac{\partial}{\partial t} \frac{\partial \Phi}{\partial Z} + u \frac{\partial}{\partial x} \frac{\partial \Phi}{\partial Z} + v \frac{\partial}{\partial y} \frac{\partial \Phi}{\partial Z} + W N_*^2 = 0, \quad (\text{P.10})$$

where $N_*^2 = (R/H)S_Z$.

having used the ideal gas equation and the definition of potential temperature, with $\kappa = R/c_p$. Evidently, S_p is an appropriate measure of static stability in pressure coordinates and it is closely related to the buoyancy frequency N , as we see in the next subsection.

The main practical difficulty with the pressure-coordinate equations is the lower boundary condition. Using

$$w \equiv \frac{Dz}{Dt} = \frac{\partial z}{\partial t} + \mathbf{u} \cdot \nabla_p z + \omega \frac{\partial z}{\partial p}, \quad (2.155)$$

and (2.149), the boundary condition of $w = 0$ at $z = z_s$ becomes

$$\frac{\partial \Phi}{\partial t} + \mathbf{u} \cdot \nabla_p \Phi - \alpha \omega = 0 \quad (2.156)$$

at $p(x, y, z_s, t)$. In theoretical studies, it is common to assume that the lower boundary is in fact a constant pressure surface and simply assume that $\omega = 0$, or that $\omega = -\alpha^{-1} \partial \Phi / \partial t$. For more realistic studies the fact that the level $z = 0$ is not a coordinate surface must be properly considered. For this reason, and especially if the lower boundary is uneven because of the presence of topography, so-called *sigma coordinates* are sometimes used, in which the vertical coordinate is chosen so that the lower boundary is itself a coordinate surface. Sigma coordinates may use height itself as a vertical measure (typical in oceanic applications) or use pressure (typical in atmospheric applications). In the latter case the vertical coordinate is $\sigma = p/p_s$ where $p_s(x, y, t)$ is the surface pressure. The difficulty of applying (2.156) is replaced by a prognostic equation for the surface pressure, derived from the mass conservation equation.

Interestingly, the pressure coordinate equations (collected together in the shaded box on the preceding page) are isomorphic to the hydrostatic, salt-free general Boussinesq equations (see the shaded box on page 74) with $z \Leftrightarrow -p$, $w \Leftrightarrow -\omega$, $\phi \Leftrightarrow \Phi$, $b \Leftrightarrow \alpha$, $\Theta \Leftrightarrow \theta$ and an equation of state $b = b(\Theta, z) \Leftrightarrow \alpha = \alpha(\theta, p)$ (and in an ideal gas $\alpha = (\theta R/p_R)(p_R/p)^{1/\gamma}$). Conservation laws that hold in one system can therefore be expected to have an analogue in the other.

2.6.3 Log-pressure coordinates

A variant of pressure coordinates arises by using *log-pressure* coordinates, in which the vertical coordinate is $Z = -H \ln(p/p_R)$ where p_R is a reference pressure (say 1000 mb) and H is a constant (for example a scale height RT_0/g where T_0 is a constant) so that Z has units of length. (Uppercase letters are conventionally used for some variables in log-pressure coordinates, and these are not to be confused with scaling parameters.) The ‘vertical velocity’ for the system is now

$$W \equiv \frac{DZ}{Dt}, \quad (2.157)$$

and the advective derivative is

$$\frac{D}{Dt} \equiv \frac{\partial}{\partial t} + \mathbf{u} \cdot \nabla_p + W \frac{\partial}{\partial Z}. \quad (2.158)$$

The horizontal momentum equation is unaltered from (2.148), although we use (2.158) to evaluate the advective derivative. It is straightforward to show that the hydrostatic equation becomes

$$\frac{\partial \Phi}{\partial Z} = \frac{RT}{H} \quad (2.159)$$

The mass continuity equation (2.152) becomes

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial W}{\partial Z} - \frac{W}{H} = 0. \quad (2.160)$$

which may be written as

$$\nabla_Z \cdot \mathbf{u} + \frac{1}{\rho_R} \frac{\partial(\rho_R W)}{\partial Z} = 0 \quad (2.161)$$

where $\nabla_Z \cdot$ is the divergence at constant Z and $\rho_R = \rho_0 \exp(-Z/H)$, so giving a form similar to the mass conservation equation in the anelastic equations. (The value of the constant ρ_0 may be set to one.)

As with pressure coordinates, it is convenient to write the thermodynamic equation in terms of temperature and not potential temperature, and in an analogous procedure to that leading to (2.153) we straightforwardly obtain

$$c_p \frac{DT}{Dt} + W \frac{RT}{H} = 0. \quad (2.162)$$

This may be written as

$$\frac{\partial T}{\partial t} + u \frac{\partial T}{\partial x} + v \frac{\partial T}{\partial y} + WS_Z = 0 \quad (2.163)$$

where

$$S_Z = \frac{\kappa T}{H} + \frac{\partial T}{\partial Z}. \quad (2.164)$$

and we may note that $S_Z = S_p p/H$. Using the hydrostatic equation we may write (2.163) as

$$\frac{\partial}{\partial t} \frac{\partial \Phi}{\partial Z} + u \frac{\partial}{\partial x} \frac{\partial \Phi}{\partial Z} + v \frac{\partial}{\partial y} \frac{\partial \Phi}{\partial Z} + WN_*^2 = 0, \quad (2.165)$$

where $N_*^2 = (R/H)S_Z$. The quantity N_* is not exactly equal to the square of the buoyancy frequency as normally defined (for an ideal gas $N^2 = (g/\theta)\partial\theta/\partial z$), but the two can be shown to be related by $N_*/N = p/(\rho gH) = RT/gH$, and are equal for an isothermal atmosphere.⁸ Integrating the hydrostatic equation between two pressure levels gives, with $\Phi = gz$,

$$z(p_2) - z(p_1) = -\frac{R}{g} \int_{p_1}^{p_2} T d \ln p. \quad (2.166)$$

Thus, the thickness of the layer is proportional to the average temperature of the layer, and at constant temperature the geometric height increases linearly with the logarithm of pressure. At a temperature of 240 K (280 K) the scale height, RT/g , is about 7 km (8.2 km). A useful rule of thumb for Earth's atmosphere (and one that holds at 240 K) is that geometric height increases by about 16 km for each factor of ten decrease in pressure, and pressures of 1000 hPa, 100 hPa, 10 hPa roughly correspond to heights of 0, 16 km 32 km and so on.

2.7 SCALING FOR HYDROSTATIC BALANCE

We first encountered hydrostatic balance in section 1.3.3; we now look in more detail at the conditions required for it to hold. Along with geostrophic balance, considered in the next section, it is one of the most fundamental balances in geophysical fluid dynamics. The corresponding states, hydrostasy and geostrophy, are not exactly realized, but their approximate satisfaction has profound consequences on the behaviour of the atmosphere and ocean.

2.7.1 Preliminaries

Consider the relative sizes of terms in (2.77c):

$$\frac{W}{T} + \frac{UW}{L} + \frac{W^2}{H} + \Omega U \sim \left| \frac{1}{\rho} \frac{\partial p}{\partial z} \right| + g. \quad (2.167)$$

For most large-scale motion in the atmosphere and ocean the terms on the right-hand side are orders of magnitude larger than those on the left, and therefore must be approximately equal. Explicitly, suppose $W \sim 1 \text{ cm s}^{-1}$, $L \sim 10^5 \text{ m}$, $H \sim 10^3 \text{ m}$, $U \sim 10 \text{ m s}^{-1}$, $T = L/U$. Then by substituting

into (2.167) it seems that the pressure term is the only one which could balance the gravitational term, and we are led to approximate (2.77c) by,

$$\frac{\partial p}{\partial z} = -\rho g. \quad (2.168)$$

This equation, which is a vertical momentum equation, is known as *hydrostatic balance*.

However, (2.168) is not always a useful equation! Let us suppose that the density is a constant, ρ_0 . We can then write the pressure as

$$p(x, y, z, t) = p_0(z) + p'(x, y, z, t), \quad \text{where} \quad \frac{\partial p_0}{\partial z} \equiv -\rho_0 g. \quad (2.169)$$

That is, p_0 and ρ_0 are in hydrostatic balance. On the f -plane, the inviscid vertical momentum equation becomes, without approximation,

$$\frac{Dw}{Dt} = -\frac{1}{\rho_0} \frac{\partial p'}{\partial z}. \quad (2.170)$$

Thus, *for constant density fluids the gravitational term has no dynamical effect*: there is no buoyancy force, and the pressure term in the horizontal momentum equations can be replaced by p' . Hydrostatic balance, and in particular (2.169), is not a useful vertical momentum equation in this case. If the fluid is stratified, we should therefore subtract off the hydrostatic pressure associated with the mean density before we can determine whether hydrostasy is a useful *dynamical* approximation, accurate enough to determine the horizontal pressure gradients. This is automatic in the Boussinesq equations, where the vertical momentum equation is

$$\frac{Dw}{Dt} = -\frac{\partial \phi}{\partial z} + b, \quad (2.171)$$

and the hydrostatic balance of the basic state is already subtracted out. In the more general equation,

$$\frac{Dw}{Dt} = -\frac{1}{\rho} \frac{\partial p}{\partial z} - g, \quad (2.172)$$

we need to compare the advective term on the left-hand side with the pressure variations arising from horizontal flow in order to determine whether hydrostasy is an appropriate vertical momentum equation. Nevertheless, if we only need to determine the pressure for use in an equation of state then we simply need to compare the sizes of the dynamical terms in (2.77c) with g itself, in order to determine whether a hydrostatic approximation will suffice.

2.7.2 Scaling and the aspect ratio

In a Boussinesq fluid we write the horizontal and vertical momentum equations as

$$\frac{D\mathbf{u}}{Dt} + \mathbf{f} \times \mathbf{u} = -\nabla_z \phi, \quad \frac{Dw}{Dt} = -\frac{\partial \phi}{\partial z} + b. \quad (2.173a,b)$$

With $\mathbf{f} = 0$, (2.173a) implies the scaling

$$\phi \sim U^2. \quad (2.174)$$

If we use mass conservation, $\nabla_z \cdot \mathbf{u} + \partial w / \partial z = 0$, to scale vertical velocity then

$$w \sim W = \frac{H}{L} U = \alpha U, \quad (2.175)$$

where $\alpha \equiv H/L$ is the aspect ratio. The advective terms in the vertical momentum equation all scale as

$$\frac{Dw}{Dt} \sim \frac{UW}{L} = \frac{U^2 H}{L^2}. \quad (2.176)$$

Using (2.174) and (2.176) the ratio of the advective term to the pressure gradient term in the vertical momentum equations then scales as

$$\frac{|Dw/Dt|}{|\partial\phi/\partial z|} \sim \frac{U^2 H/L^2}{U^2/H} \sim \left(\frac{H}{L}\right)^2. \quad (2.177)$$

Thus, the condition for hydrostasy, that $|Dw/Dt|/|\partial\phi/\partial z| \ll 1$, is:

$$\alpha^2 \equiv \left(\frac{H}{L}\right)^2 \ll 1. \quad (2.178)$$

The advective term in the vertical momentum may then be neglected. Thus, *hydrostatic balance arises from a small aspect ratio approximation*.

We can obtain the same result more formally by non-dimensionalizing the momentum equations. Using uppercase symbols to denote scaling values we write

$$\begin{aligned} (x, y) &= L(\hat{x}, \hat{y}), & z &= H\hat{z}, & \mathbf{u} &= U\hat{\mathbf{u}}, & w &= W\hat{w} = \frac{HU}{L}\hat{w}, \\ t &= T\hat{t}; \hat{t} = \frac{L}{U}\hat{t}, & \phi &= \Phi\hat{\phi} = U^2\hat{\phi}, & b &= B\hat{b} = \frac{U^2}{H}\hat{b}, \end{aligned} \quad (2.179)$$

where the hatted variables are non-dimensional and the scaling for w is suggested by the mass conservation equation, $\nabla_z \cdot \mathbf{u} + \partial w/\partial z = 0$. Substituting (2.179) into (2.173) (with $\mathbf{f} = 0$) gives us the non-dimensional equations

$$\frac{D\hat{\mathbf{u}}}{D\hat{t}} = -\nabla\hat{\phi}, \quad \alpha^2 \frac{D\hat{w}}{D\hat{t}} = -\frac{\partial\hat{\phi}}{\partial\hat{z}} + \hat{b}, \quad (2.180a,b)$$

where $D/D\hat{t} = \partial/\partial\hat{t} + \hat{u}\partial/\partial\hat{x} + \hat{v}\partial/\partial\hat{y} + \hat{w}\partial/\partial\hat{z}$ and we use the convention that when ∇ operates on non-dimensional quantities the operator itself is non-dimensional. From (2.180b) it is clear that hydrostatic balance pertains when $\alpha^2 \ll 1$.

2.7.3 ♦ Effects of stratification on hydrostatic balance

To include the effects of stratification we need to involve the thermodynamic equation, so let us first write down the complete set of non-rotating dimensional equations:

$$\frac{D\mathbf{u}}{Dt} = -\nabla_z\phi, \quad \frac{Dw}{Dt} = -\frac{\partial\phi}{\partial z} + b', \quad (2.181a,b)$$

$$\frac{Db'}{Dt} + wN^2 = 0, \quad \nabla \cdot \mathbf{v} = 0. \quad (2.182a,b)$$

We have written, without approximation, $b = b'(x, y, z, t) + \tilde{b}(z)$, with $N^2 = d\tilde{b}/dz$; this separation is useful because the horizontal and vertical buoyancy variations may scale in different ways, and often N^2 may be regarded as given. (We have also redefined ϕ by subtracting off a static component in hydrostatic balance with $\tilde{b}$.) We non-dimensionalize (2.182) by first writing

$$\begin{aligned} (x, y) &= L(\hat{x}, \hat{y}), & z &= H\hat{z}, & \mathbf{u} &= U\hat{\mathbf{u}}, & w &= W\hat{w} = \epsilon \frac{HU}{L}\hat{w}, \\ t &= T\hat{t} = \frac{L}{U}\hat{t}, & \phi &= U^2\hat{\phi}, & b' &= \Delta b\hat{b}' = \frac{U^2}{H}\hat{b}', & N^2 &= \bar{N}^2\hat{N}^2, \end{aligned} \quad (2.183)$$

where ϵ is, for the moment, undetermined, $\bar{N}$ is a representative constant value of the buoyancy frequency and Δb scales only the horizontal buoyancy variations. Substituting (2.183) into (2.181) and (2.182) gives

$$\frac{D\hat{\mathbf{u}}}{D\hat{t}} = -\nabla_z \hat{\phi}, \quad \epsilon \alpha^2 \frac{D\hat{w}}{D\hat{t}} = -\frac{\partial \hat{\phi}}{\partial \hat{z}} + \hat{b}' \quad (2.184a,b)$$

$$\frac{U^2}{\bar{N}^2 H^2} \frac{D\hat{b}'}{D\hat{t}} + \epsilon \hat{w} \bar{N}^2 = 0, \quad \nabla \cdot \hat{\mathbf{u}} + \epsilon \frac{\partial \hat{w}}{\partial \hat{z}} = 0. \quad (2.185a,b)$$

where now $D/D\hat{t} = \partial/\partial\hat{t} + \hat{\mathbf{u}} \cdot \nabla_z + \epsilon \hat{w} \partial/\partial\hat{z}$. To obtain a non-trivial balance in (2.185a) we choose $\epsilon = U^2/(\bar{N}^2 H^2) \equiv Fr^2$, where Fr is the *Froude number*, a measure of the stratification of the flow. A strong stratification corresponds to a small Froude number. From (2.183), the vertical velocity then scales as

$$W = \frac{Fr^2 UH}{L} \quad (2.186)$$

and if the flow is highly stratified the vertical velocity will be even smaller than a pure aspect ratio scaling might suggest. (There must, therefore, be some cancellation in horizontal divergence in the mass continuity equation; that is, $|\nabla_z \cdot \mathbf{u}| \ll U/L$.) With this choice of ϵ the non-dimensional Boussinesq equations may be written:

$$\frac{D\hat{\mathbf{u}}}{D\hat{t}} = -\nabla_z \hat{\phi}, \quad Fr^2 \alpha^2 \frac{D\hat{w}}{D\hat{t}} = -\frac{\partial \hat{\phi}}{\partial \hat{z}} + \hat{b}' \quad (2.187a,b)$$

$$\frac{D\hat{b}'}{D\hat{t}} + \hat{w} \bar{N}^2 = 0, \quad \nabla \cdot \hat{\mathbf{u}} + Fr^2 \frac{\partial \hat{w}}{\partial \hat{z}} = 0. \quad (2.188a,b)$$

The non-dimensional parameters in the system are the aspect ratio and the Froude number (in addition to $\bar{N}$, but by construction this is just an order one function of z). From (2.187b) condition for hydrostatic balance to hold is evidently that

$$Fr^2 \alpha^2 \ll 1, \quad (2.189)$$

so generalizing the aspect ratio condition (2.178) to a stratified fluid. Because Fr is a measure of stratification, (2.189) formalizes our intuitive expectation that the more stratified a fluid the more vertical motion is suppressed and therefore the more likely hydrostatic balance is to hold. Equation (2.189) is equivalent to

$$\frac{U^2}{\bar{N}^2 H^2} \frac{H^2}{L^2} = \frac{U^2}{L^2 \bar{N}^2} \ll 1, \quad (2.190)$$

and in a hydrostatic model the condition is always, by construction, satisfied.

Why bother with any of this scaling? Why not just say that hydrostatic balance holds when $|Dw/Dt| \ll |\partial\phi/\partial z|$? One reason is that we do not have a good idea of the value of w from direct measurements, and it may change significantly in different oceanic and atmospheric parameter regimes. On the other hand the Froude number and the aspect ratio are familiar non-dimensional parameters with a wide applicability in other contexts, and which we can control in a laboratory setting or estimate in the ocean or atmosphere. Still, when equations are scaled, ascertaining which parameters are to be regarded as given and which should be derived is often a choice, rather than being set a priori.

2.7.4 Hydrostasy in the ocean and atmosphere

Is the hydrostatic approximation in fact a good one in the ocean and atmosphere?

In the ocean

For the large-scale ocean circulation, let $N \sim 10^{-2} \text{ s}^{-1}$, $U \sim 0.1 \text{ m s}^{-1}$ and $H \sim 1 \text{ km}$. Then $Fr = U/(NH) \sim 10^{-2} \ll 1$. Thus, $Fr^2 \alpha^2 \ll 1$ even for unit aspect-ratio motion. In fact, for larger scale flow the aspect ratio is also small; for basin-scale flow $L \sim 10^6 \text{ m}$ and $Fr^2 \alpha^2 \sim 0.01^2 \times 0.001^2 = 10^{-10}$ and hydrostatic balance is an extremely good approximation.

For intense convection, for example in the Labrador Sea, the hydrostatic approximation may be less appropriate, because the intense descending plumes may have an aspect ratio (H/L) of one or greater and the stratification is very weak. The hydrostatic condition then often becomes the requirement that the Froude number is small. Representative orders of magnitude are $U \sim W \sim 0.1 \text{ m s}^{-1}$, $H \sim 1 \text{ km}$ and $N \sim 10^{-3} \text{ s}^{-1}$ to 10^{-4} s^{-1} . For these values Fr ranges between 0.1 and 1, and at the upper end of this range hydrostatic balance is violated.

In the atmosphere

Over much of the troposphere $N \sim 10^{-2} \text{ s}^{-1}$ so that with $U = 10 \text{ m s}^{-1}$ and $H = 1 \text{ km}$ we find $Fr \sim 1$. Hydrostasy is then maintained because the aspect ratio H/L is much less than unity. For larger scale synoptic activity a larger vertical scale is appropriate, and with $H = 10 \text{ km}$ both the Froude number and the aspect ratio are much smaller than one; indeed with $L = 1000 \text{ km}$ we find $Fr^2 \alpha^2 \sim 0.1^2 \times 0.01^2 = 10^{-6}$ and the flow is hydrostatic to a very good approximation indeed. However, for smaller scale atmospheric motions associated with fronts and, especially, convection, there can be little expectation that hydrostatic balance will be a good approximation.

For large-scale flows in both atmosphere and ocean, the conceptual and practical simplifications afforded by the hydrostatic approximation can hardly be overemphasized.

2.8 GEOSTROPHIC AND THERMAL WIND BALANCE

We now consider the dominant dynamical balance in the horizontal components of the momentum equation. In the horizontal plane (meaning along geopotential surfaces) we find that the Coriolis term is much larger than the advective terms and the dominant balance is between it and the horizontal pressure force. This balance is called *geostrophic balance*, and it occurs when the Rossby number is small, as we now investigate.

2.8.1 The Rossby number

The *Rossby number* characterizes the importance of rotation in a fluid.⁹ It is, essentially, the ratio of the magnitude of the relative acceleration to the Coriolis acceleration, and it is of fundamental importance in geophysical fluid dynamics. It arises from a simple scaling of the horizontal momentum equation, namely

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{v} \cdot \nabla) \mathbf{u} + \mathbf{f} \times \mathbf{u} = -\frac{1}{\rho} \nabla_z p, \quad (2.191a)$$

$$\frac{U^2/L}{fU} \quad (2.191b)$$

where U is the approximate magnitude of the horizontal velocity and L is a typical length scale over which that velocity varies. (We assume that $W/H \leq U/L$, so that vertical advection does not dominate the advection.) The ratio of the sizes of the advective and Coriolis terms is defined to be the Rossby number,

$$Ro \equiv \frac{U}{fL}. \quad (2.192)$$

If the Rossby number is small then rotation effects are important, and as the values in Table 2.1 indicate this is the case for large-scale flow in both ocean and atmosphere.

Variable	Scaling symbol	Meaning	Atmos. value	Ocean value
(x, y)	L	Horizontal length scale	10^6 m	10^5 m
t	T	Time scale	1 day (10^5 s)	10 days (10^6 s)
(u, v)	U	Horizontal velocity	10 m s $^{-1}$	0.1 m s $^{-1}$
	Ro	Rossby number, U/fL	0.1	0.01

Table 2.1 Scales of large-scale flow in atmosphere and ocean. The choices given are representative of large-scale mid-latitude eddying motion in both systems.

Another intuitive way to think about the Rossby number is in terms of time scales. The Rossby number based on a time scale is

$$Ro_T \equiv \frac{1}{fT}, \quad (2.193)$$

where T is a time scale associated with the dynamics at hand. If the time scale is an advective one, meaning that $T \sim L/U$, then this definition is equivalent to (2.192). Now, $f = 2\Omega \sin \vartheta$, where Ω is the angular velocity of the rotating frame and equal to $2\pi/T_p$ where T_p is the period of rotation (24 hours). Thus,

$$Ro_T = \frac{T_p}{4\pi T \sin \vartheta} = \frac{T_i}{T}, \quad (2.194)$$

where $T_i = 1/f$ is the ‘inertial time scale’, about three hours in mid-latitudes. Thus, for phenomena with time scales much longer than this, such as the motion of the Gulf Stream or a mid-latitude atmospheric weather system, the effects of the Earth’s rotation can be expected to be important, whereas a short-lived phenomena, such as a cumulus cloud or tornado, may be oblivious to such rotation.

2.8.2 Geostrophic balance

If the Rossby number is sufficiently small in (2.191a) then the rotation term will dominate the nonlinear advection term, and if the time period of the motion scales advectively then the rotation term also dominates the local time derivative. The only term that can then balance the rotation term is the pressure term, and therefore we must have

$$\mathbf{f} \times \mathbf{u} \approx -\frac{1}{\rho} \nabla_z p, \quad (2.195)$$

or, in Cartesian component form

$$fu \approx -\frac{1}{\rho} \frac{\partial p}{\partial y}, \quad fv \approx \frac{1}{\rho} \frac{\partial p}{\partial x}. \quad (2.196)$$

This balance is known as *geostrophic balance*, and its consequences are profound, giving geophysical fluid dynamics a special place in the broader field of fluid dynamics. We *define* the geostrophic velocity by

$$fu_g \equiv -\frac{1}{\rho} \frac{\partial p}{\partial y}, \quad fv_g \equiv \frac{1}{\rho} \frac{\partial p}{\partial x}, \quad (2.197)$$

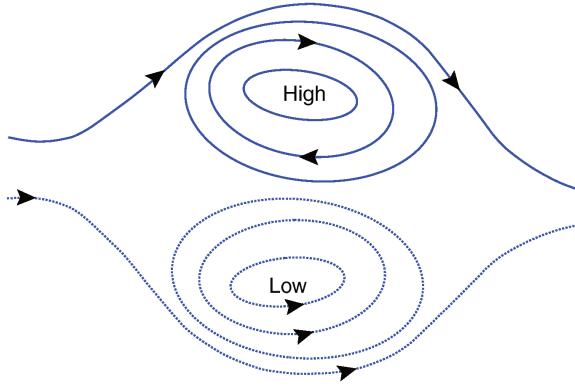


Fig. 2.5 Sketch of geostrophic flow with a positive value of the Coriolis parameter f . Flow is parallel to the lines of constant pressure (isobars). Cyclonic flow is anticlockwise around a low pressure region and anticyclonic flow is clockwise around a high. If f were negative, as in the Southern Hemisphere, (anti)cyclonic flow would be (anti)clockwise.

and for low Rossby number flow $u \approx u_g$ and $v \approx v_g$. In spherical coordinates the geostrophic velocity is

$$f u_g = -\frac{1}{\rho a} \frac{\partial p}{\partial \vartheta}, \quad f v_g = \frac{1}{a \rho \cos \vartheta} \frac{\partial p}{\partial \lambda}, \quad (2.198)$$

where $f = 2\Omega \sin \vartheta$. Geostrophic balance has a number of immediate ramifications:

- Geostrophic flow is parallel to lines of constant pressure (isobars). If $f > 0$ the flow is anticlockwise round a region of low pressure and clockwise around a region of high pressure (see Fig. 2.5).
- If the Coriolis force is constant and if the density does not vary in the horizontal the geostrophic flow is horizontally non-divergent and

$$\nabla_z \cdot \mathbf{u}_g = \frac{\partial u_g}{\partial x} + \frac{\partial v_g}{\partial y} = 0. \quad (2.199)$$

We may define the *geostrophic streamfunction*, ψ , by

$$\psi \equiv \frac{p}{f_0 \rho_0}, \quad \text{whence} \quad u_g = -\frac{\partial \psi}{\partial y}, \quad v_g = \frac{\partial \psi}{\partial x}. \quad (2.200)$$

The vertical component of vorticity, ζ , is then given by

$$\zeta = \mathbf{k} \cdot \nabla \times \mathbf{v} = \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} = \nabla_z^2 \psi. \quad (2.201)$$

- If the Coriolis parameter is not constant, then cross-differentiating (2.197) gives, for constant density geostrophic flow,

$$v_g \frac{\partial f}{\partial y} + f \nabla_z \cdot \mathbf{u}_g = 0, \quad (2.202)$$

which, using the mass continuity equation $\nabla_z \cdot \mathbf{u}_g = -\partial w / \partial z$,

$$\beta v_g = f \frac{\partial w}{\partial z}. \quad (2.203)$$

where $\beta \equiv \partial f / \partial y = 2\Omega \cos \vartheta / a$. This geostrophic vorticity balance is sometimes known as ‘Sverdrup balance’, although that expression is better restricted to the case when the vertical velocity from a wind stress, as considered in chapter 19.

2.8.3 Taylor–Proudman effect

If $\beta = 0$, then (2.203) implies that the vertical velocity is not a function of height. In fact, in that case none of the components of velocity vary with height if density is also constant. To show this, in the limit of zero Rossby number we first write the three-dimensional momentum equation as

$$\mathbf{f}_0 \times \mathbf{v} = -\nabla\phi - \nabla\chi, \quad (2.204)$$

where $\mathbf{f}_0 = 2\boldsymbol{\Omega} = 2\Omega\mathbf{k}$, $\phi = p/\rho_0$, and $\nabla\chi$ represents other potential forces. If $\chi = gz$ then the vertical component of this equation represents hydrostatic balance, and the horizontal components represent geostrophic balance. On taking the curl of this equation, the terms on the right-hand side vanish and the left-hand side becomes

$$(\mathbf{f}_0 \cdot \nabla)\mathbf{v} - \mathbf{f}_0 \nabla \cdot \mathbf{v} - (\mathbf{v} \cdot \nabla)\mathbf{f}_0 + \mathbf{v} \nabla \cdot \mathbf{f}_0 = 0. \quad (2.205)$$

But $\nabla \cdot \mathbf{v} = 0$ by mass conservation, and because $\mathbf{f}_0$ is constant both $\nabla \cdot \mathbf{f}_0$ and $(\mathbf{v} \cdot \nabla)\mathbf{f}_0$ vanish. Equation (2.205) thus reduces to

$$(\mathbf{f}_0 \cdot \nabla)\mathbf{v} = 0, \quad (2.206)$$

which, since $\mathbf{f}_0 = f_0\mathbf{k}$, implies $f_0 \partial \mathbf{v} / \partial z = 0$, and in particular we have

$$\frac{\partial u}{\partial z} = 0, \quad \frac{\partial v}{\partial z} = 0, \quad \frac{\partial w}{\partial z} = 0. \quad (2.207)$$

All three components of velocity are uniform along the axis of rotation.

A different presentation of this argument proceeds as follows. If the flow is exactly in geostrophic and hydrostatic balance then

$$\mathbf{v} = \frac{1}{f_0} \frac{\partial \phi}{\partial x}, \quad u = -\frac{1}{f_0} \frac{\partial \phi}{\partial y}, \quad \frac{\partial \phi}{\partial z} = -g. \quad (2.208a,b,c)$$

Differentiating (2.208a,b) with respect to z , and using (2.208c) yields

$$\frac{\partial v}{\partial z} = \frac{-1}{f_0} \frac{\partial g}{\partial x} = 0, \quad \frac{\partial u}{\partial z} = \frac{1}{f_0} \frac{\partial g}{\partial y} = 0. \quad (2.209)$$

Noting that the geostrophic velocities are horizontally non-divergent ($\nabla_z \cdot \mathbf{u} = 0$), and using mass continuity then gives $\partial w / \partial z = 0$, as before.

If there is a solid horizontal boundary anywhere in the fluid, for example at the surface, then $w = 0$ at that surface and thus $w = 0$ everywhere. Hence the motion occurs in planes that lie perpendicular to the axis of rotation, and the flow is effectively two dimensional. This result is known as the *Taylor–Proudman effect*, namely that for constant density flow in geostrophic and hydrostatic balance the vertical derivatives of the horizontal and the vertical velocities are zero.¹⁰ At zero Rossby number, if the vertical velocity is zero somewhere in the flow then it is zero everywhere in that vertical column; furthermore, the horizontal flow has no vertical shear, and the fluid moves like a slab. The effects of rotation have provided a *stiffening* of the fluid in the vertical.

In neither the atmosphere nor the ocean do we observe precisely such vertically coherent flow, mainly because of the effects of stratification. However, it is typical of geophysical fluid dynamics that the assumptions underlying a derivation are not fully satisfied, yet there are manifestations of it in real flow. For example, one might have naïvely expected, because $\partial w / \partial z = -\nabla_z \cdot \mathbf{u}$, that the scales of the various variables would be related by $W/H \sim U/L$. However, if the flow is rapidly rotating we expect that the horizontal flow will be in near geostrophic balance and therefore nearly divergence free; thus $\nabla_z \cdot \mathbf{u} \ll U/L$, and $W \ll HU/L$.

2.8.4 Thermal wind balance

Thermal wind balance arises by combining the geostrophic and hydrostatic approximations, and this is most easily done in the context of the anelastic (or Boussinesq) equations, or in pressure coordinates. For the anelastic equations, geostrophic balance may be written

$$-f v_g = -\frac{\partial \phi}{\partial x} = -\frac{1}{a \cos \vartheta} \frac{\partial \phi}{\partial \lambda}, \quad f u_g = -\frac{\partial \phi}{\partial y} = -\frac{1}{a} \frac{\partial \phi}{\partial \vartheta}. \quad (2.210a,b)$$

Combining these relations with hydrostatic balance, $\partial \phi / \partial z = b$, gives

$$\begin{aligned} -f \frac{\partial v_g}{\partial z} &= -\frac{\partial b}{\partial x} = -\frac{1}{a \cos \lambda} \frac{\partial b}{\partial \lambda}, \\ f \frac{\partial u_g}{\partial z} &= -\frac{\partial b}{\partial y} = -\frac{1}{a} \frac{\partial b}{\partial \vartheta}. \end{aligned} \quad (2.211a,b)$$

These equations represent *thermal wind balance*, and the vertical derivative of the geostrophic wind is the ‘thermal wind’. Equation (2.211) may be written in terms of the zonal angular momentum as

$$\frac{\partial m_g}{\partial z} = -\frac{a}{2\Omega \tan \vartheta} \frac{\partial b}{\partial y}, \quad (2.212)$$

where $m_g = (u_g + \Omega a \cos \vartheta) a \cos \vartheta$. Potentially more accurate than geostrophic balance is the so-called gradient wind balance, discussed more in section 2.9, which retains a centrifugal term in the momentum equation. The meridional momentum equation (2.50b) becomes

$$2u\Omega \sin \vartheta + \frac{u^2}{a} \tan \vartheta \approx -\frac{\partial \phi}{\partial y} = -\frac{1}{a} \frac{\partial \phi}{\partial \vartheta}. \quad (2.213)$$

For large-scale flow this only differs significantly from geostrophic balance very close to the equator. Taking the vertical derivative of (2.213) and using the hydrostatic relation $\partial \phi / \partial z = b$ gives a modified thermal wind relation, and this may be put in the simple form

$$\frac{\partial m^2}{\partial z} = -\frac{a^3 \cos^3 \vartheta}{\sin \vartheta} \frac{\partial b}{\partial y}, \quad (2.214)$$

where $m = (u + \Omega a \cos \vartheta) a \cos \vartheta$ is the annular angular momentum.

If the density or buoyancy is constant then there is no shear and (2.211) or (2.214) give the Taylor–Proudman result. But suppose that the temperature falls in the poleward direction. Then thermal wind balance implies that the (eastward) wind will increase with height — just as is observed in the atmosphere! In general, a vertical shear of the horizontal wind is associated with a horizontal temperature gradient, and this is one of the most simple and far-reaching effects in geophysical fluid dynamics. The underlying physical mechanism is illustrated in Fig. 2.6.

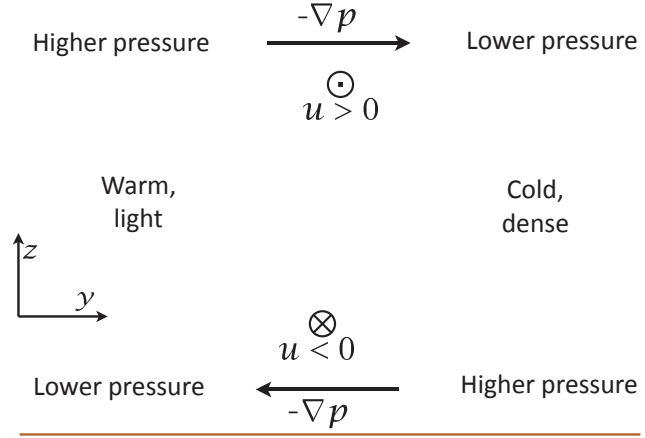
Geostrophic and thermal wind balance in pressure coordinates

In pressure coordinates geostrophic balance is just

$$\mathbf{f} \times \mathbf{u}_g = -\nabla_p \Phi, \quad (2.215)$$

where Φ is the geopotential and ∇_p is the gradient operator taken at constant pressure. If f is constant, it follows from (2.215) that the geostrophic wind is non-divergent on pressure surfaces.

Fig. 2.6 The mechanism of thermal wind. A cold fluid is denser than a warm fluid, so by hydrostasy the vertical pressure gradient is greater where the fluid is cold. Thus, pressure gradients form as shown, where ‘higher’ and ‘lower’ mean relative to the average at that height. The horizontal pressure gradients are balanced by the Coriolis force, producing (for $f > 0$) the horizontal winds shown. Only the wind *shear* is given by the thermal wind.



Taking the vertical derivative of (2.215) (that is, its derivative with respect to p) and using the hydrostatic equation, $\partial\Phi/\partial p = -\alpha$, gives the thermal wind equation

$$\mathbf{f} \times \frac{\partial \mathbf{u}_g}{\partial p} = \nabla_p \alpha = \frac{R}{p} \nabla_p T, \quad (2.216)$$

where the last equality follows using the ideal gas equation and because the horizontal derivative is at constant pressure. In component form this is

$$-f \frac{\partial v_g}{\partial p} = \frac{R}{p} \frac{\partial T}{\partial x}, \quad f \frac{\partial u_g}{\partial p} = \frac{R}{p} \frac{\partial T}{\partial y}. \quad (2.217)$$

In log-pressure coordinates, with $Z = -H \ln(p/p_R)$, thermal wind is

$$\mathbf{f} \times \frac{\partial \mathbf{u}_g}{\partial Z} = -\frac{R}{H} \nabla_Z T. \quad (2.218)$$

The effect in all these cases is the same: a horizontal temperature gradient, or a temperature gradient along an isobaric surface, is accompanied by a vertical shear of the horizontal wind.

2.8.5 ♦ Vertical velocity and hydrostatic balance

Scaling for vertical velocity

If the Coriolis parameter is constant then the horizontal component of flows that are in geostrophic balance have zero horizontal divergence ($\nabla_x \cdot \mathbf{u} = 0$) and zero vertical velocity. We can therefore expect that any flow with small Rossby number will have a ‘small’ vertical velocity. Let us make this statement more precise using the rotating Boussinesq equations, (2.173) with constant Coriolis parameter. Let $\mathbf{u} = \mathbf{u}_g + \mathbf{u}_a$ where the geostrophic flow satisfies $\mathbf{f}_0 \times \mathbf{u}_g = -\nabla\phi$. The horizontal momentum equation, with corresponding scales for each term, then becomes

$$\frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \cdot \nabla \mathbf{u} + w \frac{\partial \mathbf{u}}{\partial z} + \mathbf{f}_0 \times \mathbf{u}_a = 0, \quad (2.219)$$

$$\frac{U^2}{L} \quad \frac{U^2}{L} \quad \frac{WU}{H} \quad f_0 U_a. \quad (2.220)$$

This equation suggests a scaling for the ageostrophic flow of

$$U_a = \frac{U}{f_0 L} U = Ro U. \quad (2.221)$$

That is, the ageostrophic flow is Rossby number smaller (at least) than the geostrophic flow. To obtain a scaling for the vertical velocity we look to the mass continuity equation written in the form

$$\frac{\partial w}{\partial z} = -\nabla \cdot \mathbf{u}_a, \quad (2.222)$$

since only the ageostrophic flow has a divergence. Equations (2.221) and (2.222) suggest the scaling

$$W = Ro \frac{HU}{L}. \quad (2.223)$$

That is, the vertical velocity is order Rossby number smaller than an estimate based purely on the mass continuity equation would suggest.

If the Coriolis parameter is not constant then the geostrophic flow itself is divergent and this induces a vertical velocity, as in (2.203). The scaling for vertical velocity is now

$$W = \frac{\beta}{f} HU = Ro_\beta \frac{HU}{L}. \quad (2.224)$$

where $Ro_\beta = \beta L / f$ is the *beta Rossby number*. It is less than one for all flows except those with a truly global scale.

Scaling for hydrostatic balance

Let us non-dimensionalize the rotating Boussinesq equations, (2.173), by writing

$$\begin{aligned} (x, y) &= L(\hat{x}, \hat{y}), & z &= H\hat{z}, & \mathbf{u} &= U\hat{\mathbf{u}}, & t &= T\hat{t} = \frac{L}{U}\hat{t}, & \mathbf{f} &= f_0\hat{\mathbf{f}}, \\ w &= \frac{\epsilon HU}{L}\hat{w}, & \phi &= \Phi\hat{\phi} = f_0 UL\hat{\phi}, & b &= B\hat{b} = \frac{f_0 UL}{H}\hat{b}, \end{aligned} \quad (2.225)$$

These relations are almost the same as (2.179), except for the factor of ϵ in the scaling of w . If the Coriolis parameter is constant or nearly so then, from (2.223), $\epsilon = Ro$, whereas if the Coriolis parameter varies then $\epsilon = Ro_\beta$, as in (2.223). The scaling for ϕ and b' are suggested by geostrophic and thermal wind balance with f_0 a representative value of f . Substituting these values into (2.173) we obtain the following scaled momentum equations:

$$Ro \frac{D\hat{\mathbf{u}}}{D\hat{t}} + \hat{\mathbf{f}} \times \hat{\mathbf{u}} = -\nabla \hat{\phi}, \quad Ro \epsilon \alpha^2 \frac{D\hat{w}}{D\hat{t}} = -\frac{\partial \hat{\phi}}{\partial \hat{z}} - \hat{b}. \quad (2.226a,b)$$

where $D/D\hat{t} = \partial/\partial\hat{t} + \hat{\mathbf{u}} \cdot \nabla_z + \epsilon \hat{w} \partial/\partial\hat{z}$. There are two notable aspects to these equations. First and most obviously, when $Ro \ll 1$, (2.226a) reduces to geostrophic balance, $\hat{\mathbf{f}} \times \hat{\mathbf{u}} \approx -\nabla \hat{\phi}$. Second, the material derivative in (2.226b) is multiplied by three non-dimensional parameters, and we can understand the appearance of each as follows.

- (i) The aspect ratio dependence (α^2) arises in the same way as for non-rotating flows — that is, because of the presence of w and z in the vertical momentum equation as opposed to (u, v) and (x, y) in the horizontal equations.
- (ii) The Rossby number dependence (Ro) arises because in rotating flow the pressure gradient is balanced by the Coriolis force, which is Rossby number larger than the advective terms.
- (iii) The factor ϵ arises because in rotating flow w is smaller than u by ϵ times the aspect ratio. The factor may be the Rossby number itself, or the beta Rossby number.

The factor $Ro \epsilon \alpha^2$ is very small for large-scale flow; the reader is invited to calculate representative values. Evidently, a rapidly rotating fluid is more likely to be in hydrostatic balance than a non-rotating fluid, other conditions being equal. The combined effects of rotation and stratification are, not surprisingly, quite subtle and we leave that topic for chapter 5.

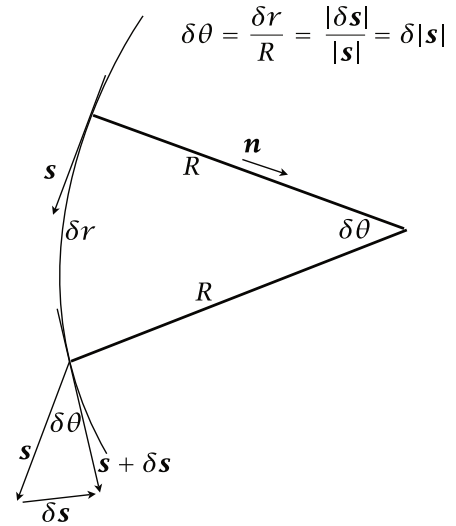


Fig. 2.7 A parcel tracing a curved path with radius of curvature R , moving a small distance δr and through a small angle $\delta\theta$.

2.9 ♦ GRADIENT WIND BALANCE

If a flow follows a curved path then our intuition suggests that it will experience a centrifugal force of some kind in addition to the Coriolis force. We can easily imagine that a parcel then experiences a three-way balance, between Coriolis, centrifugal and pressure forces, and this balance is called *gradient wind balance*. It is a more general, and more accurate, balance than geostrophic balance but it is not always as useful. To illustrate it we will keep matters simple and consider purely horizontal flow of constant density. We first introduce the notion of *natural coordinates* and show how gradient wind balance emerges straightforwardly.¹¹ We then discuss how gradient wind balance arises in the Eulerian equations of motion in a fixed coordinate system.

2.9.1 Natural Coordinates

For most purposes a coordinate system that is fixed in space, or rotating coincidentally with the Earth, is the most practically useful. However, it is sometimes useful to use a coordinate system that is moving with the local flow. To that end, and restricting our attention to horizontal flow, consider a parcel of fluid moving with velocity $\mathbf{u}$. We define a *natural coordinate system* by the set of unit vectors $\mathbf{s}$, $\mathbf{n}$ and $\mathbf{k}$, where $\mathbf{s}$ is the unit vector tangential to the flow, $\mathbf{n}$ is the horizontal unit vector normal to the flow and $\mathbf{k}$ is the unit vector in the vertical. Apart from $\mathbf{k}$, all these vectors evolve with the flow. Our goal is to split the horizontal momentum equation into components parallel to and normal to the direction of the local flow and thereby to discern the force balances in either direction. (Readers who trust their intuition may skip ahead to (2.231), which they may find obvious.)

If U is the speed of the parcel then $\mathbf{u} = U\mathbf{s}$ and so

$$\frac{d\mathbf{u}}{dt} = \mathbf{s} \frac{dU}{dt} + U \frac{d\mathbf{s}}{dt}. \quad (2.227)$$

Furthermore, if the flow of a parcel follows the curve $r(t)$ — that is, r gives the distance moved by the parcel — then $U = dr/dt$. To obtain a useful expression for $d\mathbf{s}/dt$ we note that, as in Fig. 2.7,

$$\theta \equiv \frac{\delta r}{R} = \frac{|\delta\mathbf{s}|}{|\mathbf{s}|} = |\delta\mathbf{s}|, \quad (2.228)$$

where R is the *radius of curvature* and $|\mathbf{s}| = 1$. [The radius of curvature may be evaluated geometrically as follows. Draw a line tangent to the path at some point, and draw a line perpendicular to the

tangent through that point. An infinitesimal distance along the curve, construct another perpendicular line in the same manner. The two perpendicular lines meet at the center of curvature, and the distance along one of the perpendicular lines from the center of curvature to the curve itself is the radius of curvature. By convention, the radius of curvature is positive (negative) if the parcel is curving to the left (right).]

The change of $\mathbf{s}$ is directed along $\mathbf{n}$, and so from (2.228) we infer that

$$\frac{d\mathbf{s}}{dr} = \frac{\mathbf{n}}{R}. \quad (2.229)$$

Thus, the rate of change of $\mathbf{s}$ is given by

$$\frac{d\mathbf{s}}{dt} = \frac{d\mathbf{s}}{dr} \frac{dr}{dt} = \frac{\mathbf{n}}{R} U, \quad (2.230)$$

and using this expression in (2.227), the acceleration of the fluid parcel is given by

$$\frac{d\mathbf{u}}{dt} = \mathbf{s} \frac{dU}{dt} + \mathbf{n} \frac{U^2}{R}. \quad (2.231)$$

The two terms on the right-hand side may be interpreted as being, respectively, the change in speed of the parcel along its path and the centripetal acceleration owing to the curvature of the path (if the path is a straight line then R is infinite and the centripetal acceleration is zero).

2.9.2 Application to fluids

Now consider the application of the above to a fluid in a rotating frame of reference. The total time derivatives of (2.231) may be replaced by material derivatives, and additional pressure and Coriolis forces act on the flow. The pressure-gradient force, $-\nabla\phi$, has components along and perpendicular to the flow so that

$$-\nabla\phi = -\left(\frac{\partial\phi}{\partial r}\mathbf{s} + \frac{\partial\phi}{\partial n}\mathbf{n}\right). \quad (2.232)$$

Given our convention, $\partial\phi/\partial n$ is positive if pressure increases to the left of the trajectory of the flow. The Coriolis force, $-\mathbf{f} \times \mathbf{u}$, has only a component perpendicular to the flow so that

$$-\mathbf{f} \times \mathbf{u} = -fU\mathbf{n}. \quad (2.233)$$

Because $\mathbf{n}$ is directed to the left of the flow, for positive f the Coriolis force is directed to the right of the flow. Using (2.231), (2.232) and (2.233), and neglecting friction, we write down the components of the momentum equation parallel and perpendicular to the flow, namely

$$\frac{DU}{Dt} = -\frac{\partial\phi}{\partial r}, \quad \frac{U^2}{R} + fU = -\frac{\partial\phi}{\partial n}. \quad (2.234a,b)$$

These equations tell us, at least in principle, how a fluid field will evolve from some initial conditions. The radius of curvature can only be determined from the instantaneous velocity field for steady flow: if the flow is unsteady then the radius of curvature for a parcel depends on the pressure field and differs from the radius of curvature of a streamline.

Gradient wind balance, cyclostrophic balance and inertial flow

Gradient flow (or, as it is commonly called, the gradient wind) is the flow that satisfies (2.234b). We may always define a gradient wind at a given point by using this equation, and the only forces that are neglected are frictional ones. Thus, given the pressure field ϕ we may use (2.234b) to calculate the gradient flow using the quadratic formula. Gradient wind balance is thus a better

approximation to the real flow than is geostrophic balance (which omits the centrifugal term). The gradient wind will not in general be a steady solution to the equations of motion — the flow will accelerate and the pressure field will change.

If the flow is in a straight line then the radius of curvature is infinite and (2.234b) reduces to $fU = -\partial\phi/\partial n$, which is geostrophic balance. For this to hold exactly the flow need not be steady, just in straight line. It will be a good approximation to the real flow when $fU \gg U^2/R$ or $U/fR \ll 1$, taking all quantities as positive, which is similar to a condition of low Rossby number. In contrast, cyclostrophic flow occurs when the quantity U/fR is large and the Coriolis force may be neglected, giving $U^2/R = -\partial\phi/\partial n$.

Inertial flow arises when the pressure gradient vanishes and (2.234b) reduces to $U/R + f = 0$. Now, if the pressure gradient vanishes then, from (2.234a) the speed of fluid parcels is constant. Thus, if f is constant the fluid parcels execute circles, known as inertia circles, of radius U/f and period $2\pi/f = 1 \text{ day}/(2 \sin \vartheta)$ where ϑ is latitude. Such a motion is evidently not truly inertial, for inertial motion is in a straight line and at a constant speed; other forces — gravitational and centrifugal — must act to keep the flow in the horizontal plane. Inertial motion is rarely a good approximation to flow in the atmosphere or ocean, although occasionally parcels are observed to trace approximations to inertia circles, especially in the ocean.

2.9.3 Gradient wind balance in the two-dimensional Eulerian equations

Let us now consider gradient wind balance in the Eulerian equations of motion. The basic ideas can be exposed by considering unforced constant-density two dimensional flow in a rotating reference frame, for which the equations of motion are

$$\frac{Du}{Dt} - fv = -\frac{\partial\phi}{\partial x}, \quad \frac{Dv}{Dt} + fu = -\frac{\partial\phi}{\partial y}, \quad \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0. \quad (2.235a,b,c)$$

We may introduce a stream function ψ such that $u = -\partial\psi/\partial y$ and $v = \partial\psi/\partial x$. If the flow is geostrophic then $f \times (u, v) = (-\partial\phi/\partial y, \partial\phi/\partial x)$; however, unless the Coriolis parameter f is constant the streamfunction is not proportional to the pressure field. The vorticity (considered further in chapter 4), which is defined to be the curl of the velocity, is given by

$$\boldsymbol{\omega} \equiv \nabla \times \mathbf{u} = \mathbf{k}\zeta, \quad \text{where} \quad \zeta = \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) = \frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} \equiv \nabla_z^2 \psi. \quad (2.236)$$

Let us form the evolution equations for vorticity and divergence by taking the curl and divergence of (2.235a,b). After a little algebra the vorticity equation is found to be

$$\frac{\partial \zeta}{\partial t} + \mathbf{u} \cdot \nabla (\zeta + f) = 0, \quad \text{or equivalently} \quad \frac{\partial \zeta}{\partial t} + J(\psi, \zeta + f) = 0 \quad (2.237)$$

where $J(a, b) \equiv (\partial a/\partial x)(\partial b/\partial y) - (\partial a/\partial y)(\partial b/\partial x)$ is the *Jacobian*. If $f = f(y)$ then $J(\psi, f) = \mathbf{u} \cdot \nabla f = \beta v$ where $\beta = \partial f/\partial y$. Because the velocity is divergence-free the vorticity equation above is closed; that is, it may be integrated without the need to solve any other evolution equations, and in particular without the need to solve for the pressure field. (In general the vorticity equation alone is not closed.)

If we take the divergence of (2.235a,b) then, using (2.235c) the time derivative disappears and again after a little algebra we obtain the divergence equation,

$$2J(u, v) + \nabla \cdot (f \nabla \psi) = \nabla^2 \phi. \quad (2.238)$$

In this context, (2.238) may be regarded as an equation for pressure, ϕ , given the velocity or the streamfunction. Equation (2.238) is commonly referred to as the *gradient-wind balance relation*,

and it is analogous to (2.234b) in that it generalizes and is more accurate than geostrophic balance. Equation (2.238) is *exact* for two-dimensional, incompressible, inviscid and unforced flow, even when time-dependent. If the Rossby number is small then the second term on the left-hand side of (2.238) dominates the first, and $\nabla \cdot (f \nabla \psi) \approx \nabla^2 \phi$. This is equivalent to geostrophic balance, and if f is constant the streamfunction and pressure field are proportional to each other.

In the more general case the two-dimensional divergence is not zero; that is $\partial_t(\partial_x u + \partial_y v) \neq 0$. However, if the Rossby number is small and if f is nearly constant, then the geostrophic relation implies that the two-dimensional divergence will be small compared to the two-dimensional vorticity. In this case, gradient wind balance will be a good *approximation* to the flow, and a somewhat better approximation, if not more useful, than geostrophic balance alone.

2.10 STATIC INSTABILITY AND THE PARCEL METHOD

In this and the next couple of sections we consider how a fluid might oscillate if it were perturbed away from a resting state. Our focus is on vertical displacements, and the restoring force is gravity. Given that, the simplest way to approach the problem is to consider from first principles the pressure and gravitational forces on a displaced parcel, as in Fig. ???. To this end, consider a fluid initially at rest in a constant gravitational field, and therefore in hydrostatic balance. Suppose that a small parcel of the fluid is adiabatically displaced upwards by the small distance δz , without altering the overall pressure field; that is, the fluid parcel instantly assumes the pressure of its environment. If after the displacement the parcel is lighter than its environment, it will accelerate upwards, because the upward pressure gradient force is now greater than the downward gravity force on the parcel; that is, the parcel is *buoyant* (a manifestation of Archimedes' principle) and the fluid is *statically unstable*. If on the other hand the fluid parcel finds itself heavier than its surroundings, the downward gravitational force will be greater than the upward pressure force and the fluid will sink back towards its original position and an oscillatory motion will develop. Such an equilibrium is *statically stable*. Using such simple 'parcel' arguments we will now develop criteria for the stability of the environmental profile.

2.10.1 Stability and the profile of potential density

Consider the case of a stationary fluid whose density varies with altitude. We denote this background state with a tilde, as in $\tilde{\rho}(z)$. We then displace a fluid parcel adiabatically a small distance from z to $z + \delta z$, as in Fig. 2.8. In such a displacement it is the *potential density* ρ_θ (not the actual density) that is materially conserved, because potential density takes into account the effects of pressure compressibility. Let us also use the pressure at level $z + \delta z$ as the reference level, potential density equals in situ density.

The parcel at z takes on the potential density of its environment so that $\rho_\theta(z) = \tilde{\rho}_\theta(z)$ and it preserves this as it rises, so that $\rho_\theta(z + \delta z) = \rho_\theta(z)$. But since $z + \delta z$ is the reference level, the *in situ* density of the displaced parcel, $\rho(z + \delta z)$, is equal to its potential density $\rho_\theta(z + \delta z)$, which is equal to $\tilde{\rho}_\theta(z)$. Thus, at $z + \delta z$ the environment has in situ density equal to $\tilde{\rho}_\theta(z + \delta z)$ and the parcel has in situ density equal to $\tilde{\rho}_\theta(z)$. Putting this together in a single equation we have

$$\begin{aligned} \delta \rho &= \rho(z + \delta z) - \tilde{\rho}(z + \delta z) = \rho_\theta(z + \delta z) - \tilde{\rho}_\theta(z + \delta z) \\ &= \rho_\theta(z) - \tilde{\rho}_\theta(z + \delta z) = \tilde{\rho}_\theta(z) - \tilde{\rho}_\theta(z + \delta z), \end{aligned} \quad (2.239)$$

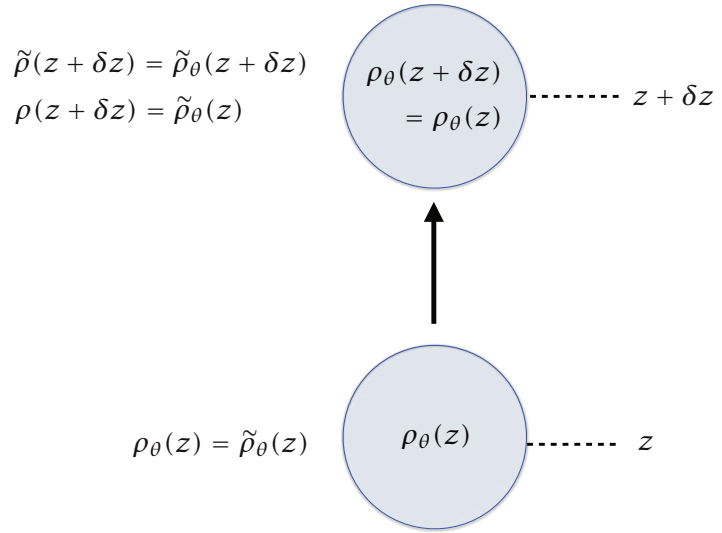
and therefore

$$\delta \rho = -\frac{\partial \tilde{\rho}_\theta}{\partial z} \delta z, \quad (2.240)$$

where the derivative on the right-hand side is the environmental gradient of potential density. If the right-hand side is positive, the parcel is heavier than its surroundings and the displacement is

Fig. 2.8 A parcel is adiabatically displaced upward from level z to $z + \delta z$. A tilde denotes the value in the environment, and variables without tildes are those in the parcel.

The parcel preserves its potential density, ρ_θ , which it takes from the environment at level z . If $z + \delta z$ is the reference level, the potential density there is equal to the actual density. The parcel's stability is determined by the difference between its density and the environmental density, as in (2.239). If the difference is positive the displacement is stable, and conversely.



stable. That is, *the stability of a parcel of fluid is determined by the gradient of the locally-referenced potential density.*

The conditions for stability are

$$\begin{aligned}
 \text{Stability : } & \frac{\partial \tilde{\rho}_\theta}{\partial z} < 0, \\
 \text{Instability : } & \frac{\partial \tilde{\rho}_\theta}{\partial z} > 0.
 \end{aligned}
 \tag{2.241a,b}$$

The equation of motion of the fluid parcel is then given by a direct application of Newton's second law, that the mass times the acceleration is given by the force acting on the parcel. The force is the above-derived buoyancy force so we have

$$\frac{\partial^2 \delta z}{\partial t^2} = \frac{g}{\rho} \left(\frac{\partial \tilde{\rho}_\theta}{\partial z} \right) \delta z = -N^2 \delta z,
 \tag{2.242}$$

where, noting that $\rho(z) = \tilde{\rho}_\theta(z)$ to within $O(\delta z)$,

$$N^2 = -\frac{g}{\tilde{\rho}_\theta} \left(\frac{\partial \tilde{\rho}_\theta}{\partial z} \right).
 \tag{2.243}$$

A parcel that is displaced in a stably stratified fluid will thus oscillate at the frequency N , known as the buoyancy frequency or Brunt–Väisälä frequency, after its discoverer. The above expression for the buoyancy frequency is a general one, true in both liquids and gases in a constant gravitational field. The quantity $\tilde{\rho}_\theta$ is the *locally-referenced* potential density of the environment. The reference level turns out not to be important for the atmosphere, but it is for the ocean particularly in the presence of salinity: parcels at the same level with the same in situ density may have different potential densities if their salinity differs. In contrast, for fresh water in a laboratory setting potential density is virtually equal to in situ density.

2.10.2 The atmosphere

Buoyancy frequency

In the atmosphere potential density is related to potential temperature by $\rho_\theta = p_R/(\theta R)$, where p_R is the reference level for potential temperature. Using this expression in (2.243) gives

$$N^2 = \frac{g}{\bar{\theta}} \left(\frac{\partial \bar{\theta}}{\partial z} \right), \quad (2.244)$$

where $\bar{\theta}$ refers to the environmental profile of potential temperature. The reference value p_R does not appear, and we are free to choose this value arbitrarily — the surface pressure is a common choice. The conditions for stability, (2.241), then correspond to $N^2 > 0$ for stability and $N^2 < 0$ for instability. On average the atmosphere is stable and in the troposphere (the lowest several kilometres of the atmosphere) the average N is about 0.01 s^{-1} , with a corresponding period, $(2\pi/N)$, of about 10 minutes. In the stratosphere (which lies above the troposphere) N^2 is a few times higher than this.

Dry adiabatic lapse rate

The negative of the rate of change of the (real) temperature in the vertical is known as the *temperature lapse rate*, or often just the lapse rate, and denoted Γ . The lapse rate corresponding to $\partial\theta/\partial z = 0$ is called the *dry adiabatic lapse rate* and denoted Γ_d . Using $\theta = T(p_0/p)^{R/c_p}$ and $\partial p/\partial z = -\rho g$ we find that the lapse rate and the potential temperature lapse rate are related by

$$\frac{T}{\theta} \frac{\partial \theta}{\partial z} = \frac{\partial T}{\partial z} + \frac{g}{c_p}, \quad (2.245)$$

so that the dry adiabatic lapse rate is given by

$$\Gamma_d = \frac{g}{c_p}, \quad (2.246)$$

as in fact we derived in (1.131). The conditions for static stability corresponding to (2.241) are thus:

$$\begin{aligned} \text{Stability : } & \frac{\partial \bar{\theta}}{\partial z} > 0, \quad \text{or} \quad -\frac{\partial \bar{T}}{\partial z} < \Gamma_d. \\ \text{Instability : } & \frac{\partial \bar{\theta}}{\partial z} < 0, \quad \text{or} \quad -\frac{\partial \bar{T}}{\partial z} > \Gamma_d. \end{aligned} \quad (2.247\text{a,b})$$

The observed lapse rate (look ahead to Fig. 15.24) is often less than 7 K km^{-1} (corresponding to a buoyancy frequency of about 10^{-2} s^{-1}) whereas a dry adiabatic lapse rate is about 10 K km^{-1} . Why the discrepancy? Why is the atmosphere so apparently stable? One reason is that the atmosphere contains water vapour and a column of air that contains water may be unstable even if its lapse rate is less than dry adiabatic. If a moist parcel rises then, as it enters a cooler environment, water vapour may condense releasing more heat and leading to more ascent; that is, a moist atmosphere may be unstable when a dry atmosphere is stable. We defer a more detailed treatment to chapter 18.

2.10.3 The liquid ocean

No simple, accurate, analytic expression is available for computing static stability in the ocean. If the ocean had no salt, then the potential density referenced to the surface would generally be a

measure of the sign of stability of a fluid column, if not of the buoyancy frequency. However, in the presence of salinity, the surface-referenced potential density is not necessarily even a measure of the sign of stability, because the coefficients of compressibility β_T and β_S vary in different ways with pressure. To see this, suppose two neighbouring fluid elements at the surface have the same potential density, but different salinities and temperatures, and displace them both adiabatically to the deep ocean. Although their potential densities referenced to the surface are still equal, we can say little about their actual densities, and hence their stability relative to each other, without doing a detailed calculation because they will each have been compressed by different amounts. It is the profile of the *locally-referenced* potential density that determines the stability.

An approximate expression for stability that is sometimes useful arises by noting that in an adiabatic displacement

$$\delta\rho_\theta = \delta\rho - \frac{1}{c_s^2}\delta p = 0. \quad (2.248)$$

If the fluid is hydrostatic $\delta p = -\rho g \delta z$ so that if a parcel is displaced adiabatically its density changes according to

$$\left(\frac{\partial\rho}{\partial z}\right)_{\rho_\theta} = -\frac{\rho g}{c_s^2}. \quad (2.249)$$

If a parcel is displaced a distance δz upwards then the density difference between it and its new surroundings is

$$\delta\rho = -\left[\left(\frac{\partial\rho}{\partial z}\right)_{\rho_\theta} - \left(\frac{\partial\tilde{\rho}}{\partial z}\right)\right]\delta z = \left[\frac{\rho g}{c_s^2} + \left(\frac{\partial\tilde{\rho}}{\partial z}\right)\right]\delta z, \quad (2.250)$$

where the tilde again denotes the environmental field. It follows that the stratification is given by

$$N^2 = -g \left[\frac{g}{c_s^2} + \frac{1}{\tilde{\rho}} \left(\frac{\partial\tilde{\rho}}{\partial z} \right) \right]. \quad (2.251)$$

This expression holds for both liquids and gases, and for ideal gases it is the same as (2.244) (as a little algebra will show, using $c_s^2 = \gamma p/\rho$). In the ocean the factor of g/c_s^2 is small but not negligible. It is a slightly destabilising factor in the sense that a density profile with an in situ density that increases with depth is not necessarily stable. Rather, a profile is stable only if the density increases with depth more rapidly than potential density gradient, equal to $-(\partial\tilde{\rho}/\partial z + g/c_s^2)$. In liquids, a good approximation is to use a reference value ρ_0 for the undifferentiated density in the denominator, whence (2.251) becomes equal to the Boussinesq expression (2.107). On average the ocean is statically stable, with typical values of N in the upper ocean being about 0.01 s^{-1} , falling to 0.001 s^{-1} in the more homogeneous abyssal ocean. These frequencies correspond to periods of about 10 and 100 minutes, respectively.

2.10.4 Gravity waves and convection using the equations of motion

The parcel approach to oscillations and stability, while simple and direct, seems divorced from the fluid-dynamical equations of motion. To remedy this, we now use the equations of motion for a stratified Boussinesq fluid to analyze the motion resulting from a small disturbance. Our treatment here is brief and introductory, with a fuller treatment given in chapter 7.

Consider a Boussinesq fluid, initially at rest, in which the buoyancy varies linearly with height. Linearizing the equations of motion about this basic state gives the linear momentum equations,

$$\frac{\partial u'}{\partial t} = -\frac{\partial\phi'}{\partial x}, \quad \frac{\partial w'}{\partial t} = -\frac{\partial\phi'}{\partial z} + b', \quad (2.252a,b)$$

the mass continuity and thermodynamic equations,

$$\frac{\partial u'}{\partial x} + \frac{\partial w'}{\partial z} = 0, \quad \frac{\partial b'}{\partial t} + w' N^2 = 0, \quad (2.253a,b)$$

where $N^2 = d\tilde{b}/dz$ is the basic state buoyancy profile, and we assume that the flow is a function only of x and z . A little algebra reduces the above equations to a single one for w' ,

$$\left[\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial z^2} \right) \frac{\partial^2}{\partial t^2} + N^2 \frac{\partial^2}{\partial x^2} \right] w' = 0. \quad (2.254)$$

Seeking solutions of the form $w' = \text{Re } W \exp[i(kx + mz - \omega t)]$ yields the dispersion relationship for gravity waves:

$$\omega^2 = \frac{k^2 N^2}{k^2 + m^2}. \quad (2.255)$$

The frequency (look ahead to Fig. 7.2) is always less than N , approaching N for small horizontal scales, $k \gg m$.

Consider two special cases. First, if we neglect pressure perturbations, as in the parcel argument, then the two equations,

$$\frac{\partial w'}{\partial t} = b', \quad \frac{\partial b'}{\partial t} + w' N^2 = 0, \quad (2.256)$$

form a closed set and give $\omega^2 = N^2$, as in the parcel argument. Second, if we make the hydrostatic approximation and omit $\partial w'/\partial t$ in (2.252b) then the dispersion relation becomes $\omega^2 = k^2 N^2 / m^2$. The frequency then grows, artifactually, without bound as the horizontal scale becomes smaller.

If the basic state density increases with height then $N^2 < 0$ and we expect this state to be unstable. Indeed, the disturbance grows exponentially according to $\exp(\sigma t)$ where $\sigma = i\omega = \pm k \tilde{N} / (k^2 + m^2)^{1/2}$, and where $\tilde{N}^2 = -N^2$. We have reproduced the result previously obtained by parcel theory, namely that if the basic state density (or more generally potential density) increases with height the flow is unstable. Most convective activity in the ocean and atmosphere is, in the end, related to an instability of this form.

2.A APPENDIX: ASYMPTOTIC DERIVATION OF THE BOUSSINESQ EQUATIONS

The Boussinesq equations are those equations that are appropriate when the density variations are very small but gravitational effects are large, and here we provide an asymptotic derivation. Two key results are that the velocity field is divergence-free, and the buoyancy must be taken as a function of z and not p in the equation of state. The first follows from fact that density variations are presumptively small, and the second follows because the lowest order balance in the vertical momentum equation is $\partial p_0 / \partial z = -\rho_0 g$, whence $p_0 = -\rho_0 g z$, and p_0 and not p should be used in the equation of state at lowest order. The following derivation, which assumes familiarity with elementary asymptotics (or which can be taken as a gentle introduction to asymptotics) mainly just formalizes these results.

Let us suppose that the density varies like $\rho(x, y, z, t) = \rho_0 + \delta\rho(x, y, z, t)$ where ρ_0 is a constant and $|\delta\rho| \ll \rho_0$. Specifically, let $\epsilon\rho_0$ be a typical magnitude for $\delta\rho$ where $\epsilon \ll 1$ so that

$$\delta\rho = (\epsilon\rho_0)\delta\hat{\rho} \quad \text{and} \quad \rho = \rho_0(1 + \epsilon\delta\hat{\rho}) \quad (2.257)$$

where a hat denotes a nondimensional quantity and $\delta\hat{\rho}$ is an $\mathcal{O}(1)$ quantity.

The dimensional vertical momentum equation, omitting rotation and viscosity for simplicity, is

$$(\rho_0 + \delta\rho) \frac{Dw}{Dt} = -\frac{\partial p}{\partial z} - (\rho_0 + \delta\rho)g. \quad (2.258)$$

Now, g is ‘big’ and variations in density are ‘small’ and so $g\delta\rho$ is taken to be the same approximate size as the advection term $\rho_0 Dw/Dt$ on the left-hand side. The term $\rho_0 g$ must then be balanced by the pressure gradient. Also, there is no necessary difference between vertical and horizontal scales and velocities. With these points in mind we nondimensionalize with the following scales

$$(u, v, w) = U(\hat{u}, \hat{v}, \hat{w}), \quad (x, y, z) = L(\hat{x}, \hat{y}, \hat{z}), \quad t = \frac{L}{U}\hat{t}, \quad p = \rho_0 \frac{U^2}{\epsilon} \hat{p}, \quad g = \frac{U^2}{\epsilon L} \hat{g}, \quad (2.259)$$

where the hatted quantities are non-dimensional and are presumptively $\mathcal{O}(1)$. Equation (2.258) becomes

$$(1 + \epsilon\delta\hat{\rho}) \frac{D\hat{w}}{D\hat{t}} = -\frac{1}{\epsilon} \frac{\partial \hat{p}}{\partial \hat{z}} - \frac{1}{\epsilon} (1 + \epsilon\delta\hat{\rho}) \hat{g}. \quad (2.260)$$

We now take ϵ as an asymptotic ordering parameter and expand the nondimensional fields as series in ϵ . Then, with subscripts denoting the asymptotic order (for non-dimensional quantities only), we have

$$\delta\hat{\rho} = \delta\hat{\rho}_0 + \epsilon\delta\hat{\rho}_1 + \epsilon^2\delta\hat{\rho}_2 \dots, \quad \hat{p} = \hat{p}_0 + \epsilon\hat{p}_1 + \epsilon^2\hat{p}_2 \dots, \quad \hat{w} = \hat{w}_0 + \epsilon\hat{w}_1 + \epsilon^2\hat{w}_2 \dots \quad (2.261)$$

and similarly for $\hat{u}$ and $\hat{v}$. If we substitute the above series into (2.260) and equate terms with the same power of ϵ , the first two orders are

$$\frac{\partial \hat{p}_0}{\partial \hat{z}} = -\hat{g}, \quad \frac{D\hat{w}_0}{D\hat{t}} = -\frac{\partial \hat{p}_1}{\partial \hat{z}} - \delta\hat{\rho}_0 g. \quad (2.262a,b)$$

Evidently the leading order pressure, $\hat{p}_0$, is hydrostatic. We may now revert to dimensional variables and (2.262a) gives $p_0(z) = -\rho_0 gz$, and (2.262b) becomes

$$\frac{Dw}{Dt} = -\frac{1}{\rho_0} \frac{\partial p'}{\partial z} - \frac{\delta\rho}{\rho_0} g \quad \text{or} \quad \frac{Dw}{Dt} = -\frac{\partial\phi}{\partial z} + b, \quad (2.263)$$

where p' is the deviation from the hydrostatic pressure, $\phi = p'/\rho_0$ and $b = -g\delta\rho/\rho_0$.

In the horizontal momentum equations only the perturbation pressure p' appears since p_0 is a function of z only. Furthermore, the density must be taken to be the constant ρ_0 (since this is $1/\epsilon$ larger than $\delta\rho$). Then, if $\mathbf{u}$ is the horizontal velocity, (u, v) , and $\mathbf{v}$ is the three-dimensional one, (u, v, w) , we obtain

$$\frac{\partial \mathbf{u}}{\partial t} + \mathbf{v} \cdot \nabla \mathbf{u} = -\frac{1}{\rho_0} \nabla_z p' \quad \text{or} \quad \frac{D\mathbf{u}}{Dt} = -\nabla_z \phi \quad (2.264)$$

where the gradients (i.e., ∇_z) on the right-hand side are horizontal, at constant z . Equations (2.263) and (2.264) combine to give

$$\frac{D\mathbf{v}}{Dt} = -\nabla\phi + b\mathbf{k}, \quad (2.265)$$

where $\mathbf{k}$ is the vertical unit vector and frictional and rotational terms may be added as needed.

The mass continuity equation is

$$\frac{D}{Dt}(\rho_0 + \delta\rho_0) + (\rho_0 + \delta\rho) \nabla \cdot \mathbf{v} = 0. \quad (2.266)$$

Since $D\rho_0/Dt = 0$ and ρ_0 is $1/\epsilon$ larger than $\delta\rho$, we see without further ado that the lowest order mass conservation equation is that of an incompressible, divergence-free fluid, namely $\nabla \cdot \mathbf{v} = 0$.

Boussinesq thermodynamics

The Boussinesq equations are completed with a thermal equation of state, an equation of evolution for the composition (e.g., S , the salinity), and an equation that provides a thermodynamic state variable Θ (e.g., entropy or internal energy). The thermal equation of state gives the density in terms of pressure, composition and the thermodynamic variable and so is of the form $b = b(p, \Theta, S)$, meaning b is a function of (p, Θ, S) . However, to be consistent with the derivation of (2.263) we should use the lowest order variables on the right-hand side of (2.267), and specifically *we should use p_0 and not p itself*. The equation of state becomes

$$b = b(p_0, \Theta, S) \quad \text{or} \quad b = b(z, \Theta, S) \quad (2.267)$$

and an example is (1.153); that equation gives the inverse density, α , in terms of z , potential temperature and salinity, and from which an expression for b immediately follows.

The thermodynamic variable itself is obtained from the first law, and the discussions of sections 1.6 and 1.7.3 generally apply. The internal energy equation is, with no external source or diffusion,

$$\frac{DI}{Dt} + p\alpha \nabla \cdot \mathbf{v} = 0. \quad (2.268)$$

The lowest order velocities are divergence-free and at lowest order we thus have $DI/Dt = 0$. (To determine the actual magnitudes of the terms, we can obtain the internal energy from the Gibbs function using $I = g + \eta T - p\alpha = g - T\partial g/\partial T - p\partial g/\partial p$, and with the seawater Gibbs function, (1.144), we obtain $I = c_{p0}T + \text{smaller terms}$. Referring to the values in table 1.2, $c_{p0}\Delta T$ is roughly comparable to $p\alpha$, so that $p\alpha \nabla \cdot \mathbf{v}$ is indeed much smaller than DI/Dt .) If we evolved I , we would then need an equation of state of the form $b = b(z, I, S)$ to complete the system.

However, as for the full (non-Boussinesq) system it is often advantageous to evolve potential enthalpy, or potential temperature or entropy, rather than internal energy. We then obtain buoyancy using an accurate equation of state but with the hydrostatic pressure instead of the full pressure. For example, and for idealized or laboratory work with fresh water, a thermodynamic equation and an approximate equation of state might use (1.128) and a simplified form of (1.153) giving

$$\frac{D\theta}{Dt} = 0 \quad \text{and} \quad b = g \frac{\delta\alpha}{\alpha_0} = g \left[\frac{gz}{c_s^2} + \beta_T(\theta - \theta_0) \right], \quad (2.269)$$

where θ is potential temperature, θ_0 is a constant reference value and c_s is the speed of sound. The potential temperature is related to the actual temperature by $\theta = T + \beta_T g \theta_0 z / c_p$, but T is not needed to evolve the system. The term in gz/c_s^2 is often small enough to be neglected, in which case the thermodynamic equation becomes an evolution of buoyancy itself, $Db/Dt = 0$.

Notes

- 1 The distinction between Coriolis force and acceleration is not always made in the literature, even after noting that the force is considered as a force per unit mass. For a fluid in geostrophic balance, one might either say that there is a balance between the pressure force and the Coriolis force, with no net acceleration, or that the pressure force produces a Coriolis acceleration. The descriptions are equivalent, because of Newton's second law, but should not be conflated.

The Coriolis effect is named after Gaspard Gustave de Coriolis (1792–1843), who discussed the eponymous force in the context of rotating mechanical systems (Coriolis 1832, 1835), but Euler was aware of the effect almost a century before. Persson (1998) provides a historical account.

- 2 Phillips (1973). A related discussion can be found in Stommel & Moore (1989).
- 3 Phillips (1966). See White (2003) for a review. In the early days of numerical modelling these equations were the most primitive — i.e., the least filtered — equations that could practically be integrated numerically. Associated with increasing computer power there is a tendency for comprehensive numerical models to use non-hydrostatic equations of motion that do not make the

shallow-fluid or traditional approximations, and it is conceivable that the meaning of the word 'primitive' may evolve to accommodate them.

- 4 The Boussinesq approximation is named for Boussinesq (1903), although similar approximations were used earlier by Oberbeck (1879, 1888). Spiegel & Veronis (1960) give a physically based derivation for an ideal gas, and Mihaljan (1962) and Gray & Giorgini (1976) provide more systematic derivations that include the effects of viscosity and diffusion and discussions of the energetics.
- 5 Young (2010).
- 6 Various versions of anelastic and pseudo-incompressible equations exist — see Batchelor (1953a), Ogura & Phillips (1962), Gough (1969), Gilman & Glatzmaier (1981), Lipps & Hemler (1982), and Durran (1989), although not all have potential vorticity and energy conservation laws (Bannon 1995, 1996; Scinocca & Shepherd 1992). The system we derive is most similar to that of Ogura & Phillips (1962) and unpublished notes by J. S. A. Green. The connection between the Boussinesq and anelastic equations is discussed by Lilly (1996) and Ingersoll (2005), and the extension to a complex equation of state and inclusion of moisture is discussed by Pauluis (2008).
- 7 A numerical model that explicitly includes sound waves must take very small timesteps in order to maintain numerical stability, in particular to satisfy the Courant–Friedrichs–Lewy (CFL) criterion. An alternative is to use implicit timestepping that effectively lets the numerics filter the sound waves. If we make the hydrostatic approximation then all sound waves except those that propagate horizontally are eliminated, and there is little numerical need to make the anelastic approximation.
- 8 Gill (1982) provides a longer discussion. Not all authors differentiate between N and N_* .
- 9 The Rossby number is named for C.-G. Rossby (see endnote on page 211), but it was also used by Kibel (1940) and is sometimes called the Kibel or Rossby–Kibel number. The notion of geostrophic balance and so, implicitly, that of a small Rossby number, predates both Rossby and Kibel.
- 10 After Taylor (1921b) and Proudman (1916). The Taylor–Proudman effect is sometimes called the Taylor–Proudman 'theorem', but it is more usefully thought of as a physical effect, with manifestations even when the conditions for its satisfaction are not precisely met. In fact, Hough (1897) seems to have been aware of the effect well before Taylor and Proudman.
- 11 This discussion owes much to that in Holton (1992). Inertial motion is discussed by Durran (1993). Jim Holton (1938–2004) made many contributions to atmospheric dynamics over the course of a distinguished career spent almost entirely at the University of Washington in Seattle. In his early career he elucidated, with Richard Lindzen of MIT, the essential mechanism of the quasi-biennial oscillation, or QBO, and he continued to make important contributions to wave–mean-flow interaction, stratosphere-troposphere interaction and stratospheric dynamics more generally throughout his career. He is also known, both to scientists and students, for his pioneering textbook *An Introduction to Dynamical Meteorology*.

Part II

WAVES, INSTABILITIES AND TURBULENCE

Part III

LARGE-SCALE ATMOSPHERIC CIRCULATION

*I'm singing in the rain, Just singing in the rain,
What a glorious feelin', I'm happy again.
I'm laughing at clouds, So dark up above,
The sun's in my heart, And I'm ready for love.*
Lyrics by Arthur Freed, music by Nacio Herb Brown, 1929.

CHAPTER 18

Water Vapour and the Tropical Atmosphere

WATER IS AN ORDINARY SUBSTANCE WITH EXTRAORDINARY EFFECTS. The most obvious is that oceans themselves are made of water, and if our planet were dry this book would perforce be much shorter (if only). Leaving aside the dynamical effects of the oceans, water covers over two-thirds of Earth's surface and because it is warm in some places and cold in others, and because the atmosphere is in motion, water evaporates into the atmosphere in one place and condenses from it elsewhere. The condensation leads to rain, one of the most talked-about aspects of weather and climate. Water also freezes to form ice, so that at any given time water exists on Earth in all three phases. Radiatively, water vapour is a greenhouse gas, meaning that it absorbs infra-red radiation that might otherwise be lost to space and so maintains the surface of the planet at a temperature over 20 K higher than an equivalent dry planet. Dynamically, the condensation of water vapour in the atmosphere releases energy, warming the air and tending to make it more unstable than otherwise and leading to convection. Further, the net transport of water vapour from low to high latitudes is effectively a meridional transport of energy.

In this chapter we focus on a small number of these issues, mainly on the kinematics and dynamics of water vapour itself and on some aspects of the dynamics of the tropical atmosphere, where the effects of water vapour are most manifest. The tropics would certainly differ from the mid-latitudes even if the atmosphere were dry — its Coriolis parameter is small among other things — so our attention there is by no means confined to the effects of water vapour. Nevertheless, tropical convection and the attendant 'radiative-convective equilibrium' are greatly influenced by the presence of water. We begin with a discussion of the thermodynamic properties of water vapour itself. We then move onto an essentially kinematic description of the factors determining the large scale distribution of relative humidity, before finally looking at convection and at tropical dynamics more generally.¹

18.1 A MOIST IDEAL GAS

Water is the compound of hydrogen and oxygen with the chemical formula H_2O , although in informal conversation water is often understood to mean only the liquid form of the compound. *Water vapour* is a gas made up of molecules of H_2O , and *ice* is the solid form of water. *Steam*, in common

parlance, is a mixture of air, water vapour and suspended droplets of water, usually at a very high temperature. Steam is formed when water vapour at temperatures above boiling point cools under contact with air and some of the water vapour condenses, forming a fine mist. Cloud and fog are also mixtures of dry air, water vapour and water droplets, but need not be at high temperature. Our focus will be on water vapour which, as we will see, can exist over a wide range of temperatures; let us first say how we quantify it and how it affects the equation of state. A number of thermodynamic derivations are also given in Appendix A (page 720). Those derivations are more systematic but less pedagogical than those below, and may appeal to some.

18.1.1 Ideal gas equation of state

The thermal equation of state for an ideal gas is conventionally written in the form

$$pV = Nk_B T = nR^* T, \quad (18.1)$$

where N is the total number of molecules in the volume V , n is the number of moles in that volume, and $N = nN_A$ where N_A is Avogadro's number. A mole is the amount of a substance that contains the same number of elementary entities, usually atoms or molecules, as there are atoms in 12 grams of carbon-12, that number being Avogadro's number ($N_A \approx 6.02 \times 10^{23}$). Two moles of a substance contains two times Avogadro's number of elementary units. The constants in the above equation are Boltzmann's constant, k_B , and the universal gas constant, R^* , where $R^* \equiv N_A k_B = 8.314 \text{ J mol}^{-1} \text{ K}^{-1}$. As noted in chapter 1, for any particular gas it is convenient to define the specific gas constant by $R = R^*/\mu$ where μ is the molar mass (mean molecular weight in kg/mol). For a single component gas we then divide (18.1) by the total mass $M = n\mu$ to obtain

$$p = \rho RT. \quad (18.2)$$

Throughout this chapter we will be concerned only with 'simple ideal gases', or 'perfect gases', for which the gas constants at constant composition are, in fact, constant.

For a multi-component ideal gas the partial pressure of each component is independent of the presence of the other components (because the volume of the molecules is negligible) and so is equal to the hypothetical pressure of that gas if it alone occupied the volume of the mixture. The total pressure is therefore the sum of the partial pressures of each gas, a dictum known as Dalton's law of partial pressures. The partial pressure of each constituent in a mixture is proportional to the number of molecules of that constituent, and therefore also proportional to the number of moles. Because of Dalton's dictum we can obtain a simple expression for the equation of state of a mixture, as follows. Denoting the constituents by subscript i , the total pressure is given by

$$p = \sum_i p_i = \sum_i \frac{1}{V} n_i R^* T = \sum_i \left(\frac{M}{V} \right) \frac{n_i \mu_i}{M \mu_i} R^* T. \quad (18.3)$$

Let us define the effective molar mass, μ_e , by

$$\frac{1}{\mu_e} = \sum_i \frac{n_i \mu_i}{M \mu_i} = \sum_i \frac{\varphi_i}{\mu_i}, \quad (18.4)$$

where $\varphi_i = (n_i \mu_i)/M$ is the mass fraction of the i 'th constituent. We then have

$$p = \rho RT, \quad \text{where} \quad R = \frac{R^*}{\mu_e} = \sum_i \varphi_i R_i, \quad (18.5a,b)$$

and $R_i = R^*/\mu_i$. The effective gas constant of the mixture is thus the mass-weighted mean of the specific gas constants of its constituents. Any given gas has a specific constant that is inversely

proportional to its molecular weight. Thus, for a given fluid density and temperature a gas with a higher molecular weight will exert a lower pressure than one that smaller molecular weight, because it will have fewer molecules per unit mass. Similar expressions apply to the heat capacities c_p and c_v , so that a heavier gas (higher molecular weight) has a smaller specific heat capacity.

18.1.2 Application to moist air

Dry air has virtually constant composition and its mean molar mass is $\mu^d = 29.0 \times 10^{-3} \text{ kg mol}^{-1}$, giving $R^d = R^*/\mu^d = 287 \text{ J kg}^{-1} \text{ K}^{-1}$. Water vapour has a molar mass of $\mu^v = 18.014 \times 10^{-3} \text{ kg mol}^{-1}$ giving $R^v = 461.5 \text{ J kg}^{-1} \text{ K}^{-1}$. The two gas constants are related by

$$\frac{R^v}{R^d} = \frac{\mu^d}{\mu^v} \equiv \frac{1}{\epsilon} \approx 1.608 \quad (18.6)$$

Now consider a mixture of dry air and water vapour.

Measures of moisture

When mixtures are present we superscripts d , v and l to denote thermodynamic quantities associated with dry air, water vapour and liquid water. The *absolute humidity* is the amount of water vapour per unit volume, with units of kg m^{-3} , or informally g m^3 . The *mixing ratio*, w , is ratio of the mass of water vapour, m^v , to that of dry air, m^d , in some volume of air and is thus

$$w \equiv \frac{m^v}{m^d} = \frac{\rho^v}{\rho^d}. \quad (18.7)$$

It is a nondimensional measure but it is often expressed in terms of grams per kilogram. In the atmosphere values range from close to zero to about 20 g kg^{-1} (2×10^{-2}) in the tropics on a humid day.

The *specific humidity*, q , is the ratio of the mass of water vapour to the total mass of air — dry air plus water vapour — and so is

$$q \equiv \frac{m^v}{m^d + m^v} = \frac{w}{1 + w} \quad \text{and} \quad w = \frac{q}{1 - q}. \quad (18.8a,b)$$

The specific humidity is just the mass concentration of water vapour in air. In most circumstances in Earth's atmosphere $m^v \ll m^d$ so that $q \approx w$, usually to an accuracy of about one percent. In much of this chapter we will ignore the differences between w and q , but this is not appropriate for all planetary atmospheres.

The partial pressure of water vapour in air, e , is the pressure exerted by water molecules and is proportional to the number of moles of water vapour in the volume. It is given by

$$e = \frac{n^v}{n^d + n^v} P = \frac{m^v/\mu^v}{m^d/\mu^d + m^v/\mu^v} P. \quad (18.9)$$

where n^v and n^d are the number of moles of water vapour and dry air in the mixture and P is the total pressure. Using (18.7) we can write (18.9) as

$$e = \frac{wP}{w + \epsilon} \quad \text{or} \quad w = \frac{\epsilon e}{P - e}. \quad (18.10)$$

In terms of q instead of w these expressions are

$$e = \frac{qP}{q + \epsilon(1 - q)} \quad \text{and} \quad q = \frac{\epsilon e}{P - e(1 - \epsilon)}, \quad (18.11)$$

In Earth's atmosphere $w \ll 1$ so that

$$e \approx w \frac{p}{\epsilon} = 1.61wp \quad \text{and} \quad q \approx w \approx \epsilon \frac{e}{p}. \quad (18.12)$$

If the mixing ratio of water vapour is 10 g kg^{-1} (a typical tropical value) and $p = 1000 \text{ hPa}$ then $e \approx 16 \text{ hPa}$.

The *relative humidity*, $\mathcal{H}$, is the ratio of the actual vapour pressure to the saturation vapour pressure, e_s , which is the maximum vapour pressure that can occur at a given temperature before condensation occurs, as will be discussed in section 18.1.4. Thus, $\mathcal{H} = e/e_s \approx q/q_s$ where q_s is the specific humidity at saturation.

18.1.3 Equation of state and virtual temperature

Using (18.5b) the effective gas constant of moist air all varies with humidity according to

$$R = \frac{m^d R^d + m^v R^v}{m^d + m^v} = (1 - q)R^d + qR^v = R^d \left[1 + q \left(\frac{1}{\epsilon} - 1 \right) \right] \quad (18.13)$$

with similar expressions for c_p and c_v . In humid air, with $q = 0.02$ say, we have $R = R^d(1 + 0.02 \times 0.61) = 467.1 \text{ J kg}^{-1} \text{ K}^{-1}$.

The heat capacity of water vapour can be estimated from its molecular properties. Water vapour is triatomic molecule with three translational and three rotational degrees of freedom. If these were the only degrees of freedom then the internal energy would be given by $I = 6R^v T/2$, whence $c_v^v \approx 3R^v = 1384 \text{ J kg}^{-1} \text{ K}^{-1}$ and $c_p^v = R^v + c_v^v = 1846 \text{ J kg}^{-1} \text{ K}^{-1}$, where c_v^v and c_p^v are the specific heat capacities for water vapour at constant volume and pressure, respectively. In fact vibrational degrees of freedom can sometimes be excited and the measured values are a little higher, namely $c_v^v = 1397 \text{ J kg}^{-1} \text{ K}^{-1}$ and $c_p^v = 1859 \text{ J kg}^{-1} \text{ K}^{-1}$ (at 273 K , increasing very slightly with temperature). The heat capacity of moist air is thus slightly higher than that of dry air, but since values of q are small the difference is only about 1%.

The variation of gas constant with humidity can be inconvenient in numerical models. A workaround is to define a so-called virtual temperature, T_v , which is the temperature that dry would need to be in order to have the same density and pressure as moist air. That is, by definition,

$$p = \rho R T = \rho R^d T_v \quad (18.14)$$

where R is given by (18.13). Using (18.13) we obtain

$$p = \rho R^d T_v, \quad \text{where} \quad T_v = T \left[1 + q \left(\frac{1}{\epsilon} - 1 \right) \right] \approx T(1 + 0.61q). \quad (18.15)$$

The virtual temperature, T_v , increases with specific humidity and if $q = 20 \text{ g kg}^{-1}$ then T_v is about 12%, or 3 K, larger than the actual temperature. Such a temperature is often used in numerical models of the atmosphere because it enables various thermodynamic equations to keep their original form, with gas constants that actually are constant.

Because the concentration of water vapour in Earth's atmosphere is so small, the variations of the heat capacities are small and constant values are often used to calculate such quantities as the potential temperature and the adiabatic lapse rate. This is not always appropriate, and the appendix on page 720 indicates how, in principle, more accurate calculations could be made.

18.1.4 Saturation vapour pressure

Vapour pressure is the partial pressure of water vapour in the atmosphere. At any given temperature, there is a maximum value of that vapour pressure beyond which condensation normally

occurs and this is known as the *saturation vapour pressure*. Why should this be so, and why don't other atmospheric gases, such as oxygen or carbon dioxide, also condense? To understand this, we have to understand the thermodynamic equilibrium between a liquid and a gas.

Equilibration of the Gibbs function

Consider a system that consists of an enclosed, insulated container partially filled with liquid, and with vapour above it. The two subsystems can exchange mass and energy, with liquid potentially evaporating into vapour and vapour condensing into the liquid. Energy is required to evaporate the liquid into a vapour to overcome the molecular forces in the liquid and this is given by $M(h^v - h^l)$, where M is the mass that has evaporated, h^v is the specific enthalpy of the vapour and h^l is the specific enthalpy of the liquid. The enthalpy of vapourization, more commonly called the *latent heat of evaporation*, is defined by the difference between the two enthalpies at a temperature T , namely,

$$L(T) \equiv h^v - h^l, \quad (18.16)$$

with L having units of J kg^{-1} . It is a function of temperature, because the enthalpy of liquid water and water vapour are both functions of temperature, and a very weak function of pressure — see appendix A (page 720) for details. For water, L diminishes almost linearly by about 10% going from 0°C to 100°C , from 2.5×10^6 to $2.26 \times 10^6 \text{ J kg}^{-1}$.

Now suppose we leave the container alone for a long time so that the liquid and vapour come into equilibrium at a temperature T . If a mass M is to evaporate from liquid into vapour then the energy required, E , can be related to the entropy difference between the liquid and vapour phases,

$$E = ML = M(h^v - h^l) = MT(\eta^v - \eta^l), \quad (18.17)$$

where η^v and η^l are the specific entropies of the vapour and liquid, and the temperature is fixed because all the energy put into the liquid is used for evaporation. Re-arranging we find

$$h^v - T\eta^v = h^l - T\eta^l \quad \text{or} \quad g^v = g^l, \quad (18.18a,b)$$

where $g^l \equiv h^l - T\eta^l$ and $g^v \equiv h^v - T\eta^v$ are the specific Gibbs functions for the liquid and vapour (section 1.5.2). That is, *the specific Gibbs functions for the liquid and water phases of a substance are the same at equilibrium*. The result follows directly from (18.17): the energy required to evaporate a mass of liquid, which is equal to the mass times the specific enthalpy difference between the vapour and the liquid, is also equal to the mass times the specific entropy difference between the vapour and liquid. The equality is true only at equilibrium, when temperature remains fixed, so that (18.18b) is an equation, not an identity.

Another way to derive the above result is to begin with the fact that the total Gibbs function, for the entire system, must remain constant. That is, if M_l and M^v are the masses of liquid and water, then

$$G = M_l g^l + M^v g^v \quad (18.19)$$

and

$$\delta G = M_l \delta g^l + M^v \delta g^v + (g^v - g^l) \delta M = 0 \quad (18.20)$$

where δM is the mass exchanged between liquid and vapour arising from a small fluctuation. Now, from (1.78) changes in Gibbs functions arise because of changes in temperature and pressure; that is, in general,

$$\delta g = -\eta \delta T + \alpha \delta p, \quad (18.21)$$

and given this (18.20) becomes

$$\delta G = M_l(-\eta^l \delta T + \alpha^l \delta p) + M^v(-\eta^v \delta T + \alpha^v \delta p) + (g^v - g^l) \delta M = 0 \quad (18.22)$$

where α_l and α^v are the specific volumes (the inverse density) of the liquid and vapour, respectively. But the temperature and pressure are fixed, and thus, in order that $\delta G = 0$ we must have that $g^v = g^l$. The reason for equality of the two Gibbs functions — as opposed to the equality of some other thermodynamic potential — stems from the fact that the Gibbs function is the only potential for which the natural variables are intensive, namely temperature and pressure. The derivation we have given exploits this directly, for we kept p and T fixed.

18.1.5 Clausius–Clapeyron equation

Now suppose that the temperature of the liquid-vapour system changes by an amount δT , leading to a change in the vapour pressure and in the specific Gibbs functions for the liquid and water vapour. Using (18.21) the change in the two Gibbs functions is given by

$$\delta g^l = -\eta^l \delta T + \alpha^l \delta p, \quad \delta g^v = -\eta^v \delta T + \alpha^v \delta p \quad (18.23)$$

and, since the two changes must be the same, $\delta g^l = \delta g^v$. Re-arranging (18.23) and taking the limit of small changes then gives

$$\frac{dp}{dT} = \frac{\eta^l - \eta^v}{\alpha^v - \alpha^l}. \quad (18.24)$$

Using (18.16) and (18.17) this equation can be written

$$\frac{dp}{dT} = \frac{h^l - h^v}{T(\alpha^v - \alpha^l)} = \frac{L}{T(\alpha^v - \alpha^l)}, \quad (18.25)$$

where to obtain the rightmost expression we use the definition of L . The quantity p is the vapour pressure of the vapour above the liquid, which we are denoting e . Furthermore, it is the *saturation* vapour pressure, e_s , because the vapour is in equilibrium with the liquid: if more vapour were added it would immediately condense. Using this notation, the *saturation vapour pressure* of a condensable gas above a liquid is given by

$$\frac{de_s}{dT} = \frac{L}{T(\alpha^v - \alpha^l)}. \quad (18.26)$$

This is the *Clausius–Clapeyron* equation, and it tells us how the equilibrium pressure of a vapour, in thermodynamic equilibrium with an adjacent liquid, varies with temperature. If for some reason the pressure of the vapour is higher than this value, and if there is an adjacent surface of liquid water, then the vapour will condense into a liquid — a common manifestation of which is the formation of clouds and rain. Evidently, since $L > 0$ and $\alpha^v > \alpha^l$, the saturation vapour pressure increases with temperature so that a falling of temperature can lead to saturation. At the temperature at which the saturation vapour pressure of a substance equals that of the ambient pressure then any liquid present will *boil*. For water at a pressure of 1000 hPa this occurs at about 100° C, with a lower temperature needed at a lower pressure, which is why it takes longer to properly boil an egg at high altitude.

The presence of a liquid surface is crucial to the derivation, and it means that the equation only applies to a condensable. For gases such as carbon dioxide or oxygen at temperatures encountered on Earth, the saturation vapour pressure is very much higher than the actual pressure at the Earth's surface and the gas never condenses; the partial pressure of the gas is then determined by the ideal gas relation and not by (18.26). On Mars, temperatures are sufficiently low that carbon dioxide (the main constituent of the Martian atmosphere) is a condensate and as much as 25% of the Martian atmosphere will condense in winter. On Titan temperatures are even lower and methane is a condensate and methane lakes cover the dystopian surface.

On Earth, the partial pressure of water vapour, however, is constrained by (18.26), meaning that the partial pressure will often reach the saturated value and the vapour will then normally condense. Condensation is not, however, guaranteed, and to see this imagine a container that contains unsaturated water vapour and no liquid water, and suppose its temperature is then lowered; the pressure of the vapour will then fall following the ideal gas law. The saturation vapour pressure falls more quickly than this and so at some temperature the vapour pressure will exceed the saturation vapour pressure, but the vapour will not automatically condense since there is no liquid present and (18.26) does not apply. The vapour is then said to be *supersaturated*. A supersaturated state is unstable and condensation will eventually occur and liquid water will form, and subsequent changes in temperature induce pressure changes that satisfy the Clausius–Clapeyron equation. Supersaturated water vapour is fairly rare in Earth’s atmosphere because there is usually no shortage of condensation nuclei.

At the other end of the temperature scale, liquid water can exist at temperatures well below freezing when there is insufficient water for the molecules to become organized into a crystalline structure, and super-cooled water droplets rather than ice then result — a common situation in clouds. If the liquid that is present is in the form of small spherical droplets then the saturation vapour pressure will differ slightly from that when the vapour is over a flat surface, because surface tension will affect the energy required for a molecule to escape from the droplet and so the latent heat of vapourization will differ. If the vapour is in contact with ice instead of water then its saturation vapour pressure will differ again, because the specific enthalpy of ice is different from that of liquid water.

Application to an ideal gas

In a mixture of ideal gases the partial pressure of one gas is unaffected by the presence of the other gases (because the volume of the gas molecules is assumed to be negligible) and in particular the saturation vapour pressure for a particular component is independent of the presence of other components. We can then, at least approximately, integrate (18.26) to see how the saturation vapour pressure of a particular component varies with temperature. Let us assume that density of the vapour is much less than that of the liquid, so that $\alpha^v \gg \alpha^l$. Given that partial pressure of the vapour satisfies the ideal gas law, namely $e^v \alpha^v = R^v T$, where R^v is the specific gas constant for the vapour, the Clausius–Clapeyron equation becomes

$$\frac{de_s}{dT} = \frac{Le_s}{R^v T^2}. \quad (18.27)$$

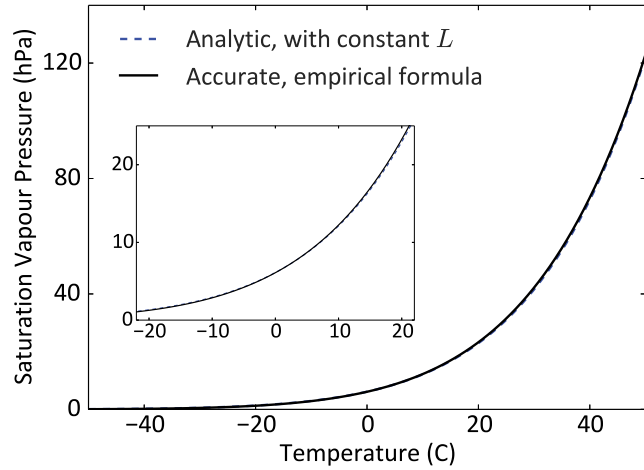
This is the form of Clausius–Clapeyron equation that is normally used in atmospheric applications. If we further assume that L is a constant then (18.27) can be integrated to give

$$e_s = e_0 \exp \left[\frac{L}{R^v} \left(\frac{1}{T_0} - \frac{1}{T} \right) \right], \quad (18.28)$$

where e_0 and T_0 are constants, for example $T_0 = 273 \text{ K}$ and $e_0 = 6.12 \text{ hPa}$. Equation (18.28) is a good approximation if the temperature range is not too wide, and as seen in Fig. 18.1 the saturation vapour pressure of water increases approximately exponentially over commonly encountered terrestrial temperatures. Note that the expression for saturation vapour pressure does not depend on the presence or otherwise of dry air.

The fact that water vapour content cannot normally exceed the saturation value distinguishes the distribution of water from other tracers in the atmosphere, even without taking the heating effects of condensation into account. Note finally that if the atmosphere were motionless it would everywhere be in thermodynamic equilibrium with the moist surface and the surface layers would be saturated, and diffusion of water vapour upwards would then saturate the rest of the atmosphere. Thus, *the relative humidity of the atmosphere is determined by its circulation*, as we now discuss.

Fig. 18.1 The saturation vapour pressure of water vapour, calculated using the analytic formula (18.28), which assumes that L is constant, and using an accurate, semi-empirical formula.² The inset is the same plot over a smaller range, and for the analytic formula we use $T_0 = 273\text{ K}$, $e_0 = 6.12\text{ hPa}$, $L = 2.44 \times 10^6\text{ J kg}^{-1}$ and $R^v = 462\text{ J kg}^{-1}\text{ K}^{-1}$.



18.2 THE DISTRIBUTION OF RELATIVE HUMIDITY

To a first approximation, the distribution of water in the atmosphere is determined by the distribution of temperature. This is because of the near-exponential dependence of absolute humidity on temperature through the Clausius–Clapeyron equation, so that variations of relative humidity of even an order of magnitude, from 10% to 100% say, are barely noticeable in the specific humidity distribution, as seen in Fig. 18.2. It is, however, the relative humidity that determines such basic quantities as rainfall, and its distribution (Fig. 18.3 and Fig. 18.4) shows a quite different picture, with the following features evident.

- High, near-saturated values close to the ground.
- Low relative humidity in the subtropics at latitudes between 15° to 40° (depending on season) in both hemispheres.
- High values near the equator extending up to the tropopause.
- Vertically near-uniform values in mid- and high latitudes, increasing with latitude close to the pole in some cases.
- Very low values over much of the stratosphere.

The gross distribution of zonally-averaged temperature can be understood, at least in a rough way, using fairly simple arguments. The incoming solar radiation at the top of the atmosphere would lead, in the absence of atmospheric motion, to a strong meridional radiative equilibrium temperature gradient, as in Fig. 14.1. The meridional transport of heat by the Hadley Cell and by mid-latitude baroclinic eddies flattens that temperature gradient, and one might crudely model this transport as a diffusion. In the vertical heat is transported upwards by convection and baroclinic eddies, which might be modelled as a relaxation back to some specified neutrally stable profile or specified isentropic slope. However, arguments of this type cannot capture some basic features of the relative humidity distribution, and diffusive arguments in particular can be quite misleading. The effects of advection, either explicitly or as represented by a stochastic process, are crucial, and in this section we consider advection-diffusion-condensation models of the general form

$$\frac{\partial q}{\partial t} + \mathbf{v} \cdot \nabla q = \nabla \cdot \kappa \nabla q - S, \quad (18.29)$$

where q is the specific humidity, $\mathbf{v}$ is a specified velocity field, κ is a diffusion coefficient and S is the condensational sink.⁴ Condensation in the atmosphere involves complicated microphysical processes but the basic effect is to remove liquid water once the volume becomes saturated and to

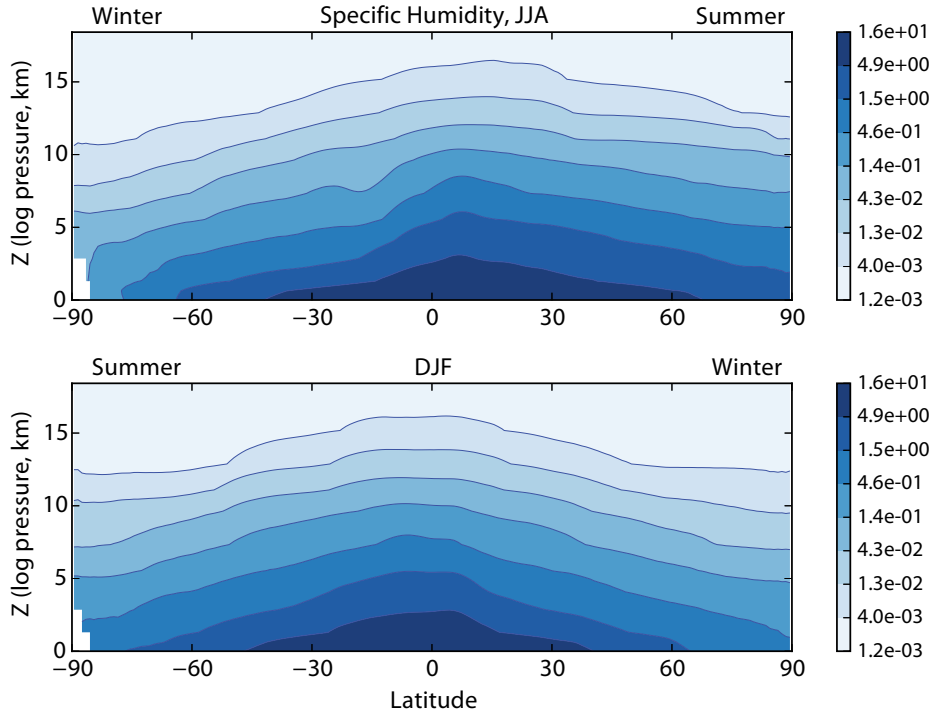


Fig. 18.2 Zonally-averaged specific humidity distribution (g/kg) in the atmosphere, as retrieved from a microwave satellite, for boreal summer in 2008 (top) and boreal winter 2008–2009 (bottom).³ Note the logarithmic scale.

largely prevent relative humidity from exceeding 100% (although supersaturation can occur locally if there are no particulates in the air onto which the water vapour may condense). A simple analytic way to represent such a condensation process is to let

$$S = \begin{cases} 0, & q \leq q_s, \\ (q - q_s)/\tau, & q > q_s, \end{cases} \quad (18.30)$$

where the time τ is much smaller than any large scale diffusion time, L^2/κ , where L is a characteristic length. Such a sink effectively prevents q from exceeding q_s except by a tiny amount. In (18.29) we may consider the advection and diffusion terms as representing larger scale processes and the sink as representing small-scale microphysical processes. We begin our exploration by omitting advection.

18.2.1 A diffusion-condensation model

If we omit advection in (18.29) we have a simple diffusion-condensation model

$$\frac{\partial q}{\partial t} = \nabla \cdot \kappa \nabla q - S. \quad (18.31)$$

Although superficially plausible, such a model is too-often unable to reproduce locally unsaturated regions. For simplicity consider the one-dimensional case satisfying

$$\frac{\partial q}{\partial t} = \kappa \frac{\partial^2 q}{\partial x^2} - S, \quad (18.32)$$

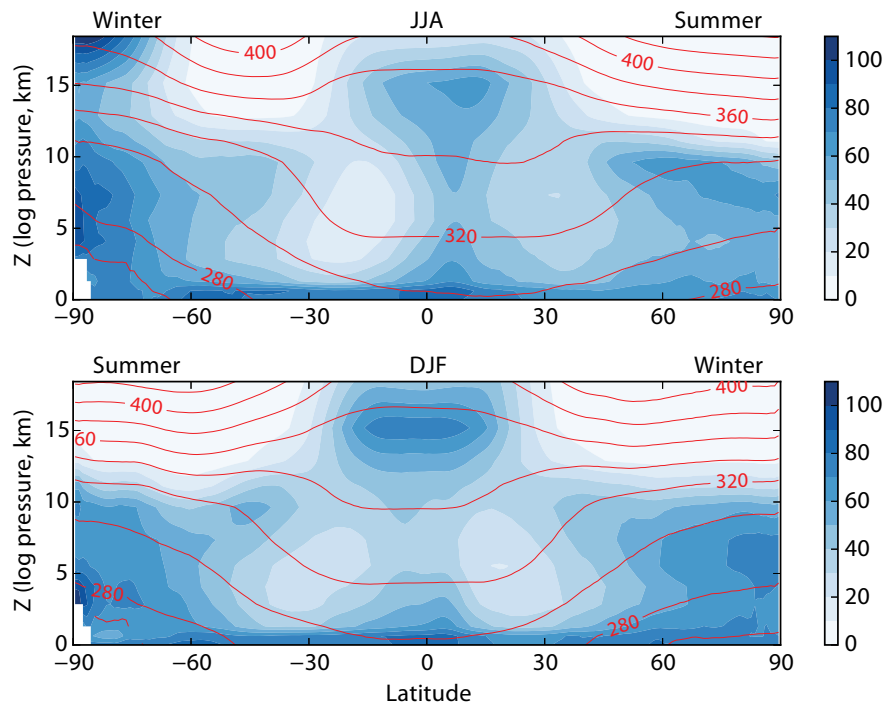


Fig. 18.3 Zonally-averaged relative humidity distribution in the atmosphere (in percent, shading), and isolines of equivalent potential temperature (contours), inferred from satellite as in Fig. 18.2. (Equivalent potential temperature is a modification of potential temperature to account for water vapour, and is approximately an adiabat; see section 18.3.2.)

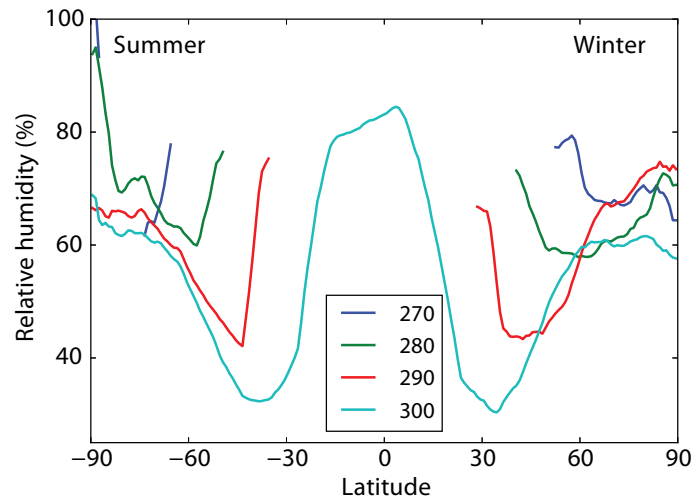


Fig. 18.4 Values of zonally-averaged relative humidity in boreal winter plotted along isolines of equivalent potential temperature (red contours in Fig. 18.3), with values of θ_{eq} as indicated in the legend.

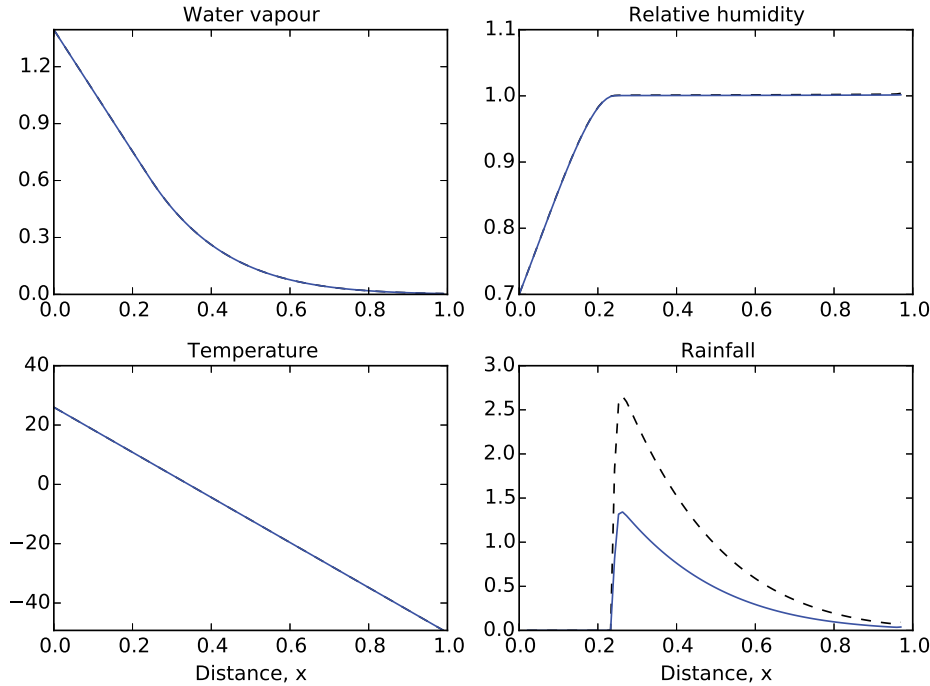


Fig. 18.5 Steady solution of the diffusion-condensation model (18.32) and (18.33), with $\mathcal{H}_b = 0.7$ at $x = 0$ and temperature ($^{\circ}\text{C}$) falling linearly from $x = 0$ as shown. Water vapour falls linearly away from the boundary at $x = 0$ until it becomes saturated at $x = 0.23$, after which the region is saturated. Two solutions are plotted, the one with dashed lines showing the solution with twice the diffusivity as that with solid lines. The two solutions are nearly identical except for the sink term, the rainfall.

with constant κ .

Regardless of the boundary conditions, any interior minimum of q_s will lead to saturation in that region. To see this, note that in any region with $\partial^2 q / \partial x^2 > 0$ and $q < q_s$ there will be a flux of water vapour into that region and that $S = 0$. Saturation must eventually occur, at which point q remains very close to the value of q_s . If $\partial^2 q_s / \partial x^2 > 0$ then the diffusive flux will maintain the saturated state. Given the monotonic dependence of q_s on temperature, this result means that, if water vapour is transported diffusively, the region of an interior minimum of temperature will become saturated.

A corollary of this result is that, unless there is a source of moisture at a boundary, a region with $\partial^2 q_s / \partial x^2 > 0$ everywhere will under many conditions lose *nearly all* its moisture. Suppose that $\partial q / \partial x = 0$ at $x = 0$ and that q_s has a maximum at $x = 0$, and that the region extends to infinity and is initially saturated. (Envision a semi-infinite domain with temperature decreasing linearly away from no-flux boundary at $x = 0$.) Moisture is transported to higher values of x , whence condensation occurs and moisture is removed. As time progresses the region of saturation moves to higher and higher values of x , but nevertheless water is continuously removed. In a finite domain, with no flux boundary conditions at either end, a small amount of moisture will remain in the system for all time because a finite amount of water vapour is needed for condensation to occur.

Now consider the more atmospherically relevant situation with a moisture source at $x = 0$, q_s decreasing monotonically away and with the other boundary either extending to infinity or being a no-flux boundary at finite x . Such a situation might represent an atmosphere sitting atop a moist

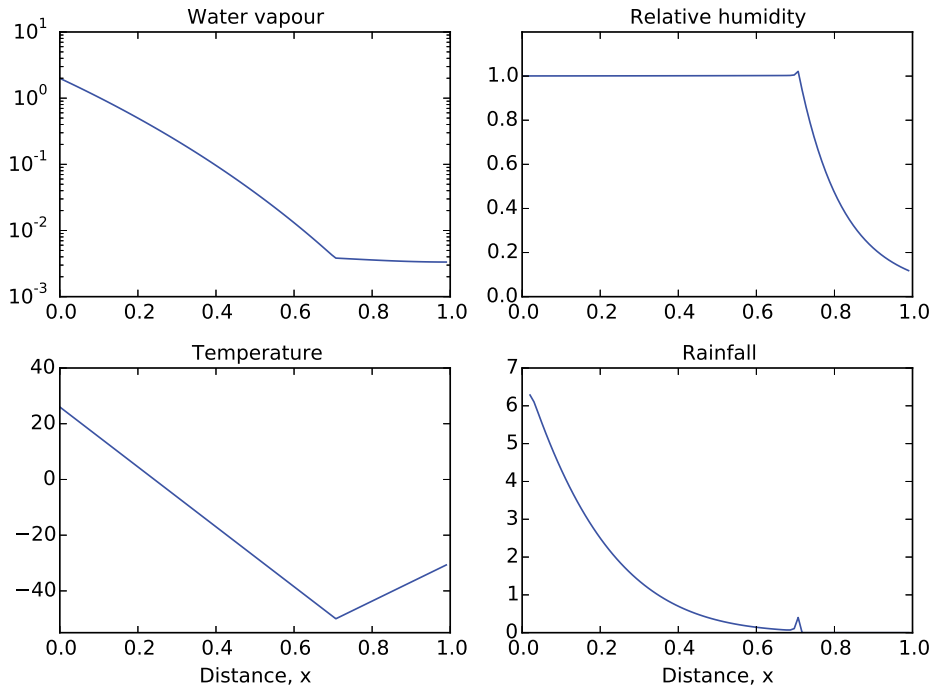


Fig. 18.6 As for Fig. 18.5, but now with $\mathcal{H}_b = 1$ and an interior temperature minimum, a ‘cold trap’, at $x = 0.7$ (see temperature panel), and water vapour has a log scale. Relative humidity falls rapidly beyond the cold trap and the rainfall there is zero.

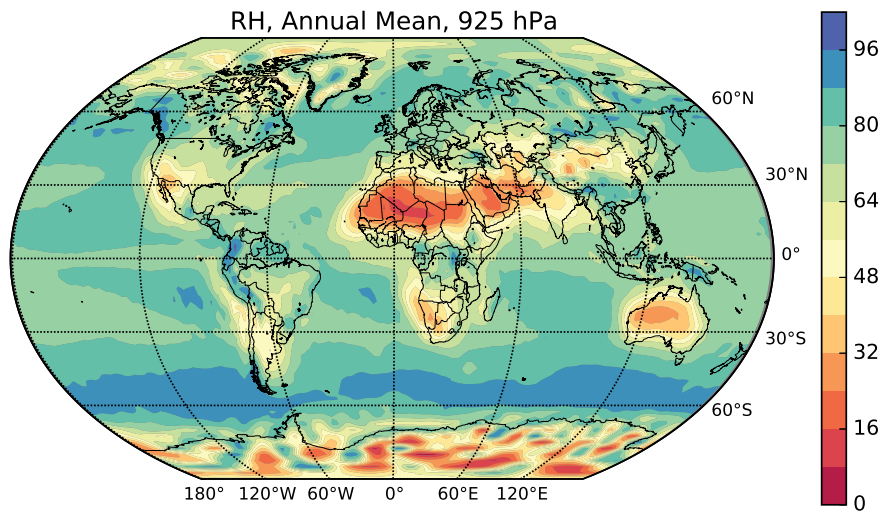


Fig. 18.7 Annually-averaged relative humidity (in percent) at 925 hPa, about 750 m above sea level. The contrast over land and ocean is apparent especially in the subtropics. (In regions of high topography values are interpolated.)

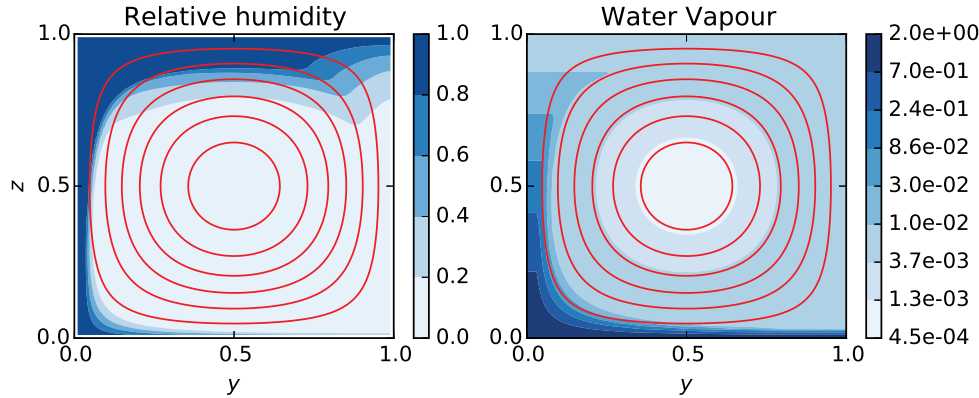


Fig. 18.8 Steady state distributions of relative humidity (filled contours, left) and water vapour (right) in a single cell as defined by the streamfunction (red contours, clockwise flow) in a closed domain. The domain boundary is saturated at the bottom, and temperature decreases linearly with height. The diffusivity $\kappa = 0.001$ and so the Peclet number, $Pe \equiv UL/\kappa \sim 1000$.

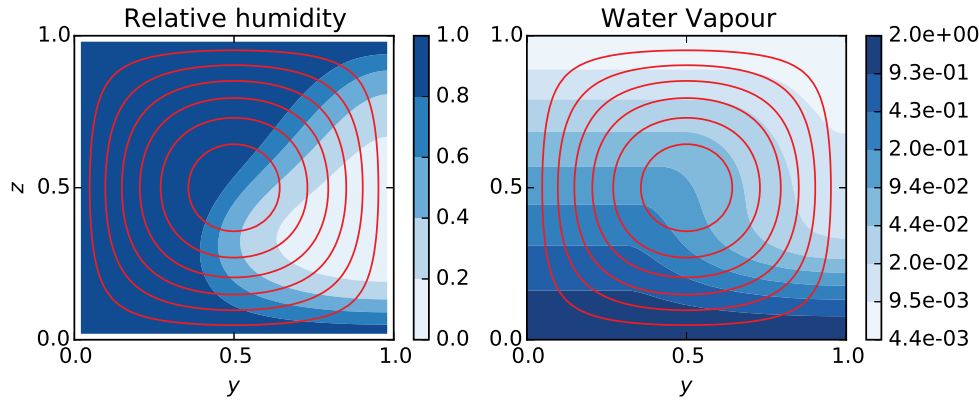


Fig. 18.9 As for Fig. 18.8, but with bigger diffusivity, $\kappa = 0.1$ and $Pe \sim 10$. Most of the domain is now much wetter.

surface with temperature decreasing with height. Consider the case

$$\begin{aligned} q &= \mathcal{H}_b q_s(T_0), & x &= 0, \\ \frac{\partial q}{\partial x} &= 0, & x &= 1, \end{aligned} \quad (18.33)$$

where $\mathcal{H}_b$ is a parameter such that if the boundary is effectively saturated then $\mathcal{H}_b = 1$, and $\mathcal{H}_b < 1$ otherwise, and we suppose that T decreases between T_0 at $x = 0$ and T_1 at $x = x_1$. Consider first the case with $\mathcal{H}_b = 1$. If T falls linearly then the value of q_s falls approximately exponentially between $x = 0$ and $x = 1$, with $\partial^2 q_s / \partial x^2 > 0$, and the steady solution of this problem is that the domain is saturated everywhere. To see this, suppose that $q < q_s$ is some region so that $S = 0$. Water vapour will then diffuse into that region until condensation begins, maintaining $q \approx q_s$ everywhere, with (if (18.30) applies, q in fact exceeding q_s by a very small amount so that the diffusion into the region is balanced by condensation. If $\mathcal{H}_b < 1$ then the region next to the surface will not be saturated and in steady state the water vapour content will decrease linearly, to satisfy $\partial^2 q / \partial x^2 = 0$, until at some value of x , x_s say, the atmosphere becomes saturated and remains so for $x > x_s$. The actual

solution is

$$q(x) = \begin{cases} \mathcal{H}_b q_s(0) + x \Delta q / x_s & x < x_s \\ q_s(x) & x \geq x_s, \end{cases} \quad (18.34)$$

where $\Delta q = (q_s(x_s) - \mathcal{H}_b q_s(0))$ and x_s is such that the flux is continuous there. A moment's thought reveals that $x_s = \Delta q / (\partial q_s / \partial x)_{x=x_s}$ and $x_s = 0$ if $\mathcal{H}_b = 1$. It is interesting that the water vapour levels in the solution (plotted in Fig. 18.5) do not depend upon κ and only very weakly on τ . In the fast condensation limit (in which τ is small compared to the diffusion time) variations of τ determine only the tiny amount by which q exceeds q_s . The amount of condensation is actually largely determined by the value of κ : large values of κ lead to a stronger diffusion of water vapour into dryer regions where it is almost immediately removed by condensation. This result illustrates the tenet that on large scales precipitation is at leading order determined by the motion of the fluid (here represented by diffusion), with variations in τ (crudely representing complex microphysical processes) being of less import. Microphysical processes are nevertheless important in many ways — a weather forecast model with poor microphysics would likely have little skill in forecasting the onset of precipitation, even if the climatology of the model were good.

Although the above model is over-simple in some respects, the dependence of the solution on $\mathcal{H}_b$ does capture the dependence of relative humidity in the lower atmosphere on the nature of the surface beneath, as seen in Fig. 18.7 showing relative humidity at 925 hPa. Over the desert regions the relative humidity is unsurprisingly low. Perhaps what is surprising is that the general dryness of the subtropics cannot be seen over the oceans — the surface moisture source simply overwhelms the drying effects of descending air (discussed more below).

A variation on the above theme introduces a temperature minimum, or ‘cold trap’, in the interior of the domain, as at $x = 0.7$ in Fig. 18.6. This configuration is a crude model of the tropopause, with temperatures increasing in the stratosphere beyond. Water vapour has to pass through the cold trap and so, since the specific humidity cannot be higher than the saturated value at the cold trap, the atmosphere will be unsaturated beyond it with relative humidity decreasing rapidly, as is seen in the real atmosphere in Fig. 18.3.

Although informative, diffusive-condensation models are fundamentally limited in what they can achieve, because of the deficiencies of diffusion in parameterizing the motion of a tracer in the presence of condensation. In particular, in the absence of a cold trap, diffusive models are prone to produce saturation everywhere. If the atmosphere obeyed (18.31) with a saturated surface, then the atmosphere would become saturated everywhere up to the tropopause, which from Fig. 18.3 is manifestly not the case. To remedy this we turn our attention to the effects of advection.

18.2.2 Advection-diffusion-condensation models

Consider now the effects of advection. Neglecting diffusion, the specific humidity of a parcel is conserved unless condensation occurs. If a moist parcel travels into a region of decreasing temperature it will eventually become saturated; however, if it passes into a region of increasing temperature then its relative humidity drops. Consider, for example, a cold trap with a temperature minimum at $x = x_{ct}$, as in the previous section. A parcel advected through the trap becomes saturated at x_{ct} , beyond which its specific humidity is constant, so that relative humidity is given by

$$\mathcal{H}(x) = \frac{q_s(T(x_{ct}))}{q_s(T(x))}. \quad (18.35)$$

That is to say, the relative humidity of a parcel is given by the value of the saturated vapour pressure at the point of last saturation, divided by the saturated vapour pressure at its current location.

Let us see the extent to which a simple advection-diffusion-condensation model can explain some of the features of the zonally-averaged relative humidity seen in Fig. 18.3. Consider a two-dimensional model in the y - z plane obeying (18.29), with a divergence-free advecting velocity, a

Water Vapour Transport and Relative Humidity

- Water vapour in the atmosphere is primarily transported by advection, much of this on large, near planetary scales but also by convection and smaller scale turbulence. Specific humidity is materially conserved in the absence of condensation and diffusion.
- If the vapour pressure exceeds the saturated value (as given by the Clausius–Clapeyron relation) condensation will occur provided condensation nuclei or liquid water are present. The condensation normally occurs much more quickly than large-scale advective processes, and this situation is known as the *fast condensation limit*.
- When dealing with large-scale flows in Earth’s atmosphere the limit is a good approximation. Levels of relative humidity are then determined mainly by advective processes rather than the microphysical details of the condensation process. Specifically, the relative humidity of a parcel is determined by the temperature at the location of last saturation, as in (18.35).
- If the advection is not fully resolved — for example if there is some small-scale turbulence in the flow — then introducing some diffusion seems natural, as is commonly done for tracers, but a large diffusivity can give unrealistic results because of the irreversible nature of condensation. Diffusion is then not a good representation of small-scale quasi-random flow and is overly prone to produce saturation.
- In Earth’s atmosphere some of the large-scale features of the relative humidity distribution may be explained as follows.
 - High levels of relative humidity close to the surface. These are due to transport from a saturated surface, especially over the ocean and moist ground.
 - High levels of relative humidity in the ascending branch of the Hadley Cell. These are due to upward advection from a nearly saturated surface. The branch is, however, not saturated on the zonal-average, because of the presence of smaller scale motion such as downdrafts that unsaturate the air.
 - A subtropical minimum of relative humidity. This largely arises because of the mean descending motion, advecting water vapour into a warmer region and decreasing its relative humidity.
 - Variable relative humidity in mid- and high latitudes, with locally strong gradients. Chaotic advection by baroclinic eddies takes moisture upwards and polewards into cooler regions where it becomes saturated, but also downwards and equatorwards so reducing relative humidity.
 - Very low levels of relative humidity in the stratosphere. The tropopause is a cold trap and so relative humidity is very low beyond it. The cold-trap effect occurs in both advective and diffusive models. In Earth’s atmosphere, the little water vapour that is in the stratosphere mainly enters advectively through the tropical tropopause.

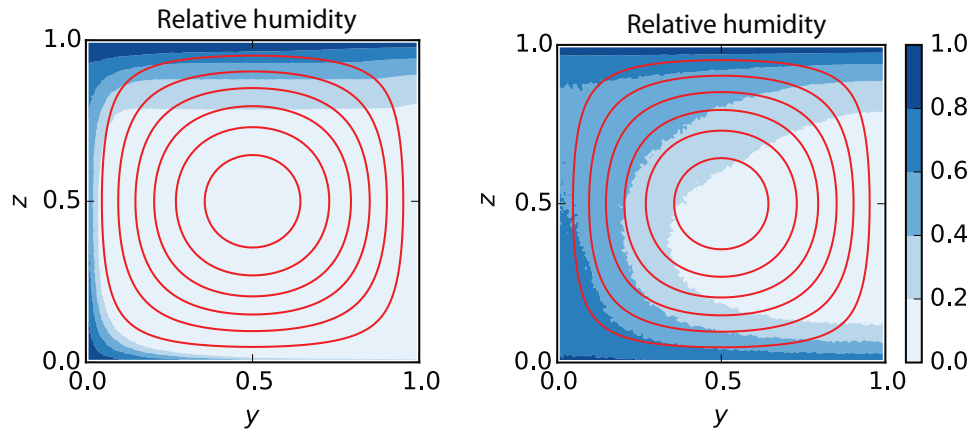


Fig. 18.10 The relative humidity produced using a stochastic advection-condensation model with the same imposed temperature, mean streamfunction (red contours) and boundary conditions as Fig. 18.8 and Fig. 18.9. The panel on the right has a larger stochastic component than the one on the left.

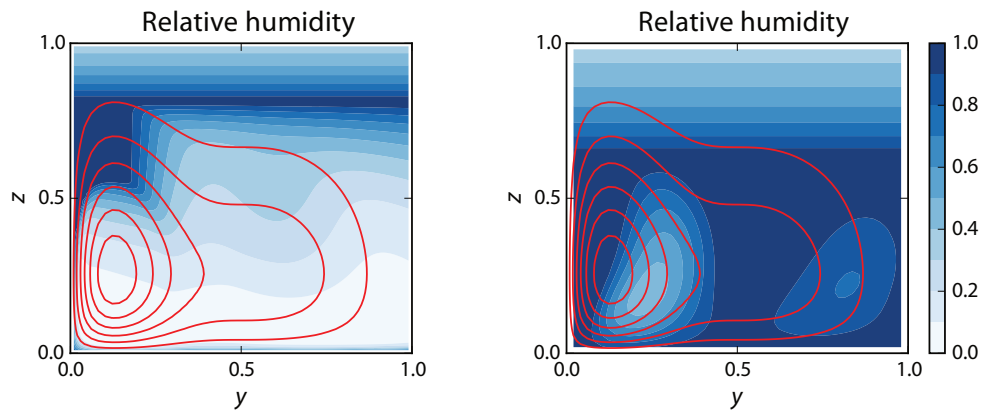


Fig. 18.11 Relative humidity (shading) for small diffusivity (left, $\kappa = 0.001$, $Pe = 1000$) and larger diffusivity (right, $\kappa = 0.05$, $Pe = 20$), and a clockwise overturning circulation as contoured. The imposed temperature falls linearly with height to a tropopause at $z = 0.8$ with $T = -50^\circ\text{C}$, then rises again to -40°C , and diminishes meridionally at the surface from 26°C at the equator ($y = 0$) to -15°C at the pole ($y = 1$).

saturated surface at $z = 0$ and no flux boundary conditions elsewhere. A simple model representing some of the features of the Hadley Cell is that of a single cell in a unit-sized square domain with a streamfunction of the form $\psi = \sin \pi y \sin \pi z$. We suppose the lower boundary is saturated, that there are no-flux boundary conditions on the other boundaries, and that temperature is uniform in latitude and falls linearly with height from 26°C to -50°C .

Solutions are illustrated in Fig. 18.8 and Fig. 18.9, the latter having diffusivity κ that is a hundred times larger. The upward branch in both cases is completely saturated. In the low diffusivity case the interior of the cell is quite dry, with a relative humidity that drops almost to zero in the centre. Increasing the diffusivity simply tends to moisten the interior, but does little to reduce the relative humidity in the upward branch of the cell. However, the upward branch of the real Hadley Cell is *not* saturated on the zonal mean (Fig. 18.3). Let us try to remedy this model failing.

Stochastic effects

The upwards branch of the Hadley Cell is not saturated because it is not steady — there is both ascending and descending motion, with the relative humidity of parcels falling as they descend so reducing the average value. We can mimic this by introducing a stochastic component into a Lagrangian model and performing ‘Monte Carlo’ simulations.⁵ The model advects infinitesimal parcels of water vapour — one million in the simulations shown in Fig. 18.10 — by a prescribed mean field plus a random component. Whenever the parcels touch the ground they become saturated and whenever their relative humidity exceeds one condensation occurs. We show results with two levels of stochastic motion, one nominally comparable to the diffusivity used in Fig. 18.8 and the other comparable to that of Fig. 18.9, all with the same temperature, boundary condition and mean velocity fields, and the results shown in Fig. 18.10 are coarse-grained time averages. Although the results with weak stochasticity do resemble those with small diffusion (compare the left-hand panel of Fig. 18.10 with Fig. 18.8), those with large stochasticity have less resemblance to those with large diffusivity. Most notably, the explicit randomness reduces the relative humidity in ascending branch without saturating the interior, a property that diffusion is unable to capture.

Relative humidity in mid-latitudes

The diffusive steady overturning model fails qualitatively when applied to an entire hemisphere, even with more realistic overturning circulation and temperature fields. To see this we construct a model in which the overturning streamfunction (representing a residual circulation) resembles that of the bottom panel of Fig. 15.22, as illustrated (red contours) in Fig. 18.11. To this we add a temperature distribution that decreases polewards and upwards until it reaches a tropopause at $z = 0.8$, then increases upwards again in the stratosphere, and we assume that the surface is saturated. With a small diffusivity (which is more realistic) the lower tropopause dries out too readily because of the sinking motion, and with larger diffusivity the atmosphere becomes more saturated but with little of the observed structure. Two features that the model *can* reproduce are the dry stratosphere, because of the tropospheric cold trap, and the subtropical minimum, a robust feature of the downwelling in the Hadley Cell.

The failings of this model arise from the fact that in mid-latitudes water vapour is not primarily advected by the overturning circulation, and diffusion is a poor model of the meridional transport. Rather, moisture is transported from the boundary layer into the free atmosphere by convection and polewards by larger-scale baroclinic eddies in a quasi-horizontal fashion, roughly along moist isentropic surfaces. Mid-latitude relative humidity is thus highly variable, as can be seen in Fig. 18.12, with sharp gradients between high and low values and with variations being strongly correlated with the parcel trajectory — note for example the swath of high relative humidity just west of the UK associated with moisture moving poleward and to a lower temperature. On the zonal average relative humidity is high in the boundary layer, diminishes as one moves upwards out of the boundary layer, then stays roughly constant as one moves polewards along a moist isentrope before increasing again at very high latitudes (Fig. 18.3 and Fig. 18.4).

We can understand some of these features using a simple advective-diffusive-condensation model, with the unsteadiness explicitly incorporated into the advection, and with no other stochastic component. Consider (18.29) in two horizontal dimensions in a channel of size (L_x, L_y) , periodic in x , with a saturated boundary at $y = 0$ and no flux at $y = L_y$, and with imposed meridional temperature gradient ($^{\circ}\text{C}$) and advecting streamfunction of the forms

$$T(y) = 20 - 30y/L_y, \quad \psi(x, y, t) = -Uy + \Psi_0 \sin(\pi y/L_y) e^{i(kx - \omega t)}. \quad (18.36)$$

We will choose $|\Psi_0| = 1$, $U = 10$, $k = 4\pi/L_x$, $\kappa = 0.005$ (so $Pe \sim 200$ on the domain scale), and a solution is shown in Fig. 18.13. The details of the solution depend on the parameters chosen, but there are couple of quite robust results.

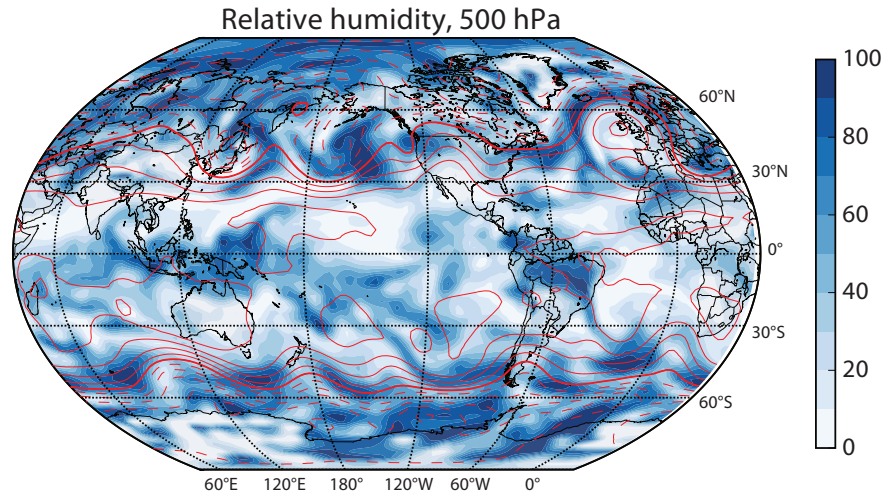


Fig. 18.12 Snapshot of relative humidity at 500 hPa on February 9, 2015 (shading, in percent), with red contours of geopotential height, from reanalysis.

- Relative humidity at a point depends on where a parcel has come from, and hence is correlated with the streamlines, and is generally higher for parcels moving to colder regions.
- Relative humidity is close to one (i.e., 100%) next to the saturated boundary, then decreases to a minimum in the interior, rising again at high latitudes (c.f., the observations shown in Fig. 18.4). The width of the saturated region near the saturated boundary diminishes as the Peclet number increases, and the mid-domain minimum arises because of the drying effects of advection, which diminish near either boundary.

There are many limitations to such a model, one being that in reality temperature itself is advected by the flow and another being that advection is three-dimensional, but exploring these effects is beyond the scope of our story.⁶ A summary of relative humidity transport is provided on page 687, but we now move on to the dynamical effects of water vapour, and in particular convection,

18.3 ATMOSPHERIC CONVECTION

As introduced in section 2.10, convection occurs when a parcel that is displaced upwards (downwards) finds itself lighter (heavier) than its surroundings, and hence becomes subject to a buoyancy force that amplifies the initial displacement. Convection is particularly prevalent in the tropics for reasons that are, at lowest order, independent of the presence of water vapour; rather, it is simply that the surface is warmer there, as sketched in Fig. 18.14. Consider, rather heuristically, a fluid that is heated by a warm surface from below and cools by radiation from above. If the surface is warmer at one end, as illustrated, a large-scale circulation will arise, transporting energy from warmer regions to cooler ones. In the absence of convection, there will be a larger vertical temperature gradient where the surface temperature is warm, and so these regions are more prone to convection. Having said this, moisture *does* affect convection in rather profound ways, one being that moisture is a *destabilizing* influence, as we now discover.

18.3.1 Generalized enthalpy and adiabatic lapse rates

A column of air that is convectively stable when dry can be made convectively unstable by the presence of water vapour, because when water vapour condenses heat is released, warming the

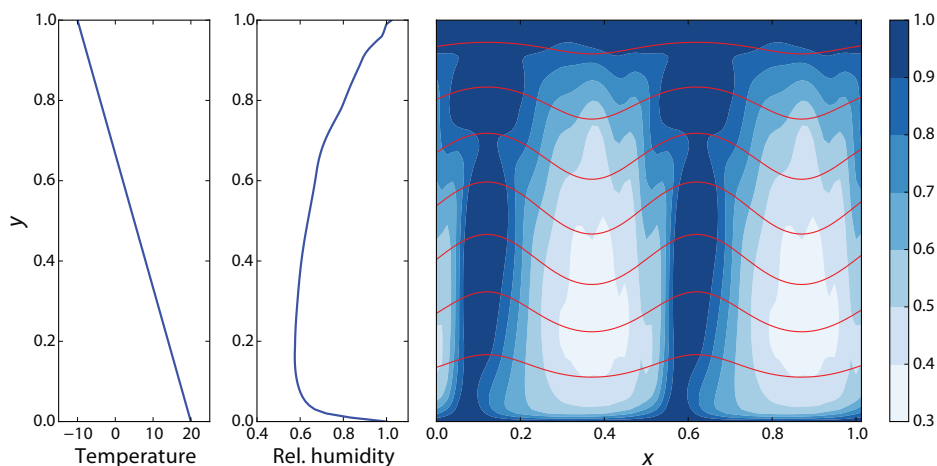


Fig. 18.13 A snapshot of relative humidity (right panel, shading) obtained by a numerical solution of the advective-diffusive-condensation model (18.29) and (18.36) in the horizontal (x - y) plane. The contours are the streamfunction. The middle panel is the zonally-averaged relative humidity and the left panel is the imposed temperature gradient ($^{\circ}\text{C}$).

parcel and potentially making it more buoyant than its surroundings. To explore this we begin with a derivation of the conditions under which a moist column of air may be unstable — essentially an extension of section 2.10 to include the effects of condensation.

An adiabatic (and convectively neutral) profile is one in which the entropy is constant. Recalling the arguments in sections 1.6 and 1.10.3, the entropy may be related to the enthalpy by writing the fundamental relation as

$$dh = Td\eta + \alpha dp, \quad (18.37)$$

and where, if the column is hydrostatically balanced, $\alpha dp = -g dz$. Thus, the condition that $d\eta = 0$ is equivalent to

$$d(h + gz) = 0. \quad (18.38)$$

That is, the generalized enthalpy or dry static stability, $h^* = h + gz$, is constant in an adiabatic,

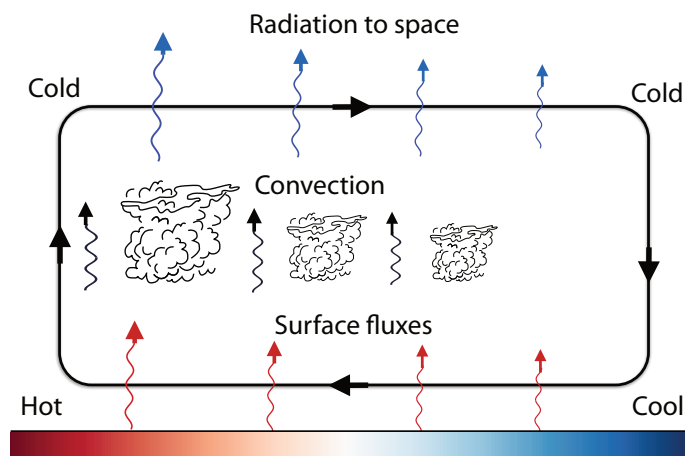


Fig. 18.14 Sketch of small scale convection embedded within a large scale circulation. The fluid is in contact with a warm surface below, and cools by radiation to space from above. The horizontal temperature gradient drives a large-scale meridional overturning circulation, embedded in which convection forms, especially at the warmer end.

hydrostatic, profile. The enthalpy of a dry parcel of ideal gas is given by $h^d = c_p T$ and thus,

$$h^* \equiv h^d + gz = c_p T + gz. \quad (18.39)$$

Since an adiabatic profile has $dh^* = 0$ we recover the dry adiabatic lapse rate (section 2.10.2), to wit

$$\Gamma^d = -\left.\frac{dT}{dz}\right|_{ad} = \frac{g}{c_p}. \quad (18.40)$$

The adiabatic lapse rate for a moist parcel is given using similar reasoning, but we need an appropriate expression for the enthalpy. A parcel of total mass M^t composed of dry air, water vapour and liquid water such that $M^t = M^v + M^l$ has an enthalpy given by

$$H = h^d M^d + h^v M^v + h^l M^l \quad (18.41)$$

where h denotes specific enthalpy and the superscripts denote dry air, water vapour and liquid water respectively. Dividing by the total mass, the specific enthalpy of the parcel is thus

$$h = (1 - q^w)h^d + q^v h^v + q^l h^l, \quad (18.42)$$

where

$$q^w = q^v + q^l, \quad q^v = \frac{M^v}{M^t}, \quad q^l = \frac{M^l}{M^t}. \quad (18.43)$$

As in (18.16) the latent heat of evaporation is defined to be $L = h^v - h^l$ so that (18.42) becomes

$$h = (1 - q^w)h^d + q^w h^l + Lq^v. \quad (18.44)$$

The generalized enthalpy is equal to this quantity plus a potential ϕ , and if that potential is equal to gz the quantity is the *moist static energy*,

$$h_m^* = (1 - q^w)h^d + q^w h^l + Lq^v + gz. \quad (18.45)$$

Since $h^d = c_p^d T$ and $h^l = c^l T$ (to a very good approximation and where c^l is the heat capacity of liquid water) we have

$$h_m^* = c_p^{dl} T + Lq^v + gz, \quad (18.46)$$

where $c_p^{dl} = (1 - q^w)c_p^d + q^w c^l$. This quantity varies with liquid water content but in Earth's atmosphere the variation is small and in the derivation below we will take c_p^{dl} to be a constant and denote it c_p . For adiabatic motion, the moist static energy is a constant, whether or not water is evaporating or condensing: the latent heat of evaporation or condensation is merely exchanged with the dry generalized enthalpy.

The saturated adiabatic lapse rate

In a moist atmosphere the moist static energy is conserved as a parcel ascends and $dh_m^*/dz = 0$. Using (18.46), an ascending parcel then has a lapse rate given by

$$c_p \frac{dT}{dz} = -L \frac{dq^v}{dz} - g. \quad (18.47)$$

If the air is moist but not saturated then an ascending parcel will follow the dry adiabatic lapse rate (because $dq^v/dz = 0$) but if it is saturated (and $q^v = q_s$) then as a parcel ascends it will cool

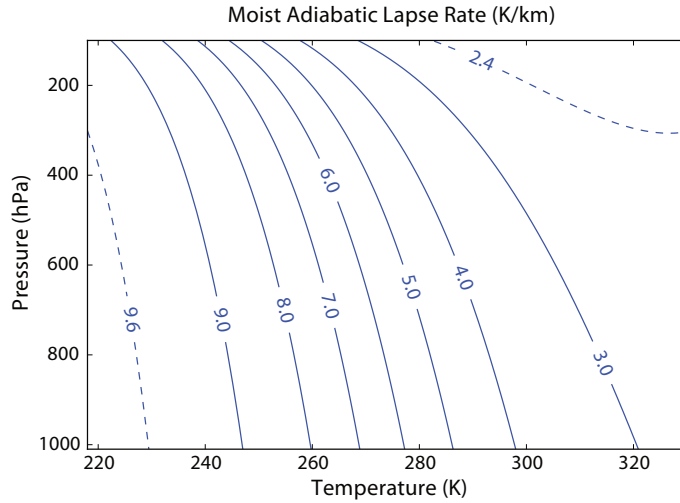


Fig. 18.15 Contours of saturated adiabatic lapse rate (K/km) as a function of pressure and temperature, calculated using (18.50) with $q_s = \epsilon e_s/p$.

and some water vapour will condense. Since $q_s \approx \epsilon e_s/p$, and so is a function of temperature and pressure, we have

$$\frac{dq_s}{dz} = \left(\frac{\partial q_s}{\partial T} \right)_p \frac{dT}{dz} + \left(\frac{\partial q_s}{\partial p} \right)_T \frac{dp}{dz} = \left(\frac{\partial q_s}{\partial T} \right)_p \frac{dT}{dz} + \left(\frac{q_s}{p} \right) \rho g, \quad (18.48)$$

using hydrostasy. Using the Clausius–Clapeyron equation and the ideal gas equation of state gives

$$\frac{dq_s}{dz} = \frac{Lq_s}{R^v T^2} \frac{dT}{dz} + \frac{gq_s}{R^d T}. \quad (18.49)$$

Using (18.47) and (18.49) we obtain an expression for the lapse rate of an adiabatically ascending saturated parcel,

$$\Gamma_s = - \left. \frac{dT}{dz} \right|_{ad} = \frac{g}{c_p} \frac{1 + Lq_s/(R^d T)}{1 + L^2 q_s/(c_p R^v T^2)}. \quad (18.50)$$

The quantity Γ_s is the *saturated adiabatic* lapse rate, plotted in Fig. 18.15. (This quantity is often called the moist adiabatic lapse rate, but that name is better given to the lapse rate of moist air that does not condense, and which differs slightly from the dry adiabatic lapse rate.) The lapse rate is a function of temperature and pressure because $q_s = \epsilon e_s/p$ and e_s is given by the solution of the Clausius–Clapeyron equation, (18.28). Values of Γ_s are typically around 6 K km^{-1} in the lower atmosphere although since dw_s/dT is an increasing function of T , Γ_s decreases with increasing temperature and can be as low as 3 K km^{-1} . The second term in the numerator of (18.50) is usually quite small (around 0.1, although it can become large at very high temperatures) but the second term in the denominator is positive and order unity. Since g/c_p is the dry adiabatic lapse rate, the saturated adiabatic lapse rate is smaller than the dry adiabatic lapse rate, as can be seen directly from (18.47) since $dq^v/dz < 0$.

The saturated adiabatic lapse rate determines the stability of a saturated profile. Using parcel theory, just as in section 2.10, a profile will be stable or unstable depending on whether the lapse rate is less than or greater than (18.50); that is

$$\text{Stability : } - \frac{\partial \bar{T}}{\partial z} < \Gamma_s, \quad \text{Instability : } - \frac{\partial \bar{T}}{\partial z} > \Gamma_s. \quad (18.51a,b)$$

where $\tilde{T}$ is the environmental temperature. The lapse of the atmosphere rarely exceeds the saturated adiabatic lapse rate, as was seen in 15.24. In the tropics and subtropics the lapse is, on average, very close to the saturated adiabat up to about 300 hPa (about 9km), whereas in midlatitudes it is considerably less, and so more stable, because of the upward transport of heat by baroclinic eddies. Convection only directly determines the lapse rate over a small fraction of the tropics where convection actually occurs, but nevertheless the average lapse rate is close to moist neutral. This is because the gravity waves emanating from convective regions adjust the tropics to have weak horizontal temperature gradient, in a process akin to geostrophic adjustment, hence maintaining approximately the same vertical profile even away from regions of active convection, as discussed more in sections 18.8 and 18.9.

18.3.2 Equivalent potential temperature

In a dry atmosphere adiabatic motion is characterized by the material conservation of potential temperature, θ , a surrogate for entropy. Thus in adiabatic flow $D\theta/Dt = 0$, and the dry adiabatic lapse rate can be characterized equivalently either as $dT/dz = -g/c_p$ or $d\theta/dz = 0$. We can construct a similar quantity for moist air. Suppose that a parcel is lifted, and so cooled, until all its moisture condenses, and that all the latent heat released goes into heating the parcel. The *equivalent potential temperature*, θ_{eq} , is the potential temperature that the parcel then achieves.⁷ If the parcel is then moved along a dry adiabat to a reference pressure p_R (commonly 1000 hPa) the actual temperature it will then have is θ_{eq} . As a consequence of the near adiabatic nature of the process, θ_{eq} is an approximate measure of the entropy of the parcel, as we will show (see also Appendix A).

We may obtain an approximate analytic expression for θ_{eq} by noting that the first law of thermodynamics, $dQ = T d\eta$, implies, by definition of potential temperature,

$$-L dq = c_p T d \ln \theta, \quad (18.52)$$

during the condensation process, where dq is the change in water vapour content. Integrating gives, by definition of equivalent potential temperature,

$$-\int_q^0 \frac{L}{c_p T} dq = \int_\theta^{\theta_{eq}} d \ln \theta. \quad (18.53)$$

Here, q is the initial amount of water vapour contained in the parcel and T is the temperature at which condensation occurs. We might imagine lifting a parcel from its initial position to a temperature T at which saturation first occurs. To remove *all* the water vapour the parcel must be lifted to great height, because all of the water vapour will only condense if the final temperature is very low, with condensation occurring continuously and so with varying temperature along the way. But if we assume that temperature is constant during condensation, and that L and c_p are also constants, then (18.53) gives

$$\theta_{eq} = \theta \exp\left(\frac{Lq}{c_p T}\right) = T \left(\frac{p_R}{p}\right)^{R/c_p} \exp\left(\frac{Lq}{c_p T}\right). \quad (18.54)$$

The equivalent potential temperature so defined is approximately conserved during condensation, the approximation arising going from (18.53) to (18.54). It is a useful expression for diagnostic purposes and in constructing theories of convection, but it is not accurate enough to use as a primary prognostic variable in a numerical model that aims to be realistic. In saturated adiabatic flow a parcel will follow a ‘moist isentrope’ (an isoline of equivalent potential temperature) more closely than a dry isentrope.

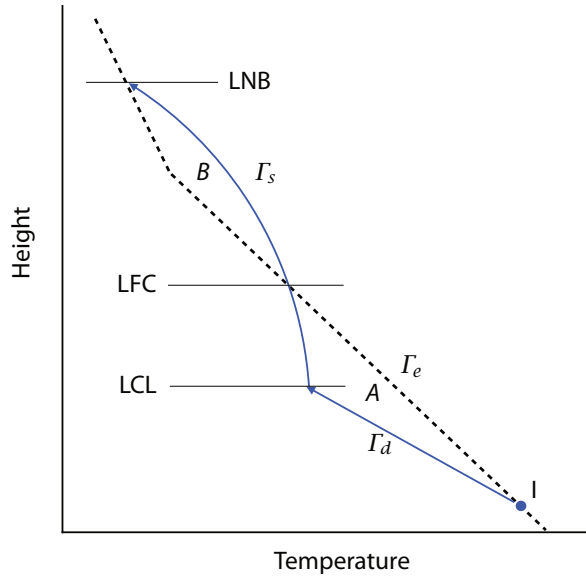


Fig. 18.16 Schematic of a conditional instability in an atmosphere with environmental lapse rate Γ_e (dashed line). A parcel at I is forced to rise, and it does so along the dry adiabat, Γ_d , until it is saturated at the lifting condensation level, LCL. It will then rise along the saturated adiabat, Γ_s , and after reaching the level of free convection (LFC) it is convectively unstable. The parcel continues to rise along the saturated adiabat, without any external forcing, until it reaches the level of neutral buoyancy (LNB). Not to scale.

To see that the equivalent potential temperature is an approximate measure of the entropy of a saturated parcel we begin with the entropy of dry air, which referring to (1.107) is

$$\eta^d = c_p \ln \theta = c_p \ln(T/T_0) - R \ln(p/p_0), \quad (18.55)$$

where T_0 and p_0 are constants. If we add to this a contribution from liquid water (see Appendix A on page 720 for a more complete treatment) then the saturation entropy, η_s , is approximately given by

$$\eta_s \approx \eta^d + \frac{Lq_s}{T} = c_p \ln(T/T_0) - R \ln(p/p_0) + \frac{Lq_s}{T}, \quad (18.56)$$

where q_s is the saturation specific humidity and final liquid water content. If we now define θ_{eq} by

$$c_p \ln(\theta_{eq}/T_0) \equiv \eta_s, \quad (18.57)$$

then using (18.56) and (18.57) we recover (18.54) with $q = q_s$ — meaning that logarithm of the equivalent potential temperature is, approximately, the entropy. To relate this result to the lapse rate, consider a parcel ascending into a colder region and condensing. The saturation moist entropy will stay the same: the last term on the right-hand side of (18.56) will fall, but the ensuing heating causes the temperature to increase. If we differentiate (18.56) with respect to z and set $\partial \eta_s / \partial z = 0$ then, using hydrostasy and a little algebra, we recover the saturated adiabatic lapse rate given by (18.50), which, therefore, has constant θ_{eq} and constant η_s .

18.4 CONVECTION IN A MOIST ATMOSPHERE

Over a wide range of temperatures and pressures the saturated adiabatic lapse rate is considerably less than the dry lapse rate (Fig. 18.15), and thus a moist atmosphere can be stable to dry convection but unstable to moist convection. If an environmental profile is unstable then convection ensues, transporting energy upward, until the atmosphere nearly stabilizes. But if the atmosphere is not saturated it is the dry adiabatic lapse rate is the relevant one (or, strictly, the adiabatic lapse rate of moist, noncondensing air, which is very slightly different). The observed environmental profile in convecting situations may thus be a combination of the dry adiabatic and saturated adiabatic profiles. An unsaturated parcel that is unstable by the dry criterion will rise and cool following a

dry adiabat, Γ_d , until it becomes saturated, above which it will rise following a saturated adiabat, Γ_s . The temperature then does not fall as rapidly as the dry case because the latent heat release warms the parcel as it rises.

These facts give moist convection a particular flavour and lead to the notion of *conditional stability*, whereby a parcel is stable to an infinitesimal perturbation but unstable to a finite perturbation. Consider an environment in which the lapse rate lies between the moist and dry rates, as in Fig. 18.16, and consider a parcel near the surface at position I, as in Fig. 18.16. Suppose the parcel is adiabatically lifted (perhaps mechanically) then its temperature profile follows the dry adiabat until it is saturated at, by definition, the *lifting condensation level* (LCL). However, the parcel is actually negatively buoyant and so would sink unless the parcel is forced to continue rising, but if it does continue to rise it will be along a saturated adiabat and eventually, at the *level of free convection* (LFC) it will become buoyant and convectively unstable. It will continue to rise until its buoyancy no longer exceed that of the environment, at the *level of neutral buoyancy* which may well be at or close to the tropopause where the temperature starts to increase again. (The height of the tropopause is not independent of the convection itself, a matter we discuss in section 18.6.) Evidently, to trigger such an instability a *finite* perturbation is needed (for example a large scale flow over a hill forcing a parcel upwards) and this is known as *conditional instability*. A parcel that is initially sitting at the LCL will only be unstable if it is saturated, which is by no means usually the case: the water has to get there somehow.⁸

Convection in the atmosphere need not be ‘conditional’, in the sense described above, nor is the atmosphere necessarily always in a conditionally unstable state. Indeed, on average the tropical atmosphere is closer to a moist profile than a dry profile. It is also important to remember that convection ultimately arises because the radiative forcing produces a vertical profile that is convectively unstable, with large-scale horizontal temperature gradients confining the convection to warmer regions, as in 18.14.

18.4.1 Energetics of convection

How much energy is available for a parcel in convection? One way is to calculate the work released by, or required for, a parcel as it moves through an environment that has a different density. The upward force per unit mass on a parcel of density ρ_p in an environment with density ρ_e is just the familiar buoyancy force,

$$F_b = g \frac{\rho_e - \rho_p}{\rho_p} = g \frac{\alpha_p - \alpha_e}{\alpha_e}. \quad (18.58)$$

Thus, the energy released as a parcel ascends from z_1 to z_2 is

$$\text{Energy} = \int_{z_1}^{z_2} g \frac{\alpha_p - \alpha_e}{\alpha_e} dz = - \int_{p_1}^{p_2} (\alpha_p - \alpha_e) dp, \quad (18.59)$$

if the environment is in hydrostatic balance. If we take the limits of integration to be the level of free convection and the level on neutral buoyancy, the energy is known as the *convective available potential energy*, or CAPE. Using the ideal gas relation, and assuming that the parcel has the same pressure as the environment, (18.59) becomes

$$\text{CAPE} = -R \int_{\text{LFC}}^{\text{LNB}} (T_p - T_e) \frac{dp}{p}. \quad (18.60)$$

If the height axis in Fig. 18.16 were log pressure then the CAPE would be proportional to the area *B*. If the limits were taken from the initial height to the LFC then the integral would be proportional

to the area A and would be negative, and is known as the ‘convective inhibition,’ or CIN. That is

$$\text{CIN} = -R \int_{p_{\text{bot}}}^{\text{LFC}} (T_p - T_e) \frac{dp}{p}, \quad (18.61)$$

and it is the amount of energy that must be supplied to initiate convection. In some accounts the definition of CAPE includes the CIN, in which case CAPE may be negative (but this is unusual). It is often the case that observed profiles of temperature in the tropics exhibit both CIN and CAPE, and a parcel in the lower atmosphere is stable and the instability is only conditional. That a finite perturbation is needed to initiate convection is a distinguishing feature of moist convection.

Given the initiation of convection, one may envision an unstable parcel of air experiencing a buoyant force and accelerating through the unstable region, gaining kinetic energy, and this energy can be quite significant. Suppose that $T_p - T_e = 1^\circ \text{C}$ and that the limits of integration are 250 hPa and 750 hPa. Then we estimate

$$\text{CAPE} \sim R(T_p - T_e) \frac{\Delta p}{p} = R \times 1 \times \frac{750 - 250}{500} = 286 \text{ J kg}^{-1}. \quad (18.62)$$

This is actually an underestimate, by a factor of a few, of the value of CAPE often found in tropical atmospheres, but nevertheless if we translate it to a vertical velocity using $w = 2\sqrt{\text{CAPE}}$ then we find $w \approx 24 \text{ m s}^{-1}$! In fact, CAPE is a significant overestimate of the kinetic energy that is imparted to a buoyant parcel: a parcel will not keep its identity because of mixing or entrainment with its surroundings and the environmental profile itself becomes altered, and even if the parcel did keep its identity it would give up some of its kinetic energy to the environment in pushing it out of the way. Nevertheless, the estimate gives the sense that atmospheric convection is a vigorous process, occurring on a fast timescale compared to large-scale dynamics. Let us suppose that the above estimate is too big by a factor of 10. The time taken for a parcel to travel half the height of the troposphere in the tropics is then approximately $7500 \text{ m} / 2.4 \text{ m s}^{-1} = 3000 \text{ s}$, or less than one hour. *Thus, the timescales of atmospheric convection are measured in hours, not days.*

18.4.2 Effects of convection

We have seen that, with even small differences in between an environmental profile and a stable profile, convection will be vigorous and act on rather fast timescales. What will the result of that convection be? Parcels will ascend until they are no longer unstable, at which point they can mix irreversibly their environment, warming the environment aloft. (Indeed the parcels will mix as they ascend, but they have more time to mix as the ascent slows.) The convection and the mixing will proceed until the environmental profile is no longer unstable, a process that, if it occurs on a short timescale, is called *convective adjustment* or, if on a longer timescale, *convective relaxation*. One might suppose that the profile to which the environment will adjust will be the saturated adiabatic or dry adiabatic lapse rates, depending on whether the atmosphere is saturated or not, although this is something of an oversimplification.⁹ The convection will also transfer moisture vertically, and if this transfer causes the profile to become saturated then condensation and precipitation may occur.

Let us denote the adjusted reference profiles of temperature and specific humidity as $T_r(z)$ and $q_r(z)$ (or equivalents in pressure). If the vertical structure of these profiles is given, the magnitude of the profile will be determined by the fact that convection will conserve enthalpy. If, for example, atmosphere is far from saturated then the reference profiles will satisfy

$$\int_{p_B}^{p_T} c_p (T_r - T_i) dp = 0 \quad \text{and} \quad \int_{p_B}^{p_T} (q_r - q_i) dp = 0, \quad (18.63)$$

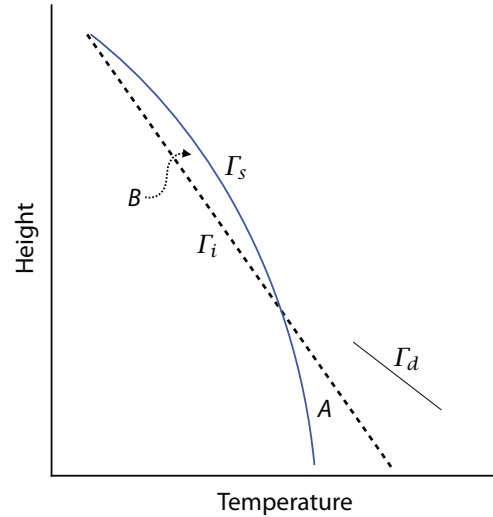


Fig. 18.17 Schematic of convective adjustment in a saturated atmosphere. The initial environment, with profile Γ_i , is unstable to moist convection and adjusts to the saturated adiabatic lapse rate Γ_s , with the dry adiabat Γ_d shown for reference. The average temperature of the final profile is such that the enthalpy is conserved, which provides a relationship between the areas A and B.

where the subscript i denotes the initial environmental profile and p_B and p_T denote the levels at the base and top of the convection. If precipitation does occur then only the total enthalpy is conserved and we have

$$\int_{p_B}^{p_T} (h_r - h_i) dp = 0. \quad (18.64)$$

where $h = c_p T + Lq$ is the (moist) specific enthalpy. One then needs a second constraint to determine the profiles of T_r and q_r separately.

The specification of these profiles, and the levels of the base and the top of the convection, is not an exact science since the notion of adjustment is an approximation. As a first estimate in a moist atmosphere one might suppose that T_r is the dry adiabat below saturation and the saturated adiabatic lapse rate above, and that the reference profile of q_r is such that the atmosphere remains saturated, or nearly so, after adjustment, and that p_B and p_T are the levels of free convection and of neutral buoyancy, respectively. Application of (18.64) then uniquely determines the final profiles of temperature and humidity, as schematically illustrated for a saturated atmosphere in Fig. 18.17. Details apart, two robust, key points should be emphasized.

- (i) To a first approximation, convection determines the *profiles* of temperature and humidity rather than the actual heating rate or the precipitation. The heating and precipitation released by convection is a function of the environmental profile that in turn is largely determined by the larger-scale processes, although convection feeds back onto this profile.
- (ii) After convection has occurred the local profile is, by construction, stable. However, this profile is not necessarily in equilibrium with the large-scale motion, or with the radiative processes that initially gave rise to the unstable profile. This disequilibrium may cause the profile to evolve further, possibly giving rise to more convection.

The above state of affairs — a putative initial profile set up by radiation and/or large-scale dynamics, modified quickly by convection, leading to a further slow evolution and convection — is called *quasi-equilibrium*.

18.4.3† Quasi-equilibrium

Convective quasi-equilibrium is a posited state in which the forcing of a convectively unstable profile by large-scale dynamics and/or radiation is statistically balanced by convection.¹⁰ The large-scale forcing may change, and the system may evolve, but in quasi-equilibrium the convection is

Convection, Quasi-Equilibrium and Radiative-Convective Equilibrium

- Atmospheric convection ultimately arises because the radiative forcing tends to produce temperature profiles, and hence buoyancy profiles, that are statically unstable. Large-scale advection, moving cold air over warm air, also leads to convection.
- Convection is most prevalent in the tropics because it is here that the surface is warmest, and the radiative-equilibrium lapse rate would be most unstable (section 18.5). This would be true even in the absence of moisture, but moisture is most important in the tropics because it is warmest, and because water is imported via the Hadley Cell.
- The presence of moisture may cause the critical lapse rate for convection to be lower than that for dry air, because of the release of heat when water vapour condenses. The effect only arises if a parcel is saturated. Thus, a column of air may be *conditionally unstable* if it requires a finite perturbation to raise a parcel to a level where it saturates (Fig. 18.16).
- Atmospheric convection tends to occur on much faster timescales than those associated with the large-scale circulation. Once convection occurs the lapse rate is closely constrained to the critical lapse rate in the regions of convection.
- In a process analogous to geostrophic adjustment, internal waves propagating away from convection maintain weak horizontal temperature gradients and ensure that the lapse rate does not deviate too far from saturated adiabatic anywhere in the tropics (Fig. 15.24). In midlatitudes, in contrast, energy is also transported upward by large-scale baroclinic eddies and the lapse rate is lower (more stable).
- When the large-scale dynamical or radiative forcing changes, the convective flux changes to constrain the lapse rate, and the production of CAPE by the large scale is *approximately* balanced by the relaxation of CAPE by convection. This state of affairs is called quasi-equilibrium, and its occurrence is really a hypothesis that appears to be fairly well-satisfied, if the larger timescales are sufficiently long, in Earth's tropical atmosphere.
- Quasi-equilibrium, if and where closely satisfied, leads to considerable simplifications regarding the dynamics of large-scale flow for it de-emphasizes the need to explicitly calculate vertical energy fluxes because the lapse rate is constrained. Equations resembling the shallow-water equations, with a small number of vertical modes, then become a decent approximation for large-scale baroclinic flow, with diabatic effects forcing the first baroclinic mode.
- A radiative-convective equilibrium state arises when the radiative-equilibrium state is convectively unstable state. Convection then balances the radiative forcing, producing at first approximation a convectively active state up to some finite height, with a radiative equilibrium state beyond (Fig. 18.18).
- The height to which the constrained lapse rate extends is determined by the need to maintain an overall radiative balance — the incoming solar radiation must equal the outgoing infrared radiation — and this is the leading-order determinant of the tropopause height (section 18.6.2). A similar calculation could be applied in mid-latitudes, with a lapse rate determined as much or more by baroclinic instability as by convection.
- Quasi-equilibrium is a useful idealization when convection occurs quickly and efficiently, but it does not mean that the large scale always controls the convection. There are many situations when there is no causal separation between convection and the large scale — indeed the convection may create its own large scale flow, the MJO being one example.

assumed to occur on a faster timescale than the other processes and the fluid quickly enters into a statistical equilibrium state. Even as the large-scale evolves, the vertical profile at each step is largely determined by convection. That such a state exists is an assumption, but in Earth's tropical atmosphere it seems a fairly good one when dealing with timescales of longer than a few days, because convective timescales are relatively short and so convection is able to significantly alter the stratification. The assumption of quasi-equilibrium is marginally satisfied over the diurnal cycle.

The most obvious consequence of quasi-equilibrium is that the temperature profile is constrained to be close to being neutrally stable, which generally means a saturated-adiabatic profile except possibly in some regions in the lower atmosphere where the appropriate profile may dry adiabatic, and in the boundary layer where properties are well mixed. A second important consequence concerns CAPE, convective available potential energy. Quasi-equilibrium implies that the CAPE is released by convection at about the same rate as it is generated by the forcing. Thus, at the same time as radiative forcing may be creating an unstable profile with a finite amount of CAPE, convection is destroying that potential energy, largely by converting it to kinetic energy where it is dissipated. The amount of CAPE in a profile may (and does) vary on slow timescales, the variation being produced by the weak imbalance between production and destruction; observations indicate it can vary by a factor of a few over the course of many days, and the environmental profile may also depart from the saturated adiabat away from convectively active regions. However, it turns out that the *average* stratification over the tropics will not deviate too much from moist neutral because internal waves extend the moist-neutral profile in the convective regions to the cloud-free regions, and horizontal pressure and temperature gradients are constrained to be relatively small. This 'weak temperature gradient' effect is described in section 18.8.

Finally, albeit a little less tangibly, quasi-equilibrium gives rise to a *point of view* of tropical dynamics, that convection does not act as a primary determinant of the heat source for the large-scale circulation. Rather, convection controls the temperature and water vapour profiles, with the heating associated with the convection being determined by the requirement that such a profile be maintained, and thus ultimately arising from the larger scale forcing. (This process is analogous to the production of condensation in the fast- condensation models of section 18.2, in which the condensation criteria determine the profile of water vapour but not the amount of condensation.) The resulting environmental profile then lies on the edge of stability, with a non-zero CAPE because of the finite time-scale and finite efficiency of convection and with the intensity of convection being just what is required to maintain that profile. The value of CAPE then evolves slowly, compared to convective timescales, if and as external factors change.

Having said this, quasi-equilibrium should not be thought of as being exact or even especially profound — it is a way to gain an understanding of the structure of the tropics but is not a universal recipe. There are many situations — the diurnal cycle, aspects of boundary layer development, weak convection — in which the quasi-equilibrium assumptions do not hold. Furthermore, the recipe may mislead if misapplied: even when quasi-equilibrium holds in so far as the buoyancy profile is constrained to be moist neutral, it should not be thought that convection and the associated condensation is always a quasi-passive process. That is, the scale separation between convective events and the large-scale does not mean that the large-scale always 'controls' the convection and the convection merely 'feeds back' on the large scale. For example, a localised source of convection near the equator may itself create a large-scale flow pattern similar to that of the Matsuno–Gill problem (section 8.5), and this effect may be at the heart of the Madden–Julian oscillation (section 18.10). Here one might say the convection controls the large-scale and the large-scale feeds back on the convection! Also, in the moist model of the Hadley Cell (section 14.2.7) the release of latent heat in the upward branch serves to change the horizontal distribution of heating and greatly intensify the overturning circulation. The convection should not be thought of as the primary driver of the Hadley Cell, but it has a lowest-order effect.

Caveats aside, quasi-equilibrium is a very useful concept and in following sections we look at two particularly important consequences. First (albeit after an introduction to radiative equilib-

rium) we discuss radiative-convective equilibrium, and we then use quasi-equilibrium to simplify the equations of motion for the larger-scale circulation.

18.5 RADIATIVE EQUILIBRIUM

In order to understand the effects of convection we must first determine what the profile of temperature would be in its absence — the *radiative-equilibrium* state. The electromagnetic radiation in the Earth's atmosphere may usefully be divided into two types, solar (or shortwave) radiation and infra-red (or longwave) radiation. We will assume the atmosphere is semi-grey (i.e., grey in the infra-red) and, in some instances, transparent to solar radiation. Neither assumption is quantitatively good, but they capture the essence. In Appendix B (page 724) we show that the upward, U , and downward, D , streams of longwave radiation then satisfy the radiative-transfer equations,

$$\frac{dD_L}{d\tau} = B - D_L, \quad \frac{dU_L}{d\tau} = U_L - B. \quad (18.65a,b)$$

Here, $B = \sigma T^4$ is the radiative flux emitted by a black body, τ is the optical depth and σ is Stefan's constant. The optical depth is related to the geometric height z by $d\tau = -e_L dz$ where e_L is the longwave emissivity. The net flux of longwave radiation is $N_L = U_L - D_L$ and the longwave heating is proportional to the net flux divergence, $-\partial N_L / \partial z$.

18.5.1 Solutions

Formal solution to radiative transfer equations

If the temperature profile is known then B is known and (18.65) is a pair of first order differential equations in the two unknowns U and D . The solution requires two boundary conditions and in atmospheric problems these might be provided at the top of the atmosphere, for example by requiring that the downward infrared radiation be zero (i.e., $D_L(\tau = 0)$), and by specifying the upward radiation (i.e., let $U_L(\tau = 0) = U_0$ where U_0 is given). Alternatively, the upward radiation at the bottom of the atmosphere could be specified if the ground temperature were known. The solution in physical space is found by specifying the form of $\tau(z)$; that is, by specifying the emissivity. To obtain the solution we multiply (18.65) by the integrating factors $\exp(\tau)$ and $\exp(-\tau)$ to give

$$\frac{d}{d\tau}(D_L e^\tau) = B e^\tau, \quad \frac{d}{d\tau}(U_L e^{-\tau}) = -B e^{-\tau}. \quad (18.66a,b)$$

Integrating between 0 and τ' straightforwardly gives

$$D_L(\tau') = e^{-\tau'} \left[D_L(0) - \int_0^{\tau'} B(\tau) e^\tau d\tau \right], \quad U_L(0) = U_L(\tau') e^{-\tau'} + \int_0^{\tau'} B(\tau) e^{-\tau} d\tau. \quad (18.67a,b)$$

There are other ways to write the solutions that may be appropriate depending on the boundary conditions, but in any case the solutions are in general *nonlocal*, for they depend on the temperature along the path. The terms in (18.67) represent the attenuation of radiation as it travels along its path as well as the cumulative emission. However, in some important special cases we can get a local solution, as we now see.

Radiative equilibrium solution

A *radiative equilibrium* state has, by definition, no radiative heating. If the atmosphere is transparent to solar radiation then the condition implies that the vertical divergence of the longwave radiation is zero:

$$\frac{\partial(U_L - D_L)}{\partial z} = 0 \quad \text{implying} \quad \frac{\partial(U_L - D_L)}{\partial \tau} = 0. \quad (18.68a,b)$$

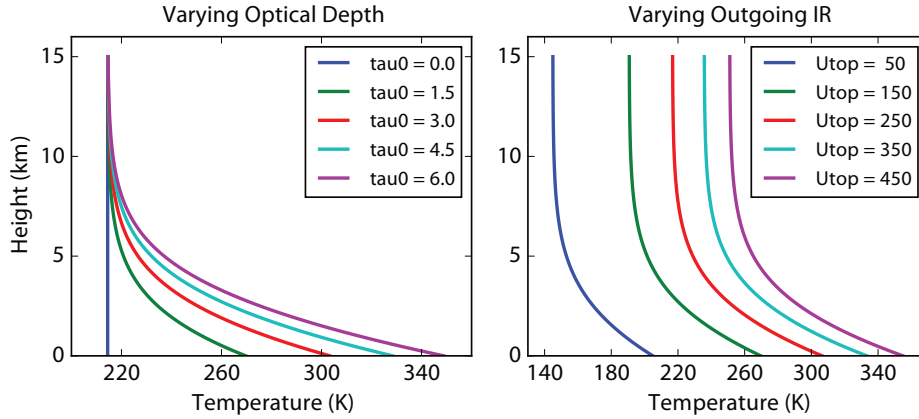


Fig. 18.18 Radiative equilibrium temperature calculated using (18.73), with $H_a = 2$ km. Left: incoming solar radiation is 240 W m^{-2} and surface optical depth varies from 0 to 6. Right: the optical depth is 3.0 and outgoing IR radiation varies from 50 to 450 W m^{-2} .

This condition is normally *not* satisfied in the atmosphere because the air is in motion. If it were satisfied then (18.65) and (18.68b) form three equations in three unknowns, U_L , D_L , and B , and a solution can be found as follows.

Consider an atmosphere with net incoming solar radiation S_{net} and suppose the planet is in radiative equilibrium with the incoming solar radiation balanced by outgoing infra-red radiation. That is, $U_{Lt} \equiv U_L(\tau = 0) = S_{net}$ where U_{Lt} is the net outgoing longwave radiation (OLR) at the top of the atmosphere. The downward infrared radiation at the top of the atmosphere is zero, so that the boundary conditions on the radiative transfer equations at the top of the atmosphere are

$$D_L = 0, \quad U_L = U_{Lt} \quad \text{at} \quad \tau = 0. \quad (18.69)$$

To obtain a solution we rewrite (18.65) as

$$\frac{\partial}{\partial \tau}(U_L - D_L) = U_L + D_L - 2B, \quad \frac{\partial}{\partial \tau}(U_L + D_L) = U_L - D_L. \quad (18.70a,b)$$

and a little algebra reveals that a solution of these equations that satisfies (18.68b) is

$$D_L = \frac{\tau}{2} U_{Lt}, \quad U_L = \left(1 + \frac{\tau}{2}\right) U_{Lt}, \quad B = \left(\frac{1 + \tau}{2}\right) U_{Lt}, \quad (18.71a,b,c)$$

as can be easily verified by substitution back into the equations.

It remains to explicitly relate τ to z , and one approximate recipe is to suppose that τ has an exponential profile,

$$\tau(z) = \tau_0 \exp(-z/H_a), \quad (18.72)$$

where τ_0 is the optical depth at $z = 0$ and H_a is the scale height of the absorber. In the Earth's atmosphere the optical depth is determined by the concentrations of water vapour (primarily) and carbon dioxide (secondarily) and τ_0 (the scaled optical depth) typically varies between 2 and 4, depending on the water vapour content of the atmosphere, and $H_a \approx 2$ km, this being a typical scale height for water vapour. From (18.71c) the temperature then varies as

$$T^4 = U_{Lt} \left(\frac{1 + \tau_0 e^{-z/H_a}}{2\sigma} \right), \quad (18.73)$$

which is illustrated in Fig. 18.18. The following aspects of the solution deserve mention.

- (i) Temperature increases rapidly with height near the ground.
- (ii) The upper atmosphere, where τ is small, is nearly isothermal.
- (iii) The temperature at the top of the atmosphere, T_t is given by

$$\sigma T_t^4 = \frac{U_{Lt}}{2} \quad (18.74)$$

Thus, if we define the emitting temperature, T_e , to be such that $\sigma T_e^4 = U_{Lt}$, then $T_t = T_e/2^{1/4} < T_e$; that is, the temperature at the top of the atmosphere is *lower* than the emitting temperature.

- (iv) Related to the previous point, $B_t/U_{Lt} = 1/2$. That is, the upwards long wave flux at the top of the atmosphere is twice that which would arise if there were a black surface at a temperature T_t . The reason is that there is radiation coming from all heights in the atmosphere.

Interestingly, in obtaining a solution we have not imposed a boundary condition at the ground — in fact there is no ground at all in this problem! What happens if we add one? That is, suppose that we declare that there is a black surface at some height, say $z = 0$, and we require that the atmosphere remain in radiative equilibrium with the same temperature profile. What temperature does the ground take? From (18.71) the upward irradiance and temperature at any height z are related by

$$U_L(z) = \left(\frac{2 + \tau(z)}{1 + \tau(z)} \right) \sigma T^4(z). \quad (18.75)$$

At $z = 0$ the ground will have to supply upwards radiation equal to that given by (18.75), and therefore its temperature, T_g is given by

$$\sigma T_g^4 = \left(\frac{2 + \tau_0}{1 + \tau_0} \right) \sigma T_s^4, \quad (18.76)$$

where T_s is the temperature of the fluid adjacent to the ground (the ‘surface temperature’). That is, $T_g > T_s$ and there is a temperature discontinuity at the ground, especially if optical depth is small. In fact, in very still and clear conditions a very rapid change of temperature near the ground can sometimes be observed, but usually the presence of conduction and convection, as we discuss below, ensure that T_g and T_s are equal.

We note that in the limit in which $\tau = 0$ in the upper atmosphere we have

$$D_L = 0, \quad U_L = U_{Lt}, \quad B = \frac{U_{Lt}}{2}. \quad (18.77)$$

That is, the upper atmosphere is isothermal, there is no downwelling irradiance and the upward flux is constant. The upper-atmosphere temperature, T_{ua} say, and the emitting temperature are related by $T_{ua} = T_e/2^{1/4}$.

The above results imply that at a given optical depth, the temperature difference between the surface and the ground increases with temperature. From (18.73) we have that

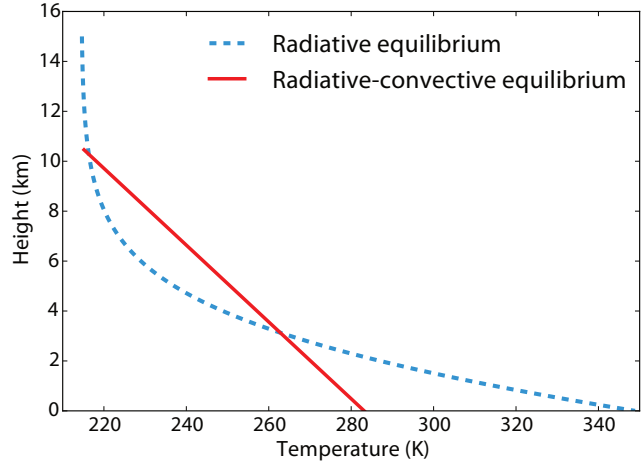
$$T_t^4 = \frac{U_{Lt}}{2\sigma}, \quad T_s^4 = \frac{U_{Lt}(1 + \tau_0)}{2\sigma} \quad (18.78)$$

whence

$$T_s^4 - T_t^4 = T_t^4 \tau_0 \quad \text{or} \quad \Delta T \approx \frac{\tau_0}{4} T_t, \quad (18.79)$$

where $\Delta T = T_s - T_t$ and we assume $\Delta T \ll T_s, T_t$. Thus, in the absence of other effects, higher temperatures lead to larger average lapse rates, and so are potentially more conducive to convection.

Fig. 18.19 Radiative and radiative-convective equilibrium profiles. The initial radiative equilibrium temperature adjusts to a specified profile (in this instance a constant lapse rate) that extends to a finite height, beyond which radiative equilibrium holds. This height (the tropopause) is determined by the requirement of radiative balance, and the overall adjustment process is not adiabatic. The curves are results of numerical calculations with $\tau_0 = 6$, $\Gamma = 6.5 \text{ K km}^{-1}$ and outgoing radiation of 240 W m^{-2} (see section 18.6.2).



18.6 RADIATIVE-CONVECTIVE EQUILIBRIUM

In the radiative equilibrium solution the temperature gradient near the ground varies so rapidly that $-\partial T/\partial z$ may exceed even the dry adiabatic lapse rate. If so, the radiative equilibrium solution is convectively unstable and convection will be triggered and energy (and moisture) will be redistributed throughout the column. What is the resulting temperature profile?

18.6.1 ♦ The general case

We first approach the problem without making any simplifying quasi-equilibrium assumptions, supposing only that we have a balance between longwave and solar radiation and the effects of convection. Let us assume that upward and downward shortwave fluxes, U_S and D_S , obey the two-stream radiative equations with no thermal emission and no scattering, namely

$$\frac{dD_S}{d\tau_S} = -D_S, \quad \frac{dU_S}{d\tau_S} = U_S. \quad (18.80a,b)$$

Here, τ_S is the shortwave optical depth, which is related to the physical height by $d\tau_S = -e_S dz$, where e_S is the shortwave emissivity. and we can then write the radiative equations as

$$\frac{dD_S}{d\tau_L} = -\frac{e_S}{e_L} D_S, \quad \frac{dU_S}{d\tau} = \frac{e_S}{e_L} U_S, \quad (18.81a,b)$$

The solar heating is proportional to the convergence of the net solar fluxes, $-\partial N_S/\partial z$ where $N_S = U_S - D_S$. The convection also provides a local heating, and this is proportional to the convergence of the upwards enthalpy flux, $-\partial \mathcal{H}/\partial z$. Thus, in a steady state,

$$\frac{\partial}{\partial z}(N_S + N_L + \mathcal{H}) = 0 \quad \text{implying} \quad \frac{\partial}{\partial \tau}(N_S + N_L + \mathcal{H}) = 0. \quad (18.82a,b)$$

Using (18.70a) we write (18.82b) as

$$U_L + D_L - 2B + \frac{1}{e_L} Q_S + \frac{\partial \mathcal{H}}{\partial \tau} = 0, \quad (18.83)$$

where Q_S is the solar heating given by $Q_S = -\partial N_S/\partial z = e_L^{-1} \partial N_S/\partial \tau$. If we had a theory for the enthalpy fluxes $\mathcal{H}$ then (18.83), in conjunction with the Schwarzschild equations and knowledge of the solar radiation and atmospheric emissivity, would determine the temperature profile with height. We have no such theory, but we can still make progress.

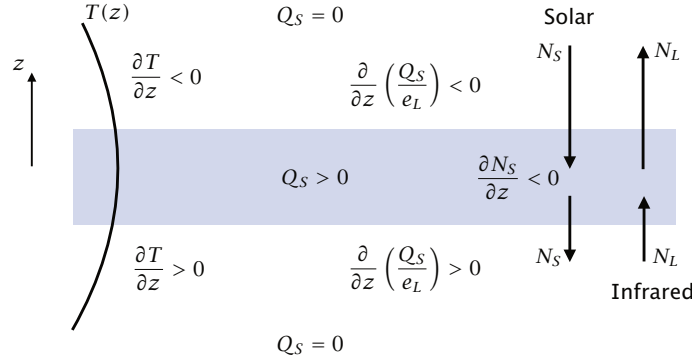


Fig. 18.20 Effects of a band of solar radiation absorption. Shading marks the absorbing region where $\partial N_s/\partial z < 0$, with consequences for the heating, Q_s , and temperature profile, $T(z)$, determined by (18.86b), as indicated.

Effects of solar fluxes

Suppose first (and taking a diversion from convection) that the enthalpy fluxes are zero, much as in the stratosphere. We then have a radiative equilibrium profile then satisfies

$$\frac{\partial}{\partial z}(N_L + N_s) = 0, \quad (18.84)$$

implying

$$\frac{\partial N_L}{\partial z} = Q_s \quad \text{or} \quad \frac{\partial N_L}{\partial \tau} = -\frac{Q_s}{e_L}. \quad (18.85a,b)$$

If we differentiate (18.85b) with respect to τ and use (18.70a,b) we obtain

$$2\frac{\partial B}{\partial \tau} = N_L + \frac{\partial}{\partial \tau}\left(\frac{Q_s}{e_L}\right) \quad \text{or} \quad 8\sigma T^3 \frac{\partial T}{\partial z} = -e_L N_L + \frac{\partial}{\partial z}\left(\frac{Q_s}{e_L}\right). \quad (18.86a,b)$$

The middle of the stratosphere in Earth's atmosphere contains a layer of ozone that absorbs solar radiation. Thus, $\partial N_s/\partial z < 0$ and Q_s is positive. The net longwave radiation N_L is positive in this region, as there is much more upwelling radiation than downwelling, and this tends to produce a negative vertical temperature gradient, but the effect is weak because e_L is small. However, the solar radiation can have a strong effect because e_L appears in the *denominator* of the term involving Q_s . Just below the heating region $\partial Q_s/\partial z$ is positive and this produces a positive value of $\partial T/\partial z$, with the converse above the ozone layer. Thus, region of solar heating corresponds to a maximum of temperature, or at least a region of small vertical temperature gradient, with negative values of $\partial^2 T/\partial z^2$, as schematically illustrated in Fig. 18.20. The result is intuitively reasonable, but (18.86) provides the explicit solution. A non-intuitive result that follows from (18.86b) concerns the effects of an increasing infra-red emissivity, such as happens with global warming. Suppose that $\partial T/\partial z > 0$, as in the stratosphere because of the presence of a layer of ozone, and suppose further that e_L then increases, because of an increase concentration of greenhouse gases. Changes in the longwave term are small, because e_L is small, but changes in the solar term are large, and these cause $\partial T/\partial z$ to diminish. That is, an increase in the concentration of greenhouse gases will cause stratospheric temperatures to *fall*.¹¹ The result arises because of the balance, in this case, between solar heating and infrared cooling, as in (18.85a) and Fig. 18.20. If the emissivity increases there will still be the same infrared loss to space, to achieve radiative balance, and if the emissivity is higher the loss can be achieved at a lower temperature.

Effects of enthalpy fluxes

Now consider the effects of an enthalpy flux, supposing that the atmosphere is transparent to solar radiation. The equilibrium condition, (18.83), becomes

$$U_L + D_L - 2B + \frac{\partial \mathcal{H}}{\partial \tau} = 0. \quad (18.87)$$

Differentiating with respect to τ and using (18.70b) gives

$$2 \frac{\partial B}{\partial \tau} = N_L + \frac{\partial^2 \mathcal{H}}{\partial \tau^2}. \quad (18.88)$$

Now, from (18.82a) we must have $N_S + N_L + \mathcal{H} = 0$ for all z , if the atmosphere is in an overall radiative balance and if $\mathcal{H} = 0$ at the atmosphere's top. If the solar flux is constant with height then, also for all z ,

$$N_L = S_{net} - \mathcal{H}, \quad (18.89)$$

where S_{net} is the net incoming solar radiation, a positive quantity. Using this expression (18.88) becomes

$$2 \frac{\partial B}{\partial \tau} = S_{net} - \mathcal{H} + \frac{\partial^2 \mathcal{H}}{\partial \tau^2} \quad \text{or} \quad \frac{8\sigma T^3}{e_L} \frac{\partial T}{\partial z} = -S_{net} + \mathcal{H} - \frac{1}{e_L} \frac{\partial}{\partial z} \left(\frac{1}{e_L} \frac{\partial \mathcal{H}}{\partial z} \right). \quad (18.90a,b)$$

These expression explicitly tell us how the lapse rate is affected by enthalpy fluxes. If we knew $\mathcal{H}$ we could integrate (18.90) to give us the temperature at every level, and if $\mathcal{H} = 0$ the equations are equivalent to the derivative of (18.71c). Enthalpy fluxes are usually positive (i.e., upwards) in the atmosphere and so they tend increase $\partial T/\partial z$ and reduce the lapse rate; that is, they tend to warm the upper atmosphere and cool the lower atmosphere and the surface. Since we do not have a quantitative theory for these fluxes let us invoke some ideas of quasi-equilibrium and specify the resulting lapse rate rather than the fluxes themselves.

18.6.2 Convective adjustment and the height of the tropopause

Let us assume that convection occurs with sufficient efficiency so that it will establish a convectively neutral lapse rate, Γ say, which for simplicity we here assume is constant. We may also assume that the convection equalizes the ground temperature and the surface temperature (the temperature layer of air immediately above the ground). Such a model simplifies the problem enormously, for we avoid completely the problem of solving the radiative transfer equations. However, radiation is still present, and we must still satisfy an overall radiative balance, and this determines the height to which the convection extends — which is, effectively, the height of the tropopause. In the first instance we can suppose that the convection occurs quickly and adiabatically, so that the re-arrangement of the temperature profile conserves energy. However, such a profile will not necessarily be a solution of the radiative transfer equations (18.67) and the profile must adjust until a solution is found. If the lapse rate is fixed, the only degree of freedom is the height to which convection reaches — that is, the height of the tropopause — above which radiative equilibrium holds.

The computation of this height follows a straightforward algorithm. If the top of the convecting layer occurs at a height, H_T , at which the optical depth is small, then outgoing long-wave radiation there will be approximately equal to the outgoing radiation at the top of the atmosphere, $U_L(\tau = 0)$, which is equal to the incoming solar radiation and therefore known. This gives the tropopause temperature (from (18.76), namely $T_T^4 = U_L(\tau = 0)/2\sigma$) and therefore, if we know H_T and the lapse rate, the temperature at all heights. We can then calculate the upwards radiative flux using a variant of (18.67b), which we may write as

$$U_L(z = 0) = U_L(z = H_T) e^{\tau_s} + \int_0^{H_T} B(\tau) e^{\tau_s - \tau} \frac{d\tau}{dz} dz, \quad (18.91)$$

where τ_s is the optical depth at the surface. However, for an arbitrary tropopause height the upwelling radiation at the bottom, $U_L(z = 0)$, as given by (18.91), will *not* equal σT_s^4 , which it must do if radiative balance is to be satisfied. The height of the tropopause must therefore adjust, and an iterative algorithm for finding the equilibrium solution goes as follows.

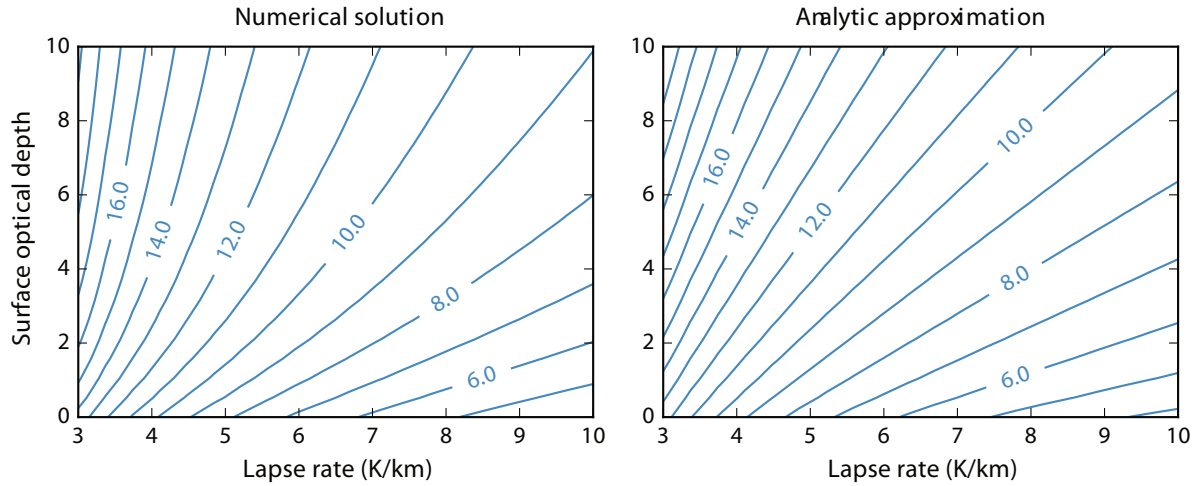


Fig. 18.21 Contours of tropopause height (km) as a function of lapse rate and surface optical depth in a grey atmosphere, calculated numerically by iterating the Schwarzschild equations (left) and using an analytic approximation (right), with $H_a = 2\text{ km}$ and outgoing longwave radiation of 242 W m^{-2} , or tropopause temperature of 215 K .

- (i) Obtain the radiative equilibrium temperature profile.
- (ii) Make a guess for the height of the tropopause, and using the given lapse rate obtain the temperature all the way down to the ground.
- (iii) Integrate the radiative transfer equations, (18.91), down from the top. The outgoing radiative balance is achieved this way but there is no balance at the surface if temperature is continuous. That is, $\sigma T_s^4 \neq U_L(z=0)$.
- (iv) Change the height of the tropopause, find another solution, and iterate until the surface radiative balance is properly achieved.

A profile of temperature so calculated, along with the radiative-equilibrium temperature, are shown in Fig. 18.19, and solutions as a function of lapse rate and temperature are given in Fig. 18.21.

18.6.3 Approximate analytic solution

In Appendix C (page 725) we show that an approximate analytic solution of the above algorithm for the height of the tropopause, H_T , is

$$8\Gamma H_T^2 - C H_T T_T - \tau_s H_a T_T = 0. \quad (18.92)$$

or

$$H_T = \frac{1}{16\Gamma} \left(C T_T + \sqrt{C^2 T_T^2 + 32\Gamma \tau_s H_a T_T} \right), \quad (18.93)$$

where $C = \log 4 \approx 1.4$, Γ is the lapse rate, T_T is the temperature at the tropopause, τ_s is the surface optical depth and H_a is the scale height of the main infra-red absorber. For Earth's atmosphere, $H_a \approx 2\text{ km}$, $\tau_s \approx 5$ and $\Gamma \approx 6.5\text{ K km}^{-1}$. All three terms in (18.92) are then approximately the same size and (18.93) gives $H_T \approx 11\text{ km}$. We can now be a little more precise about what it means for an atmosphere to be optically thin or thick. Using (18.93) and approximating $C^2 = 2$ we see that the optically thick limit arises when

$$\tau_s H_a \gg \frac{T_T}{16\Gamma} \quad \text{whence} \quad H_T \approx \sqrt{\frac{T_T \tau_s H_a}{8\Gamma}}. \quad (18.94)$$

The optically thin case has

$$\tau_s H_a \ll \frac{T_T}{16\Gamma} \quad \text{whence} \quad H_T \approx \frac{CT_T}{8\Gamma}. \quad (18.95)$$

With parameters appropriate for Earth's atmosphere both of the above limits give estimates in the range 6–15 km, and they are additive effects. Plots of the tropopause height as a function of lapse rate and optical depth, calculated using (18.93), as well as numerical solutions using the algorithm of the previous subsection, are given in Fig. 18.21, and the agreement is fairly good in Earth's parameter regime. A robust result is that as the lapse rate diminishes ($-\partial T/\partial z$ becomes smaller) the tropopause height increases, essentially to maintain the same tropopause temperature.

The main quantitative deficiency of this argument is that the atmosphere is not grey — the absorption of radiation is a function of wavelength. A less severe approximation is to suppose that the infra-red radiation occurs in two bands — a 'window' region and the remainder. In the window region the atmosphere is fairly transparent, meaning that a fraction ($\sim 1/4$ or $1/3$) of the radiation emitted by the surface goes straight to space. This has the effect of slightly decoupling the tropopause temperature from the effective emitting temperature. Nonetheless, the general arguments leading to (18.93) remain valid and the way in which the tropopause height varies with optical depth and stratification will carry through in the more realistic case.¹²

Effects of lateral energy transport and application to mid-latitudes.

Although we have set the above calculation in the context of convection, it applies in any situation where the lapse rate can be specified; the argument would equally well apply if the lapse rate were set by baroclinic instability. We can also apply a modified argument even when there is a horizontal transport of energy, for such a transport effectively just acts to change the radiative emitting temperature. If the lateral convergence of energy and the incoming solar radiation are known then the amount of radiation that the column must emit to space in order to maintain energy balance is easily calculated. For example, if there is a convergence of energy at high latitudes because of the transport of energy by baroclinic eddies, then the column needs to emit less infra-red radiation to space that it would otherwise in order to maintain radiative equilibrium. In actuality, in mid-latitudes the lapse rate cannot be regarded as being specified independently of the lateral heat transport, and the determination of the lapse rate and the tropopause height become intertwined. The reader is referred back to section 15.5 to follow this argument.

18.7 VERTICALLY-CONSTRAINED EQUATIONS OF MOTION FOR LARGE SCALES

Let us now segue toward the large scale dynamics of the tropics. We first show how having a nearly constant lapse rate constrains the vertical degrees of freedom, so reducing the equations of motion to something akin to the shallow water equations, and the readers may wish to skim 3.4 before continuing.

18.7.1 Reduction of vertical degrees of freedom

The assumption that convection maintains a moist-adiabatic profile everywhere constrains the vertical structure of the horizontal temperature gradient and, at least in so far as the momentum dynamics are linear, reduces the equations of motion governing the large-scale to a set similar to the shallow-water equations.¹³ To see this we begin with hydrostasy in pressure coordinates applied to the fluctuating fields,

$$\frac{\partial \phi'}{\partial p} = -\alpha', \quad (18.96)$$

where ϕ' and α' are variations in the geopotential and the specific volume. The latter be taken to be a function of the state variables pressure and saturation entropy, η (dropping the subscript s). Variations in α at constant pressure thus obey

$$\alpha' = \left(\frac{\partial \alpha}{\partial \eta} \right)_p \eta' = \left(\frac{\partial T}{\partial p} \right)_\eta \eta', \quad (18.97)$$

using one of Maxwell's equations (page 19). If we integrate (18.96) in the vertical and use (18.97) we obtain

$$\phi'(x, y, p, t) = - \int \alpha \, dp = \int \left(\frac{\partial T}{\partial p} \right)_\eta \eta' \, dp = \eta' \int dT, \quad (18.98)$$

where we can take η' out of the integral *because it is constant if the atmospheric has a saturated adiabatic profile*. Integrating gives

$$\phi(x, y, p, t) - \bar{\phi}(x, y, t) = (\bar{T}(x, y, t) - T(x, y, p, t)) \eta(x, y, t), \quad (18.99)$$

where we have written the constant of integration such that $\bar{\phi}$ and $\bar{T}$ are the vertical means, in pressure coordinates, of ϕ and T . Following conventional usage, we call the vertically integrated components 'barotropic' and the deviations 'baroclinic'.

The temperature difference $(\bar{T}(x, y, t) - T(x, y, p, t))$ in (18.99) is determined by the saturated adiabatic lapse rate and in the horizontal this varies quite weakly with the temperature variations found in the tropics. Thus, the horizontal variation of the terms on the right-hand side are dominated by variations in entropy and to a good approximation we can write

$$\nabla \phi(x, y, p, t) = \nabla \bar{\phi}(x, y, t) + M(p) \nabla \eta(x, y, t), \quad (18.100)$$

where $M(p) = \bar{T}(x, y, t) - T(x, y, p, t)$ and which *we take to be a function of pressure alone*. The profile of $M(p)$ has a single node in the vertical, and this constrains the horizontal velocity to have a similar variation in the vertical, and the vertical integral of $M(p)$ vanishes.

Velocity decomposition

First consider the vertical velocity. In pressure coordinates the mass continuity equation is

$$\frac{\partial \omega}{\partial p} = -\nabla_p \cdot \mathbf{u} \quad (18.101)$$

where $\bar{\mathbf{u}} = (u, v)$. Let $\mathbf{u} = \bar{\mathbf{u}} + \mathbf{u}^*$, where $\bar{\mathbf{u}}$ and $\mathbf{u}^*$ are the barotropic and baroclinic components of $\mathbf{u}$, and ω is the vertical (pressure) velocity. We suppose that the fluid is confined between the ground at $p = p_g$ and a rigid tropopause at p_t , with $\omega = 0$ at both. The baroclinic components of the velocity will, as a consequence of their definition, vanish when integrated over the troposphere. An integration of (18.101) over the troposphere implies the barotropic flow is divergence-free:

$$\nabla_p \cdot \bar{\mathbf{u}} = 0. \quad (18.102)$$

The vertical velocity is thus related to the baroclinic flow by,

$$\frac{\partial \omega}{\partial p} = -\nabla_p \cdot \mathbf{u}^*. \quad (18.103)$$

The horizontal flow itself is given using the momentum equation which, using (18.100), we write as

$$\frac{\partial \mathbf{u}}{\partial t} + \mathbf{v} \cdot \nabla \mathbf{u} + \mathbf{f} \times \mathbf{u} = -\nabla \bar{\phi} + M(p) \nabla_p \eta. \quad (18.104)$$

where $\mathbf{v}$ is the three-dimensional velocity and we omit forcing and viscous terms, and ∇ without a subscript denotes horizontal derivative. The above equation suggests that it will be a good approximation to suppose that the horizontal velocity has the same vertical structure as $M(p)$ and we let

$$\mathbf{u}(x, y, p, t) = \mathbf{u}_0(x, y, t) + m(p)\mathbf{u}_1(x, y, t), \quad (18.105)$$

where $\mathbf{u}_0 (= \bar{\mathbf{u}})$ and $\mathbf{u}_1$ are the ‘barotropic’ and ‘first baroclinic’ components of the flow, and $m(p) = M(p)/N$ where N is a normalizing coefficient, say $[\int M^2 dp / (p_t - p_g)]^{1/2}$, so that $m(p)$ is a dimensionless basis function. The reader will appreciate the similarity of this approach to that of section 3.4, now with the quasi-equilibrium constraints determining the form of the vertical eigenfunctions for us.

Momentum equations

Neglecting the advective term the horizontal momentum equation decomposes exactly into separate barotropic and baroclinic equations which, omitting frictional terms, are

$$\frac{\partial \mathbf{u}_0}{\partial t} + \mathbf{f} \times \mathbf{u}_0 = -\nabla \phi_0, \quad \frac{\partial \mathbf{u}_1}{\partial t} + \mathbf{f} \times \mathbf{u}_1 = -\nabla \phi_1, \quad (18.106a,b)$$

where $\phi_0 = \bar{\phi}$ and $\phi_1 = N\eta$. These have the same form as the linearized shallow water equations. We may obtain a diagnostic equation for the barotropic pressure by taking the divergence of (18.106a), whence the time derivatives disappear because $\nabla \cdot \mathbf{u}_0 = 0$. Alternatively, take the curl of (18.106a) to give the linear barotropic vorticity equation,

$$\frac{\partial \zeta_0}{\partial t} + \beta v_0 = 0. \quad (18.107)$$

This equation is closed because, since the flow is divergence-free, there exists a streamfunction ψ such that $(u_0, v_0) = (-\partial\psi/\partial y, \partial\psi/\partial x)$ and $\zeta_0 = \nabla^2 \psi$.

If we add nonlinearity back in then the baroclinic and barotropic modes interact. Thus, the terms $\mathbf{u}_1 \cdot \nabla \mathbf{u}_1$ and $\mathbf{u}_0 \cdot \nabla \mathbf{u}_0$ both project onto the barotropic flow, and the terms $\mathbf{u}_1 \cdot \nabla \mathbf{u}_0$ and $\mathbf{u}_0 \cdot \nabla \mathbf{u}_1$ project onto the baroclinic flow, in a way that is analogous to that occurring in two-layer quasi-geostrophic flow (section 12.2.2). A baroclinic-baroclinic interaction also arises because of vertical advection and we obtain

$$\frac{\partial \mathbf{u}_0}{\partial t} + \mathbf{u}_0 \cdot \nabla \mathbf{u}_0 + D_0(\mathbf{u}_1, \mathbf{u}_1) + \mathbf{f} \times \mathbf{u}_0 = -\nabla \phi_0, \quad (18.108a)$$

$$\frac{\partial \mathbf{u}_1}{\partial t} + \mathbf{u}_1 \cdot \nabla \mathbf{u}_0 + \mathbf{u}_0 \cdot \nabla \mathbf{u}_1 + D_1(\mathbf{u}_1, \mathbf{u}_1) + \mathbf{f} \times \mathbf{u}_1 = -\nabla \phi_1, \quad (18.108b)$$

where D_0 and D_1 are nonlinear (largely advective) operators whose exact form is not of concern here. The barotropic flow is divergence free and this determines the pressure ϕ_0 , but ϕ_1 is as yet undetermined.

Thermodynamics

To close the equation for baroclinic flow we use an equation for the entropy or for the equivalent potential temperature in conjunction with (18.100). Since entropy is assumed invariant with height there is no vertical dependence and the equation might be written in the general form,

$$\frac{\partial \eta}{\partial t} + \mathbf{u} \cdot \nabla \eta = S, \quad (18.109)$$

where S are various source and sink terms of both heat and moisture. This term hides a multitude of sins, for it potentially includes radiative, condensational, evaporative and turbulent flux terms

with complex flow-dependent specifications, possibly needing separate equations for dry air and moisture. However, since entropy is (it is assumed) constant with height, we need a thermodynamic equation only at one level, which we might take to be near the surface, or we may use a vertical average.

In summary, the constraints on the vertical temperature structure imposed by convection lead to a relatively simple vertical structure of all the dynamical fields, and as a consequence a heat source will primarily force the first baroclinic mode. In the linear approximation the baroclinic flow is uncoupled from the barotropic flow. The procedure provides some justification for using shallow water-like equations in tropical dynamics, as in chapter 8, with the velocity fields to be interpreted as being those of the first baroclinic mode, an approach that is especially useful if the dynamics are predominantly linear. More generally, it is perhaps useful to think of this type of quasi-equilibrium model in a similar light as quasi-geostrophy: neither are quantitatively accurate for flow evolution (we would use neither model for an accurate weather forecast) but they may be able to provide insight where the full equations are too complex.

The equations derived above are still unbalanced, and are simpler than the primitive equations only in their constrained vertical structure. Let us try to go a little further and see if we can incorporate any balances in the horizontal that might simplify matters.

18.8 SCALING AND BALANCED DYNAMICS FOR LARGE-SCALE FLOW IN THE TROPICS

In earlier chapters we looked at flow with small Rossby number and, by performing a scale analysis, determined what the dominant balance of terms was in the equations of motion. This allowed us to derive the quasi-geostrophic equations, which are the basis for much of the theory of mid-latitude motion. Is such a program possible for the tropics? The answer is, ‘well, in part’. It is possible to some reduced sets of equations for the tropics, and that will be the main topic of this section. However, these reduced sets have not proven nearly as useful for the tropics as quasi-geostrophy has been for the mid-latitudes, in part because it is harder to make relevant equations that are simple, or simple equations that are relevant. In any case, let us begin with the stratified primitive equations, without explicitly invoking a constrained vertical structure.

18.8.1 Balanced, adiabatic flow

We will present a scaling for tropical flow side-by-side with the corresponding scaling for mid-latitude flow.¹⁴ We begin with the hydrostatic primitive equations for adiabatic, frictionless flow, which, reprising (5.15b), may be written as,

$$\frac{D\mathbf{u}}{Dt} + \mathbf{f} \times \mathbf{u} = -\nabla_z \phi, \quad \frac{\partial \phi}{\partial z} = b, \quad (18.110a,b)$$

$$\frac{Db}{Dt} + N^2 w = 0, \quad \nabla \cdot (\bar{\rho} \mathbf{v}) = 0. \quad (18.110c,d)$$

These are nominally the anelastic equations in height co-ordinates in our standard notion, but an entirely equivalent derivation could use pressure co-ordinates, as these have similar form. The reader will recall that $b = g \delta \theta / \theta_0$ is the buoyancy and $N^2 = d\tilde{b}/dz$ where $\tilde{b}(z)$ is a reference stratification. We will suppose that the basic variables scale according to

$$(x, y) \sim L, \quad z \sim H, \quad (u, v) \sim U, \quad w \sim W, \quad t \sim \frac{L}{U}, \quad \phi \sim \Phi, \quad b \sim B, \quad f \sim f_0. \quad (18.111)$$

The quantity B is representative of horizontal variations in buoyancy. Vertical variations scale differently, hence their separate representation in (18.110c). By choosing the time t to scale advectively we are implicitly eliminating gravity waves. The nondimensional numbers that will arise are

the Rossby, Burger and Richardson numbers,

$$Ro = \frac{U}{f_0 L}, \quad Bu = \left(\frac{L_d}{L} \right)^2 = \left(\frac{NH}{f_0 L} \right)^2, \quad Ri = \left(\frac{NH}{U} \right)^2, \quad (18.112)$$

and evidently

$$Bu = Ri \times Ro^2. \quad (18.113)$$

The Rossby number Ro is generally small in mid-latitudes for large-scale flow, but in the tropics it is $\mathcal{O}(1)$ or larger. The Richardson number (which is the inverse square of a Froude number) is usually large in both midlatitudes and tropics, *except* in regions of active convection where N is very small. If we take $N = 10^{-2} \text{ s}^{-1}$, $H = 10^4 \text{ m}$ and $U = 10 \text{ m s}^{-1}$ then $Ri = 100$. In fact, for large-scale flow the Richardson number is usually sufficiently large that $1/(Ro Ri)$ is small in both mid-latitudes and tropics.

The difference between the tropics and mid-latitudes is apparent from the dominant balance in the momentum equation. At small Rossby number we have the familiar geostrophic balance and with hydrostatic balance we obtain the scaling:

$$\mathbf{f} \times \mathbf{u} \approx -\nabla_z \phi, \quad \frac{\partial \phi}{\partial z} = b, \quad \implies \quad \Phi = f_0 UL, \quad B = \frac{f_0 UL}{H}. \quad (18.114)$$

In the tropics the advective term, or the advectively-scaled time derivative, balances the pressure gradient meaning that $D\mathbf{u}/Dt \sim \nabla_z \phi$ and, since we still have $\partial \phi / \partial z = b$ we find

$$\Phi = U^2, \quad B = \frac{U^2}{H}. \quad (18.115)$$

If U is of similar magnitude in the tropics and mid-latitudes (and in the absence of a dynamical analysis this is an assumption), then *variations of pressure and temperature are smaller in the tropics than in mid-latitudes*. This is an important and not-quite obvious result and it is the essence of the *weak temperature gradient approximation*, discussed further in the next section. If we were to carry through the derivation in pressure co-ordinates (see the shaded box on page 81) with geopotential and temperature as the variables we would find

$$\text{Mid-latitudes: } \Phi = f_0 UL, \quad T_s = \frac{f_0 UL}{R}, \quad (18.116a)$$

$$\text{Tropics: } \Phi = U^2, \quad T_s = \frac{U^2}{R}, \quad (18.116b)$$

where Φ is now the scaling for geopotential, T_s is the scaling for temperature and R is the ideal gas constant. For $U = 10 \text{ m s}^{-1}$, $L = 10^6 \text{ m}$ and, for mid-latitudes only, $f_0 = 10^{-4} \text{ s}^{-1}$, we obtain

$$\text{Mid-latitudes: } \Phi \sim 1000 \text{ m}^2 \text{ s}^{-2}, \quad T_s \sim 3 \text{ K} \quad (18.117a)$$

$$\text{Tropics: } \Phi \sim 100 \text{ m}^2 \text{ s}^{-2}, \quad T_s \sim 0.3 \text{ K}. \quad (18.117b)$$

The implications of these results are illustrated in Fig. 18.22, which shows a snapshot of observed contours of geopotential height, temperature and zonal wind; the large-scale variability of geopotential and temperature is evidently much smaller in the tropics than mid-latitudes.

Vertical velocity

The obvious scaling for the vertical velocity is that suggested by the mass continuity equation, namely $W = UH/L$. However, as we know from chapter 5, the vertical velocity may be much less

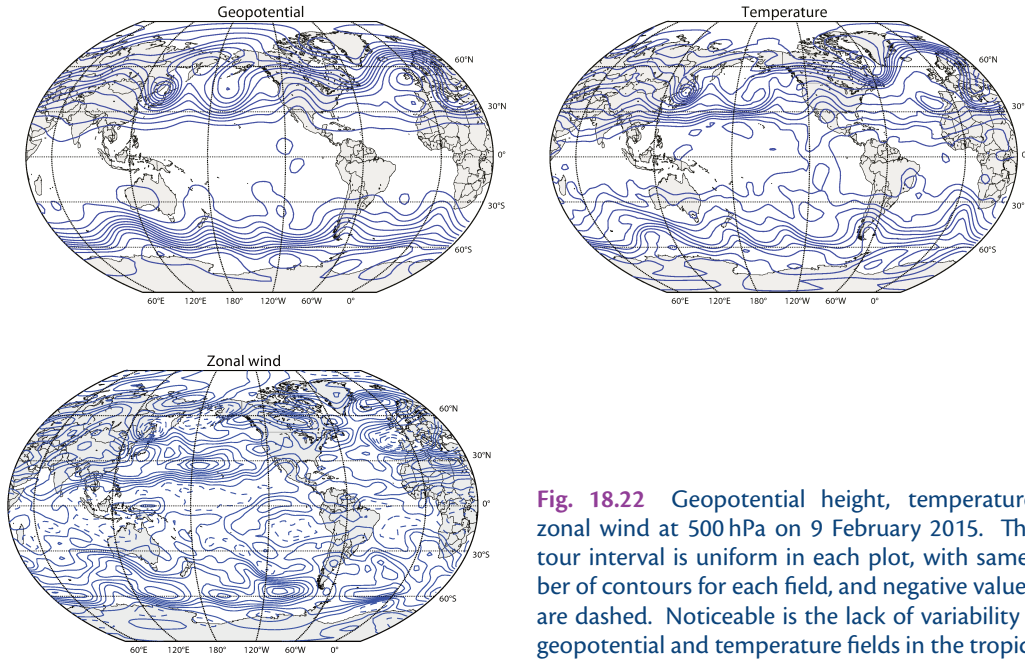


Fig. 18.22 Geopotential height, temperature and zonal wind at 500 hPa on 9 February 2015. The contour interval is uniform in each plot, with same number of contours for each field, and negative values for u are dashed. Noticeable is the lack of variability of the geopotential and temperature fields in the tropics.

than this estimate in a stratified, rotating fluid, and here we start with the thermodynamic equation for adiabatic flow,

$$\mathbf{u} \cdot \nabla b + wN^2 = 0, \quad \implies \quad W = \frac{UB}{LN^2} \quad (18.118)$$

Using (18.114) and (18.115) gives

$$\text{Mid-latitudes: } W = \frac{f_0 U^2}{HN^2} = \frac{f_0 L}{U} \frac{U^2}{N^2 H^2} \frac{UH}{L} = (Ro \ Ri)^{-1} \left(\frac{UH}{L} \right), \quad (18.119a)$$

$$\text{Tropics: } W = \frac{U^3}{LHN^2} = \frac{U^2}{N^2 H^2} \frac{UH}{L} = Ri^{-1} \left(\frac{UH}{L} \right), \quad (18.119b)$$

where $Ri \equiv N^2 H^2 / U^2$ is the Richardson number, with typical values of $\mathcal{O}(10)$ – $\mathcal{O}(100)$ for large-scale flow. Thus, again perhaps non-intuitively, the vertical velocity is, for adiabatic flow, *smaller* in the tropics than in mid-latitudes, by order of a mid-latitude Rossby number. In midlatitudes the scaling for W is more commonly written as

$$W = \frac{f_0 U^2}{HN^2} = Ro \frac{L^2}{L_d^2} \frac{UH}{L} = \frac{Ro}{Bu} \frac{UH}{L}, \quad (18.120)$$

so that for scales comparable to the mid-latitude deformation radius (i.e., with $Bu \sim 1$) the vertical velocity is order Rossby-number smaller than mass-continuity scaling suggests.

Vorticity

Cross-differentiating the horizontal momentum equation gives, as in (4.66) but without a baroclinic term, the vertical component of the vorticity equation with associated scalings,

$$\frac{D}{Dt}(\zeta + f) = -(\zeta + f) \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right) + \left(\frac{\partial u}{\partial z} \frac{\partial w}{\partial y} - \frac{\partial v}{\partial z} \frac{\partial w}{\partial x} \right), \quad (18.121a)$$

$$\text{Tropics:} \quad \left(\frac{U}{L} \right)^2 \sim \left(\frac{U}{L} + \frac{1}{Ro} \frac{U}{L} \right) \left(\frac{1}{Ri} \frac{U}{L} \right) \left(\frac{U}{H} \right) \left(\frac{1}{Ri} \frac{UH}{L^2} \right), \quad (18.121b)$$

$$\text{Mid-latitudes:} \quad \left(\frac{U}{L} \right)^2 \sim \left(\frac{U}{L} + \frac{1}{Ro} \frac{U}{L} \right) \left(\frac{Ro}{Bu} \frac{U}{L} \right) \left(\frac{U}{H} \right) \left(\frac{Ro}{Bu} \frac{UH}{L^2} \right). \quad (18.121c)$$

In the tropical case with $Ro = \mathcal{O}(1)$ or larger *all* the terms on the right-hand side are much smaller than the terms on the left-hand side, whereas in the midlatitude case vortex stretching by the Coriolis term, $f(\partial u/\partial x + \partial v/\partial y)$, is the same order (because f is big and the divergence is small). Thus, in the tropical case the vorticity equation simplifies severely and at lowest order becomes the two-dimensional vorticity equation,

$$\frac{D}{Dt}(\zeta + f) = 0. \quad (18.122)$$

The large-scale velocity is, at this order, purely rotational and is given by a streamfunction ψ such that

$$\nabla^2 \psi = \zeta, \quad (u, v) = \left(-\frac{\partial \psi}{\partial y}, \frac{\partial \psi}{\partial x} \right). \quad (18.123)$$

This equation nominally holds independently at each vertical level, although the assumptions that give rise to it assume that the depth scale is large. The pressure is not needed to step forward (18.123) but it may be obtained diagnostically. To do so we go back to the horizontal velocity equation written in the vector-invariant form,

$$\frac{\partial \mathbf{u}}{\partial t} - \mathbf{u} \times (\zeta + f) = -\nabla \left(\phi + \frac{1}{2} \mathbf{u}^2 \right), \quad (18.124)$$

neglecting the vertical and divergent and velocities. Taking the divergence gives the so-called non-linear balance or gradient-wind equation,

$$\nabla^2 \phi = \nabla \cdot \left[(f + \zeta) \nabla \psi - \frac{1}{2} \nabla (\nabla \psi)^2 \right], \quad (18.125)$$

which is similar to (2.238) and expresses a balance between the pressure gradient, Coriolis and centrifugal forces.

18.8.2 A few remarks

The above derivations and results require some comment.

- The relative weakness of large-scale horizontal gradients of pressure (or geopotential) and temperature in the tropics, for a given velocity field, is a robust result of the scaling analysis and borne out in observations.
- The smallness of the vertical velocity requires that the Richardson number, $N^2 H^2 / U^2$ be large; that is, stratification is strong. This is only true in regions that are not actively convecting; in convective regions N may be small.

- Relatedly, the scaling does not take into account diabatic sources, which may be expected to be particularly important in tropical regions.
- Equally tellingly, (18.123) tells us nothing about the vertical structure and so, unlike quasi-geostrophy in mid-latitudes, is not a sufficiently complete equation to be a useful prognostic, or even diagnostic, equation for tropical motion.

Let us now look at how diabatic effects might affect large-scale motion.

18.9† SCALING AND BALANCE FOR LARGE-SCALE FLOW WITH DIABATIC SOURCES

In the tropics we might expect that heat sources, for example condensational heating, would be important to the extent that they should be explicitly included in any development of reduced equations. Let us see if and how this is possible, using the shallow water equations for illustration. Using the standard nonlinear shallow water equations for the tropical atmosphere is rather ad hoc (and not justified by quasi-equilibrium except in the linear case) but our intention here is primarily illustrative. The underlying physical assumption is that in an air column adiabatic cooling associated with vertical motion balances diabatic heating, which in the shallow water equations becomes a balance between the heating and the divergence.

18.9.1 Diabatic balanced shallow water equations

On an f -plane and in conventional notation the equations may be written in vorticity-divergence form as

$$\frac{\partial h}{\partial t} + \nabla \cdot (\mathbf{u}h) = Q, \quad (18.126a)$$

$$\frac{\partial \zeta}{\partial t} + \nabla \cdot [\mathbf{u}(\zeta + f_0)] = -r\zeta, \quad (18.126b)$$

$$\frac{\partial \delta}{\partial t} + \nabla^2 \left(\frac{1}{2} \mathbf{u}^2 + gh \right) - \mathbf{k} \cdot \nabla \times [\mathbf{u}(\zeta + f_0)] = -r\delta. \quad (18.126c)$$

where $\delta = \partial u / \partial x + \partial v / \partial y$ is the divergence, Q is the mass or heating source, r is a frictional coefficient and other notation is standard. The height field is a proxy for both pressure and temperature and we will assume that horizontal gradients are weak, by which we mean that the dominant balance in (18.126a) is characterized by the scaling

$$H\Delta = Q_0 \quad (18.127)$$

where H is the mean thickness of the layer, Δ is a scaling for the divergence and Q_0 is the magnitude of the heating. We then choose the velocity scale, U and the vorticity scale, Z , to be

$$U = \frac{Q_0 L}{H}, \quad Z = \Delta = \frac{Q_0}{H}. \quad (18.128)$$

Finally, the magnitude of horizontal deviations in the height field, $\mathcal{H}$, are determined from the divergence equation. Depending on whether rotation is or is not important we deduce

$$\mathcal{H} = \frac{f_0 U L}{g} = \frac{Q_0 f_0 L^2}{gH}, \quad \text{or} \quad \mathcal{H} = \frac{U^2}{g} = \frac{Q_0^2 L^2}{gH^2}. \quad (18.129a,b)$$

The height field may then be separated into a mean and deviation, $h = H + \eta$, where $\eta = \mathcal{H}\hat{\eta}$. These scalings involve the heating in an essential way and so are fundamentally different from the adiabatic scaling of the previous section.

Using the above scalings, with rotation, (18.126) may be written in nondimensional form as,

$$Bu^{-1} \left[\frac{1}{f_0 T} \frac{\partial \hat{\eta}}{\partial \hat{t}} + Ro \nabla \cdot (\hat{\mathbf{u}} \hat{\eta}) \right] + \hat{\delta} = \hat{Q}, \quad (18.130a)$$

$$\frac{1}{f_0 T} \frac{\partial \hat{\zeta}}{\partial \hat{t}} + \nabla \cdot [\mathbf{u}(\hat{\zeta} + \hat{f}_0)] = -\frac{r}{f_0} \hat{\zeta}, \quad (18.130b)$$

$$\frac{1}{f_0 T} \frac{\partial \hat{\delta}}{\partial \hat{t}} + \nabla^2 \left(\frac{1}{2} \hat{\mathbf{u}}^2 + \hat{\eta} \right) - \mathbf{k} \cdot \nabla \times [\hat{\mathbf{u}}(\hat{\zeta} + \hat{f}_0)] = -\frac{r}{\hat{f}_0} \hat{\delta}. \quad (18.130c)$$

where T is the scaling for time, $Bu = (L_d/L)^2$ with $L_d = \sqrt{gH}/f_0$, and $\hat{f}_0 = 1$. The Rossby number is given by $Ro = Q_0/f_0 H$ and is not necessarily small. In the nonrotating case a similar set of equations can be derived but with somewhat different coefficients.

Reduced equations

Let us suppose that the mass source determines the divergence in (18.130a). The condition for this is that

$$\max \left(\frac{1}{f_0 T}, Ro \right) Bu^{-1} \ll 1, \quad (18.131)$$

which means that the scale of motion cannot be too large and the time scale cannot be too short. If (18.131) is satisfied then the dimensional height equation, (18.126a), becomes

$$\nabla \cdot \mathbf{u} = \frac{Q}{H}. \quad (18.132a)$$

This value of the divergence is used in (18.126b) and (18.126c) which, retaining all terms since none are obviously small, become

$$\frac{\partial \zeta}{\partial t} + \mathbf{u} \cdot \nabla (\zeta + f_0) + (\zeta + f)Q = -r\zeta, \quad (18.132b)$$

$$g\nabla^2 h = \mathbf{k} \cdot \nabla \times [\mathbf{u}(\zeta + f_0)] - \frac{\partial Q}{\partial t} - r\delta - \nabla^2 \frac{\mathbf{u}^2}{2}. \quad (18.132c)$$

The equation set (18.132) has but one prognostic equation, namely (18.132b), and so is truly balanced and may be thought of as a generalization of (18.122) and (18.125) to the case with nonzero heating. The divergence equation is a nonlinear balance equation, similar to (18.125), except now with a diabatic term on the right-hand side. The divergent flow itself is computed using the height equation, by an assumed balance between adiabatic cooling and diabatic heating. The relationship between velocity and geopotential (or pressure) is the same as in the adiabatic case, because this arises through the momentum equation. Thus, even in the presence of a heating, gradients of geopotential and temperature remain relatively weak, a result that ultimately arises from the smallness of the Coriolis parameter. The arguments that lead to (18.132), along with the scaling of section 18.8, provide us with the *weak temperature-gradient approximation*.¹⁵ The importance of the result lies in what it implies about the response of the atmosphere to a localized heating: without making any linear approximation, the equations provide a scaling for the response of the velocity, and suggest that the response may become spread out over a sufficient area to keep the temperature gradients small.

Gravity wave adjustment and weak temperature gradients

Although convection certainly constrains the lapse rate where it is occurring, it does not directly do so elsewhere. Nevertheless, the observed tropical atmosphere has an *average* lapse rate close

to the saturated adiabat. Why should this be so? The reason is that a process akin to geostrophic adjustment (section 3.9) brings the atmosphere close to a state with weak horizontal temperature gradients. Suppose that a particular column undergoes moist convection and, perhaps in a matter of hours, adjusts to a moist neutral profile. Away from the cloud the buoyancy profile will in general be different from that, and so there will be unbalanced horizontal pressure (and hence temperature) gradients. Gravity waves, initiated by the motion surrounding the convection ('compensating subsidence'), will spread from the cloud and will adjust the environmental buoyancy to the same profile as that of the convecting region. The timescale for the adjustment is determined by the time that gravity waves take to propagate horizontally between convecting regions. Internal gravity wave speeds are typically about 10 m s^{-1} or somewhat faster and so will spread a distance 50 km from a cloud in a couple of hours, and the adjustment time on this scale will be of this order. This buoyancy adjustment time is much less than the time it would take a passive tracer to homogenize between clouds by advective or mixing processes. The actual process of gravity wave initiation and subsequent adjustment is complex and beyond our scope, and the reader is referred to the literature.¹⁶

18.9.2 ♦ Weak temperature gradient and stratified flow

The weak temperature gradient approximation is formally independent of any quasi-equilibrium argument, so in this section we briefly discuss its application to the stratified, three-dimensional equation of motion. The thermodynamic equation in pressure coordinates (see equation (P.5) on page 81) may be written,

$$\frac{\partial T}{\partial t} + \mathbf{u} \cdot \nabla T + \omega \frac{\partial s}{\partial p} = Q, \quad (18.133)$$

where $s = T + gz/c_p$ is the dry static energy divided by c_p and Q represents heating terms. In the weak temperature gradient approximation this equation becomes

$$\omega \frac{\partial s}{\partial p} = Q. \quad (18.134)$$

If the stratification s is known (e.g., moist neutral) the above equation becomes a diagnostic for the vertical velocity, namely $\omega(p) = Q/\partial_p s$. Even with this approximation the remaining stratified equations (e.g., momentum equation) are somewhat complex, with both nonlinearity and continuous vertical structure.

If we are willing in addition to make the quasi-equilibrium assumption then further simplifications are possible because all the fields are assumed to have a simple vertical structure and the flow may be described, as before, by a system similar to the shallow water equations. The vertical and horizontal velocities are related by the mass continuity equation, (18.101). The barotropic flow has zero horizontal divergence so that the vertical (pressure) velocity and the horizontal are related by

$$\omega(x, y, p, t) = - \int_{p_s}^p \nabla \cdot \mathbf{u} \, dp = -n(p) \nabla \cdot \mathbf{u}_1, \quad (18.135)$$

where $n(p) = \int m(p) \, dp$ and we take $\omega = 0$ at the bottom boundary. Thus, we may write $\omega = n(p)\omega_1(x, y, t)$ and $\omega_1 = -\nabla \cdot \mathbf{u}_1$. If we divide (18.134) and vertically integrate we obtain

$$\bar{s} \nabla \cdot \mathbf{u}_1 = \bar{Q}, \quad (18.136)$$

where $\bar{Q}$ is a vertically integrated heating term, weighted by $1/n(p)$. This equation is the direct analog of (18.132a) and it provides a predictive equation for the divergence of the baroclinic flow if the heating and stratification are known. Moisture may be added to the mix, in which case a moist static energy appears, and the effects of evaporation and condensation must be included in

Q. The precise form of (18.136) and other particulars of implementation will depend on the basis functions chosen for the vertical structure, but the form is generically $\nabla \cdot \mathbf{u}_1 = [Q]/[s]$, where $[Q]$ is a measure of heating and $[s]$ a measure of stratification.¹⁷ Equation (18.136) may be used to close the momentum equation (18.108b). If we take the curl of that equation we obtain a vorticity equation analogous to (18.132b), and if we take its divergence we obtain a diagnostic equation for $\nabla^2 \phi_1$ analogous to (18.132c). The combination of a weak temperature gradient approximation and shallow-water-like equations arising from a constrained vertical stratification may be as close as the tropical atmosphere allows us to get to tractable and understandable equations of motion.

18.10† CONVECTIVELY COUPLED GRAVITY WAVES AND THE MJO

We now look at some of the manifestations of the theoretical development of this chapter and chapter 8, and three phenomena suggest themselves, namely the Walker circulation, monsoons, and the Madden–Julian Oscillations (MJO). We leave monsoons for another day, in part because the subject is large and its theoretical development is a moving target. And we defer discussion of the Walker circulation — an atmospheric overturning circulation largely in the zonal (i.e., y – z) plane — to chapter 22 because the phenomenon is closely tied to the equatorial ocean. So let us briefly discuss the MJO, although we cannot give it a wholly crisp explanation because at the time of writing none exists.¹⁸

What is the MJO? It is a pattern of precipitation and winds that takes shape across the western tropical Indian Ocean and drifts eastward at about 4 – 8 m s^{-1} into the western Pacific before dying out over the cooler waters of the eastern tropical Pacific. It often recurs roughly every 30–60 days or so, although it does not oscillate like a conventional wave. Rather, it is more like a somewhat coherent, drifting pattern several thousand kilometres across, consisting of a wet, rainy region of ascending air flanked by dryer regions on either side.

18.10.1 The observations

The MJO has a characteristic spectral signature that can be obtained from satellite measurements of cloud brightness. Figure 18.23 shows the symmetric (Northern Hemisphere plus Southern Hemisphere) and antisymmetric power spectrum (that is, the intensity of the field, Fourier analysed in zonal wavenumber and frequency) of tropical satellite brightness temperature, after filtering away some background noisiness.¹⁹ Power spectra of other fields, including velocity fields from re-analysis, show similar features. The main shaded regions in the figure fall nicely on the theoretical dispersion relations for Rossby, Kelvin and mixed Rossby-gravity waves as derived in section 8.2. This is perhaps somewhat startling to see, but evidently equatorial waves do exist!

The theoretical curves match the observations best if the unadorned gravity wave speed, c , is in region of 10 to 20 m s^{-1} . This speed is given by $c = \sqrt{gH_e}$ where H_e is the equivalent depth (section 3.4.2), so theoretical values match the observations with $H_e \approx 10$ – 50 m , perhaps with the best match around 25 m , somewhat smaller than the first equivalent depth computed for the atmosphere in section 3.4.²¹ Part of the difference may come from the fact that the tropopause is not a rigid lid, and part from the fact that the presence of moisture may reduce the effective static stability of the atmosphere below that implied by the value of N^2 computed using dry potential temperature ($g/\theta \partial \theta / \partial z$). In a saturated atmosphere the (smaller) value of $g/\theta_e \partial \theta_e / \partial z$ may be more relevant for the generation and propagation of gravity waves. The Rossby wave speed is given by $c_R \approx -\beta k / [(2m+1)\beta/c + k^2]$, so it too depends, albeit more weakly, on c .

In addition to the relatively well understood gravity and Rossby waves, there appears to be a prominent spectral signature in Fig. 18.23 at a small positive zonal wavenumber (between 1 and 5 in the left-hand panel) and a timescale of about 40 days, and this is a signature of the MJO. If we look in physical space, we get the sense that the MJO resembles a drifting Matsuno–Gill pattern more than an oscillating wave — compare Fig. 18.24 with Fig. 8.11 on page 325. Consistently, the

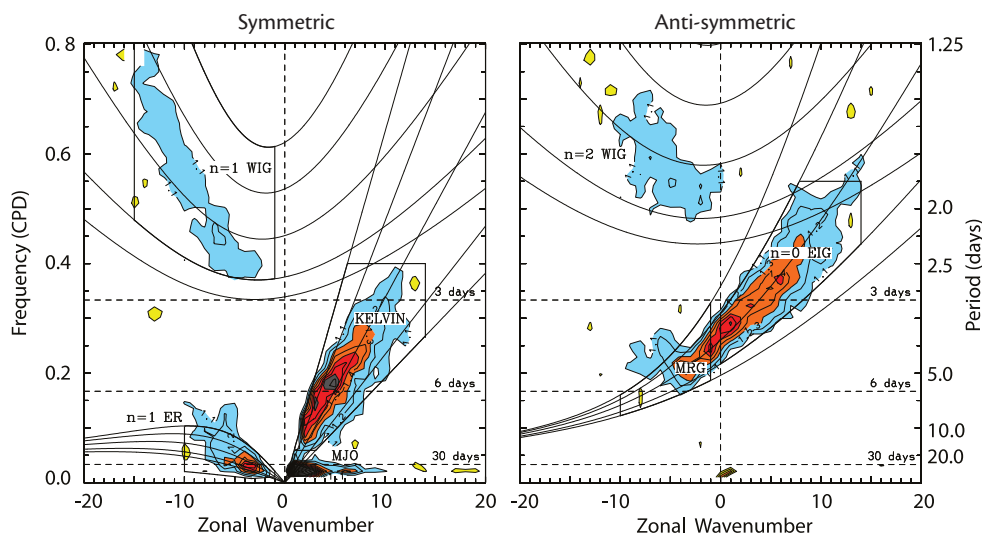


Fig. 18.23 A power spectrum of cloud brightness (colouring) from 15°S to 15°N , measured from satellite and Fourier-transformed into frequency-wavenumber space. Left panel is the symmetric spectrum (Northern plus Southern Hemispheres) and right panel the anti-symmetric spectrum. The solid lines are the corresponding dispersion relations with multiple equivalent depths ranging from 8 m to 90 m, with the best fit at about 25 m. Compare with Fig. 8.6 and Fig. 8.7.²⁰

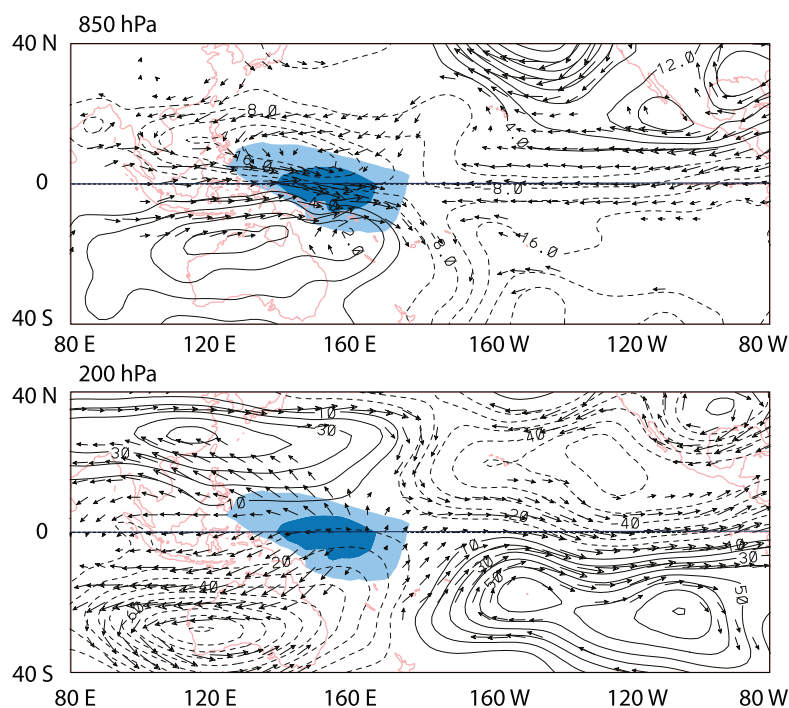


Fig. 18.24 Composite of an observed pattern during an MJO period. Shading indicates anomalously low outgoing IR radiation (less than 16 and 32 W m^{-2} below normal) and indicates the presence of high clouds and precipitation. The solid lines are streamfunctions and the arrows indicate velocities.²²

observed vertical structure largely has a first baroclinic mode structure, as expected if the lapse rate is constrained to be nearly constant by convection — note the oppositely directed velocity fields in Fig. 18.24, just as in Fig. 8.12. The shaded region corresponds to a region of heating in the Matsuno–Gill model and without too much imagination one may see a Kelvin wave response to its east, along the equator, and off-equatorial Rossby lobes to the west over Australia and South East Asia. A defining characteristic of the MJO is that this pattern drifts eastward at a speed of a few metres per second, much lower than the Kelvin wave speed at that equivalent depth.

18.10.2 † The mechanism

Precisely why this pattern should move eastward remains unclear, and there is little certainty as to the underlying mechanism except in so far as it likely has to do with the interaction of moist convection with the large-scale circulation — perhaps moist convection *producing* a large-scale circulation that then modulates the convection. Thus, a locally warm region near the equator (initiated by a sea-surface temperature anomaly, for example) will produce a pattern that evolves towards that of the Matsuno–Gill solution (Fig. 8.11), with a low-level convergence of moisture and condensation amplifying the initial pattern. Furthermore, the convergence will produce a dryer region on either side of the heat source (as observed). However, the timescales of condensation are fast and the process is unsteady, and it is unlikely that the condensation can produce a self-sustained stationary pattern — in the initial value problem it takes several days for a heating anomaly to settle into a steady Matsuno–Gill pattern in the tropical atmosphere.²³

Thus, instead of a truly steady pattern, the convection will initiate an eastward moving Kelvin wave, along with slower westward moving Rossby waves trapped nearer the source. Now, the initial moisture convergence arises because the Kelvin wave draws air in from the west, and the Rossby wave similarly draws air from the east. If the Kelvin waves moves east the region of low-level moisture convergence, and thus the region of condensational heating, will also move eastward, and the process begins again. However, the convergence region will move eastward much more slowly than the Kelvin wave, since it is the advective convergence that provides the heat source that is the source of both the Kelvin and Rossby waves. A Kelvin wave that breaks free of the convergence will eventually decay since it will not produce the moisture convergence to feed itself. Rather, the MJO is a self-sustained interaction between moisture convergence and the forced-dissipative Kelvin and Rossby waves it produces, with the requirement for moisture convergence greatly slowing the Kelvin wave. Radiative effects tied to the variable cloud field also almost certainly play a role in the way the pattern is damped.

Although wavelike, in this picture the MJO is not a conventional linear wave with a dispersion relation; rather it is related to the translation of a forced quasi-steady pattern with a longer timescale than either of those waves themselves, with the forcing *maintained by the pattern it itself produces*. Evidently, this will be a delicate balance for the interaction of moist convection with Kelvin and Rossby waves is a complex process, so it is not surprising that even the most sophisticated numerical models have trouble reproducing the phenomenon properly, even if the underlying mechanism can be described in relatively simple terms. Note also that the convection is not in quasi-equilibrium with a more slowly and independently evolving external environment — the convection helps produce its own external environment and the two evolve in synchrony.

18.A APPENDIX A: MOIST THERMODYNAMICS AND THE GIBBS FUNCTION

Many of the thermodynamic quantities of interest for moist air can be obtained from knowledge of the Gibbs function, in a manner analogous to that followed for dry air in the appendix on page 47. The benefits are that the approach is systematic, explicit formulae for thermodynamic variables can be given, and approximations can be made consistently as needed. Here we outline the methodology and provide some examples.²⁴ We use superscripts d , v and l to denote dry air, water

vapour, and liquid water, respectively, and subscripts of g denote derivatives with other variables held constant, except that a subscript 0 always denotes a constant, for any quantity. Quantities with multiple superscripts are mixtures of the quantities, but the superscripts are occasionally omitted. A subscript s denotes the saturated value.

Given (1.215), the specific Gibbs functions for dry and water vapour are, respectively,

$$g^d(p, T) = R^d T \ln(p^d/p_0) + c_p^d T [1 - \ln(T/T_0)], \quad (18.137a)$$

$$g^v(p, T) = R^v T \ln(p^v/p_0) + c_p^v T [1 - \ln(T/T_0)] - L_0^v \frac{T}{T_0} + L_0^v, \quad (18.137b)$$

where p^d and p^v are the partial pressures of dry air and water vapour. A constant, L_0^v , appears in the expression for the Gibbs function of water vapour because we will be concerned with water in its liquid phase. We take the analogous constant to be zero for dry air.

The two fluids are at the same temperature, T and the total pressure, p , is given by the sum of the partial pressures. Let q be the mass fraction of moist air in the mixture (i.e., the specific humidity), and let ϵ be the ratio of the molecular weights of water vapour and dry air, μ^v/μ^d . The two partial pressures are given by

$$p^v = \frac{qp}{q + \epsilon(1 - q)}, \quad p^d = \frac{\epsilon(1 - q)p}{q + \epsilon(1 - q)}, \quad (18.138)$$

satisfying $p = p^d + p^v$, and p^v is often denoted e in the literature. The Gibbs function for the mixture is

$$g^{dv}(p, T, q) = (1 - q)g^d + qg^v, \quad (18.139)$$

whence

$$\begin{aligned} g^{dv}(p, T, q) = & (qc_p^v + (1 - q)c_p^d) T [1 - \ln(T/T_0)] \\ & + T \left[qR^v \ln \left(\frac{qp/p_0}{q + \epsilon(1 - q)} \right) + (1 - q)R^d \ln \left(\frac{\epsilon(1 - q)p/p_0}{q + \epsilon(1 - q)} \right) \right] \\ & + qL_0^v \left(1 - \frac{T}{T_0} \right). \end{aligned} \quad (18.140)$$

This expression is symmetric in dry air and water vapour, and valid for any mass fraction of moisture in the air (given that $q \ln q \rightarrow 0$ as $q \rightarrow 0$). We will also need the Gibbs function for liquid water, which for our purposes may be written

$$g^l(p, T) = c^l T [1 - \ln(T/T_0)] - \eta_0^l T + \alpha^l p + g_0^l. \quad (18.141)$$

This expression is a simplified version of (1.144) with slightly different notation, with c^l being the heat capacity and α^l the inverse density.

Density and the thermal equation of state

The inverse density of the gas mixture is given by

$$\alpha^{dv} = \left(\frac{\partial g}{\partial p} \right)_{q, T} = \frac{T}{p} [qR^v + (1 - q)R^d] \quad (18.142)$$

or equivalently $p = \rho RT$ where $R = qR^v + (1 - q)R^d$ is the specific gas 'constant' for the mixture (which of course varies with q).

Entropy

The entropy is given by $\eta = -(\partial g / \partial T)_{p,q}$, giving

$$\eta^{dv} = (qc_p^v + (1-q)c_p^d) \ln T + \left[qR^v \ln \left(\frac{qp/p_0}{q + \epsilon(1-q)} \right) + (1-q)R^d \ln \left(\frac{\epsilon(1-q)p/p_0}{q + \epsilon(1-q)} \right) \right] + \frac{qL_0^v}{T_0}. \quad (18.143)$$

The entropy of the mixture may also be written as

$$\eta^{dv} = (1-q)\eta^d + q\eta^v, \quad (18.144)$$

where the specific entropies of dry air and water vapour are

$$\begin{aligned} \eta^d &= c_p^d \ln(T/T_0) - R^d \ln(p/p_0) - R^d \ln(1 - p^v/p), \\ \eta^v &= c_p^v \ln(T/T_0) - R^v \ln(p/p_0) - R^v \ln(1 - p^d/p) + L_0^v. \end{aligned} \quad (18.145)$$

The third terms on right-hand sides are called the entropies of mixing.

The entropy of liquid water is given by

$$\eta^l = \eta_0^l + c^l \ln(T/T_0). \quad (18.146)$$

The entropy of a mixture of dry air, liquid water and water vapour is given by

$$\eta^{dvl} = (1 - q^{vl})\eta^d + q^v\eta^v + q^l\eta^l \quad (18.147)$$

where q^v and q^l are the mass concentrations of vapour and liquid in the total mixture and $q^{vl} = q^v + q^l$.

Heat capacities

The heat capacity at constant pressure of moist air (i.e., dry air and water vapour, with no liquid content) is given by

$$c_p^{dv} \equiv T \left(\frac{\partial \eta}{\partial T} \right)_{p,q} = qc_p^v + (1-q)c_p^d. \quad (18.148)$$

That is, it is the mass weighted heat capacity of water vapour and dry air, as expected. The heat capacity at constant volume is given by

$$c_v^{dv} \equiv T \left(\frac{\partial \eta}{\partial T} \right)_{\alpha,q} = \frac{T(g_{pT}^2 - g_{pp}g_{TT})}{g_{pp}} = qc_v^v + (1-q)c_v^d. \quad (18.149)$$

where $c_v^v = c_p^v - R^v$ and $c_v^d = c_p^d - R^d$, and details are left to the reader. Alternatively, eliminate p in favour of α in (18.143) using (18.142) and then directly evaluate $T(\partial \eta / \partial T)_{\alpha,q}$.

Potential temperature

The potential temperature is the temperature that a parcel has if moved adiabatically and at constant composition to a reference pressure, and so satisfies

$$\eta(p_R, \theta, q) = \eta(p, T, q). \quad (18.150)$$

For moist air we use (18.143) for the entropy and (18.150) becomes

$$\begin{aligned} c_p \ln \theta - \left[qR^v \ln \left(\frac{qp_R/p_0}{q + \epsilon(1-q)} \right) + (1-q)R^d \ln \left(\frac{\epsilon(1-q)p_R/p_0}{q + \epsilon(1-q)} \right) \right] \\ = c_p \ln T - \left[qR^v \ln \left(\frac{qp/p_0}{q + \epsilon(1-q)} \right) + (1-q)R^d \ln \left(\frac{\epsilon(1-q)p/p_0}{q + \epsilon(1-q)} \right) \right]. \end{aligned} \quad (18.151)$$

With a couple of lines of algebra this expression simplifies to

$$c_p \ln \theta - [qR^v + (1 - q)R^d] \ln(p_R/p) = c_p \ln T \quad \text{or} \quad \theta = T \left(\frac{p_R}{p} \right)^{R/c_p}, \quad (18.152)$$

where $R = qR^v + (1 - q)R^d$ and c_p is given by (18.148).

Latent heat of evaporation and condensation

The latent heat of evaporation (or condensation) is the amount of energy that must be supplied to evaporate a unit mass of liquid to a vapour, or equivalently the amount of energy released when water vapour condenses. It is therefore equal to the difference between the specific enthalpy of water vapour and liquid water at a given temperature and also known as the enthalpy of vapourization. Using (18.141) the entropy and enthalpy of liquid water are given by

$$\eta^l = - \left(\frac{\partial g^l}{\partial T} \right)_p = \eta_0^l + c^l \ln(T/T_0), \quad h^l = g^l + T\eta = g_0^l + \alpha^l p + c^l T. \quad (18.153)$$

Using (18.137b) the enthalpy of water vapour is given by

$$h^v = g - T \left(\frac{\partial g^v}{\partial T} \right)_p = c_p^v T + L_0^v. \quad (18.154)$$

The enthalpy of vapourization, L , is therefore given by

$$L = h^v - h^l = L_0^v - (g_0^l + \alpha^l p) + (c_p^v - c^l)T. \quad (18.155)$$

The pressure term is negligibly small: $\alpha^l p \sim 100 \text{ J kg}^{-1}$ whereas, by experiment, $L = 2.501 \times 10^6 \text{ J kg}^{-1}$ at 0°C . Using this value for L_0^v (and setting $g_0^l = 0$), and with $c_p^v = 1859 \text{ J kg}^{-1} \text{ K}^{-1}$, $c^l = 4218 \text{ J kg}^{-1} \text{ K}^{-1}$, we obtain $L \approx (2.501 \times 10^6 - 2359T) \text{ J kg}^{-1}$, with T in Celsius. The temperature dependence of L arises because we have to expend the same amount of energy evaporating water at (say) 10°C and then raising the vapour temperature to 20°C as we do in raising the water temperature from 10° to 20° and then evaporating it.

Using the latent heat provides a convenient expression for the entropy of a saturated mixture of dry air, water vapour and liquid water. If the mixture is in equilibrium then $q^v = q_s^v$ where q_s^v is given by the Clausius–Clapeyron relation and is a function of p and T . Furthermore, the differences in entropy between vapour and liquid water are then related to the latent heat of vapourization by

$$T(\eta^v - \eta^l) = h^v - h^l = L, \quad (18.156)$$

and (18.147) becomes

$$\eta_s^{dvl} = (1 - q^{vl})\eta^d + q^{vl}\eta^l + \frac{Lq_s^v}{T}. \quad (18.157)$$

Equivalent potential temperature

There are various definitions of equivalent potential temperature, θ_{eq} , with different names and small quantitative differences.⁷ For qualitative uses the differences don't really matter as they are small, and for an exact calculation one rarely needs to use equivalent potential temperature anyway (entropy is better). Nevertheless, it is a commonly used thermodynamic variable. The definition used in the main text was that equivalent potential temperature, θ_{eq} , is the potential temperature a parcel reaches after it is lifted adiabatically and at constant composition to level at which it becomes saturated, and then all the water vapour is condensed and all of the latent heat released is used to

heat the parcel. To obtain an analytic expression for θ_{eq} so defined we assume the condensation occurs at constant temperature and so obtain (18.54), namely

$$\theta_{eq}^p = \theta \exp\left(\frac{Lq}{c_p T}\right) = T \left(\frac{p_R}{p}\right)^{R/c_p} \exp\left(\frac{Lq}{c_p T}\right). \quad (18.158)$$

Here we give it a superscript p and call it the pseudo-adiabatic equivalent potential temperature. It is related to entropy, but it is not a true measure of it. One problem is that the above process is not physically realizable. In order to condense all the water vapour the temperature must be taken to absolute zero (or in practice we must lift the parcel to a great height where the temperature is very low), but we cannot simply condense all the water at a constant temperature. Second, we have neglected the contribution of liquid water. To see this, begin with an expression for the entropy of a mixture of dry air, water vapour and liquid water in equilibrium, with an entropy given by (18.157). An equivalent potential temperature, θ_{eq} , may then be defined by

$$(1 - q^{vl})c_p^d \ln(\theta_{eq}/T_0) = (1 - q^{vl})\eta^d + q^{vl}\eta^l + \frac{Lq_s^v}{T}. \quad (18.159)$$

Using the expressions for η^d and η^l given in (18.145) the above expression becomes

$$(1 - q^{vl})c_p^d \ln\left(\frac{\theta_{eq}}{T_0}\right) = (1 - q^{vl})c_p^d \left[\ln\left(\frac{\theta^d}{T_0}\right) - \frac{R^d}{c_p^d} \ln\left(1 - \frac{p^v}{p}\right) \right] + q^{vl}c^l \ln\left(\frac{T}{T_0}\right) + \frac{Lq_s^v}{T}. \quad (18.160)$$

Solving this for θ_{eq} gives

$$\theta_{eq} = \theta^d \exp\left(\frac{Lw_s}{c_p^d T}\right) \left(\frac{T}{T_0}\right)^{w^{vl}c^l/c_p^d} \left(1 - \frac{p^v}{p}\right)^{-R^d/c_p^d}, \quad (18.161)$$

where $w = q/(1 - q^{vl})$ and $\theta^d = T(p_R/p)^{R^d/c_p^d}$. Equation (18.161) is a true measure of the entropy of a moist parcel, albeit a convoluted one. It differs slightly from (18.54) because θ_{eq}^p is not obtained by an adiabatic process, since the entropy of the liquid water is lost during condensation. If the water content is small ($q \ll 1$, $p^v/p \ll 1$) then (18.161) reduces to the more commonly used expression (18.158), with $q = q_s^v$.

Various other temperature-like quantities may be defined, notably the dew-point temperature, the wet-bulb temperature and wet-bulb potential temperature. Such quantities have intuitive and human appeal, and wet-bulb temperature is directly measurable. On the other hand, there is little objective reason to ever use a ‘potential’ temperature in a quantitative calculation — entropy itself is nearly always a more straightforward alternative. This comment also applies to seawater.

18.B APPENDIX B: EQUATIONS OF RADIATIVE TRANSFER

Consider a beam of radiation propagating through a thin slab of gas. Some of the incoming radiation may be absorbed, some may be scattered, and the slab may emit radiation of its own. Scattering is the change in direction of the radiation, so that it may reduce or — if radiation from other directions is scattered into the beam — amplify the beam’s intensity. The difference between the incoming and outgoing radiation is

$$dI_\nu \equiv I_\nu^{\text{out}} - I_\nu^{\text{in}} = -d\tau_\nu I_\nu + dJ_\nu. \quad (18.162)$$

In this expression I_ν is the spectral radiance (power per unit area, per unit solid angle, per unit frequency interval) of the radiation, the term $d\tau_\nu I_\nu$ is the extinction (the absorption plus the radiation scattered away) and dJ_ν is the emission plus the scattering into the beam. The quantity $d\tau_\nu$ is

the nondimensional *optical depth*; it may be written as $d\tau_v = k_v \rho ds$ where ds is the slab thickness, ρ is its density and k_v is the extinction coefficient, a property of the gas in question. The minus sign on $d\tau_v$ is appropriate when τ increases in the direction of the beam, and all of the above quantities depend on the frequency, ν , of the radiation.

Suppose there is no scattering, which is a good approximation for infra-red radiation. The emission of radiation is, in thermal equilibrium, then given by the Planck function, B_v , multiplied by the optical depth. Equation (18.162) becomes

$$dI_v = -d\tau_v(I_v - B_v) \quad \text{or} \quad \frac{dI_v}{d\tau_v} = -(I_v - B_v). \quad (18.163)$$

This equation is the foundation of much of radiative transfer. If radiation is propagating in all directions we must integrate over solid angle to obtain the upward and downward spectral irradiances (power per unit area per unit frequency interval). This is a complicated procedure in general but, to a good approximation for infra-red radiation in Earth's atmosphere, the simple upshot is the multiplication of the optical depth by a geometric, order-one (for example, 5/3) factor of γ — because most of the radiation is passing slantwise through the medium — and the multiplication of B by π because of the integration over a hemisphere, giving the two-stream approximation.²⁵ Equation (18.163) becomes

$$\frac{dF_v}{d\tau_v^*} = -F_v - \pi B_v, \quad (18.164)$$

where F_v is the spectral irradiance along a vertical path along which τ increases and τ_v^* is the 'scaled' optical depth given by $\tau^* = \gamma\tau_v$. We will drop the asterisk on τ_v^* and we will absorb the factor π into the definition of B_v .

In Earth's atmosphere it is common to choose τ increasing downwards, from 0 at the top of the atmosphere, although no physical result depends on this choice. The downwards (D_v) and upwards (U_v) irradiances are then

$$\frac{dD_v}{d\tau_v} = B_v - D_v, \quad \frac{dU_v}{d\tau_v} = U_v - B_v. \quad (18.165a,b)$$

These are the *two-stream equations*, without scattering, commonly known as the *Schwarzschild equations*. The downward and upward fluxes are uncoupled, because of the absence of scattering.

If there is no dependence of the optical depth on frequency then the medium is said to be *grey* and we may integrate (18.165) over frequency to give

$$\frac{dD}{d\tau} = B - D, \quad \frac{dU}{d\tau} = U - B, \quad (18.166a,b)$$

where U and D are the upward and downward (total) irradiances, $B = \sigma T^4$ (all three with units of W m^{-2}), and $\sigma = 5.6704 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$ is Stefan's constant. The grey assumption is not accurate for Earth's atmosphere. Nevertheless, for conceptual or approximate calculations it is often useful to suppose that the atmosphere is grey in the infra-red, in which case (18.166) applies to infra-red radiation with separate equations (that might include scattering or reflection, but that might also be grey) for solar radiation; this is the *semi-grey* approximation.

18.C APPENDIX C: ANALYTIC APPROXIMATION OF TROPOPAUSE HEIGHT

Here we provide an approximate, analytic, expression for the height of the tropopause, given the optical depth and lapse rate. The idea is to solve for a self-consistent radiative-convective state, with a specified lapse rate extending upward to a tropopause and then transitioning to a radiative-equilibrium state, with the tropopause height being determined by the requirement of overall radiative balance.²⁶

Instead of trying to use the formal solutions to the Schwarzschild equations, it is easier to approximately solve the radiative-transfer equations for the upward long wave irradiance, U_L , *ab initio*. We make one other approximation, that the value of B/U_L varies linearly from the tropopause (where its value is 0.5) to its value at the surface (where $B/U_L = 1$). Thus,

$$\frac{B}{U_L} = 1 - \frac{z}{2H_T}. \quad (18.167)$$

Numerical calculations suggest this is a decent approximation. Proceeding, we write (18.65b) as

$$\frac{d \log U_L}{d\tau} = 1 - \frac{B}{U_L} = \frac{z}{2H_T}. \quad (18.168)$$

Using $\tau(z) = \tau_s \exp(-z/H_a)$ we obtain

$$\frac{d \log U_L}{dz} = -\frac{z}{2H_T H_a} \tau_s \exp(-z/H_a). \quad (18.169)$$

We can integrate this expression by parts to obtain a value of the upwelling radiation at the tropopause $U_L(H_T)$, namely

$$\log \left(\frac{U_L(H_T)}{U_L(0)} \right) = -\frac{\tau_s}{2H_T} \int_0^{H_T} \exp(-z/H_a) dz \approx -\frac{\tau_s H_a}{2H_T}. \quad (18.170)$$

for $H_T \gg H_a$. This is an expression for the upwelling longwave radiation, $U_L(H_T)$, and the unknowns in the equation are $U_L(0)$, the upwelling radiation at the surface, and the tropopause height, H_T . the value of $U_L(0)$ is given by the surface temperature, which is a function of the tropopause temperature, T_T , the specified lapse rate, Γ , and the tropopause temperature, T_T . Inverting the argument, if we are given $U_L(H_T)$ and T_T then we can calculate H_T .

Let us assume that the stratosphere is optically thin, in which case $U_L(H_T)$ is the outgoing longwave radiation, equal to $2\sigma T_T^4$, and this known because it equals the incoming solar radiation. Thus, we can in principle now solve (18.170) for H_T . To do this note that the upwelling radiation at the surface is given by $U_L(0) = \sigma T_g^4 = \sigma T_s^4$ where $T_s = T_T + \Gamma H_T$. The left-hand side of (18.170) then becomes

$$\log \left(\frac{2\sigma T_T^4}{\sigma T_s^4} \right) = \log 2 + 4 \log \frac{T_T}{T_s} = \log 2 + 4 \log \left(\frac{T_T}{T_T + \Gamma H_T} \right) \approx \log 2 - \frac{4\Gamma H_T}{T_T}. \quad (18.171)$$

The rightmost terms in (18.171) and (18.170) are approximately equal so that

$$\log 2 - \frac{4\Gamma H_T}{T_T} = -\frac{\tau_s H_a}{2H_T} \quad \text{or} \quad 8\Gamma H_T^2 - C H_T T_T - \tau_s H_a T_T = 0. \quad (18.172)$$

where $C = 2 \log 2 \approx 1.39$. The solution of this equation is

$$H_T = \frac{1}{16\Gamma} \left(C T_T + \sqrt{C^2 T_T^2 + 32\Gamma \tau_s H_a T_T} \right). \quad (18.173)$$

The tropopause height given by this equation is fairly close to the actual solution of the radiative-convective equations, obtained by numerically integrating the Schwarzschild equations and iterating to obtain the correct tropopause height following the algorithm of section 18.6.2, as seen in Fig. 18.21. The analytic approximation may be further improved with a bit of effort, but even in the form above it captures the essential aspects of the true solution.

Notes

- 1 I am grateful to Adam Sobel for a number of conversations and notes that informed the sections on convection and quasi-equilibrium. Will Beeson and his colleagues in Chicago also provided a number of useful comments. Brian Mapes provided a detailed critique of this chapter and although I was unable to address all of his concerns his point of view was salutary.
- 2 I use a formula given by Bolton (1980), which is a variant on the original Magnus formula, *aka* the August–Roche–Magnus formula or the Tetens formula — see Lawrence (2005) for discussion.
- 3 Measurements come from the hybrid advanced microwave sounding unit, Atmospheric Infrared Sounder (AIRS), as in Sherwood *et al.* (2010).
- 4 To read more about models of this form see Pierrehumbert *et al.* (2007), with extensions and applications by O’Gorman *et al.* (2011), Sukhatme & Young (2011), Tsang & Vanneste (2015) and others.
- 5 The simulations here used a model developed by Dr. Yue-Kin Tsang, and I am grateful to him for discussions and help.
- 6 The interested reader might start with Sherwood *et al.* (2010) or Schneider *et al.* (2010) and go forward and back from there.
- 7 This definition of θ_{eq} is sometimes called the ‘pseudo-equivalent potential temperature’, because the condensation product, liquid water, is assumed to fall out of the air parcel and the process is ‘pseudo-adiabatic’, not adiabatic. Other names with slightly different definitions exist (Betts 1973, Emanuel 1994, Ambaum 2010).
- 8 The notion of conditional instability has been with us for many years — it appears in Haurwitz (1941) for example — and an influential form was introduced Ooyama (1963) (see Ooyama 1982) and Charney & Eliassen (1964). They proposed models of a cooperative mechanism between the convection and larger scale flow, with the convection producing a convergence at low levels, leading to ascent, more latent heat release and convection and so on. This mechanism and variations about it became known as ‘Conditional Instability of the Second Kind’, or CISK, to distinguish it from more conventional conditional instability (‘of the first kind’) which does not involve such a feedback with the large scale. The CISK mechanism tends to produce vortical updraughts and may be important for hurricane growth, but this remains a topic of some debate (e.g., Raymond 1995, Emanuel 1994, Smith 1997). Other theories of hurricanes tend to de-emphasize the CISK mechanism in favour of model involving a feedback between wind speed and evaporation, called wind-induced surface heat exchange or WISHE, with stronger winds giving more evaporation, leading to saturation and thence convection. The convection then mainly serves to establish a moist adiabatic lapse rate (as in the quasi-equilibrium ideas of section 18.4.3) which ties the boundary layer to a warm core extending upward (e.g., Craig & Gray 1996).
- 9 Convective adjustment is a great simplification over what actually occurs, but nevertheless it a useful concept and the basis of many early parameterization schemes for numerical models of the atmosphere. Convective adjustment was introduced into modelling by Manabe & Strickler (1964) and a theoretical and observational discussion of the general problem was given by Ludlam (1966). A popular variation (a relaxation rather than an adjustment) was proposed by Betts (1986) and Betts & Miller (1986). These days (c. 2017) GCMs rarely use simple convective adjustment or relaxation schemes, although the underlying ideas endure.
- 10 The nature of a ‘quasi-equilibrium’ between convection and large-scale forcing was made explicit by Betts (1973) and Arakawa & Schubert (1974), with precedents to be found in Scorer & Ludlam (1951) and Ludlam (1966). Many papers have since followed, with, for example, an extended discussion in Emanuel *et al.* (1994), a counterpoint, examples and further discussion in Mapes (1997, 1998, 2000), and application to the boundary layer in Raymond (1997). More references can be found in the reviews by Arakawa (2004) and Emanuel (2007), and debates continue about applicability and efficacy.
- 11 This behaviour is seen in many comprehensive General Circulation Models of the atmosphere, going back to Manabe & Wetherald (1980) and beyond. The explanation given here follows Vallis *et al.* (2015) but has earlier roots.

- 12 Numerical calculations of the radiative constraint with more realistic treatments of radiation were carried out by Thuburn & Craig (2000).
- 13 We roughly follow Emanuel (1987). Neelin & Zeng (2000) and Zeng *et al.* (2000) give details of how a functional reduced model, including diabatic and frictional terms, may be constructed. Lindzen & Nigam (1987) have another, albeit related, take on the problem.
- 14 The tropical scaling was presented by Charney (1963).
- 15 The derivation of the weak temperature-gradient approximation given here is more-or-less that of Sobel *et al.* (2001), and may be regarded as an extension of Charney's (1963) ideas to include diabatic effects. A number of authors previously used the approximation, implicitly or explicitly, in one form or another (e.g., Neelin 1988, Browning *et al.* 2000), and various extensions and rigour have been added by Majda & Klein (2003) and others.
- 16 See Bretherton & Smolarkiewicz (1989) and Mapes (1997) and go from there.
- 17 Details are described in Bretherton & Sobel (2002) and Neelin & Zeng (2000).
- 18 A review of convectively coupled equatorial waves is provided by Kiladis *et al.* (2009). The MJO was first described by Madden & Julian (1971, 1972), and a review may be found in Zhang (2005). Schubert & Masarik (2006) discuss aspects of a moving Matsuno–Gill pattern and its relation to the MJO, and Raymond & Fuchs (2009), Majda & Stechmann (2009) and Sobel & Maloney (2013), among others, offer theoretical models of the MJO.
- 19 Diagrams such as these are known as Wheeler–Kiladis diagrams, after Wheeler & Kiladis (1999). Filtering the noise is required to obtain clean plots and requires some attention.
- 20 Adapted from Kiladis *et al.* (2009).
- 21 Dias & Kiladis (2014) discuss reasons why the value is smaller than might be expected.
- 22 Adapted from Kiladis *et al.* (2005).
- 23 Heckley & Gill (1984).
- 24 Feistel *et al.* (2010) and Thuburn (2017) give more derivations. I am grateful to John Thuburn for a number of conversations on this matter.
- 25 There are various versions of the two-stream approximation; see Goody & Yung (1995) or Pierrehumbert (2010).
- 26 The contents of this appendix are the results of a collaboration with Prof. Pablo Zurita-Gotor.

Part IV

LARGE-SCALE OCEANIC CIRCULATION

References

- Abarbanel, H. D. I. & Young, W. R., Eds., 1987. *General Circulation of the Ocean*. Springer-Verlag, 291 pp.
- Abbe, C., 1901. The physical basis of long-range weather forecasts. *Mon. Wea. Rev.*, **29**, 551–561.
- Ablowitz, M. J., 2011. *Nonlinear dispersive waves: asymptotic analysis and solitons*. Vol. 47, Cambridge University Press, 311 pp.
- Abramowitz, M. & Stegun, I. A., 1965. *Handbook of Mathematical Functions*. Dover Publications, 1046 pp.
- Allen, J. S., 1993. Iterated geostrophic intermediate models. *J. Phys. Oceanogr.*, **23**, 2447–2461.
- Allen, J. S., Barth, J. A. & Newberger, P. A., 1990a. On intermediate models for barotropic continental shelf and slope flow fields. Part I: formulation and comparison of exact solutions. *J. Phys. Oceanogr.*, **20**, 1017–1042.
- Allen, J. S., Barth, J. A. & Newberger, P. A., 1990b. On intermediate models for barotropic continental shelf and slope flow fields. Part II: comparison of numerical model solutions in doubly periodic domains. *J. Phys. Oceanogr.*, **20**, 1043–1082.
- Allen, J. S., Holm, D. D. & Newberger, P. A., 2002. Extended-geostrophic Euler–Poincaré models for mesoscale oceanographic flow. In J. Norbury & I. Roulstone, Eds., *Large-scale Atmosphere–Ocean Dynamics I*. Cambridge University Press.
- Ambaum, M. H. P., 2010. *Thermal Physics of the Atmosphere*. Wiley, 239 pp.
- Andrews, D. G., 1987. On the interpretation of the Eliassen–Palm flux divergence. *Quart. J. Roy. Meteor. Soc.*, **113**, 323–338.
- Andrews, D. G., 2010. *An Introduction to Atmospheric Physics*. Cambridge University Press, 237 pp.
- Andrews, D. G., Holton, J. R. & Leovy, C. B., 1987. *Middle Atmosphere Dynamics*. Academic Press, 489 pp.
- Andrews, D. G. & McIntyre, M. E., 1976. Planetary waves in horizontal and vertical shear: the generalized Eliassen–Palm relation and the mean zonal acceleration. *J. Atmos. Sci.*, **33**, 2031–2048.
- Andrews, D. G. & McIntyre, M. E., 1978. Generalized Eliassen–Palm and Charney–Drazin theorems for waves on axisymmetric mean flows in compressible atmospheres. *J. Atmos. Sci.*, **35**, 175–185.
- Angell, J. K. & Korshover, J., 1964. Quasi-biennial variations in temperature, total ozone, and tropopause height. *J. Atmos. Sci.*, **21**, 479–492.
- Arakawa, A., 2004. The cumulus parameterization problem: Past, present, and future. *J. Climate*, **17**, 13, 2493–2525.
- Arakawa, A. & Schubert, W. H., 1974. Interaction of a cumulus cloud ensemble with the large-scale environment, part i. *J. Atmos. Sci.*, **31**, 3, 674–701.
- Arbic, B. K., Flierl, G. R. & Scott, R. B., 2007. Cascade inequalities for forced-dissipated geostrophic turbulence. *J. Phys. Oceanogr.*, **37**, 6, 1470–1487.
- Arnold, V. I., 1965. Conditions for nonlinear stability of stationary plane curvilinear flows of an ideal fluid. *Dokl. Akad. Nauk SSSR*, **162**, 975–978. Engl. transl.: *Sov. Math.* **6**, 773–777 (1965).
- Arnold, V. I., 1966. On an a priori estimate in the theory of hydrodynamic stability. *Izv. Vyssh. Uchebn. Zaved. Math.*, **54**, 3–5. Engl. transl.: *Am Mat. Soc. Transl. Ser.*, **2**, **79**, 267–289 (1969).

- Assmann, R., 1902. Über die Existenz eines wärmeren Luftstromes in der Höhe von 10 bis 15 km. (On the existence of a warmer airflow at heights from 10 to 15 km). *Sitzber. Königl. Preuss. Akad. Wiss. Berlin*, **24**, 495–504.
- Aubin, D. & Dahan Dalmedico, A., 2002. Writing the history of dynamical systems and chaos: *longue durée* and revolution, disciplines and cultures. *Historia Mathematica*, **29**, 1–67.
- Baldwin, M. P., Gray, L. J., Dunkerton, T. J., Hamilton, K. *et al.*, 2001. The quasi-biennial oscillation. *Rev. Geophys.*, **39**, 179–229.
- Ball, J. M. & James, R. D., 2002. The scientific life and influence of Clifford Ambrose Truesdell III. *Arch. Rational Mech. Anal.*, **161**, 1–26.
- Bannon, P. R., 1995. Potential vorticity conservation, hydrostatic adjustment, and the anelastic approximation. *J. Atmos. Sci.*, **52**, 2301–2312.
- Bannon, P. R., 1996. On the anelastic equation for a compressible atmosphere. *J. Atmos. Sci.*, **53**, 3618–3628.
- Bartello, P. & Warn, T., 1996. Self-similarity of decaying two-dimensional turbulence. *J. Fluid Mech.*, **326**, 357–372.
- Batchelor, G. K., 1953a. The conditions for dynamical similarity of motions of a frictionless perfect-gas atmosphere. *Quart. J. Roy. Meteor. Soc.*, **79**, 224–235.
- Batchelor, G. K., 1953b. *The Theory of Homogeneous Turbulence*. Cambridge University Press, 197 pp.
- Batchelor, G. K., 1959. Small-scale variation of convected quantities like temperature in turbulent fluid. Part 1: general discussion and the case of small conductivity. *J. Fluid Mech.*, **5**, 113–133.
- Batchelor, G. K., 1967. *An Introduction to Fluid Dynamics*. Cambridge University Press, 615 pp.
- Batchelor, G. K., 1969. Computation of the energy spectrum in homogeneous two-dimensional turbulence. *Phys. Fluids Suppl.*, **12**, II–233–II–239.
- Batchelor, G. K., Howells, I. D. & Townsend, A. A., 1959. Small-scale variation of convected quantities like temperature in turbulent fluid. Part 2: the case of large conductivity. *J. Fluid Mech.*, **5**, 134–139.
- Batchelor, G. K. & Townsend, A. A., 1956. Turbulent diffusion. In G. K. Batchelor & R. M. Davies, Eds., *Surveys in Mechanics*, pp. 352–399. Cambridge University Press.
- Battisti, D. S., 1988. Dynamics and thermodynamics of a warming event in a coupled tropical atmosphere-ocean model. *J. Atmos. Sci.*, **45**, 20, 2889–2919.
- Baumert, H. Z., Simpson, J. & Sündermann, J., Eds., 2005. *Marine Turbulence: Theories, Observations and Models*. Cambridge University Press, 630 pp.
- Bell, M. J., 2015a. Meridional overturning circulations driven by surface wind and buoyancy forcing. *J. Phys. Oceanogr.*, **45**, 11, 2701–2714.
- Bell, M. J., 2015b. Water mass transformations driven by ekman upwelling and surface warming in subpolar gyres. *J. Phys. Oceanogr.*, **45**, 9, 2356–2380.
- Bender, C. M. & Orszag, S. A., 1978. *Advanced Mathematical Methods for Scientists and Engineers*. McGraw-Hill, 593 pp.
- Berrisford, P., Marshall, J. C. & White, A. A., 1993. Quasi-geostrophic potential vorticity in isentropic coordinates. *Quart. J. Roy. Meteor. Soc.*, **50**, 778–782.
- Betts, A. K., 1973. Non-precipitating convection and its parameterization. *Quart. J. Roy. Meteor. Soc.*, **99**, 178–196.
- Betts, A. K., 1986. A new convective adjustment scheme. Part I: Observational and theoretical basis. *Quart. J. Roy. Meteor. Soc.*, **112**, 677–691.
- Betts, A. K. & Miller, M. J., 1986. A new convective adjustment scheme. Part II: Single column test using GATE wave, BOMEX, ATEX and Arctic air mass data sets. *Quart. J. Roy. Meteor. Soc.*, **112**, 693–709.
- Birner, T., 2006. Fine-scale structure of the extratropical tropopause region. *J. Geophys. Res.*, **111**, D04104.
- Birner, T., Dörnbrack, A. & Schumann, U., 2002. How sharp is the tropopause at midlatitudes? *Geophys. Res. Lett.*, **29**, 1700.
- Bjerknes, J., 1919. On the structure of moving cyclones. *Geofys. Publ.*, **1** (2), 1–8.
- Bjerknes, J., 1937. Die Theorie der aussertropischen Zyklonenbildung (The theory of extra-tropical cyclone formation). *Meteor. Zeitschr.*, **12**, 460–466.
- Bjerknes, J., 1959. Atlantic air–sea interaction. In *Advances in Geophysics*, Vol. 10, pp. 1–82. Academic Press.
- Bjerknes, J., 1969. Atmospheric teleconnections from the equatorial Pacific. *Mon. Wea. Rev.*, **97**, 163–172.

- Bjerknes, V., 1898a. Über die Bildung von Cirkulationsbewegungen und Wirbeln in reibungslosen Flüssigkeiten (On the generation of circulation and vortices in inviscid fluids). *Skr. Nor. Vidensk.-Akad. 1: Mat.-Naturvidensk. Kl.*, **5**, 3–29.
- Bjerknes, V., 1898b. Über einen hydrodynamischen Fundamentalsatz und seine Anwendung besonders auf die Mechanik der Atmosphäre und des Weltmeeres (On a fundamental principle of hydrodynamics and its application particularly to the mechanics of the atmosphere and the world's oceans). *Kongl. Sven. Vetensk. Akad. Handlingar*, **31**, 1–35.
- Bjerknes, V., 1902. Circulation relativ zu der Erde (Circulation relative to the Earth). *Meteor. Z.*, **37**, 97–108.
- Bjerknes, V., 1904. Das Problem der Wettervorhersage, betrachtet vom Standpunkte der Mechanik und der Physik (The problem of weather forecasting as a problem in mathematics and physics). *Meteor. Z.*, January, 1–7. Engl. transl.: Y. Mintz, in Shapiro and Grønas (1999), pp. 1–7.
- Blumen, W., 1968. On the stability of quasi-geostrophic flow. *J. Atmos. Sci.*, **25**, 929–933.
- Boccaletti, G., Pacanowski, R. C., Philander, S. G. H. & Fedorov, A. V., 2004. The thermal structure of the upper ocean. *J. Phys. Oceanogr.*, **34**, 888–902.
- Boer, G. J. & Shepherd, T. G., 1983. Large-scale two-dimensional turbulence in the atmosphere. *J. Atmos. Sci.*, **40**, 164–184.
- Boffetta, G. & Ecke, R. E., 2012. Two-dimensional turbulence. *Ann. Rev. Fluid Mech.*, **44**, 427–451.
- Bohren, C. F., Albrecht, B. A. & Schroeder, D. V., 2000. Atmospheric thermodynamics. *American Journal of Physics*, **68**, 12, 1159–1160.
- Bolton, D., 1980. The computation of equivalent potential temperature. *Mon. Wea. Rev.*, **108**, 7, 1046–1053.
- Booker, J. R. & Bretherton, F. P., 1967. The critical layer for internal gravity waves in a shear flow. *J. Fluid Mech.*, **27**, 03, 513–539.
- Boussinesq, J., 1903. Théorie analytique de la chaleur (Analytic theory of heat). *Tome, Paris, Gauthier-Villars*, **II**, 170–172.
- Boyd, J. P., 1976. The noninteraction of waves with the zonally averaged flow on a spherical Earth and the interrelationships of eddy fluxes of energy, heat and momentum. *J. Atmos. Sci.*, **33**, 2285–2291.
- Boyd, J. P., 1978. The effects of latitudinal shear on equatorial waves. Part 1. Theory and method. *J. Atmos. Sci.*, **35**, 2236–2258.
- Boyd, J. P., 1980. The nonlinear equatorial Kelvin wave. *J. Phys. Oceanogr.*, **10**, 1, 1–11.
- Branscome, L. E., 1983. The Charney baroclinic stability problem: approximate solutions and modal structures. *J. Atmos. Sci.*, **40**, 1393–1409.
- Bretherton, C. S. & Schär, C., 1993. Flux of potential vorticity substance: a simple derivation and a uniqueness property. *J. Atmos. Sci.*, **50**, 1834–1836.
- Bretherton, C. S. & Smolarkiewicz, P. K., 1989. Gravity waves, compensating subsidence and detrainment around cumulus clouds. *J. Atmos. Sci.*, **46**, 740–759.
- Bretherton, C. S. & Sobel, A. H., 2002. A simple model of a convectively coupled Walker circulation using the weak temperature gradient approximation. *J. Climate*, **15**, 2907–2920.
- Bretherton, C. S. & Sobel, A. H., 2003. The Gill model and the weak temperature gradient approximation. *J. Atmos. Sci.*, **60**, 451–460.
- Bretherton, F. P., 1964. Low frequency oscillations trapped near the equator. *Tellus*, **16**, 181–185.
- Bretherton, F. P., 1966a. Baroclinic instability and the short wavelength cut-off in terms of potential vorticity. *Quart. J. Roy. Meteor. Soc.*, **92**, 335–345.
- Bretherton, F. P., 1966b. Critical layer instability in baroclinic flows. *Quart. J. Roy. Meteor. Soc.*, **92**, 325–334.
- Bretherton, F. P., 1969. Momentum transport by gravity waves. *Quart. J. Roy. Meteor. Soc.*, **95**, 404, 213–243.
- Brewer, A. W., 1949. Evidence for a world circulation provided by the measurements of helium and water vapour distribution in the stratosphere. *Quart. J. Roy. Meteor. Soc.*, **75**, 251–363.
- Brillouin, L., 1926. La mécanique ondulatoire de Schrödinger; une méthode générale de résolution par approximations successives (The wave mechanics of Schrödinger: a general method of solution by successive approximation). *Comptes Rendus*, **183**, 24–26.
- Browning, G., Kreiss, H. & Schubert, W., 2000. The role of gravity waves in slowly varying in time tropospheric motions near the equator. *J. Atmos. Sci.*, **57**, 24, 4008–4019.
- Bryan, F., 1986. High-latitude salinity effects and interhemispheric thermohaline circulations. *Nature*, **323**, 301–304.

- Bryden, H. & Brady, E. C., 1985. Diagnostic study of the three-dimensional circulation of the upper equatorial Pacific Ocean. *J. Phys. Oceanogr.*, **15**, 1255–1273.
- Buchanan, J. Y., 1886. On the similarities in the physical geography of the great oceans. *Proc. Roy. Geogr. Soc.*, **8**, 753–770.
- Bühler, O., 2009. *Waves and mean flows*. Cambridge Univ. Press, 370 pp. ISBN: 978-0521866361.
- Burger, A., 1958. Scale considerations of planetary motions of the atmosphere. *Tellus*, **10**, 195–205.
- Burke, A., Stewart, A. L., Adkins, J. F., Ferrari, R. *et al.*, 2015. The glacial mid-depth radiocarbon bulge and its implications for the overturning circulation. *Paleoceanog.*, **30**, 7, 1021–1039.
- Callen, H. B., 1985. *Thermodynamics and an Introduction to Thermostatistics*. John Wiley & Sons, 493 pp.
- Cane, M. A., 1979a. The response of an equatorial ocean to simple wind stress patterns: I. Model formulation and analytic results. *J. Mar. Res.*, **37**, 232–252.
- Cane, M. A., 1979b. The response of an equatorial ocean to simple wind stress patterns: II. Numerical results. *J. Mar. Res.*, **37**, 355–398.
- Cane, M. A. & Zebiak, S. E., 1985. A theory for El Niño and the Southern Oscillation. *Science*, **228**, 1084–1087.
- Carillo, C. N., 1892. Desertacion sobre las corrientes y estudios de la corriente Peruana de Humboldt. (Dissertation on currents and studies of the Peruvian Humboldt current). *Bol. Soc. Geogr. Lima*, **11**, 72–110.
- Carlini, F., 1837. Ricerche sulla convergenza della serie che serve alla soluzione del problema di Keplero (Research on the convergence of series for the solution of Kepler's problem). Milan.
- Cessi, P., 2001. Thermohaline circulation variability. In *Conceptual Models of the Climate, Woods Hole Program in Geophysical Fluid Dynamics* (2001). Also available from <http://gfd.whoi.edu/proceedings/2001/PDFvol2001.html>.
- Cessi, P. & Fantini, M., 2004. The eddy-driven thermocline. *J. Phys. Oceanogr.*, **34**, 2642–2658.
- Cessi, P. & Young, W. R., 1992. Multiple equilibria in two-dimensional thermohaline circulation. *J. Fluid Mech.*, **241**, 291–309.
- Chai, J., 2016. Understanding geostrophic turbulence in a hierarchy of models. Ph.D thesis, Princeton University.
- Chandrasekhar, S., 1961. *Hydrodynamic and Hydromagnetic Stability*. Oxford University Press, 652 pp. Reprinted by Dover Publications, 1981.
- Chang, E. K. M. & Orlanski, I., 1994. On energy flux and group velocity of waves in baroclinic flows. *J. Atmos. Sci.*, **51**, 24, 3823–3828.
- Chang, P., Ji, L., Li, H. & Flügel, M., 1996. Chaotic dynamics versus stochastic processes in el nino-southern oscillation in coupled ocean-atmosphere models. *Physica D*, **98**, 2, 301–320.
- Chapman, D. C., Malanotte-Rizzoli, P. & Hendershott, M., 1989. Wave motions in the ocean. Unpublished notes based on lectures by Myrl Hendershott.
- Chapman, S. & Lindzen, R. S., 1970. *Atmospheric Tides*. Gordon and Breach, 200 pp.
- Charlton, A. J. & Polvani, L. M., 2007. A new look at stratospheric sudden warmings. Part I: Climatology and modeling benchmarks. *J. Climate*, **20**, 449–469.
- Charlton, A. J., Polvani, L. M., Perlwitz, J., Sassi, F. *et al.*, 2007. A new look at stratospheric sudden warmings. Part II: Evaluation of numerical model simulations. *J. Climate*, **20**, 470–488.
- Charney, J. G., 1947. The dynamics of long waves in a baroclinic westerly current. *J. Meteor.*, **4**, 135–162.
- Charney, J. G., 1948. On the scale of atmospheric motion. *Geofys. Publ. Oslo*, **17** (2), 1–17.
- Charney, J. G., 1955. The Gulf Stream as an inertial boundary layer. *Proc. Nat. Acad. Sci.*, **41**, 731–740.
- Charney, J. G., 1960. Non-linear theory of a wind-driven homogeneous layer near the equator. *Deep-Sea Res.*, **6**, 303–310.
- Charney, J. G., 1963. A note on large-scale motions in the tropics. *J. Atmos. Sci.*, **20**, 6, 607–609.
- Charney, J. G., 1971. Geostrophic turbulence. *J. Atmos. Sci.*, **28**, 1087–1095.
- Charney, J. G. & Drazin, P. G., 1961. Propagation of planetary scale disturbances from the lower into the upper atmosphere. *J. Geophys. Res.*, **66**, 83–109.
- Charney, J. G. & Eliassen, A., 1949. A numerical method for predicting the perturbations of the mid-latitude westerlies. *Tellus*, **1**, 38–54.
- Charney, J. G. & Eliassen, A., 1964. On the growth of the hurricane depression. *J. Atmos. Sci.*, **21**, 68–75.

- Charney, J. G., Fjørtoft, R. & Neumann, J. V., 1950. Numerical integration of the barotropic vorticity equation. *Tellus*, **2**, 237–254.
- Charney, J. G. & Stern, M. E., 1962. On the stability of internal baroclinic jets in a rotating atmosphere. *J. Atmos. Sci.*, **19**, 159–172.
- Charnock, H., Green, J., Ludlam, F., Scorer, R. & Sheppard, P., 1966. Dr. E. T. Eady, B. A. (Obituary). *Quart. J. Roy. Meteor. Soc.*, **92**, 591–592.
- Chasnov, J. R., 1991. Simulation of the inertial-conductive subrange. *Phys. Fluids A*, **3**, 1164–1168.
- Chelton, D. B., de Szoeke, R. A., Schlax, M. G., Naggar, K. E. & Siwertz, N., 1998. Geographical variability of the first-baroclinic Rossby radius of deformation. *J. Phys. Oceanogr.*, **28**, 433–460.
- Christiansen, B., 1999. Stratospheric vacillations in a general circulation model. *J. Atmos. Sci.*, **56**, 1858–1872.
- Christiansen, B., 2000. Chaos, quasiperiodicity, and interannual variability: studies of a stratospheric vacillation model. *J. Atmos. Sci.*, **57**, 3161–3173.
- Clarke, A. J., 2007. *Dynamics of El Niño and the Southern Oscillation*. Academic Press, 308 pp.
- Clarke, A. J., 2008. *An Introduction to the Dynamics of El Niño and the Southern Oscillation*. Elsevier, 308 pp.
- Clarke, A. J., Van Gorder, S. & Colantuono, G., 2007. Wind stress curl and ENSO discharge/recharge in the equatorial Pacific. *J. Phys. Oceanogr.*, **37**, 4, 1077–1091.
- Colin de Verdière, A., 1980. Quasi-geostrophic turbulence in a rotating homogeneous fluid. *Geophys. Astrophys. Fluid Dyn.*, **15**, 213–251.
- Colin de Verdière, A., 1989. On the interaction of wind and buoyancy driven gyres. *J. Mar. Res.*, **47**, 595–633.
- Conkright, M. E., Antonov, J., Baranova, O., Boyer, T. P. et al., 2001. World ocean database 2001, vol. 1. In S. Levitus, Ed., *NOAA Atlas NESDIS 42*, pp. 167. US Government Printing Office, Washington D. C.
- Coriolis, G. G., 1832. Mémoire sur le principe des forces vives dans les mouvements relatifs des machines (On the principle of kinetic energy in the relative movement of machines). *J. Ec. Polytech*, **13**, 268–301.
- Coriolis, G. G., 1835. Mémoire sur les équations du mouvement relatif des systèmes de corps (On the equations of relative motion of a system of bodies). *J. Ec. Polytech*, **15**, 142–154.
- Corrsin, S., 1951. On the spectrum of isotropic temperature fluctuations in an isotropic turbulence. *J. Appl. Phys.*, **22**, 469–473. Erratum: *J. Appl. Phys.*, **22**, 1292, (1951).
- Craig, G. C. & Gray, S. L., 1996. CISK or WISHE as the mechanism for tropical cyclone intensification. *J. Atmos. Sci.*, **53**, 23, 3528–3540.
- Cressman, G. P., 1996. The origin and rise of numerical weather prediction. In J. R. Fleming, Ed., *Historical Essays on Meteorology 1919–1995*, pp. 617. American Meteorological Society.
- Cromwell, T., 1953. Circulation in a meridional plane in the central equatorial Pacific. *J. Mar. Res.*, **12**, 196–213.
- Danielsen, E. F., 1990. In defense of Ertel's potential vorticity and its general applicability as a meteorological tracer. *J. Atmos. Sci.*, **47**, 2353–2361.
- Danilov, S. & Gryanik, V., 2002. Rhines scale and spectra of the β -plane turbulence with bottom drag. *Phys. Rev. E*, **65**, 067301–1–067301–3.
- Danilov, S. & Gurarie, D., 2001. Quasi-two-dimensional turbulence. *Usp. Fiz. Nauk.*, **170**, 921–968.
- Davidson, P., 2015. *Turbulence: An Introduction for Scientists and Engineers*. Oxford University Press, 688 pp. ISBN:978-0198722595.
- Davies-Jones, R., 2003. Comments on “A generalization of Bernoulli's theorem”. *J. Atmos. Sci.*, **60**, 2039–2041.
- Davis, R. E., de Szoeke, R., Halpern, D. & Niiler, P., 1981. Variability in the upper ocean during MILE. Part I: The heat and momentum balances. *Deep-Sea Res.*, **28**, 1427–1452.
- De Morgan, A., 1872. *A Budget of Paradoxes*. Thoemmes Continuum, 814 pp.
- de Szoeke, R. & Bennett, A. F., 1993. Microstructure fluxes across density surfaces. *J. Phys. Oceanogr.*, **24**, 2254–2264.
- de Szoeke, R. A., 2000. Equations of motion using thermodynamic coordinates. *J. Phys. Oceanogr.*, **30**, 2814–2829.
- de Szoeke, R. A., 2004. An effect of the thermobaric nonlinearity of the equation of state: A mechanism for sustaining solitary Rossby waves. *J. Phys. Oceanogr.*, **34**, 2042–2056.
- Dee, D., Uppala, S., Simmons, A., Berrisford, P. et al., 2011. The ERA-interim reanalysis: Configuration and performance of the data assimilation system. *Quart. J. Roy. Meteor. Soc.*, **137**, 656, 553–597.

- Defant, A., 1921. Die Zirkulation der Atmosphäre in den gemäßigten Breiten der Erde. Grundzüge einer Theorie der Klimaschwankungen (The circulation of the atmosphere in the Earth's midlatitudes. Basic features of a theory of climate fluctuations). *Geograf. Ann.*, **3**, 209–266.
- Dewar, W. K. & Huang, R. X., 1995. Fluid flow in loops driven by freshwater and heat fluxes. *J. Fluid Mech.*, **297**, 153–191.
- Dewar, W. K., Samelson, R. S. & Vallis, G. K., 2005. The ventilated pool: a model of subtropical mode water. *J. Phys. Oceanogr.*, **35**, 137–150.
- Dias, J. & Kiladis, G. N., 2014. Influence of the basic state zonal flow on convectively coupled equatorial waves. *Geophys. Res. Lett.*, **41**, 19, 6904–6913.
- Dickinson, R. E., 1968. Planetary Rossby waves propagating vertically through weak westerly wind wave guides. *J. Atmos. Sci.*, **25**, 984–1002.
- Dickinson, R. E., 1969. Theory of planetary wave–zonal flow interaction. *J. Atmos. Sci.*, **26**, 73–81.
- Dickinson, R. E., 1980. Planetary waves: theory and observation. In *Orographic Effects on Planetary Flows*, Number 23 in GARP Publication Series. World Meteorological Organization.
- Dijkstra, H. A., 2008. *Dynamical Oceanography*. Springer, 407 pp.
- Dima, I. & Wallace, J. M., 2003. On the seasonality of the Hadley Cell. *J. Atmos. Sci.*, **60**, 1522–1527.
- Dobson, G. M. B., 1956. Origin and distribution of the polyatomic molecules in the atmosphere. *Proc. Roy. Soc. Lond. A*, **236**, 187–193.
- Döös, K. & Coward, A., 1997. The Southern Ocean as the major upwelling zone of the North Atlantic. *Int. WOCE Newsletter* 27, 3–4.
- Drazin, P. G. & Reid, W. H., 1981. *Hydrodynamic Stability*. Cambridge University Press, 527 pp.
- Drijfhout, S. S. & Hazeleger, W., 2001. Eddy mixing of potential vorticity versus temperature in an isopycnic ocean model. *J. Phys. Oceanogr.*, **31**, 481–505.
- Dritschel, D. & McIntyre, M., 2008. Multiple jets as PV staircases: The Phillips effect and the resilience of eddy-transport barriers. *J. Atmos. Sci.*, **65**, 855–874.
- Dunkerton, T., Hsu, C.-P. & McIntyre, M. E., 1981. Some eulerian and lagrangian diagnostics for a model stratosphere warming. *JAS*, **38**, 819–843.
- Dunkerton, T. J., 1980. A Lagrangian-mean theory of wave, mean-flow interaction with applications to non-acceleration and its breakdown. *Rev. Geophys. Space Phys.*, **18**, 387–400.
- Dunkerton, T. J., 1982. Shear zone asymmetry in the observed and simulated quasi-biennial oscillation. *J. Atmos. Sci.*, **38**, 461–469.
- Dunkerton, T. J., 1997. The role of gravity waves in the quasi-biennial oscillation. *JGR*, **102**, 26053–26076.
- Dunkerton, T. J., Delisi, D. P. & Baldwin, M. P., 1988. Distribution of major stratospheric warmings in relation to the quasi-biennial oscillation. *Geophys. Res. Lett.*, **15**, 136–139.
- Durrán, D. R., 1989. Improving the anelastic approximation. *J. Atmos. Sci.*, **46**, 1453–1461.
- Durrán, D. R., 1990. Mountain waves and downslope winds. *Meteor. Monogr.*, **23**, 59–81.
- Durrán, D. R., 1993. Is the Coriolis force really responsible for the inertial oscillations? *Bull. Am. Meteor. Soc.*, **74**, 2179–2184.
- Durrán, D. R., 2015. Lee waves and mountain waves. *Encycl. Atmos. Sci.*, 2nd edn, **4**, 95–102.
- Durst, C. S. & Sutcliffe, R. C., 1938. The effect of vertical motion on the “geostrophic departure” of the wind. *Quart. J. Roy. Meteor. Soc.*, **64**, 240.
- Dutton, J. A., 1986. *The Ceaseless Wind: An Introduction to the Theory of Atmospheric Motion*. Dover Publications, 617 pp.
- Eady, E. T., 1949. Long waves and cyclone waves. *Tellus*, **1**, 33–52.
- Eady, E. T., 1950. The cause of the general circulation of the atmosphere. In *Cent. Proc. Roy. Meteor. Soc.* (1950), pp. 156–172.
- Eady, E. T., 1954. The maintenance of the mean zonal surface currents. *Proc. Toronto Meteor. Conf.* 1953, **138**, 124–128. Royal Meteorological Society.
- Eady, E. T., 1957. The general circulation of the atmosphere and oceans. In D. R. Bates, Ed., *The Earth and its Atmosphere*, pp. 130–151. New York, Basic Books.
- Eady, E. T. & Sawyer, J. S., 1951. Dynamics of flow patterns in extra-tropical regions. *Quart. J. Roy. Meteor. Soc.*, **77**, 531–551.

- Ebdon, R. A., 1960. Notes on the wind flow at 50 mb in tropical and subtropical regions in January 1957 and in 1960. *Quart. J. Roy. Meteor. Soc.*, **86**, 540–542.
- Eden, C. & Willebrand, J., 1999. Neutral density revisited. *Deep Sea Res., Part II*, **46**, 33–54.
- Edmon, H. J., Hoskins, B. J. & McIntyre, M. E., 1980. Eliassen–Palm cross sections for the troposphere. *J. Atmos. Sci.*, **37**, 2600–2616.
- Egger, J., 1976. Linear response of a two-level primitive equation model to forcing by topography. *Mon. Wea. Rev.*, **104**, 351–364.
- Ekman, V. W., 1905. On the influence of the Earth's rotation on ocean currents. *Arch. Math. Astron. Phys.*, **2**, 1–52.
- Eliassen, A. & Palm, E., 1961. On the transfer of energy in stationary mountain waves. *Geofys. Publ.*, **22**, 1–23.
- Eluszkiewicz, J., Crisp, D., Grainger, R. G., Lambert, A. *et al.*, 1997. Sensitivity of the residual circulation diagnosed from the UARS data to the uncertainties in the input fields and to the inclusion of aerosols. *J. Atmos. Sci.*, **54**, 1739–1757.
- Emanuel, K., 2007. Quasi-equilibrium dynamics of the tropical atmosphere. In T. Schneider & A. Sobel, Eds., *The Global Circulation of the Atmosphere: Phenomena, Theory, Challenges*, pp. 143–185. Princeton University Press.
- Emanuel, K. A., 1987. An air-sea interaction model of intraseasonal oscillations in the tropics. *J. Atmos. Sci.*, **44**, 16, 2324–2340.
- Emanuel, K. A., 1994. *Atmospheric Convection*. Oxford University Press, 580 pp.
- Emanuel, K. A., 2011. Edward Norton Lorenz, 1917–2008. In *Biog. Memoirs*, pp. 1–28. Nat'l. Acad. Sci.
- Emanuel, K. A., Neelin, J. D. & Bretherton, C. S., 1994. On large-scale circulations in convecting atmospheres. *Quart. J. Roy. Meteor. Soc.*, **120**, 1111–1143.
- Er-El, J. & Peskin, R., 1981. Relative diffusion of constant-level balloons in the Southern Hemisphere. *J. Atmos. Sci.*, **38**, 2264–2274.
- Ertel, H., 1942a. Ein neuer hydrodynamischer Wirbelsatz (A new hydrodynamic eddy theorem). *Meteorol. Z.*, **59**, 277–281.
- Ertel, H., 1942b. Über des Verhältnis des neuen hydrodynamischen Wirbelsatzes zum Zirkulationssatz von V. Bjerknes (On the relationship of the new hydrodynamic eddy theorem to the circulation theorem of V. Bjerknes). *Meteorol. Z.*, **59**, 385–387.
- Ertel, H. & Rossby, C.-G., 1949a. Ein neuer Erhaltungssatz der Hydrodynamik (A new conservation theorem of hydrodynamics). *Sitzungsber. d. Deutschen Akad. Wissenschaften Berlin*, **1**, 3–11.
- Ertel, H. & Rossby, C.-G., 1949b. A new conservation theorem of hydrodynamics. *Geofis. Pura Appl.*, **14**, 189–193.
- Fang, M. & Tung, K. K., 1996. A simple model of nonlinear Hadley circulation with an ITCZ: analytic and numerical solutions. *J. Atmos. Sci.*, **53**, 1241–1261.
- Fang, M. & Tung, K. K., 1999. Time-dependent nonlinear Hadley circulation. *J. Atmos. Sci.*, **56**, 1797–1807.
- Farrell, B., 1984. Modal and non-modal baroclinic waves. *J. Atmos. Sci.*, **41**, 668–673.
- Farrell, B. F. & Ioannou, P. J., 1996. Generalized stability theory. Part I: autonomous operators. *J. Atmos. Sci.*, **53**, 2025–2040.
- Farrell, B. F. & Ioannou, P. J., 2008. Formation of jets by baroclinic turbulence. *J. Atmos. Sci.*, **65**, 11, 3353–3375.
- Feistel, R., 2008. A Gibbs function for seawater thermodynamics for -6°C to 80°C and salinity up to 120 g kg^{-1} . *Deep Sea Res. Part I*, **55**, 12, 1639–1671.
- Feistel, R., Wright, D., Kretzschmar, H.-J., Hagen, E. *et al.*, 2010. Thermodynamic properties of sea air. *Oce. Sci.*, **6**, 91–141.
- Ferrari, R., Jansen, M. F., Adkins, J. F., Burke, A. *et al.*, 2014. Antarctic sea ice control on ocean circulation in present and glacial climates. *Proceedings of the National Academy of Sciences*, **111**, 24, 8753–8758.
- Ferrari, R., McWilliams, J. C., Canuto, V. M. & Dubovikov, M., 2008. Parameterization of eddy fluxes near ocean boundaries. *J. Climate*, **21**, 12, 2770–2789.
- Ferrel, W., 1856a. An essay on the winds and currents of the ocean. *Nashville J. Med. & Surg.*, **11**, 287–301.
- Ferrel, W., 1856b. The problem of the tides. *Astron. J.*, **4**, 173–176.
- Ferrel, W., 1858. The influence of the Earth's rotation upon the relative motion of bodies near its surface. *Astron. J.*, **V**, No. 13 (109), 97–100.

- Ferrel, W., 1859. The motion of fluids and solids relative to the Earth's surface. *Math. Monthly*, **1**, 140–148, 210–216, 300–307, 366–373, 397–406.
- Fjørtoft, R., 1950. Application of integral theorems in deriving criteria for laminar flows and for the baroclinic circular vortex. *Geophys. Publ.*, **17**, 1–52.
- Fjørtoft, R., 1953. On the changes in the spectral distribution of kinetic energy for two-dimensional nondivergent flow. *Tellus*, **5**, 225–230.
- Fleming, E. L., Chandra, S., Schoeberl, M. R. & Barnett, J. J., 1988. Monthly mean global climatology of temperature, wind, geopotential height, and pressure for 0–120 km. Technical report, NASA/Goddard Space Flight Center, Greenbelt, MD. NASA Tech. Memo. 100697.
- Fofonoff, N., 1959. Interpretation of oceanographic measurements – thermodynamics. In *Physical and Chemical Properties of Sea Water*, Vol. 600. Nat. Acad. Sci., Nat. Res. Council., Publ.
- Fofonoff, N. P., 1954. Steady flow in a frictionless homogeneous ocean. *J. Mar. Res.*, **13**, 254–262.
- Fofonoff, N. P. & Montgomery, R. B., 1955. The equatorial undercurrent in the light of the vorticity equation. *Tellus*, **7**, 518–521.
- Fox-Kemper, B. & Pedlosky, J., 2004. Wind-driven barotropic gyre I: Circulation control by eddy vorticity fluxes to an enhanced removal region. *J. Mar. Res.*, **62**, 169–193.
- Franklin, W. S., 1898. Review of P. Duhem, *Traité Elementaire de Mécanique Chimique fondée sur la Thermodynamique*, Two volumes. Paris, 1897. *Phys. Rev.*, **6**, 170–175.
- Friedman, R. M., 1989. *Appropriating the Weather: Vilhelm Bjerknes and the Construction of a Modern Meteorology*. Cornell University Press, 251 pp.
- Frierson, D. M. W., Lu, J. & Chen, G., 2007. Width of the Hadley cell in simple and comprehensive general circulation models. *Geophys. Res. Lett.*, **34**, L18804.
- Fu, L. L. & Flierl, G. R., 1980. Nonlinear energy and enstrophy transfers in a realistically stratified ocean. *Dyn. Atmos. Oceans*, **4**, 219–246.
- Gage, K. S. & Nastrom, G. D., 1986. Theoretical interpretation of atmospheric wavenumber spectra of wind and temperature observed by commercial aircraft during GASP. *J. Atmos. Sci.*, **43**, 729–740.
- Galanti, E. & Tziperman, E., 2000. ENSO's phase locking to the seasonal cycle in the fast-SST, fast-wave, and mixed-mode regimes. *J. Atmos. Sci.*, **57**, 2936–2950.
- Galperin, B., Sukoriansky, S. & Dikovskaya, N., 2010. Geophysical flows with anisotropic turbulence and dispersive waves: flows with a β -effect. *Ocean Dynamics*, **60**, 2, 427–441.
- Galperin, B., Sukoriansky, S., Dikovskaya, N., Read, P. *et al.*, 2006. Anisotropic turbulence and zonal jets in rotating flows with a β -effect. *Nonlinear Proc. Geophys.*, **13**, 83–98.
- Garcia, R. R., 1987. On the mean meridional circulation of the stratosphere. *J. Atmos. Sci.*, **44**, 2599–3609.
- Gardiner, C. W., 1985. *Handbook of Stochastic Methods*. Springer-Verlag, 442 pp.
- Gent, P. R. & McWilliams, J. C., 1990. Isopycnal mixing in ocean circulation models. *J. Phys. Oceanogr.*, **20**, 150–155.
- Gent, P. R., Willebrand, J., McDougall, T. J. & McWilliams, J. C., 1995. Parameterizing eddy-induced transports in ocean circulation models. *J. Phys. Oceanogr.*, **25**, 463–474.
- Gierasch, P. J., 1975. Meridional circulation and the maintenance of the Venus atmospheric rotation. *J. Atmos. Sci.*, **32**, 1038–1044.
- Gill, A. E., 1971. The equatorial current in a homogeneous ocean. *Deep-Sea Res.*, **18**, 421–431.
- Gill, A. E., 1975. Models of equatorial currents. In *Proceedings of Numerical Models of Ocean Circulation*, pp. 181–203. National Academy of Science.
- Gill, A. E., 1980. Some simple solutions for heat induced tropical circulation. *Quart. J. Roy. Meteor. Soc.*, **106**, 447–462.
- Gill, A. E., 1982. *Atmosphere–Ocean Dynamics*. Academic Press, 662 pp.
- Gill, A. E. & Clarke, A. J., 1974. Wind-induced upwelling, coastal currents and sea-level changes. *Deep-Sea Res.*, **21**, 325–345.
- Gill, A. E., Green, J. S. A. & Simmons, A. J., 1974. Energy partition in the large-scale ocean circulation and the production of mid-ocean eddies. *Deep-Sea Res.*, **21**, 499–528.
- Gille, S. T., 1997. The Southern Ocean momentum balance: evidence for topographic effects from numerical model output and altimeter data. *J. Phys. Oceanogr.*, **27**, 2219–2232.

- Gilman, P. A. & Glatzmaier, G. A., 1981. Compressible convection in a rotating spherical shell. I. Anelastic equations. *Astrophys. J. Suppl. Ser.*, **45**, 335–349.
- Giorgetta, M. A., Manzini, E. & Roeckner, E., 2002. Forcing of the quasi-biennial oscillation from a broad spectrum of atmospheric waves. *Geophys. Res. Lett.*, **29**, 1245.
- Gnanadesikan, A., 1999. A simple predictive model for the structure of the oceanic pycnocline. *Science*, **283**, 2077–2079.
- Godske, C. L., Bergeron, T., Bjerknes, J. & Budgaard, R. C., 1957. *Dynamic Meteorology and Weather Forecasting*. American Meteorological Society, 864 pp.
- Goody, R. M. & Yung, Y. L., 1995. *Atmospheric radiation: theoretical basis*. Oxford University Press, 544 pp.
- Gough, D. O., 1969. The anelastic approximation for thermal convection. *J. Atmos. Sci.*, **216**, 448–456.
- Graham, F. S. & McDougall, T. J., 2013. Quantifying the nonconservative production of conservative temperature, potential temperature and entropy. *J. Phys. Oceanogr.*, **43**, 838–862.
- Grant, H. L., Hughes, B. A., Vogel, W. M. & Moilliet, A., 1968. The spectrum of temperature fluctuation in turbulent flow. *J. Fluid Mech.*, **344**, 423–442.
- Grant, H. L., Stewart, R. W. & Moilliet, A., 1962. Turbulent spectra from a tidal channel. *J. Fluid Mech.*, **12**, 241–268.
- Gray, D. D. & Giorgini, A., 1976. The validity of the Boussinesq approximation for liquids and gases. *Int. J. Heat and Mass Transfer*, **19**, 5, 545–551.
- Gray, L. J., 2010. Stratospheric equatorial dynamics. In *The Stratosphere: Dynamics, Transport, and Chemistry*, Geophys. Monogr. Ser. Vol. 190 (2010).
- Gray, L. J., Crooks, S., C. Pascoe & Palmer, M., 2001. Solar and QBO influences on the timing of stratospheric sudden warmings. *J. Atmos. Sci.*, **61**, 2777–2796.
- Gray, L. J., Phipps, S. J., Dunkerton, T. J., Baldwin, M. P. *et al.*, 2001. A data study of the influence of the upper stratosphere on northern hemisphere stratospheric warmings. *Quart. J. Roy. Meteor. Soc.*, **127**, 1985–2003.
- Greatbatch, R. J., 1998. Exploring the relationship between eddy-induced transport velocity, vertical momentum transfer, and the isopycnal flux of potential vorticity. *J. Phys. Oceanogr.*, **28**, 422–432.
- Green, G., 1837. On the motion of waves in a variable canal of small depth and width. *Trans. Camb. Phil. Soc.*, **6**, 457–462.
- Green, J. S. A., 1960. A problem in baroclinic stability. *Quart. J. Roy. Meteor. Soc.*, **86**, 237–251.
- Green, J. S. A., 1970. Transfer properties of the large-scale eddies and the general circulation of the atmosphere. *Quart. J. Roy. Meteor. Soc.*, **96**, 157–185.
- Green, J. S. A., 1977. The weather during July 1976: some dynamical considerations of the drought. *Weather*, **32**, 120–128.
- Green, J. S. A., 1999. *Atmospheric Dynamics*. Cambridge University Press, 213 pp.
- Greenspan, H., 1962. A criterion for the existence of inertial boundary layers in oceanic circulation. *Proc. Nat. Acad. Sci.*, **48**, 2034–2039.
- Gregg, M. C., 1998. Estimation and geography of diapycnal mixing in the stratified ocean. In J. Imberger, Ed., *Physical Processes in Lakes and Oceans*, pp. 305–338. American Geophysical Union.
- Griffies, S. M., 1998. The Gent–McWilliams skew flux. *J. Phys. Oceanogr.*, **28**, 831–841.
- Griffies, S. M., 2004. *Fundamentals of Ocean Climate Models*. Princeton University Press, 518 pp.
- Grose, W. & Hoskins, B., 1979. On the influence of orography on large-scale atmospheric flow. *J. Atmos. Sci.*, **36**, 223–234.
- Hadley, G., 1735. Concerning the cause of the general trade-winds. *Phil. Trans. Roy. Soc.*, **29**, 58–62.
- Haine, T. W. N. & Marshall, J., 1998. Gravitational, symmetric, and baroclinic instability of the ocean mixed layer. *J. Phys. Oceanogr.*, **28**, 634–658.
- Hallberg, R. & Gnanadesikan, A., 2001. An exploration of the role of transient eddies in determining the transport of a zonally reentrant current. *J. Phys. Oceanogr.*, **31**, 3312–3330.
- Hamilton, K., Wilson, R. J. & Hemler, R. S., 2001. Spontaneous QBO-like oscillations simulated by the GFDL SKYHI general circulation mode. *J. Atmos. Sci.*, **58**, 3271–3292.
- Hamilton, K. P., 1998. Dynamics of the tropical middle atmosphere: a tutorial review. *Atmosphere–Ocean*, **36**, 319–354.

- Haney, R. L., 1971. Surface thermal boundary condition for ocean circulation models. *J. Phys. Oceanogr.*, **1**, 241–248.
- Harnik, N. & Heifetz, E., 2007. Relating overreflection and wave geometry to the counterpropagating Rossby wave perspective: Toward a deeper mechanistic understanding of shear instability. *J. Atmos. Sci.*, **64**, 7, 2238–2261.
- Haurwitz, B., 1941. *Dynamic meteorology*. Haurwitz press, 380 pp. Reprinted in 2007.
- Hayes, M., 1977. A note on group velocity. *Proc. Roy. Soc. Lond. A*, **354**, 533–535.
- Haynes, P., 2005. Stratospheric dynamics. *Ann. Rev. Fluid Mech.*, **37**, 263–293.
- Haynes, P. H., 2015. Critical layers. *Encycl. Atmos. Sci.*, 2nd edn, **2**, 317–323.
- Haynes, P. H. & McIntyre, M. E., 1987. On the evolution of vorticity and potential vorticity in the presence of diabatic heating and frictional or other forces. *J. Atmos. Sci.*, **44**, 828–841.
- Haynes, P. H. & McIntyre, M. E., 1990. On the conservation and impermeability theorem for potential vorticity. *J. Atmos. Sci.*, **47**, 2021–2031.
- Heckley, W. & Gill, A., 1984. Some simple analytical solutions to the problem of forced equatorial long waves. *Quart. J. Roy. Meteor. Soc.*, **110**, 463, 203–217.
- Heifetz, E. & Caballero, R., 2014. An alternative view on the role of the β -effect in the rossby wave propagation mechanism. *Tellus A*, **66**, 1–6.
- Held, I. M., 1982. On the height of the tropopause and the static stability of the troposphere. *J. Atmos. Sci.*, **39**, 412–417.
- Held, I. M., 1983. Stationary and quasi-stationary eddies in the extratropical troposphere: theory. In B. Hoskins & R. P. Pearce, Eds., *Large-Scale Dynamical Processes in the Atmosphere*, pp. 127–168. Academic Press.
- Held, I. M., 1985. Pseudomomentum and the orthogonality of modes in shear flows. *J. Atmos. Sci.*, **42**, 2280–2288.
- Held, I. M., 2000. The general circulation of the atmosphere. In *Woods Hole Program in Geophysical Fluid Dynamics* (2000), pp. 66.
- Held, I. M. & Hou, A. Y., 1980. Nonlinear axially symmetric circulations in a nearly inviscid atmosphere. *J. Atmos. Sci.*, **37**, 515–533.
- Held, I. M. & Larichev, V. D., 1996. A scaling theory for horizontally homogeneous, baroclinically unstable flow on a beta-plane. *J. Atmos. Sci.*, **53**, 946–952.
- Held, I. M., Tang, M. & Wang, H., 2002. Northern winter stationary waves: theory and modeling. *J. Climate*, **15**, 2125–2144.
- Helmholtz, H., 1858. Über Integrale der hydrodynamischen Gleichungen welche den Wirbelbewegungen entsprechen (On the integrals of the hydrodynamic equations that correspond to eddy motion). *J. Reine Angew. Math.*, **25**, 25–55. Engl. transl.: C. Abbe, *Smithson. Misc. Collect.*, no. 34, pp. 78–93, Smithsonian Institution, Washington D. C., 1893.
- Helmholtz, H., 1868. Über discontinuirliche Flüssigkeitsbewegungen (On discontinuous liquid motion). *Monats. Königl. Preuss. Akad. Wiss. Berlin*, **23**, 215–228. Engl. trans.: F. Guthrie: On discontinuous movements of fluids. *Phil. Mag.*, **36**, 337–346 (1868).
- Hendershott, M., 1987. Single layer models of the general circulation. In H. Abarbanel & W. R. Young, Eds., *General Circulation of the Ocean*, pp. 202–267. Springer-Verlag.
- Henning, C. C. & Vallis, G. K., 2004. The effect of mesoscale eddies on the main subtropical thermocline. *J. Phys. Oceanogr.*, **34**, 2428–2443.
- Henning, C. C. & Vallis, G. K., 2005. The effects of mesoscale eddies on the stratification and transport of an ocean with a circumpolar channel. *J. Phys. Oceanogr.*, **35**, 880–896.
- Hide, R., 1969. Dynamics of the atmospheres of major planets with an appendix on the viscous boundary layer at the rigid boundary surface of an electrically conducting rotating fluid in the presence of a magnetic field. *J. Atmos. Sci.*, **26**, 841–853.
- Hilborn, R. C., 2004. Sea gulls, butterflies, and grasshoppers: a brief history of the butterfly effect in nonlinear dynamics. *Am. J. Phys.*, **72**, 425–427.
- Hirst, A. C., 1986. Unstable and damped equatorial modes in simple coupled ocean-atmosphere models. *J. Atmos. Sci.*, **43**, 6, 606–632.

- Hockney, R., 1970. The potential calculation and some applications. In *Methods of Computational Physics*, Vol. 9, pp. 135–211. Academic Press.
- Hogg, N., 2001. Quantification of the deep circulation. In G. Siedler, J. Church, & J. Gould, Eds., *Ocean Circulation and Climate: Observing and Modelling the Global Ocean*, pp. 259–270. Academic Press.
- Hoinka, K. P., 1997. The tropopause: discovery, definition and demarcation. *Meteorol. Z.*, **6**, 281–303.
- Holland, W. R., Keffer, T. & Rhines, P. B., 1984. Dynamics of the oceanic circulation: The potential vorticity field. *Nature*, **308**, 698–705.
- Holloway, G. & Hendershott, M. C., 1977. Stochastic closure for nonlinear Rossby waves. *J. Fluid Mech.*, **82**, 747–765.
- Holm, D. D., Marsden, J. E., Ratiu, T. & Weinstein, A., 1985. Nonlinear stability of fluid and plasma equilibria. *Phys. Rep.*, **123**, 1–116.
- Holmes, M. H., 2013. *Introduction to Perturbation Methods*. 2nd edn. Springer, 436 pp.
- Holton, J. R., 1974. Forcing of mean flows by stationary waves. *J. Atmos. Sci.*, **31**, 942–945.
- Holton, J. R., 1992. *An Introduction to Dynamic Meteorology*. 3rd edn. Academic Press, 507 pp.
- Holton, J. R., Haynes, P. R., McIntyre, M. E., Douglass, A. R. *et al.*, 1995. Stratosphere-troposphere exchange. *Rev. Geophys.*, **33**, 403–439.
- Holton, J. R. & Lindzen, R. S., 1972. An updated theory for the quasi-biennial cycle of the tropical stratosphere. *J. Atmos. Sci.*, **29**, 1076–1080.
- Holton, J. R. & Mass, C., 1976. Stratospheric vacillation cycles. *J. Atmos. Sci.*, **33**, 2218–2215.
- Holton, J. R. & Tan, H.-C., 1980. The influence of the equatorial quasi-biennial oscillation on the global circulation at 50 mb. *J. Atmos. Sci.*, **37**, 2200–2208.
- Holton, J. R. & Tan, H.-C., 1982. The quasi-biennial oscillation in the northern hemisphere lower stratosphere. *J. Meteor. Soc. Japan*, **60**, 140–18.
- Hoskins, B. J. & Karoly, D. J., 1981. The steady linear response of a spherical atmosphere to thermal and orographic forcing. *J. Atmos. Sci.*, **38**, 1179–1196.
- Hough, S. S., 1897. On the application of harmonic analysis to the dynamical theory of the tides. Part I: On Laplace's "Oscillations of the first species", and on the dynamics of ocean currents. *Phil. Trans. (A)*, **189 (IX)**, 201–258.
- Hough, S. S., 1898. On the application of harmonic analysis to the dynamical theory of the tides. Part II: On the general integration of Laplace's dynamical equations. *Phil. Trans. (A)*, **191 (V)**, 139–186.
- Huang, R. X., 1986. Solutions of the ideal fluid thermocline with continuous stratification. *J. Phys. Oceanogr.*, **16**, 1, 39–59.
- Huang, R. X., 1998. Mixing and available potential energy in a Boussinesq ocean. *J. Phys. Oceanogr.*, **28**, 669–678.
- Huang, R. X., 1999. Mixing and energetics of the oceanic thermohaline circulation. *J. Phys. Oceanogr.*, **29**, 727–746.
- Huang, R. X., 2010. *Ocean Circulation*. Cambridge University Press, 791 pp. ISBN: 978-0-521-85228-9.
- Hughes, C. W., 2002. Sverdrup-like theories of the Antarctic Circumpolar Current. *J. Mar. Res.*, **60**, 1–17.
- Hughes, C. W. & de Cuevas, B., 2001. Why western boundary currents in realistic oceans are inviscid: a link between form stress and bottom pressure torques. *J. Phys. Oceanogr.*, **31**, 2871–2885.
- Hughes, G. O. & Griffiths, R. W., 2008. Horizontal convection. *Ann. Rev. Fluid Mech.*, **185–2008**, 40.
- Ierley, G. R. & Ruehr, O. G., 1986. Analytic and numerical solutions of a nonlinear boundary value problem. *Stud. Appl. Math.*, **75**, 1–36.
- Ilicak, M. & Vallis, G. K., 2012. Simulations and scaling of horizontal convection. *Tellus A*, **64**, 1–17.
- Il'in, A. M. & Kamenkovich, V. M., 1964. The structure of the boundary layer in the two-dimensional theory of ocean currents (in Russian). *Okeanologiya*, **4 (5)**, 756–769.
- Ingersoll, A. P., 2005. Boussinesq and anelastic approximations revisited: potential energy release during thermobaric instability. *J. Phys. Oceanogr.*, **35**, 1359–1369.
- IOC, SCOR & IAPSO, 2010. The international thermodynamic equation of seawater – 2010: Calculation and use of thermodynamic properties. Technical report, Intergovernmental Oceanographic Commission, Manuals and Guides No. 56, UNESCO (English).
- Iwayama, T., Shepherd, T. G. & Watanabe, T., 2002. An 'ideal' form of decaying two-dimensional turbulence. *J. Fluid Mech.*, **456**, 183–198.

- Jackett, D. R. & McDougall, T. J., 1997. A neutral density variable for the world's oceans. *J. Phys. Oceanogr.*, **28**, 237–263.
- Jackson, L., Hughes, C. W. & Williams, R. G., 2006. Topographic control of basin and channel flows: the role of bottom pressure torques and friction. *J. Phys. Oceanogr.*, **36**, 1786–1805.
- Jansen, M. & Ferrari, R., 2012. Macroturbulent equilibration in a thermally forced primitive equation system. *J. Atmos. Sci.*, **69**, 2, 695–713.
- Jansen, M. & Ferrari, R., 2013. Equilibration of an atmosphere by adiabatic eddy fluxes. *J. Atmos. Sci.*, **70**, 9, 2948–2962.
- Jeffreys, H., 1924. On certain approximate solutions of linear differential equations of the second order. *Proc. London Math. Soc.* (2), **23**, 428–436.
- Jeffreys, H., 1926. On the dynamics of geostrophic winds. *Quart. J. Roy. Meteor. Soc.*, **51**, 85–104.
- Jeffreys, H. & Jeffreys, B. S., 1946. *Methods of Mathematical Physics*. Cambridge University Press, 728 pp.
- Jin, F.-F., 1997a. An equatorial ocean recharge paradigm for ENSO. Part I: Conceptual model. *J. Atmos. Sci.*, **54**, 7, 811–829.
- Jin, F.-F., 1997b. An equatorial ocean recharge paradigm for ENSO. Part II: A stripped-down coupled model. *J. Atmos. Sci.*, **54**, 7, 830–847.
- Johnson, G. C. & Bryden, H. L., 1989. On the size of the Antarctic Circumpolar Current. *Deep-Sea Res.*, **36**, 39–53.
- Jones, D. B. A., Schneider, H. R. & McElroy, M. B., 1998. Effects of the quasi-biennial oscillation on the zonally averaged transport of tracer. *J. Geophys. Res.*, **103**, 11235–11249.
- Jones, W. L., 1967. Propagation of internal gravity waves in fluids with shear flow and rotation. *J. Fluid Mech.*, **30**, 03, 439–448.
- Jucker, M., 2014. Scientific visualisation of atmospheric data with ParaView. *J. Open Res. Software*, **2**, e4.
- Jucker, M., Fueglistaler, S. & Vallis, G. K., 2013. Maintenance of stratospheric structure in an idealized general circulation model. *J. Atmos. Sci.*, **70**, 3341–3358.
- Jucker, M., Fueglistaler, S. & Vallis, G. K., 2014. Stratospheric sudden warmings in an idealized GCM. *J. Geophys. Res. (Atmospheres)*, **119**, 11054–11064.
- Juckes, M. N., 2000. The static stability of the midlatitude troposphere: the relevance of moisture. *J. Atmos. Sci.*, **57**, 3050–3057.
- Juckes, M. N., 2001. A generalization of the transformed Eulerian-mean meridional circulation. *Quart. J. Roy. Meteor. Soc.*, **127**, 147–160.
- Kalnay, E., 1996. The NCEP/NCAR 40-year reanalysis project. *Bull. Amer. Meteor. Soc.*, **77**, 437–471.
- Karsten, R., Jones, H. & Marshall, J., 2002. The role of eddy transfer in setting the stratification and transport of a circumpolar current. *J. Phys. Oceanogr.*, **32**, 39–54.
- Keffer, T., 1985. The ventilation of the world's oceans: maps of potential vorticity. *J. Phys. Oceanogr.*, **15**, 509–523.
- Kessler, W. S., Johnson, G. C. & Moore, D. W., 2003. Sverdrup and nonlinear dynamics of the Pacific equatorial currents. *J. Phys. Oceanogr.*, **33**, 994–1008.
- Kevorkian, J. & Cole, J. D., 2011. *Multiple Scale and Singular Perturbation Methods*. Springer-Verlag, 648 pp.
- Kibel, I., 1940. Priloozhenie k meteorogi uravnenii mekhaniki baroklinnoi zhidkosti (Application of baroclinic fluid dynamic equations to meteorology). *SSSR Ser. Geogr. Geofiz.*, **5**, 627–637.
- Kiladis, G. N., Straub, K. H. & Haertel, P. T., 2005. Zonal and vertical structure of the Madden-Julian oscillation. *J. Atmos. Sci.*, **62**, 8, 2790–2809.
- Kiladis, G. N., Wheeler, M. C., Haertel, P. T., Straub, K. H. & Roundy, P. E., 2009. Convectively coupled equatorial waves. *Rev. Geophys.*, **47**, RG2003.
- Killworth, P. D., 1987. A continuously stratified nonlinear ventilated thermocline. *J. Phys. Oceanogr.*, **17**, 1925–1943.
- Killworth, P. D., 1997. On the parameterization of eddy transfer. Part I: theory. *J. Marine Res.*, **55**, 1171–1197.
- Killworth, P. D. & McIntyre, M. E., 1985. Do Rossby-wave critical layers absorb, reflect, or over-reflect? *J. Fluid Mech.*, **161**, 449–492.
- Kimoto, M. & Ghil, M., 1993. Multiple flow regimes in the northern hemisphere winter. Part 1: methodology and hemispheric regimes. *J. Atmos. Sci.*, **50**, 2625–2643.

- Kolmogorov, A. N., 1941. The local structure of turbulence in incompressible viscous fluid for very large Reynolds numbers. *Dokl. Acad. Sci. USSR*, **30**, 299–303.
- Kolmogorov, A. N., 1962. A refinement of previous hypotheses concerning the local structure of turbulence in a viscous incompressible fluid at high Reynolds numbers. *J. Fluid Mech.*, **13**, 82–85.
- Kraichnan, R., 1967. Inertial ranges in two-dimensional turbulence. *Phys. Fluids*, **10**, 1417–1423.
- Kraichnan, R. & Montgomery, D., 1980. Two-dimensional turbulence. *Rep. Prog. Phys.*, **43**, 547–619.
- Kramers, H. A., 1926. Wellenmechanik und halbzahlige Quantisierung (Wave mechanics and semi-integral quantization). *Zeit. für Physik A*, **39**, 828–840.
- Kundu, P., Allen, J. S. & Smith, R. L., 1975. Modal decomposition of the velocity field near the Oregon coast. *J. Phys. Oceanogr.*, **5**, 683–704.
- Kundu, P. & Cohen, I. M., 2002. *Fluid Mechanics*. Academic Press, 730 pp.
- Kuo, H.-I., 1949. Dynamic instability of two-dimensional nondivergent flow in a barotropic atmosphere. *J. Meteorol.*, **6**, 105–122.
- Kuo, H.-I., 1951. Vorticity transfer as related to the development of the general circulation. *J. Meteorol.*, **8**, 307–315.
- Kushner, P. J., 2010. Annular modes of the troposphere and stratosphere. In *The Stratosphere: Dynamics, Transport, and Chemistry*, Geophys. Monogr. Ser (2010).
- Kushnir, Y., Robinson, W. A., Bladé, I., Hall, N. M. J. et al., 2002. Atmospheric GCM response to extratropical SST anomalies: synthesis and evaluation. *J. Climate*, **15**, 2233–2256.
- Labitzke, K., Kunze, M. & Brönnimann, S., 2006. Sunspots, the qbo and the stratosphere in north polar regions — 20 years later. *Meteor. Z.*, **15**, 355–363.
- LaCasce, J. H. & Ohlmann, C., 2003. Relative dispersion at the surface of the Gulf of Mexico. *J. Mar. Res.*, **65**, 285–312.
- Lait, L. R., 1994. An alternative form for potential vorticity. *J. Atmos. Sci.*, **51**, 1754–1759.
- Lanczos, C., 1970. *The Variational Principles of Mechanics*. University of Toronto Press, Reprinted by Dover Publications 1980, 418 pp.
- Landau, L. D., 1944. On the problem of turbulence. *Dokl. Akad. Nauk SSSR*, **44**, 311–314.
- Landau, L. D. & Lifshitz, E. M., 1987. *Fluid Mechanics* (Course of Theoretical Physics, v. 6). 2nd edn. Pergamon Press, 539 pp.
- Larichev, V. D. & Held, I. M., 1995. Eddy amplitudes and fluxes in a homogeneous model of fully developed baroclinic instability. *J. Phys. Oceanogr.*, **25**, 2285–2297.
- Latif, M. & Barnett, T., 1996. Decadal climate variability over the North Pacific and North America: dynamics and predictability. *J. Climate*, **10**, 219–239.
- Lau, K.-M., 1981. Oscillations in a simple equatorial climate system. *J. Atmos. Sci.*, **38**, 2, 248–261.
- Lawrence, M. G., 2005. The relationship between relative humidity and the dewpoint temperature in moist air: A simple conversion and applications. *Bull. Am. Meteor. Soc.*, **86**, 2, 225–233.
- LeBlond, P. H. & Mysak, L. A., 1980. *Waves in the Ocean*. Elsevier, 616 pp.
- Ledwell, J., Watson, A. & Law, C., 1998. Mixing of a tracer released in the pycnocline. *J. Geophys. Res.*, **103**, 21499–21529.
- Lee, M.-M., Marshall, D. P. & Williams, R. G., 1997. On the eddy transfer of tracers: advective or diffusive? *J. Mar. Res.*, **55**, 483–595.
- Lee, T. D., 1951. Difference between turbulence in a two-dimensional fluid and in a three-dimensional fluid. *J. Appl. Phys.*, **22**, 524.
- Leetmaa, A., Niiler, P. & Stommel, H., 1977. Does the Sverdrup relation account for the mid-Atlantic circulation? *J. Mar. Res.*, **35**, 1–10.
- Leith, C. E., 1968. Diffusion approximation for two-dimensional turbulence. *Phys. Fluids*, **11**, 671–672.
- Lesieur, M., 1997. *Turbulence in Fluids: Third Revised and Enlarged Edition*. Kluwer, 515 pp.
- Levinson, N., 1950. The 1st boundary value problem for $\epsilon \Delta U + A(x, y)U_x + B(x, y)U_y + C(x, y)U = D(x, t)$ for small epsilon. *Ann. Math.*, **51**, 429–445.
- Lewis, R., Ed., 1991. *Meteorological Glossary*. 6th edn. Her Majesty's Stationery Office, 335 pp.
- Lighthill, J., 1978. *Waves in Fluids*. Cambridge University Press, 504 pp.
- Lighthill, M. J., 1965. Group velocity. *J. Inst. Math. Appl.*, **1**, 1–28.

- Lighthill, M. J., 1969. Dynamic response of the Indian Ocean to onset of the southwest monsoon. *Phil. Trans. Roy. Soc. Lond. A*, **265**, 1159, 45–92.
- Lilly, D. K., 1969. Numerical simulation of two-dimensional turbulence. *Phys. Fluid Suppl. II*, **12**, 240–249.
- Lilly, D. K., 1996. A comparison of incompressible, anelastic and Boussinesq dynamics. *Atmos. Res*, **40**, 143–151.
- Limpasuvan, V., Thompson, D. W. J. & Hartmann, D. L., 2000. The life cycle of the Northern Hemisphere sudden stratospheric warmings. *J. Climate*, **17**, 2584–2596.
- Lindborg, E., 1999. Can the atmospheric kinetic energy spectrum be explained by two-dimensional turbulence? *J. Fluid Mech.*, **388**, 259–288.
- Lindborg, E. & Alvelius, K., 2000. The kinetic energy spectrum of the two-dimensional enstrophy turbulence cascade. *Phys. Fluids*, **12**, 945–947.
- Lindzen, R. S. & Farrell, B., 1980. The role of the polar regions in global climate, and a new parameterization of global heat transport. *Mon. Wea. Rev.*, **108**, 2064–2079.
- Lindzen, R. S. & Holton, J. R., 1968. A theory of the quasi-biennial oscillation. *J. Atmos. Sci.*, **25**, 1095–1107.
- Lindzen, R. S. & Hou, A. Y., 1988. Hadley circulation for zonally averaged heating centered off the equator. *J. Atmos. Sci.*, **45**, 2416–2427.
- Lindzen, R. S., Lorenz, E. N. & Plazman, G. W., Eds., 1990. *The Atmosphere — a Challenge: the Science of Jule Gregory Charney*. American Meteorological Society, 321 pp.
- Lindzen, R. S. & Nigam, S., 1987. On the role of sea surface temperature gradients in forcing low-level winds and convergence in the tropics. *J. Atmos. Sci.*, **44**, 17, 2418–2436.
- Liouville, J., 1837. Sur le développement des fonction ou parties de fonction en séries (On the development of functions of parts of functions in series). *J. Math. Pures Appl.*, **2**, 16–35.
- Lipps, F. B. & Hemler, R. S., 1982. A scale analysis of deep moist convection and some related numerical calculations. *J. Atmos. Sci.*, **39**, 2192–2210.
- Longuet-Higgins, M. S., 1964. Planetary waves on a rotating sphere, I. *Proc. Roy. Soc. Lond. A*, **279**, 446–473.
- Longuet-Higgins, M. S., 1968. The eigenfunctions of Laplace's tidal equations over a sphere. *Proc. Roy. Soc. Lond. A*, **262**, 511–607.
- Lorenz, E. N., 1955. Available potential energy and the maintenance of the general circulation. *Tellus*, **7**, 157–167.
- Lorenz, E. N., 1963. Deterministic nonperiodic flow. *J. Atmos. Sci.*, **20**, 130–141.
- Lorenz, E. N., 1967. *The Nature and the Theory of the General Circulation of the Atmosphere*. WMO Publications, Vol. 218, World Meteorological Organization.
- Lozier, S., Owens, W. B. & Curry, R. G., 1996. The climatology of the North Atlantic. *Prog. Oceanogr.*, **36**, 1–44.
- Ludlam, F. H., 1966. Cumulus and cumulonimbus convection. *Tellus*, **18**, 687–698.
- Lumpkin, R. & Speer, K., 2007. Global ocean meridional overturning. *J. Phys. Oceanogr.*, **37**, 10, 2550–2562.
- Luyten, J. R., Pedlosky, J. & Stommel, H., 1983. The ventilated thermocline. *J. Phys. Oceanogr.*, **13**, 292–309.
- Madden, R. A. & Julian, P. R., 1971. Detection of a 40–50 day oscillation in the zonal wind in the tropical Pacific. *J. Atmos. Sci.*, **28**, 5, 702–708.
- Madden, R. A. & Julian, P. R., 1972. Description of global-scale circulation cells in the tropics with a 40–50 day period. *J. Atmos. Sci.*, **29**, 6, 1109–1123.
- Majda, A. J. & Klein, R., 2003. Systematic multiscale models for the tropics. *J. Atmos. Sci.*, **60**, 2, 393–408.
- Majda, A. J. & Stechmann, S. N., 2009. The skeleton of tropical intraseasonal oscillations. *Proc. Nat. Acad. Sci.*, **106**, 21, 8417–8422.
- Maltrud, M. E. & Vallis, G. K., 1991. Energy spectra and coherent structures in forced two-dimensional and beta-plane turbulence. *J. Fluid Mech.*, **228**, 321–342.
- Manabe, S. & Stouffer, R. J., 1988. Two stable equilibria of a coupled ocean–atmosphere model. *J. Climate*, **1**, 841–866.
- Manabe, S. & Strickler, R. F., 1964. Thermal equilibrium of the atmosphere with a convective adjustment. *J. Atmos. Sci.*, **21**, 361–385.
- Manabe, S. & Wetherald, R. T., 1980. On the distribution of climate change resulting from an increase in CO₂ content of the atmosphere. *J. Atmos. Sci.*, **37**, 99–118.
- Mapes, B. E., 1997. Equilibrium vs. activation control of large-scale variations of tropical deep convection.

- In R. K. Smith, Ed., *The Physics and Parameterization of Moist Atmospheric Convection*, pp. 321–358. Springer.
- Mapes, B. E., 1998. The large-scale part of tropical mesoscale convective system circulations: A linear vertical spectral band model. *J. Meteor. Soc. Japan*, **76**, 1, 29–55.
- Mapes, B. E., 2000. Convective inhibition, subgrid-scale triggering energy, and stratiform instability in a toy tropical wave model. *J. Atmos. Sci.*, **57**, 10, 1515–1535.
- Marcus, P. S., 1993. Jupiter's Great Red Spot and other vortices. *Ann. Rev. Astron. Astrophys.*, **31**, 523–573.
- Margules, M., 1903. Über die Energie der Stürme (On the energy of storms). *Jahrb. Kais.-kön. Zent. für Met. und Geodynamik, Vienna*, 26 pp. Engl. transl.: C. Abbe, *Smithson. Misc. Collect.*, no. 51, pp. 533–595, Smithsonian Institution, Washington D. C., 1910.
- Marotzke, J., 1989. Instabilities and multiple steady states of the thermohaline circulation. In D. L. T. Anderson & J. Willebrand, Eds., *Oceanic Circulation Models: Combining Data and Dynamics*, pp. 501–511. NATO ASI Series, Kluwer.
- Marshall, D. P., Maddison, J. R. & Berloff, P. S., 2012. A framework for parameterizing eddy potential vorticity fluxes. *J. Phys. Oceanogr.*, **42**, 4, 539–557.
- Marshall, D. P., Williams, R. G. & Lee, M.-M., 1999. The relation between eddy-induced transport and isopycnic gradients of potential vorticity. *J. Phys. Oceanogr.*, **29**, 1571–1578.
- Marshall, J. & Plumb, R. A., 2013. *Atmosphere, Ocean and Climate dynamics: An Introductory Text*. Academic Press, 344 pp.
- Marshall, J. C., 1981. On the parameterization of geostrophic eddies in the ocean. *J. Phys. Oceanogr.*, **11**, 257–271.
- Marshall, J. C. & Nurser, A. J. G., 1992. Fluid dynamics of oceanic thermocline ventilation. *J. Phys. Oceanogr.*, **22**, 583–595.
- Marshall, J. C. & Radko, T., 2003. Residual-mean solutions for the Antarctic Circumpolar Current and its associated overturning circulation. *J. Phys. Oceanogr.*, **22**, 2341–2354.
- Marshall, J. C. & Schott, F., 1999. Open-ocean convection: observations, theory, and models. *Rev. Geophys.*, **37**, 1–64.
- Matsuno, T., 1966. Quasi-geostrophic motions in the equatorial area. *J. Meteor. Soc. Japan*, **44**, 25–43.
- Matsuno, T., 1971. A dynamical model of the sudden stratospheric warming. *J. Atmos. Sci.*, **28**, 1479–1494.
- Maze, G. & Marshall, J., 2011. Diagnosing the observed seasonal cycle of Atlantic subtropical mode water using potential vorticity and its attendant theorems. *J. Phys. Oceanogr.*, **41**, 10, 1986–1999.
- McCarthy, M. C. & Talley, L. D., 1999. Three-dimensional isoneutral potential vorticity structure in the Indian Ocean. *J. Geophys. Res. (Oceans)*, **104**, C6, 13251–13267.
- McCreary, J. P., 1981. A linear stratified ocean model of the equatorial undercurrent. *Phil. Trans. Roy. Soc. Lond.*, **A298**, 603–635.
- McCreary, J. P., 1985. Modeling equatorial ocean circulation. *Ann. Rev. Fluid Mech.*, **17**, 359–407.
- McCreary, J. P. & Lu, P., 1994. Interaction between the subtropical and equatorial ocean circulations: the subtropical cell. *J. Phys. Oceanogr.*, **24**, 466–497.
- McDougall, T. J., 1987. Neutral surfaces. *J. Phys. Oceanogr.*, **17**, 11, 1950–1964.
- McDougall, T. J., 1998. Three-dimensional residual mean theory. In E. P. Chassignet & J. Verron, Eds., *Ocean Modeling and Parameterization*, pp. 269–302. Kluwer Academic.
- McDougall, T. J., 2003. Potential enthalpy: a conservative oceanic variable for evaluating heat content and heat fluxes. *J. Phys. Oceanogr.*, **33**, 945–963.
- McIntosh, P. C. & McDougall, T. J., 1996. Isopycnal averaging and the residual mean circulation. *J. Phys. Oceanogr.*, **26**, 1655–1660.
- McIntyre, M. E. & Norton, W. A., 1990. Dissipative wave–mean interactions and the transport of vorticity or potential vorticity. *J. Fluid Mech.*, **212**, 403–435.
- McIntyre, M. E. & Norton, W. A., 2000. Potential vorticity inversion on a hemisphere. *J. Atmos. Sci.*, **57**, 1214–1235.
- McIntyre, M. E. & Shepherd, T. G., 1987. An exact local conservation theorem for finite-amplitude disturbances to nonparallel shear flows, with remarks on Hamiltonian structure and on Arnol'd's stability theorems. *J. Fluid Mech.*, **181**, 527–565.

- McKee, W. D., 1973. The wind-driven equatorial circulation in a homogeneous ocean. *Deep-Sea Res.*, **20**, 889–899.
- McPhaden, M. J., Timmermann, A., Widlansky, M. J., Balmaseda, M. A. & Stockdale, T. N., 2015. The curious case of the El Niño that never happened: A perspective from 40 years of progress in climate research and forecasting. *Bull. Am. Meteor. Soc.*, **96**, 1647–1665.
- McWilliams, J. C., 1984. The emergence of isolated coherent vortices in turbulent flow. *J. Fluid Mech.*, **146**, 21–43.
- Mihaljan, J. M., 1962. A rigorous exposition of the Boussinesq approximations applicable to a thin layer of fluid. *Astrophysical J.*, **136**, 1126–1133.
- Millero, F. J., Feistel, R., Wright, D. G. & McDougall, T. J., 2008. The composition of standard seawater and the definition of the reference-composition salinity scale. *Deep Sea Res., Part I*, **55**, 1, 50–72.
- Moffatt, H. K., 1983. Transport effects associated with turbulence with particular attention to the influence of helicity. *Rep. Progress Phys.*, **46**, 621–664.
- Morel, P. & Larcheveque, M., 1974. Relative dispersion of constant-level balloons in the 200 mb general circulation. *J. Atmos. Sci.*, **31**, 2189–2196.
- Mundt, M., Vallis, G. K. & Wang, J., 1997. Balanced models for the large- and meso-scale circulation. *J. Phys. Oceanogr.*, **27**, 1133–1152.
- Munk, W. H., 1950. On the wind-driven ocean circulation. *J. Meteorol.*, **7**, 79–93.
- Munk, W. H., 1966. Abyssal recipes. *Deep-Sea Res.*, **13**, 707–730.
- Munk, W. H. & Palmén, E., 1951. Note on dynamics of the Antarctic Circumpolar Current. *Tellus*, **3**, 53–55.
- Munk, W. H. & Wunsch, C., 1998. Abyssal recipes II: energetics of tidal and wind mixing. *Deep-Sea Res.*, **45**, 1976–2009.
- Namias, J., 1959. Recent seasonal interaction between North Pacific waters and the overlying atmospheric circulation. *J. Geophys. Res.*, **64**, 631–646.
- Neelin, J. D., 1988. A simple model for surface stress and low-level flow in the tropical atmosphere driven by prescribed heating. *Quart. J. Roy. Meteor. Soc.*, **114**, 481, 747–770.
- Neelin, J. D. & Zeng, N., 2000. A quasi-equilibrium tropical circulation model—formulation. *J. Atmos. Sci.*, **57**, 1741–1766.
- Newell, A. C., 1969. Rossby wave packet interactions. *J. Fluid Mech.*, **35**, 255–271.
- Newman, P. A., Coy, L., Pawson, S. & Lait, L. R., 2016. The anomalous change in the QBO in 2015–2016. *Geophys. Res. Lett.*, 1–4. doi:10.1002/2016GL070373.
- Nicholls, S., 1985. Aircraft observations of the Ekman layer during the joint air-sea interaction experiment. *Quart. J. Roy. Meteor. Soc.*, **111**, 391–426.
- Nikurashin, M. & Vallis, G. K., 2011. A theory of deep stratification and overturning circulation in the ocean. *J. Phys. Oceanogr.*, **41**, 485–502.
- Nikurashin, M. & Vallis, G. K., 2012. A theory of the interhemispheric meridional overturning circulation and associated stratification. *J. Phys. Oceanogr.*, **42**, 1652–1667.
- Nof, D., 2003. The Southern Ocean's grip on the northward meridional flow. In *Progress in Oceanography*, Vol. 56, pp. 223–247. Pergamon.
- Novikov, E. A., 1959. Contributions to the problem of the predictability of synoptic processes. *Izv. An. SSSR Ser. Geophys.*, **11**, 1721. Eng. transl.: *Am. Geophys. U. Transl.*, 1209–1211.
- Nycander, J., 2011. Energy conversion, mixing energy, and neutral surfaces with a nonlinear equation of state. *J. Phys. Oceanogr.*, **41**, 28–41.
- Nycander, J. & Roquet, F., 2015. The nonlinear equation of state of sea water and the global water mass distribution. *Geophys. Res. Lett.*, **42**, 7714–7721. In press.
- Oberbeck, A., 1879. Über die Wärmeleitung der Flüssigkeiten bei Berücksichtigung der Strömungen infolge von Temperaturdifferenzen (On the thermal conduction of liquids taking into account flows due to temperature differences). *Ann. Phys. Chem., Neue Folge*, **7**, 271–292.
- Oberbeck, A., 1888. Über die Bewegungerscheinungen der Atmosphäre (On the phenomena of motion in the atmosphere). *Sitzb. K. Preuss. Akad. Wiss.*, **7**, 383–395 and 1129–1138. Engl. trans.: in B. Saltzman, Ed., *Theory of Thermal Convection*, Dover, 162–183.
- Obukhov, A. M., 1941. Energy distribution in the spectrum of turbulent flow. *Izv. Akad. Nauk. SSR, Ser. Geogr. Geofiz.*, **5**, 453–466.

- Obukhov, A. M., 1949. Structure of the temperature field in turbulent flows. *Izv. Akad. Nauk. SSR, Ser. Geogr. Geofiz.*, **13**, 58–63.
- Obukhov, A. M., 1962. On the dynamics of a stratified liquid. *Dokl. Akad. Nauk SSSR*, **145**, 1239–1242. Engl. transl.: *Soviet Physics–Dokl.* **7**, 682–684.
- Oetzel, K. & Vallis, G. K., 1997. Strain, vortices, and the enstrophy inertial range in two-dimensional turbulence. *Phys. Fluids*, **9**, 2991–3004.
- O’Gorman, P. A., Lamquin, N., Schneider, T. & Singh, M. S., 2011. The relative humidity in an isentropic advection–condensation model: Limited poleward influence and properties of subtropical minima. *J. Atmos. Sci.*, **68**, 3079–3093.
- O’Gorman, P. A. & Pullin, D. I., 2005. Effect of Schmidt number of the velocity–scaler cospectrum in isotropic turbulence with a mean scalar gradient. *J. Fluid Mech.*, **532**, 111–140.
- Ogura, Y. & Phillips, N. A., 1962. Scale analysis of deep and shallow convection in the atmosphere. *J. Atmos. Sci.*, **19**, 173–179.
- Olbers, D., 1998. Comments on “On the Obscurantist Physics of ‘Form Drag’ in Theorizing about the Circumpolar Current”. *J. Phys. Oceanogr.*, **28**, 8, 1647–1654.
- Olbers, D., Borowski, D., Völker, C. & Wolff, J.-O., 2004. The dynamical balance, transport and circulation of the Antarctic Circumpolar Current. *Antarctic Science*, **16**, 439–470.
- Olbers, D., Willebrand, J. & Eden, C., 2012. *Ocean Dynamics*. Springer, 704 pp. ISBN: 978-3642234491.
- Ollitrault, M., Gabillet, C. & Colin de Verdière, A., 2005. Open ocean regimes of relative dispersion. *J. Fluid Mech.*, **533**, 381–407.
- Onsager, L., 1931. Reciprocal relations in irreversible processes. II. *Phys. Rev.*, **38**, 2265–2279.
- Onsager, L., 1949. Statistical hydrodynamics. *Nuovo Cim. (Suppl.)*, **6**, 279–287.
- Ooyama, K., 1963. A dynamical model for the study of tropical cyclone development. New York University, 26pp, unpublished manuscript.
- Ooyama, K. V., 1982. Conceptual evolution of the theory and modeling of the tropical cyclone. *J. Meteor. Soc. Japan*, **60**, 1, 369–380.
- Osprey, S. M., Butchart, N., Knight, J. R., Scaife, A. *et al.*, 2016. An unexpected disruption of the atmospheric quasi-biennial oscillation. *Science*, **353**, 6306, 1424–1427.
- Paldor, N., Rubin, S. & Mariono, A. J., 2007. A consistent theory for linear waves of the shallow-water equations on a rotating plane in midlatitudes. *J. Atmos. Sci.*, **37**, 115–128.
- Paldor, N. & Sigalov, A., 2011. An invariant theory of the linearized shallow water equations with rotation and its application to a sphere and a plane. *Dyn. Atmos. Oceans*, **51**, 26–44.
- Palmer, T. N., 1981. Diagnostic study of a wavenumber-2 stratospheric sudden warming in a transformed Eulerian-mean formalism. *J. Atmos. Sci.*, **38**, 844–855.
- Palmer, T. N., 1997. A nonlinear dynamical perspective on climate prediction. *J. Climate*, **12**, 575–591.
- Palmer, T. N., 2009. Edward Lorenz, 1917–2008. *Biogr. Mem. Fell. R. Soc.*, **55**, 139–155.
- Paparella, F. & Young, W. R., 2002. Horizontal convection is non-turbulent. *J. Fluid Mech.*, **466**, 205–214.
- Pascoe, C. L., Gray, L. J., Crooks, S. A., Juckes, M. N. & Baldwin, M. P., 2005. The quasi-biennial oscillation: Analysis using ERA-40 data. *J. Geophys. Res.*, **110**, D08105.
- Pauluis, O., 2007. Sources and sinks of available potential energy in a moist atmosphere. *J. Atmos. Sci.*, **64**, 2627–2641.
- Pauluis, O., 2008. Thermodynamic consistency of the anelastic approximation for a moist atmosphere. *J. Atmos. Sci.*, **65**, 2719–2729.
- Pedlosky, J., 1964. The stability of currents in the atmosphere and ocean. Part I. *J. Atmos. Sci.*, **21**, 201–219.
- Pedlosky, J., 1987a. *Geophysical Fluid Dynamics*. 2nd edn. Springer-Verlag, 710 pp.
- Pedlosky, J., 1987b. An inertial theory of the equatorial undercurrent. *J. Phys. Oceanogr.*, **17**, 1978–1985.
- Pedlosky, J., 1996. *Ocean Circulation Theory*. Springer-Verlag, 453 pp.
- Pedlosky, J., 2003. *Waves in the Ocean and Atmosphere: Introduction to Wave Dynamics*. Springer-Verlag, 260 pp.
- Peixoto, J. P. & Oort, A. H., 1992. *Physics of Climate*. American Institute of Physics, 520 pp.
- Peltier, W. R. & Stuhne, G., 2002. The upscale turbulence cascade: shear layers, cyclones and gas giant bands. In R. P. Pearce, Ed., *Meteorology at the Millennium*, pp. 43–61. Academic Press.

- Persson, A., 1998. How do we understand the Coriolis force? *Bull. Am. Meteor. Soc.*, **79**, 1373–1385.
- Philander, S. G., 1990. *El Niño, La Niña, and the Southern Oscillation*. Academic Press, 289 pp.
- Phillips, N. A., 1954. Energy transformations and meridional circulations associated with simple baroclinic waves in a two-level, quasi-geostrophic model. *Tellus*, **6**, 273–286.
- Phillips, N. A., 1956. The general circulation of the atmosphere: a numerical experiment. *Quart. J. Roy. Meteor. Soc.*, **82**, 123–164.
- Phillips, N. A., 1963. Geostrophic motion. *Rev. Geophys.*, **1**, 123–176.
- Phillips, N. A., 1966. The equations of motion for a shallow rotating atmosphere and the traditional approximation. *J. Atmos. Sci.*, **23**, 626–630.
- Phillips, N. A., 1973. Principles of large-scale numerical weather prediction. In P. Morel, Ed., *Dynamic Meteorology*, pp. 1–96. Riedel.
- Phillips, O., 1972. Turbulence in a strongly stratified fluid—is it unstable? In *Deep Sea Research and Oceanographic Abstracts*, Vol. 19 (1972), pp. 79–81. Elsevier.
- Pierini, S. & Vulpiani, A., 1981. Nonlinear stability analysis in multi-layer quasigeostrophic system. *J. Phys. A*, **14**, L203–L207.
- Pierrehumbert, R. T., 2010. *Principles of Planetary Climate*. Cambridge University Press, 652 pp.
- Pierrehumbert, R. T., Brogniez, H. & Roca, R., 2007. On the relative humidity of the atmosphere. In T. Schneider & A. Sobel, Eds., *The Global Circulation of the Atmosphere: Phenomena, Theory, Challenges*, pp. 143–185. Princeton University Press.
- Pierrehumbert, R. T. & Swanson, K. L., 1995. Baroclinic instability. *Ann. Rev. Fluid Mech.*, **27**, 419–467.
- Plumb, R., 1981. Instability of the distorted polar night vortex: A theory of stratospheric warmings. *J. Atmos. Sci.*, **38**, 11, 2514–2531.
- Plumb, R. & McEwan, A., 1978. The instability of a forced standing wave in a viscous stratified fluid: A laboratory analogue of the Quasi-Biennial Oscillation. *J. Atmos. Sci.*, **35**, 10, 1827–1839.
- Plumb, R. A., 1977. The interaction of two internal waves with the mean flow: Implications for the theory of the Quasi-Biennial Oscillation. *J. Atmos. Sci.*, **34**, 1847–1858.
- Plumb, R. A., 1979. Eddy fluxes of conserved quantities by small-amplitude waves. *J. Atmos. Sci.*, **36**, 1699–1704.
- Plumb, R. A., 1984. The quasi-biennial oscillation. In J. R. Holton & T. Matsuno, Eds., *Dynamics of the Middle Atmosphere*, pp. 217–251. Terra Scientific Publishing.
- Plumb, R. A., 1990. A nonacceleration theorem for transient quasi-geostrophic eddies on a three-dimensional time-mean flow. *J. Atmos. Sci.*, **47**, 1825–1836.
- Plumb, R. A., 2002. Stratospheric transport. *J. Meteor. Soc. Japan*, **80**, 793–809.
- Plumb, R. A. & Bell, R. C., 1982. An analysis of the quasi-biennial oscillation on an equatorial beta-plane. *Quart. J. Roy. Meteor. Soc.*, **108**, 335–352.
- Poincaré, H., 1893. *Théorie des Tourbillons (Theory of Vortices [literally, Swirls])*. Georges Carré, Éditeur. Reprinted by Éditions Jacques Gabay, 1990, 211 pp.
- Poincaré, H., 1908. *Science and Method*. T. Nelson and Sons. Engl. transl.: F. Maitland. Reprinted in *The Value of Science: Essential Writings of Henri Poincaré*, Ed. S. J. Gould, Random House, 584 pp.
- Polzin, K. L., Toole, J. M., Ledwell, J. R. & Schmidt, R. W., 1997. Spatial variability of turbulent mixing in the abyssal ocean. *Science*, **276**, 93–96.
- Pope, S. B., 2000. *Turbulent flows*.
- Price, J. F., Weller, R. A. & Schudlich, R. R., 1987. Wind-driven ocean currents and Ekman transport. *Science*, **238**, 1534–1538.
- Proudman, J., 1916. On the motion of solids in liquids. *Proc. Roy. Soc. Lond. A*, **92**, 408–424.
- Queney, P., 1948. The problem of air flow over mountains: A summary of theoretical results. *BAMS*, **29**, 16–26.
- Quinn, W. H., 1974. Monitoring and predicting El Niño invasions. *J. Appl. Meteor.*, **13**, 825–830.
- Quon, C. & Ghil, M., 1992. Multiple equilibria in thermosolutal convection due to salt-flux boundary conditions. *J. Fluid Mech.*, **245**, 449–484.
- Randall, D. A., 2015. *An introduction to the global circulation of the atmosphere*. Princeton University Press, 456 pp.

- Randel, W. J., Wu, F., Swinbank, R., Nash, J. & O'Neill, A., 1999. Global QBO circulation derived from UKMO stratospheric analyses. *J. Atmos. Sci.*, **56**, 457–474.
- Rayleigh, Lord, 1880. On the stability, or instability, of certain fluid motions. *Proc. London Math. Soc.*, **11**, 57–70.
- Rayleigh, Lord, 1894. *The Theory of Sound, Volume II*. 2nd edn. Macmillan, 522 pp. Reprinted by Dover Publications, 1945.
- Rayleigh, Lord, 1912. On the propagation of waves through a stratified medium, with special reference to the question of reflection. *Proc. Roy. Soc. Lond. A*, **86**, 207–226.
- Raymond, D. J., 1995. Regulation of moist convection over the west Pacific warm pool. *J. Atmos. Sci.*, **52**, 22, 3945–3959.
- Raymond, D. J., 1997. Boundary layer quasi-equilibrium. In R. K. Smith, Ed., *The Physics and Parameterization of Moist Atmospheric Convection*, pp. 387–397. Springer.
- Raymond, D. J. & Fuchs, Ž., 2009. Moisture modes and the Madden-Julian oscillation. *J. Climate*, **22**, 11, 3031–3046.
- Read, P. L., 2001. Transition to geostrophic turbulence in the laboratory, and as a paradigm in atmospheres and oceans. *Surveys Geophys.*, **33**, 265–317.
- Reed, R. J., 1960. The structure and dynamics of the 26-month oscillation. Paper presented at the 40th anniversary meeting of the Am. Meteor. Soc., Boston.
- Reed, R. J., Campbell, W. J., Rasmussen, L. A. & Rogers, D. G., 1961. Evidence of a downward-propagating annual wind reversal in the equatorial stratosphere. *J. Geophys. Res.*, **66**, 813–818.
- Reif, F., 1965. *Fundamentals of Statistical and Thermal Physics*. McGraw-Hill, 651 pp.
- Rhines, P. B., 1975. Waves and turbulence on a β -plane. *J. Fluid. Mech.*, **69**, 417–443.
- Rhines, P. B., 1977. The dynamics of unsteady currents. In E. A. Goldberg, I. N. McCane, J. J. O'Brien, & J. H. Steele, Eds., *The Sea*, Vol. 6, pp. 189–318. J. Wiley and Sons.
- Rhines, P. B. & Holland, W. R., 1979. A theoretical discussion of eddy-driven mean flows. *Dyn. Atmos. Oceans*, **3**, 289–325.
- Rhines, P. B. & Young, W. R., 1982a. Homogenization of potential vorticity in planetary gyres. *J. Fluid Mech.*, **122**, 347–367.
- Rhines, P. B. & Young, W. R., 1982b. A theory of wind-driven circulation. I. Mid-ocean gyres. *J. Mar. Res. (Suppl.)*, **40**, 559–596.
- Richardson, L. F., 1920. The supply of energy from and to atmospheric eddies. *Proc. Roy. Soc. Lond. A*, **97**, 354–373.
- Richardson, L. F., 1922. *Weather Prediction by Numerical Process*. Cambridge University Press, 236 pp. Reprinted by Dover Publications.
- Richardson, L. F., 1926. Atmospheric diffusion on a distance-neighbour graph. *Proc. Roy. Soc. Lond. A*, **110**, 709–737.
- Richardson, P. L., 1983. Eddy kinetic-energy in the North Atlantic from surface drifters. *J. Geophys. Res.*, **88**, NC7, 4355–4367.
- Riehl, H. & Fultz, D., 1957. Jet stream and long waves in a steady rotating-dishpan experiment: structure of the circulation. *Quart. J. Roy. Meteor. Soc.*, **82**, 215–231.
- Rintoul, S. R., Hughes, C. & Olbers, D., 2001. The Antarctic Circumpolar Current system. In G. Siedler, J. Church, & J. Gould, Eds., *Ocean Circulation and Climate*, pp. 271–302. Academic Press.
- Ripa, P., 1981. On the theory of nonlinear wave–wave interactions among geophysical waves. *J. Fluid Mech.*, **103**, 87–115.
- Robinson, A. R., 1966. An investigation into the wind as the cause of the equatorial undercurrent. *J. Mar. Res.*, **24**, 179–204.
- Robinson, A. R., Ed., 1984. *Eddies in Marine Science*. Springer-Verlag, 609 pp.
- Robinson, A. R. & McWilliams, J. C., 1974. The baroclinic instability of the open ocean. *J. Phys. Oceanogr.*, **4**, 281–294.
- Robinson, A. R. & Stommel, H., 1959. The oceanic thermocline and the associated thermohaline circulation. *Tellus*, **11**, 295–308.
- Rooth, C., 1982. Hydrology and ocean circulation. *Prog. Oceanogr.*, **11**, 131–149.

- Roquet, F., Madec, G., Brodeau, L. & Nycander, J., 2015. Defining a simplified yet 'realistic' equation of state for seawater. *J. Phys. Oceanogr.*, **45**, 2564–2579.
- Roquet, F., Madec, G., McDougall, T. J. & Barker, P. M., 2015. Accurate polynomial expressions for the density and specific volume of seawater using the TEOS-10 standard. *Oce. Model.*, **90**, 29–43.
- Rossby, C.-G., 1936. Dynamics of steady ocean currents in the light of experimental fluid dynamics. *Papers Phys. Oceanogr. Meteor.*, **5**, 1–43.
- Rossby, C.-G., 1938. On the mutual adjustment of pressure and velocity distributions in certain simple current systems, II. *J. Mar. Res.*, **5**, 239–263.
- Rossby, C.-G., 1939. Relations between variation in the intensity of the zonal circulation and the displacements of the semi-permanent centers of action. *J. Marine Res.*, **2**, 38–55.
- Rossby, C.-G., 1940. Planetary flow patterns in the atmosphere. *Quart. J. Roy. Meteor. Soc.*, **66**, suppl., 68–87.
- Rossby, C.-G., 1949. On the nature of the general circulation of the lower atmosphere. In G. P. Kuiper, Ed., *The Atmospheres of the Earth and Planets*, pp. 16–48. University of Chicago Press.
- Rossby, H. T., 1965. On thermal convection driven by non-uniform heating from below: an experimental study. *Deep-Sea Res.*, **12**, 9–16.
- Ruddick, B., McDougall, T. & Turner, J., 1989. The formation of layers in a uniformly stirred density gradient. *Deep-Sea Res.*, **36**, 4, 597–609.
- Rudnick, D. L. & Weller, R. A., 1993. Observations of superinertial and near-inertial wind-driven flow. *J. Phys. Oceanogr.*, **23**, 2351–2359.
- Ruelle, D. & Takens, F., 1971. On the nature of turbulence. *Commun. Math. Phys.*, **20**, 167–192.
- Salmon, R., 1980. Baroclinic instability and geostrophic turbulence. *Geophys. Astrophys. Fluid Dyn.*, **10**, 25–52.
- Salmon, R., 1983. Practical use of Hamilton's principle. *J. Fluid Mech.*, **132**, 431–444.
- Salmon, R., 1990. The thermocline as an internal boundary layer. *J. Mar. Res.*, **48**, 437–469.
- Salmon, R., 1998. *Lectures on Geophysical Fluid Dynamics*. Oxford University Press, 378 pp.
- Saltzman, B., 1962. Finite amplitude free convection as an initial value problem. *J. Atmos. Sci.*, **19**, 329–341.
- Samelson, R. M., 1999a. Geostrophic circulation in a rectangular basin with a circumpolar connection. *J. Phys. Oceanogr.*, **29**, 3175–3184.
- Samelson, R. M., 1999b. Internal boundary layer scaling in 'two-layer' solutions of the thermocline equations. *J. Phys. Oceanogr.*, **29**, 2099–2102.
- Samelson, R. M., 2004. Simple mechanistic models of middepth meridional overturning. *J. Phys. Oceanogr.*, **34**, 2096–2103.
- Samelson, R. M., 2011. *The Theory of Large-Scale Ocean Circulation*. Cambridge University Press, 193 pp.
- Samelson, R. M. & Tziperman, E., 2001. Instability of the chaotic ENSO: the growth-phase predictability barrier. *J. Atmos. Sci.*, **58**, 23, 3613–3625.
- Samelson, R. M. & Vallis, G. K., 1997. Large-scale circulation with small diapycnal diffusion: the two-thermocline limit. *J. Mar. Res.*, **55**, 223–275.
- Sandström, J. W., 1908. Dynamische Versuche mit Meerwasser (Dynamical experiments with seawater). *Annal. Hydrogr. Marit. Meteorol.*, **36**, 6–23.
- Sandström, J. W., 1916. Meteorologische Studien im Schwedischen Hochgebirge (Meteorological studies in the Swedish high mountains). *Goteborgs Kungl. Vetenskaps-och Vitterhets-Samhalles, Handlingar*, **27**, 1–48.
- Sarachik, E. S. & Cane, M. A., 2010. *The El Niño-Southern Oscillation Phenomenon*. Cambridge University Press, 369 pp.
- Sarkisyan, A. & Ivanov, I., 1971. Joint effect of baroclinicity and relief as an important factor in the dynamics of sea currents. *Izv. Akad. Nauk Atmos. Ocean. Phys.*, **7**, 116–124.
- Scaife, A., Butchart, N., Warner, C. D. & Swinbank, R., 2002. Impact of a spectral gravity wave parameterization on the stratosphere in the Met Office Unified Model. *J. Atmos. Sci.*, **59**, 1473–1489.
- Scaife, A. & James, I. N., 2000. Response of the stratosphere to interannual variability of tropospheric planetary waves. *Quart. J. Roy. Meteor. Soc.*, **126**, 275–297.
- Schär, C., 1993. A generalization of Bernoulli's theorem. *J. Atmos. Sci.*, **50**, 1437–1443.
- Schmidtke, S., Johnson, G. C. & Lyman, J. M., 2013. MIMOC: a global monthly isopycnal upper-ocean climatology with mixed layers. *J. Geophys. Res. (Oceans)*, **118**, 4, 1658–1672.

- Schmitz, W. J., 1995. On the interbasin-scale thermohaline circulation. *Rev. Geophysics*, **33**, 151–173.
- Schneider, E. K., 1977. Axially symmetric steady-state models of the basic state for instability and climate studies. Part II: nonlinear calculations. *J. Atmos. Sci.*, **34**, 280–297.
- Schneider, T., Held, I. & Garner, S. T., 2003. Boundary effects in potential vorticity dynamics. *J. Atmos. Sci.*, **60**, 1024–1040.
- Schneider, T., O’Gorman, P. A. & Levine, X. J., 2010. Water vapor and the dynamics of climate changes. *Rev. Geophys.*, **48**, RG3001.
- Schneider, T. & Sobel, A., Eds., 2007. *The Global Circulation of the Atmosphere*. Princeton University Press.
- Schneider, T. & Walker, C. C., 2006. Self-organization of atmospheric macroturbulence into critical states of weak nonlinearity. *J. Atmos. Sci.*, **63**, 1569–1586.
- Schoeberl, M. R., 1978. Stratospheric warmings: observations and theory. *Revs. Geophys. Space Phys.*, **16**, 521–538.
- Schubert, W. H., Hausman, S. A., Garcia, M., Ooyama, K. V. & Kuo, H.-C., 2001. Potential vorticity in a moist atmosphere. *J. Atmos. Sci.*, **58**, 3148–3157.
- Schubert, W. H. & Masarik, M. T., 2006. Potential vorticity aspects of the MJO. *Dyn. Atmos. Oceans*, **42**, 1, 127–151.
- Schubert, W. H., Ruprecht, E., Hertenstein, R., Nieto Ferreira, R. *et al.*, 2004. English translations of twenty-one of Ertel’s papers on geophysical fluid dynamics. *Meteor. Z.*, **13**, 527–576.
- Scinocca, J. F. & Shepherd, T. G., 1992. Nonlinear wave-activity conservation laws and Hamiltonian structure for the two-dimensional anelastic equations. *J. Atmos. Sci.*, **49**, 5–28.
- Scorer, R. S. & Ludlam, F. H., 1951. Bubble theory of penetrative convection. *Q. J. Mech. Appl. Maths*, **3**, 107–112.
- Scott, R. B., 2001. Evolution of energy and enstrophy containing scales in decaying, two-dimensional turbulence with friction. *Phys. Fluids*, **13**, 2739–2742.
- Scott, R. K. & Dritschel, D. G., 2012. The structure of zonal jets in geostrophic turbulence. *J. Fluid Mech.*, **711**, 576–598.
- Scott, R. K. & Haynes, P. H., 1998. Internal interannual variability of the extratropical stratospheric circulation: The low latitude flywheel. *Quart. J. Roy. Meteor. Soc.*, **124**, 2149–2173.
- Scott, R. K. & Haynes, P. H., 2000. Internal vacillations in stratosphere-only models. *J. Atmos. Sci.*, **57**, 2333–2350.
- Shapiro, M. & Grønås, S., Eds., 1999. *The Life Cycles of Extratropical Cyclones*. American Meteorological Society, 359 pp.
- Shepherd, T. G., 1983. Mean motions induced by baroclinic instability in a jet. *Geophys. Astrophys. Fluid Dyn.*, **27**, 35–72.
- Shepherd, T. G., 1987. A spectral view of nonlinear fluxes and stationary-transient interaction in the atmosphere. *J. Atmos. Sci.*, **44**, 1166–1179.
- Shepherd, T. G., 1990. Symmetries, conservation laws, and Hamiltonian structure in geophysical fluid dynamics. *Adv. Geophys.*, **32**, 287–338.
- Shepherd, T. G., 1993. A unified theory of available potential energy. *Atmosphere–Ocean*, **31**, 1–26.
- Sherwood, S. C., Roca, R., Weckwerth, T. M. & Andronova, N. G., 2010. Tropospheric water vapor, convection and climate. *Rev. Geophys.*, **48**, RG2001.
- Shutts, G. J., 1983. Propagation of eddies in diffluent jet streams: eddy vorticity forcing of blocking flow fields. *Quart. J. Roy. Meteor. Soc.*, **109**, 737–761.
- Siedler, G., Church, J. & Gould, J., 2001. *Ocean Circulation and Climate: Observing and Modelling the Global Ocean*. Academic Press, 715 pp.
- Silberstein, L., 1896. O tworzeniu się wirow, w płynie doskonałym (On the creation of eddies in an ideal fluid). *W Krakowie Nakładem Akademii Umiejętności (Proc. Cracow Acad. Sci.)*, **31**, 325–335.
- Simmonds, J. G. & Mann, J. E., 1998. *A First Look at Perturbation Theory*. Dover Publications, 139 pp.
- Simmons, A. & Hoskins, B., 1978. The life-cycles of some nonlinear baroclinic waves. *J. Atmos. Sci.*, **35**, 414–432.
- Sjoberg, J. P. & Birner, T., 2012. Transient tropospheric forcing of sudden stratospheric warmings. *J. Atmos. Sci.*, **69**, 11, 3420–3432.

- Smagorinsky, J., 1953. The dynamical influences of large-scale heat sources and sinks on the quasi-stationary mean motions of the atmosphere. *Quart. J. Roy. Meteor. Soc.*, **79**, 342–366.
- Smagorinsky, J., 1969. Problems and promises of deterministic extended range forecasting. *Bull. Am. Meteor. Soc.*, **50**, 286–311.
- Smith, K. S., Boccaletti, G., Henning, C. C., Marinov, I. *et al.*, 2002. Turbulent diffusion in the geostrophic inverse cascade. *J. Fluid Mech.*, **469**, 13–48.
- Smith, K. S. & Vallis, G. K., 1998. Linear wave and instability properties of extended range geostrophic models. *J. Atmos. Sci.*, **56**, 1579–1593.
- Smith, K. S. & Vallis, G. K., 2001. The scales and equilibration of mid-ocean eddies: freely evolving flow. *J. Phys. Oceanogr.*, **31**, 554–571.
- Smith, K. S. & Vallis, G. K., 2002. The scales and equilibration of mid-ocean eddies: forced-dissipative flow. *J. Phys. Oceanogr.*, **32**, 1669–1721.
- Smith, L. M. & Waleffe, F., 1999. Transfer of energy to two-dimensional large scales in forced, rotating three-dimensional turbulence. *Phys. Fluids*, **11**, 1608–1622.
- Smith, R. B., 1979. The influence of mountains on the atmosphere. In B. Saltzman, Ed., *Advances in Geophysics*, vol. 21, pp. 87–230. Academic Press.
- Smith, R. K., 1997. On the theory of CISK. *Quart. J. Roy. Meteor. Soc.*, **123**, 538, 407–418.
- Sobel, A. & Maloney, E., 2013. Moisture modes and the eastward propagation of the MJO. *J. Atmos. Sci.*, **70**, 1, 187–192.
- Sobel, A. H., Nilsson, J. & Polvani, L., 2001. The weak temperature gradient approximation and balanced tropical moisture waves. *J. Atmos. Sci.*, **58**, 3650–3665.
- Spall, M. A., 2000. Generation of strong mesoscale eddies by weak ocean gyres. *J. Mar. Res.*, **58**, 97–116.
- Spiegel, E. A. & Veronis, G., 1960. On the Boussinesq approximation for a compressible fluid. *Astrophys. J.*, **131**, 442–447. (Correction: *Astrophys. J.*, **135**, 655–656).
- Squire, H., 1933. On the stability of three-dimensional disturbances of viscous flow between parallel walls. *Proc. Roy. Soc. Lond. A*, **142**, 621–628.
- Srinivasan, K. & Young, W., 2012. Zonostrophic instability. *J. Atmos. Sci.*, **69**, 5, 1633–1656.
- Stammer, D., 1997. Global characteristics of ocean variability estimated from regional TOPEX/Poseidon altimeter measurements. *J. Phys. Oceanogr.*, **27**, 1743–1769.
- Starr, V. P., 1948. An essay on the general circulation of the Earth's atmosphere. *J. Meteor.*, **78**, 39–43.
- Starr, V. P., 1968. *Physics of Negative Viscosity Phenomena*. McGraw-Hill, 256 pp.
- Stechmann, S. N. & Ogrosky, H. R., 2014. The Walker circulation, diabatic heating, and outgoing longwave radiation. *Geophys. Res. Lett.*, **41**, 9097–9105.
- Steers, J. A., 1962. *An Introduction to the Study of Map Projections*. Univ. of London Press, 288 pp.
- Stern, M. E., 1963. Trapping of low frequency oscillations in an equatorial boundary layer. *Tellus*, **15**, 246–250.
- Stevens, D. P. & Ivchenko, V. O., 1997. The zonal momentum balance in an eddy-resolving general-circulation model of the southern ocean. *Quart. J. Roy. Meteor. Soc.*, **123**, 929–951.
- Stewart, G. R., 1941. *Storm*. Random House, 349 pp.
- Stewartson, K., 1977. The evolution of the critical layer of a Rossby wave. *Geophys. Astrophys. Fluid Dyn.*, **9**, 1, 185–200.
- Stips, A., 2005. Dissipation measurement: theory. In H. Z. Baumert, J. Simpson, & J. Sündermann, Eds., *Marine Turbulence*. Cambridge University Press.
- Stommel, H., 1958. The abyssal circulation. *Deep-Sea Res.*, **5**, 80–82.
- Stommel, H., 1960. Wind-drift near the equator. *Deep Sea Research Part*, **6**, 298–302.
- Stommel, H., 1961. Thermohaline convection with two stable regimes of flow. *Tellus*, **13**, 224–230.
- Stommel, H. & Arons, A. B., 1960. On the abyssal circulation of the world ocean—I. Stationary planetary flow patterns on a sphere. *Deep-Sea Res.*, **6**, 140–154.
- Stommel, H., Arons, A. B. & Faller, A. J., 1958. Some examples of stationary planetary flow patterns in bounded basins. *Tellus*, **10**, 179–187.
- Stommel, H. & Moore, D. W., 1989. *An Introduction to the Coriolis Force*. Columbia University Press, 297 pp.

- Stommel, H. & Webster, J., 1963. Some properties of the thermocline equations in a subtropical gyre. *J. Mar. Res.*, **44**, 695–711.
- Stone, P. H., 1972. A simplified radiative-dynamical model for the static stability of rotating atmospheres. *J. Atmos. Sci.*, **29**, 405–418.
- Stone, P. H., 1978. Baroclinic adjustment. *J. Atmos. Sci.*, **35**, 561–571.
- Stone, P. H. & Nemet, B., 1996. Baroclinic adjustment: a comparison between theory, observations, and models. *J. Atmos. Sci.*, **53**, 1663–1674.
- Straub, D. N., 1993. On the transport and angular momentum balance of channel models of the Antarctic Circumpolar Current. *J. Phys. Oceanogr.*, **23**, 776–782.
- Suarez, M. & Schopf, P., 1988. A delayed action oscillator for ENSO. *J. Atmos. Sci.*, **45**, 323–3287.
- Sukhatme, J. & Young, W. R., 2011. The advection–condensation model and water-vapour probability density functions. *Quart. J. Roy. Meteor. Soc.*, **137**, 1561–1572.
- Sukoriansky, S., Dikovskaya, N. & Galperin, B., 2007. On the arrest of inverse energy cascade and the Rhines scale. *J. Atmos. Sci.*, **64**, 9, 3312–3327.
- Sutcliffe, R. C., 1939. Cyclonic and anticyclonic development. *Quart. J. Roy. Meteor. Soc.*, **65**, 518–524.
- Sutcliffe, R. C., 1947. A contribution to the problem of development. *Quart. J. Roy. Meteor. Soc.*, **73**, 370–383.
- Sutherland, B., 2010. *Internal Gravity Waves*. Cambridge University Press, 394 pp. ISBN: 978-0521839150.
- Swallow, J. C. & Worthington, V., 1961. An observation of a deep countercurrent in the western North Atlantic. *Deep-Sea Res.*, **8**, 1–19.
- Tailleux, R., 2016. Neutrality versus materiality: A thermodynamic theory of neutral surfaces. *Fluids*, **1**, 4, 32.
- Takahashi, M., 1996. Simulation of the stratospheric quasi-biennial oscillation using a general circulation model. *Geophys. Res. Lett.*, **23**, 6, 661–664.
- Talley, L., 1988. Potential vorticity distribution in the North Pacific. *J. Phys. Oceanogr.*, **18**, 89–106.
- Talley, L. D., Pickard, G., Emery, W. J. & Swift, J. H., 2011. *Descriptive physical oceanography: An introduction*. Academic press, 555 pp. ISBN: 978-0-7506-4552-2.
- Taylor, G. I., 1921a. Diffusion by continuous movements. *Proc. London Math. Soc.*, **2** (20), 196–211.
- Taylor, G. I., 1921b. Experiments with rotating fluids. *Proc. Roy. Soc. Lond. A*, **100**, 114–121.
- Tennekes, H. & Lumley, J. L., 1972. *A First Course in Turbulence*. The MIT Press, 330 pp.
- Tesserenc De Bort, L. P., 1902. Variations de la température de l'air libre dans la zone comprise 8 km et 13 km d'altitude (Variations in the temperature of the free air in the zone between 8 km and 13 km of altitude). *C. R. Hebd. Séances Acad. Sci.*, **134**, 987–989.
- Thompson, A. F., Stewart, A. L. & Bischoff, T., 2016. A multibasin residual-mean model for the global overturning circulation. *J. Phys. Oceanogr.*, **46**, 9, 2583–2604.
- Thompson, A. F. & Young, W. R., 2006. Scaling baroclinic eddy fluxes: vortices and energy balance. *J. Phys. Oceanogr.*, **36**, 720–738.
- Thompson, P. D., 1957. Uncertainty of initial state as a factor in the predictability of large scale atmospheric flow patterns. *Tellus*, **9**, 275–295.
- Thompson, R. O. R. Y., 1971. Why there is an intense eastward current in the North Atlantic but not in the South Atlantic. *J. Phys. Oceanogr.*, **1**, 235–237.
- Thompson, R. O. R. Y., 1980. A prograde jet driven by Rossby waves. *J. Atmos. Sci.*, **37**, 1216–1226.
- Thomson, J., 1892. Bakerian lecture. On the grand currents of atmospheric circulation. *Phil. Trans. Roy. Soc. Lond. A*, **183**, 653–684.
- Thomson, W. (Lord Kelvin), 1869. On vortex motion. *Trans. Roy. Soc. Edinburgh*, **25**, 217–260.
- Thomson, W. (Lord Kelvin), 1871. Hydrokinetic solutions and observations. *Phil. Mag. and J. Science*, **42**, 362–377.
- Thomson, W. (Lord Kelvin), 1879. On gravitational oscillations of rotating water. *Proc. Roy. Soc. Edinburgh*, **10**, 92–100.
- Thorncroft, C. D., Hoskins, B. J. & McIntyre, M. E., 1993. Two paradigms of baroclinic-wave life-cycle behaviour. *Quart. J. Roy. Meteor. Soc.*, **119**, 17–55.
- Thorpe, A. J., Volkert, H. & Ziemianski, M. J., 2003. The Bjerknes' circulation theorem: a historical perspective. *Bull. Am. Meteor. Soc.*, **84**, 471–480.

- Thual, O. & McWilliams, J. C., 1992. The catastrophe structure of thermohaline convection in a two-dimensional fluid model and a comparison with low-order box models. *Geophys. Astrophys. Fluid Dyn.*, **64**, 67–95.
- Thuburn, J., 2017. Use of the Gibbs thermodynamic potential to express the equation of state in atmospheric models. *Quart. J. Roy. Meteor. Soc.*, submitted.
- Thuburn, J. & Craig, G. C., 1997. GCM tests of theories for the height of the tropopause. *J. Atmos. Sci.*, **54**, 869–882.
- Thuburn, J. & Craig, G. C., 2000. Stratospheric influence on tropopause height: the radiative constraint. *J. Atmos. Sci.*, **57**, 17–28.
- Tobias, S. & Marston, J., 2013. Direct statistical simulation of out-of-equilibrium jets. *Phys. Rev. Lett.*, **110**, 10, 104502.
- Toggweiler, J. R. & Samuels, B., 1995. Effect of Drake Passage on the global thermohaline circulation. *Deep-Sea Res.*, **42**, 477–500.
- Toggweiler, J. R. & Samuels, B., 1998. On the ocean's large-scale circulation in the limit of no vertical mixing. *J. Phys. Oceanogr.*, **28**, 1832–1852.
- Toole, J. M., Polzin, K. L. & Schmitt, R. W., 1994. Estimates of diapycnal mixing in the abyssal ocean. *Science*, **264**, 1120–1123.
- Tréguier, A. M., Held, I. M. & Larichev, V. D., 1997. Parameterization of quasi-geostrophic eddies in primitive equation ocean models. *J. Phys. Oceanogr.*, **29**, 567–580.
- Trenberth, K. E., 1997. The definition of El Niño. *Bull. Am. Meteor. Soc.*, **78**, 2771–2777.
- Trenberth, K. E. & Caron, J. M., 2001. Estimates of meridional atmosphere and ocean heat transports. *J. Climate*, **14**, 3433–3443.
- Tritton, D. J., 1988. *Physical Fluid Dynamics*. Oxford University Press, 519 pp.
- Truesdell, C., 1951. Proof that Ertel's vorticity theorem holds in average for any medium suffering no tangential acceleration on the boundary. *Geofis Pura Appl.*, **19**, 167–169.
- Truesdell, C., 1954. *The Kinematics of Vorticity*. Indiana University Press, 232 pp.
- Tsang, Y.-K. & Vanneste, J., 2015. Advection-condensation of water vapor in a model of coherent stirring. Unpublished.
- Tudhope, A. W., Chilcott, C. P., McCulloch, M. T., Cook, E. R. *et al.*, 2001. Variability in the El Niño Southern Oscillation through a glacial-interglacial cycle. *Science*, **291**, 1511–1517.
- Tung, K. K., 1979. A theory of stationary long waves. Part III: quasi-normal modes in a singular wave guide. *Mon. Wea. Rev.*, **107**, 751–774.
- Tziperman, E., Stone, L., Cane, M. A. & Jarosh, H., 1994. El Niño chaos: Overlapping of resonances between the seasonal cycle and the Pacific ocean-atmosphere oscillator. *Science*, **264**, 5155, 72–73.
- Valdes, P. J. & Hoskins, B. J., 1988. Baroclinic instability of the zonally averaged flow with boundary layer damping. *J. Atmos. Sci.*, **45**, 1584–1593.
- Vallis, G. K., 1982. A statistical dynamical climate model with a simple hydrology cycle. *Tellus*, **34**, 211–227.
- Vallis, G. K., 1985. Instability and flow over topography. *Geophys. Astrophys. Fluid Dyn.*, **34**, 1–38.
- Vallis, G. K., 1988a. Conceptual models of El Niño and the Southern Oscillation. *J. Geophys. Res.*, **93**, 13979–13991.
- Vallis, G. K., 1988b. Numerical studies of eddy transport properties in eddy-resolving and parameterized models. *Quart. J. Roy. Meteor. Soc.*, **114**, 183–204.
- Vallis, G. K., 1996. Potential vorticity and balanced equations of motion for rotating and stratified flows. *Quart. J. Roy. Meteor. Soc.*, **122**, 291–322.
- Vallis, G. K., 2000. Large-scale circulation and production of stratification: effects of wind, geometry and diffusion. *J. Phys. Oceanogr.*, **30**, 933–954.
- Vallis, G. K. & Maltrud, M. E., 1993. Generation of mean flows and jets on a beta plane and over topography. *J. Phys. Oceanogr.*, **23**, 1346–1362.
- Vallis, G. K., Zurita-Gotor, P., Cairns, C. & Kidston, J., 2015. The response of the large-scale structure of the atmosphere to global warming. *Quart. J. Roy. Meteor. Soc.*, **141**, 1479–1501.
- Vanneste, J. & Shepherd, T. G., 1998. On the group-velocity property for wave-activity conservation laws. *J. Atmos. Sci.*, **55**, 1063–1068.
- Verkley, W. T. M. & van der Velde, I. R., 2010. Balanced dynamics in the tropics. *QJRM*, **136**, 41–49.

- Veronis, G., 1960. An approximate theoretical analysis of the equatorial undercurrent. *Deep-Sea Res.*, **6**, 318–327.
- Veronis, G., 1966a. Wind-driven ocean circulation – Part 1: linear theory and perturbation analysis. *Deep-Sea Res.*, **13**, 17–29.
- Veronis, G., 1966b. Wind-driven ocean circulation – Part 2: numerical solutions of the non-linear problem. *Deep-Sea Res.*, **13**, 30–55.
- Veronis, G., 1969. On theoretical models of the thermocline circulation. *Deep-Sea Res.*, **31** Suppl., 301–323.
- Veryard, R. G. & Ebdon, R. A., 1961. Fluctuations in tropical stratospheric winds. *Meteor. Mag.*, **90**, 125–143.
- Visbeck, M., Marshall, J., Haine, T. & Spall, M., 1997. Specification of eddy transfer coefficients in coarse-resolution ocean circulation models. *J. Phys. Oceanogr.*, **27**, 381–402.
- Walker, C. & Schneider, T., 2005. Response of idealized Hadley circulations to seasonally varying heating. *Geophys. Res. Lett.*, **32**, L06813. doi:10.1029/2004GL022304.
- Wallace, J. M., 1973. General circulation of the tropical lower stratosphere. *Rev. Geophys.*, **11**, 191–222.
- Wallace, J. M., 1983. The climatological mean stationary waves: observational evidence. In B. Hoskins & R. P. Pearce, Eds., *Large-Scale Dynamical Processes in the Atmosphere*, pp. 27–63. Academic Press.
- Wallace, J. M. & Hobbs, P. V., 2006. *Atmospheric Science: An Introductory Survey*. 2nd edn. Elsevier, 483 pp.
- Wallace, J. M. & Holton, J. R., 1968. A diagnostic numerical model of the quasi-biennial oscillation. *J. Atmos. Sci.*, **25**, 280–292.
- Warn, T., Bokhove, O., Shepherd, T. G. & Vallis, G. K., 1995. Rossby number expansions, slaving principles, and balance dynamics. *Quart. J. Roy. Meteor. Soc.*, **121**, 723–739.
- Warn, T. & Warn, H., 1976. On the development of a Rossby wave critical level. *J. Atmos. Sci.*, **33**, 10, 2021–2024.
- Warren, B. A., 1981. Deep circulation of the world ocean. In B. A. Warren & C. Wunsch, Eds., *Evolution of Physical Oceanography*, pp. 6–41. The MIT Press.
- Warren, B. A., 1999. Approximating the energy transport across oceanic sections. *J. Geophys. Res.*, **104**, 7915–7920.
- Warren, B. A., 2006. The first law of thermodynamics in a salty ocean. *Progress in Oceanography*, **70**, 2, 149–167.
- Warren, B. A., LaCasce, J. H. & Robbins, P. E., 1996. On the obscurantist physics of form drag in theorizing about the Circumpolar Current. *J. Phys. Oceanogr.*, **26**, 2297–2301.
- Wasow, W., 1944. Asymptotic solution of boundary value problems for the differential equation $\Delta U + \lambda(\partial/\partial x)U = \lambda f(x, y)$. *Duke Math J.*, **11**, 405–415.
- Webb, D. J. & Suginohara, N., 2001. Vertical mixing in the ocean. *Nature*, **409**, 37.
- Weinstock, R., 1952. *Calculus of Variations*. McGraw-Hill. Reprinted by Dover Publications, 1980, 328 pp.
- Welander, P., 1959. An advective model of the ocean thermocline. *Tellus*, **11**, 309–318.
- Welander, P., 1968. Wind-driven circulation in one- and two-layer oceans of variable depth. *Tellus*, **20**, 1–15.
- Welander, P., 1971a. Some exact solutions to the equations describing an ideal-fluid thermocline. *J. Mar. Res.*, **29**, 60–68.
- Welander, P., 1971b. The thermocline problem. *Phil. Trans. Roy. Soc. Lond. A*, **270**, 415–421.
- Welander, P., 1973. Lateral friction in the ocean as an effect of potential vorticity mixing. *Geophys. Fluid Dyn.*, **5**, 101–120.
- Welander, P., 1986. Thermohaline effects in the ocean circulation and related simple models. In J. Willebrand & D. L. T. Anderson, Eds., *Large-scale Transport Processes in Oceans and Atmospheres*, pp. 163–200. Reidel.
- Wentzel, G., 1926. Eine Verallgemeinerung der Quantenbedingungen für die Zwecke der Wellenmechanik (A generalization of the quantum conditions for the purposes of wave mechanics). *Zeit. für Physic A*, **38**, 518–529.
- Wheeler, M. & Kiladis, G. N., 1999. Convectively coupled equatorial waves: Analysis of clouds and temperature in the wavenumber-frequency domain. *J. Atmos. Sci.*, **56**, 3, 374–399.
- White, A. A., 1977. Modified quasi-geostrophic equations using geometric height as vertical co-ordinate. *Quart. J. Roy. Meteor. Soc.*, **103**, 383–396.
- White, A. A., 2002. A view of the equations of meteorological dynamics and various approximations. In J.

- Norbury & I. Roulstone, Eds., *Large-Scale Atmosphere-Ocean Dynamics I*, pp. 1–100. Cambridge University Press.
- White, A. A., 2003. The primitive equations. In J. Holton, J. Pyle, & J. Curry, Eds., *Encyclopedia of Atmospheric Science*, pp. 694–702. Academic Press.
- Whitehead, J. A., 1975. Mean flow generated by circulation on a beta-plane: An analogy with the moving flame experiment. *Tellus*, **27**, 358–364.
- Whitehead, J. A., 1995. Thermohaline ocean processes and models. *Ann. Rev. Fluid Mech.*, **27**, 89–113.
- Whitham, G. B., 1974. *Linear and Nonlinear Waves*. Wiley-Interscience, 656 pp.
- Williams, G. P., 1978. Planetary circulations: 1. Barotropic representation of Jovian and terrestrial turbulence. *J. Atmos. Sci.*, **35**, 1399–1426.
- Wittenberg, A. T., 2009. Are historical records sufficient to constrain ENSO simulations? *Geophys. Res. Lett.*, **36**, L12702. doi:10.1029/2009GL038710.
- Wolfe, C. L. & Cessi, P., 2010. What sets the middepth stratification of eddying ocean models? *J. Phys. Oceanogr.*, **40**, 7, 1520–1538.
- Wolfe, C. L. & Cessi, P., 2011. The adiabatic pole-to-pole overturning circulation. *J. Phys. Oceanogr.*, **41**, 9, 1795–1810.
- World Meteorological Organization, 1957. Definition of the tropopause. *WMO Bulletin*, **6**, 136.
- Wunsch, C., 2002. What is the thermohaline circulation? *Science*, **298**, 1179–1180.
- Wunsch, C., 2015. *Modern Observational Physical Oceanography*. Princeton University Press, 481 pp. ISBN: 978-0-691-15882-2.
- Wunsch, C. & Ferrari, R., 2004. Vertical mixing, energy, and the general circulation of the oceans. *Ann. Rev. Fluid Mech.*, **36**, 281–314.
- Wunsch, C. & Roemmich, D., 1985. Is the North Atlantic in Sverdrup balance? *J. Phys. Oceanogr.*, **15**, 1876–1880.
- Wyrtki, K., 1952. Der Einfluss des Windes auf den mittleren Wasserstand Der Nordsee und ihren Wasserhaushalt (the influence of the wind on the mean sea level and water budget of the north sea). *Dtsch. Hydrogr. Z.*, **5**, 21–27.
- Wyrtki, K., 1975. El Niño—The dynamic response of the equatorial Pacific ocean to atmospheric forcing. *J. Phys. Oceanogr.*, **5**, 4, 572–584.
- Wyrtki, K., 1985. Water displacements in the Pacific and the genesis of El Niño cycles. *J. Geophys. Res.*, **90**, C4, 7129–7132.
- Wyrtki, K., Magsaard, L. & Hager, J., 1976. Eddy energy in oceans. *J. Geophys. Res.*, **81**, 15, 2641–2646.
- Xu, X., Rhines, P. B., Chassignet, E. P. & Schmitz, W. J., 2015. Spreading of Denmark Strait overflow water in the western subpolar North Atlantic: insights from eddy-resolving simulations with a passive tracer. *J. Phys. Oceanogr.*, **45**, 12, 2913–2932.
- Yaglom, A. M., 1994. A. N. Kolmogorov as a fluid mechanician and founder of a school in turbulence research. *Ann. Rev. Fluid Mech.*, **26**, 1–22.
- Yanai, M. & Maruyama, T., 1966. Stratospheric wave disturbances in the tropical stratosphere. *J. Meteor. Soc. Japan*, **44**, 291–294.
- Yoden, S., 1987. Bifurcation properties of a stratospheric vacillation model. *J. Atmos. Sci.*, **44**, 1723–1733.
- Yoden, S., 1990. An illustrative model of seasonal and interannual variations of the stratospheric circulation. *J. Atmos. Sci.*, **47**, 1845–1853.
- Young, W. R., 2010. Dynamic enthalpy, conservative temperature, and the seawater Boussinesq approximation. *J. Phys. Oceanogr.*, **40**, 5, 394–400.
- Young, W. R., 2012. An exact thickness-weighted average formulation of the Boussinesq equations. *J. Phys. Oceanogr.*, **42**, 5, 692–707.
- Young, W. R. & Rhines, P. B., 1982. A theory of the wind-driven circulation II. Gyres with western boundary layers. *J. Mar. Res.*, **40**, 849–872.
- Zeblak, S. E. & Cane, M. A., 1987. A model El Niño Southern Oscillation. *Mon. Wea. Rev.*, **115**, 2262–78.
- Zeng, N., Neelin, J. D. & Chou, C., 2000. A quasi-equilibrium tropical circulation model—implementation and simulation. *J. Atmos. Sci.*, **57**, 11, 1767–1796.
- Zhang, C., 2005. Madden-Julian oscillation. *Rev. Geophys.*, **43**, 1–36.

- Zhang, R. & Vallis, G. K., 2007. The role of the bottom vortex stretching on the path of the North Atlantic western boundary current and on the northern recirculation gyre. *J. Phys. Oceanogr.*, **37**, 2053–2080.
- Zurita-Gotor, P. & Lindzen, R., 2007. Theories of baroclinic adjustment and eddy equilibration. In T. Schneider & A. Sobel, Eds., *The Global Circulation of the Atmosphere: Phenomena, Theory, Challenges*. Princeton University Press.
- Zurita-Gotor, P. & Vallis, G. K., 2009. Equilibration of baroclinic turbulence in primitive equations and quasi-geostrophic models. *J. Atmos. Sci.*, **66**, 837–863.
- Zurita-Gotor, P. & Vallis, G. K., 2011. Dynamics of mid-latitude tropopause height in an idealized model. *J. Atmos. Sci.*, **68**, 823–838.

Index

Bold face denotes a primary entry or an extended discussion.

- Abbe, Cleveland, 442
- Absolute humidity, 675
- Abyssal ocean circulation, **818–829**
 - wind driven, 829
- ACC, **836–846**
 - adiabatic model of, 843
 - and mesoscale eddies, 837
 - form drag in, 844
 - momentum balance, 838, 839, 841
- Acoustic-gravity waves, **293–296**
- Adiabatic lapse rate
 - dry, 99
 - moist, 692
- Adiabatic lapse rate, 48
- Advection-diffusion-condensation models, 686
- Advective derivative, 4
- Anelastic approximation, **75–78**
- Anelastic equations, 76, 78
 - energetics of, 78
- Angular momentum, 67
 - spherical coordinates, 67
- Antarctic Circumpolar Current, **836–846**
- Antisymmetric turbulent diffusivity, 491
- APE, 137
- Arnold stability conditions, 406
- Asymptotic models
 - conservation properties of, 187
 - quasi-geostrophy, 188
- Atmospheric stratification, **572–580**
- Auto-barotropic fluid, 14
- Available potential energy, **137–140**
 - Boussinesq fluid, 138
 - ideal gas, 139
- Balanced dynamics
 - tropics, 711
- Baroclinic adjustment, 576
- Baroclinic circulation theorem, 151
- Baroclinic eddies, **464–471**
 - effect on Hadley Cell, 528
 - in atmosphere, 465
 - in ocean, 468
 - magnitude and scale, 464
- Baroclinic eddy diffusivities, 494
- Baroclinic Fluid, 14
- Baroclinic instability, **335, 347–367**
 - beta effect in continuous model, 369
 - beta effect in two-layer model, 360
 - Eady problem, 351
 - effect of stratosphere, 372
 - energetics of, 367
 - high-wavenumber cut-off, 343, 360, 410, 459
 - in ocean, 373
 - interacting edge waves, 363
 - linear QG equations, 349
 - mechanism of, 347, 363
 - minimum shear, 361
 - necessary conditions for, 350, 408, 409
 - neutral curve in two-layer problem, 362
 - non-uniform shear and stratification, 372
 - sloping convection, 347
 - two-layer problem, 356
- Baroclinic lifecycle, 466
- Baroclinic lifecycles
 - in atmosphere, 465
 - in ocean, 468, 470
- Baroclinic term, 146
- Baroclinic triads, 457
- Barotropic fluid, 14, 20
- Barotropic instability, **335**
- Barotropic jet, **540–549**
 - and Rossby waves, 542
 - and the EP flux, 547

- numerical example, 548
- Barotropic triads, 457
- Batchelor scale, 439
- Batchelor spectrum, 439
- Bernoulli function, 44
- Bernoulli's theorem, 44
 - and potential vorticity flux, 167
- Beta effect, 154, 155
 - in two-dimensional turbulence, 445
- Beta plane vorticity equation, 156
- Beta scale, 446, 447
- Beta-plane approximation, 69
- Beta-Rossby number, 559, 735
- Bjerknes, Jacob, 907
- Bjerknes, Vilhelm, 168
- Bjerknes-Silberstein circulation theorem, 152
- Boiling point, 678
- Bolus velocity, 500, 503
- Bottom pressure stress, 755
- Boundary layers
 - Ekman, 201
- Boussinesq approximation, 70–75
- Boussinesq equations, 71
 - asymptotic derivation, 101
 - energetics of, 74
 - potential vorticity conservation, 162
 - relation to pressure coordinates, 82
 - strong and weak versions, 78
 - summary, 74
- Box ocean models, 813–818
 - many boxes, 817
 - two boxes, 814
- Breaking waves, 599
- Bretherton's boundary layer, 191
- Brewer–Dobson circulation, 631–643, 644, 670
- Brunt–Väisälä frequency, 97
- Buoyancy frequency, 73, 97
 - ideal gas, 99
 - ocean, 99
- Buoyancy-driven ocean circulation, 801, 813
- Burger number, 173
- Cabelling, 38
- CAPE, 696
- Centrifugal force, 57
- Chaos, 433, 435
 - a brief history, 443
- Charney problem, 369
- Charney, Jule, 376
- Charney–Drazin condition, 599, 601, 603
- Charney–Eliassen problem, 609
- Charney–Green number, 370
- Charney–Stern–Pedlosky criterion for instability, 351
- Chemical potential, 16
- CIN, 696
- Circular reference, *see* Reference, circular
- Circulation, 143–153
- Circulation theorem, 149, 147–156
 - baroclinic, 151
 - barotropic fluid, 149
 - beta effect, 154
 - hydrostatic flow, 153
 - rotating flow, 152
- CISK, 727
- Clausius–Clapeyron equation, 678–679
 - ideal gas, 679
- Closure problem of turbulence, 413
- Compensating subsidence, 716
- Compressible flow, 40
- Concentration and mixing ratio, 10
- Condensation-diffusion models, 681
- Condensation-diffusion-advection models, 686
- Conditional instability, 696
- Conservative Temperature, 35
- Conservative temperature, 39
- Conservative tracers, 10
- Convection, 699
 - energetics of, 696
 - moist, 695
- Convective adjustment, 697, 706
- Convective instability, 261
- Convective quasi-equilibrium, 698
- Conventional equation of state, 13
- Coriolis acceleration, 57
- Coriolis force, 58, 57–58
- Coriolis, Gaspard Gustave de, 103
- Critical layer, 594
- Critical layers
 - gravity wave, 597
- Critical levels, 634
- Critical line, 593, 594, 607
- Cyclostrophic balance, 91, 95, 96
- Deacon Cell, 833, 841
- Deacon cell, 834
- Deformation radius, 124, 125
- Delayed oscillator model of El Niño, 903
- Diffusion
 - equation of, 473
 - turbulent, 473
- Diffusion-condensation models, 681
- Diffusive Fluxes, 490
- Diffusive thermocline, 780–785, 805
- Diffusive transport, 473
- Diffusivity tensors, 490
- Dispersion relation, 215, 216, 219
 - Rossby waves, 228
- Divergence equation, 96
- Downward control, 648
- Dry adiabatic lapse rate, 29, 48, 99

- Dumbbell in beta-plane turbulence, 448
- Eady problem, **351–356**, 357
 - eddy effect on mean flow, 399
 - secondary circulation, 399
 - with beta, 371
- Eady, Eric, 376
- Eddy diffusion, 475
 - two-dimensional, 483
- Eddy transport
 - and the TEM, 506
 - velocity, 500
- Edge waves, 341, 343
 - Eady problem, 365
 - in shear flows, 340
- Effective gravity, 59
- Egg, boiling, 678
- Ekman layers, **201–211**
 - integral properties of, 204
 - momentum balance, 203
 - observed, 209
 - stress in, 201
- Ekman number, 202
- Ekman spiral, 206, 209
- El Niño, **886–903**
 - toy models, 898
- Eliassen–Palm flux, **383–387**, 568
 - and barotropic jets, 547
 - and form drag, 397
 - observed, 568
 - primitive equations, 581
 - spherical coordinates, 581
- Eliassen–Palm relation, 384
- Energetics
 - of quasi-geostrophic equations, 198
- Energy budget, 42
 - constant density fluid, 42
 - variable density fluid, 43
 - viscous effects, 44
- Energy conservation
 - Boussinesq equations, 74
 - primitive equations, 65
 - shallow water equations, 122
- Energy flux, 234
 - Bernoulli function, 44
 - Rossby waves, 234–236
- Energy inertial range
 - in two-dimensional turbulence, 429
- Energy transfer in two-dimensional flow, 424
- ENSO, 886
- Enstrophy inertial range, 424, 429
 - passive tracer in, 439
- Enstrophy transfer in two-dimensional flow, 424
- Enthalpy, 16, 17, 45, 49
 - dynamic, 37
 - fluxes in convection, 705
 - generalized, 45
 - ideal gas, 21
 - of vapourization, 677
 - potential, 35, 37, 49
 - potential, generalized, 46
- Enthalpy of vapourization, 723
- Entropy, 14, 24, 47
 - moist air, 722
- Equation of State, **13–14**
- Equation of state
 - conventional or thermal, 13
 - fundamental, 14, 15, 19
 - ideal gas, 13
 - moist air, 674
 - seawater, 13, 33
- Equations of motion, **3**
 - for tropics, 711
 - rotating frame, 55
 - tropical atmosphere, 708
- Equatorial undercurrent, **865**, 867, 874
 - ideal fluid model, **876–885**
 - local model, **865–876**
- Equatorial waves
 - stratosphere, 634
- Equivalent potential temperature, 680, **694**, 723
- Equivalent topography, 617
- Euler, Leonard, 52
- Eulerian and Lagrangian, 3
- Eulerian derivative, 4
- Eulerian viewpoint, 4
- Exner function, 136
- f-plane approximation, 69
- Ferrel Cell, **533–536**, 571
 - eddy fluxes in, 535
 - surface flow in, 535
- Ferrel, William, 537
- Field or Eulerian viewpoint, 4
- First law of thermodynamics, 15, **49**
- Fjørtoft's criterion for instability, 346, 408
- Fluid element, 4
- Fofonoff model, 752
- Force
 - centrifugal, 57
 - Coriolis, 58
- Form drag, 119
 - and Eliassen Palm flux, 397
 - at ocean bottom, 755
 - in ACC, 839, 844
- Form stress, **119–120**
- Four-thirds law, 482
- Free energy, 17
- Free-slip condition, 741
- Frequency, 216
- frictional–geostrophic balance, 202

- Froude number, 86, 173, 494
- Frozen in property of vorticity, 147
- Fundamental equation of state, 14, 15, 19
 - ideal gas, 25, **47–49**
- Fundamental postulate of thermodynamics, 14
- Fundamental thermodynamic relation, 16
- Gas constant, 674
- Generalized enthalpy, 46
- Gent-McWilliams scheme, 499
- Geopotential surfaces, 60
- Geostrophic adjustment, **127–134**
 - energetics of, 130
 - Rossby problem, 128
- Geostrophic balance, **87–93**, 95, 118
 - a variational perspective, 133
 - frictional, 202
 - in shallow water equations, 118
 - pressure coordinates, 91
- Geostrophic contours, 764
- Geostrophic scaling, 171
 - in continuously stratified equations, 174
 - in shallow water equations, 171
- Geostrophic turbulence, **445**
 - Larichev–Held model, 460
 - stratified, 454
 - two layers, 455
 - two-dimensional, beta-plane, 445
- Gibbs function, 16–18
 - equilibration, 677
 - for seawater, 32
 - ideal gas, 25, **47**
 - moist air, 720
- Gradient wind balance, 91, 96, **94–97**
 - in Eulerian equations, 96
- Gravity waves, 100, **101**
 - acoustic, 293
 - critical line, 594
 - hydrostatic, 261
 - stratosphere, 634
- Green and Stone turbulent transport, 494
- Group velocity, **220–224**, **240**
 - internal waves, 264
 - property for wave activity, 384
- Group velocity property, **242–246**
- Gyres, 731
- Hadley Cell, **516–533**
 - angular-momentum-conserving model, 516
 - effects of eddies on, 528, 532
 - effects of moisture on, 522
 - poleward extent, 516
 - radiative equilibrium solution, 523
 - seasonal effects and hemispheric asymmetry, 526
 - shallow water model of, 524
 - strength of, 516
- Hadley, George, 537
- Haney boundary condition, 802, 858
- Heat capacity, 21
- Held–Hou model of Hadley Cell, 516
- Helmholtz function, 18
- Hermite Polynomials, **330**
- Hide’s theorem, 523, 537
- Holton, Jim, 104
- Homentropic fluid, 20, 53
- Homogenization of a tracer, **487–489**
- Horizontal convection, **802–813**
 - maintenance of, 808
- Hurricanes, 727
- Hydrostasy, 12
 - accuracy, 86
 - scaling for, 83, 84
- Hydrostatic approximation
 - accuracy, 86
 - in deriving primitive equations, 64
- Hydrostatic balance, 12, **87**
 - effects of rotation, 92
 - effects of stratification, 85
 - scaling for, 84, 93, **83–93**
- Hydrostatic equations
 - potential vorticity conservation, 163
- Hydrostatic internal waves, 261
- Ideal gas, 20
 - buoyancy frequency, 99
 - enthalpy, 21
 - equation of state, 13
 - fundamental equation of state, 25, 47
 - heat capacity, 21
 - simple and general, 20
 - thermodynamics of, 23
- Impermeability of potential vorticity, 165
- Incompressible flow, 40, **40–42**
 - conditions for, 41
- Inertial flow, 96
- Inertial oscillations, 126
- Inertial range, **418**, 423
 - 3D, 418
 - energy in 2D, 429
 - energy in 3D, 419
 - enstrophy, 429
 - two-dimensional turbulence, 427
- Inertial western boundary currents, **747**
- Inertial-diffusive range, 441
- Inflection point criterion, 346
- Instability
 - baroclinic, 335, 347
 - barotropic, 335
 - Kelvin–Helmholtz, 335
 - necessary conditions in baroclinic flow, 350

- necessary conditions in shear flow, 345
- parallel shear flow, 337
- Intermediate models, 186
- Intermittency, 423
- Internal energy, 20, 48
- Internal thermocline, **780–785**
- Internal waves, 100, **259–261**
 - energetics, 267
 - group velocity, 264, 268
 - polarization properties, 261
 - polarization relations, 261
 - rays, 273
 - reflection, 268
 - stratosphere, 634
 - topographic generation, 283
- Inverse cascade, **424, 429**
 - energy-entropy argument, 425
 - similarity theory, 426
 - vorticity elongation, 424
- Inversion, 147
 - of vorticity, 147
- Inviscid western boundary currents, **753–758**
- Isentropic coordinates, **134–137**
 - and quasi-geostrophy, 196
 - Boussinesq fluid, 135
 - ideal gas, 136
- Isopycnal coordinates, 135
- JEBAR, joint effect of baroclinicity and relief, 757
- Jets, **448–453, 540–549**
 - and the pseudomomentum budget, 544
 - and the vorticity budget, 541
 - atmospheric, 540
 - eddy-driven, 540
 - in beta-plane turbulence, 448
 - numerical simulation of, 449
- Joint effect of baroclinicity and relief, 757
- Joint effect of beta and friction, 449
- Jump conditions, 339
- JWKB approximation, 247, 623
- K41 theory, 416, 418
- Kelvin waves, 126, 636
- Kelvin's circulation theorem, **149**
- Kelvin–Helmholtz instability, **335, 340**
- Kinematic stress, 202
- Kinematic viscosity, 12
- Kinematics
 - of waves, 215
- Kolmogorov scale, 420
- Kolmogorov theory, **416–422**
- Kolmogorov theory of turbulence, 418
- Kolmogorov, A. N., 441
- Lagrange, Joseph-Louis, 52
- Lagrangian derivative, 4
- Lagrangian viewpoint, 3, 4
- Lamb waves, 295
- Lapse rate, **99**
 - adiabatic, of density, 27
 - adiabatic, of temperature, 29
 - dry adiabatic, 29, 99
 - dry ideal gas, 99
 - of seawater, 34
 - saturated, 692
- Larichev–Held model of geostrophic turbulence, 460
- Latent heat, 723
- Latent heat of evaporation, 677
- Level of free convection, 696
- Lifecycle of baroclinic waves, 466
 - in atmosphere, 465
 - in ocean, 468, 470
- Lifting condensation level, 695, 696
- Lindzen–Hou model of Hadley Cell, 526
- Liouville–Green approximation, 247
- Locality in turbulence, 423
- Log-pressure coordinates, 82
- Lorenz equations, 433
- Lorenz, Edward, 141, 142
- LPS model, 786
- Luyten–Pedlosky–Stommel model, 786
- M equation, 780
 - one-dimensional model, 781
- Mach number, 42
- Macro-turbulence, 445
- Madden–Julian oscillation, 718
- Main thermocline, 774
- Margules relation, 119
- Mass continuity, **7–10**
 - Eulerian derivation, 7
 - in a rotating frame, 58
 - in Boussinesq equations, 72
 - Lagrangian derivation, 9
- Mass continuity equation
 - shallow water, 107
- Material derivative, **4–7**
 - finite volume, 5
 - fluid property, 5
- Material viewpoint, 4
- Maxwell relations, **16–17, 19**
- Mercator coordinates, 621
- Meridional overturning circulation, 801
 - atmospheric, Eulerian, 514
 - of atmosphere, 514
 - of ocean, 801
 - wind-driven, ocean, 829
- Mid-latitude atmospheric circulation, **549–571**
- Minimum shear for baroclinic instability, 361
- Mixing length theory, **484–487**
- Mixing ratio, 675

- Mixing ratio and concentration, 10
- MJO, 718
- MOC, 801
 - stratosphere, 644
 - wind-driven, ocean, 829
- Moist convection, **695–701**
- Moist thermodynamics, 720
- Moisture
 - effect on potential vorticity conservation, 160
 - effects on Hadley Cell, 522
- Momentum equation, **11–13**
 - in a rotating frame, 58
 - shallow water, 106
 - vector invariant form, 66
- monsoons, 718
- Montgomery potential, 136
- Mountain waves, 283
- Multi-layer QG equations, 185
- Munk wind-driven model, 740
 - properties of, 741
- Natural coordinates, 94
- Navier, Claude, 53
- Necessary conditions for baroclinic instability, 408, 409
- Necessary conditions for instability, **403–411**
 - baroclinic flow, 350
 - Charney–Stern–Pedlosky criterion, 350, 404
 - Fjørtoft’s criterion, 346, 408
 - Rayleigh–Kuo criterion, 346, 404
 - relation to eddy fluxes, 564
 - shear flow, 345
 - use of pseudoenergy, 406
 - use of pseudomomentum, 403
- Neutral density, 774, 799
- neutral density, 32
- No-slip condition, 741
- Non-acceleration result, 379, 638
- Non-acceleration theorem, **394–399**
- Non-dimensionalization, **46–47**
 - in rotating flow, 171
- Non-homentropic term, 146
- Oblate spheroid, 59, 60
- Observations
 - MJO, 718
 - abyssal ocean, 829
 - Atlantic Ocean, 803
 - atmospheric meridional overturning circulation, 514, 572
 - atmospheric stratification, mean, 572
 - atmospheric wind and temperature, 513
 - deep ocean circulation, 801
 - deep western boundary current, 827
 - Ekman layers, 209
 - Eliassen–Palm flux, 568, 570
 - Eliassen–Palm flux divergence, 570
 - Global Ocean, 735
 - global ocean currents, 732
 - main thermocline, 774
 - North Atlantic, 827
 - North Atlantic currents, 733
 - ocean stratification, 802
 - oceanic meridional overturning circulation, 801
 - of the atmosphere, 511
 - potential vorticity, ocean, 770
 - reanalysis, 536
 - relative humidity, 682
 - surface winds, 514
 - zonally averaged atmosphere, 515
 - zonally averaged zonal wind, 570
- Ocean circulation
 - abyssal, 821
 - laboratory model of, 818
 - scaling for buoyancy-driven, 813
 - wind-driven, 733
 - wind-driven abyssal, 829
- Ocean currents, 732
- Ocean gyres, 731
- Omega equation, 192
- Outcropping, 788
- Parabolic cylinder functions, 330
- Parcel Method, **99, 695**
- Parcel Method for static instability, 97
- Passive tracer, **437–441**
 - in three dimensions, 439
 - in two dimensions, 439
 - spectra of, 437
- Perfect gas, 47
- Phase speed, 216, **216–219**
- Phase velocity, 219
- Phillips instability problem, 356
- Piecewise linear flows, 338
- Plane waves, 216
- Planetary waves, 585
- Planetary-geostrophic equations, **176–180**
 - for shallow water flow, 176
 - for stratified flow, 178
- Planetary-geostrophic potential vorticity
 - equation, 178, 179
 - shallow water, 178
 - stratified, 179
- Poincaré waves, 124, 125, 244
- Poincaré, Henri, 141
- Polar vortex, 629
- Polarization properties of internal waves, 261
- Polarization relations, 261
- Polytropic fluid, 14
- Potential enthalpy, 35

- Potential density, 24, 25, 30, 32
 - and static instability, 97
 - of liquids, 27
- Potential enthalpy, 46, 49
- Potential temperature, 24, 28, 30, 49
 - equivalent, 694
 - ideal gas, 25
 - moist air, 722
 - of liquids, 28
 - seawater, 35
- Potential vorticity, 143, **156–168**
 - and Bernoulli's theorem, 167
 - and the frozen-in property, 158
 - Boussinesq equations, 162
 - concentration, 165
 - conservation of, 156
 - diffusion of, 505
 - for baroclinic fluids, 157
 - for barotropic fluids, 157
 - homogenization of, 793
 - hydrostatic equations, 163
 - impermeability of isentropes, 165
 - mixing, 575, 576
 - ocean observations, 770
 - on isentropic surfaces, 164
 - planetary-geostrophic, 178
 - quasi-geostrophic, 190
 - relation to circulation, 157
 - salinity effects, 160
 - shallow water, 120, 161, 182
 - staircase, 451–453
 - substance, 165
- Potential vorticity flux, 566
- Potential vorticity fluxes, atmospheric, 566
- Potential vorticity homogenization, 487
- Potential vorticity transport
 - and tropospheric stratification, 574
- Prandtl number, 439, 804
- Predictability, **433–437**
 - of Lorenz equations, 433
 - of turbulence, 435
 - of weather, 437
- Pressure, 11, 16
- Pressure coordinates, **79, 81**
 - and quasi-geostrophy, 192
 - relation to Boussinesq equations, 82
- Primitive equations, **64**
 - potential vorticity conservation, 163
 - vector form, 65
- Pseudomomentum
 - and hydrodynamic stability, 403
- Pseudoenergy, 406
 - and hydrodynamic instability, 406
 - and wave activity, 407
- Pseudomomentum, 384
 - a wave activity, 384
 - and zonal jets, 544
- Pseudomomentum equation, 384
- QBO, **651–663**
 - essentials, 655
- Quasi-biennial oscillation, **651–663**
- Quasi-equilibrium, **698, 699**
- Quasi-geostrophic
 - wave–mean-flow interaction, 380
- Quasi-geostrophic potential vorticity
 - equation, 190
 - relation to Ertel PV, 195
- Quasi-geostrophic turbulence, 454
- Quasi-geostrophy, **180–195**
 - asymptotic derivation, 188
 - buoyancy advection at surface, 191
 - continuously stratified, 187
 - energetics, 198
 - in isentropic coordinates, 196
 - informal derivation, 193
 - multi-layer, 185–186
 - pressure coordinates, 192
 - shallow water, 180
 - sheet at boundary, 191
 - single layer, 180
 - stratified equations, **187–194**
 - two-layer, 184–185
 - two-level, 194
- Radiation condition, 543
- Radiative equilibrium temperature, 511
- Radiative equilibrium, 701–703
- Radiative transfer, 724
- Radiative-convective equilibrium, 704–706
- Radius of deformation, 124, 125
- Random walk, 476
- Ray theory, **224–226, 621**
- Ray tracing, 621
- Rayleigh number, 804
- Rayleigh's criterion for instability, 345
- Rayleigh's equation, 338
- Rayleigh–Kuo criterion, 346, 404
- Rays, 226, 273
 - equatorial, 314
 - in internal waves, 273
- Reanalysis, **536, 536**
- Recharge-discharge oscillator, 901
- Reduced gravity equations, **110–112**
- Reference, circular, *see* Circular reference
- Reflection
 - internal waves, 268
 - Rossby waves, 237
- Refractive index, 599
- refractive index, 587
- Relative humidity, 675, 676, **680–690**

- in mid-latitudes, 689
- Relative vorticity, 153
- Residual circulation, 388
 - and thickness-weighted circulation, 391
 - atmospheric, mid-latitude, 571
 - atmospheric, observations of, 572
 - stratospheric, 642
- Resonance of stationary waves, 610
- Reynolds number, 46
- Reynolds stress, 415
- Reynolds, Osborne, 53
- Rhines length, 447
- Rhines scale, 446
- Rhines–Young model, 761
- Richardson number, 494
- Richardson’s four-thirds law, 482
- Richardson, Lewis Fry, 442
- Rigid body rotation, 144
- Rigid lid, 108, 111
- Rossby number, **87**
- Rossby wave trains, 611
- Rossby waves, **226–240**, 585
 - and barotropic jets, 542
 - and ray tracing, 621
 - and turbulence, 446
 - barotropic, 227
 - breaking, 599, 642
 - continuously stratified, 231, 599
 - critical layers, 594
 - dispersion relation, 228
 - energy flux, **234–236**
 - finite deformation radius, 229
 - group velocity property, 384
 - horizontal propagation, 588
 - mechanism of, 229
 - meridional propagation, 621
 - momentum transport in, 542
 - planetary geostrophic, 302
 - propagation, 585
 - reflection, 237
 - topographic, 602
 - two layers, 230
 - vertical propagation, 599, 603, 605, 606
- Rossby, Carl-Gustav, 212
- Rotating frame, **55–59**
- Salinity, 13, 33
 - effect on potential vorticity conservation, 160
 - in box models, 814
- Salt, 13
- Sandström’s effect, 809
- saturated adiabatic lapse rate, 692
- Saturation vapour pressure, 676
- Scale height
 - atmosphere, 42
 - density, 27
 - temperature, 28
- Scale height, atmosphere, 83
- Scaling, **46–47**
 - geostrophic, 171
 - in rotating shallow water equations, 171
 - in rotating stratified equations, 174
- Schwarzschild equations, 724
- Seawater, 13, **32**
 - adiabatic lapse rate, 34
 - equation of state, 13, 33, 35
 - heat capacity, 34
 - potential temperature, 34
 - thermodynamic properties, 32
- Shadow zone, 791
- Shallow water
 - quasi-geostrophic equations, 180
- Shallow water equations
 - multi-layer, 112
 - potential vorticity conservation, 161
 - reduced gravity, 110
 - rotation effects, 121
- Shallow water model of Hadley Cell, 524
- Shallow water systems, **105–123**
 - conservation properties of, 120
 - potential vorticity in, 120
- Shallow water waves, **123–127**
- Shallow-fluid approximation, 65
- Sideways convection, **802–813**
 - conditions for maintenance, 809
 - energy budget, 808
 - limit of small diffusivity, 811
 - maintenance of, 808
 - mechanical forcing of, 812
 - phenomenology, 807
- Sigma coordinates, 82
- Singing in the rain, 673
- Single-particle diffusivity, 478
- Skew diffusion, 491
- Skew flux, 491
- Skew fluxes, 490
- Sloping convection, 347
- Solenoidal term, 146
- Solenoids, 146, 151
- Sound waves, 39–40
- Southern Ocean, 836
- Southern Oscillation, 887
- Specific heat capacities, 20
- Specific humidity, 675
- Spectra of passive tracers, 437
- Spherical coordinates, **59–68**
 - centrifugal force in, 59
- Squire’s theorem, 375
- Stacked shallow water equations, 112
- Staircases of potential vorticity, 451–453
- Standard atmosphere, 572

- State estimate, 536
- Static Instability, **97–99**, **695**
- Stationary phase, 223
- Stationary waves, **609–624**
 - adequacy of linear theory, 614
 - and ray tracing, 621
 - Green function, 613
 - in a single-layer, 609
 - meridional propagation, 621
 - one-dimensional wave trains, 611
 - resonant response, 610
 - thermal forcing of, 615
- Stokes, George, 53
- Stommel box models, 813
- Stommel wind-driven model, **733**
 - boundary layer solution, 737
 - properties of, 741
 - quasi-geostrophic formulation, 736
 - the nonlinear problem, 745
- Stommel, Henry, 758
- Stommel–Arons model, **821**
 - single-hemisphere, 821
 - two-hemisphere, 826
- Stommel–Arons–Faller laboratory model, 818
- Stommel–Arons–Faller model, 818
- Stratification
 - in midlatitudes, 578
 - of the atmosphere, 572
- Stratified geostrophic turbulence, 454
- Stratosphere, 514, 572, **627**
 - polar vortex, 629
 - sudden warming of, 631
- Stratospheric dynamics, **627–669**
- Stratospheric sudden warmings, 663
- Stress
 - Ekman layer, 201
 - kinematic, 202
- Stretching, 149
- Sudden warming, 631
- Sudden warmings, 663
- Super-rotation, 523
- Surf zone, 629
- Surface drifters, 483
- Surface westerlies, 540
- Surface winds, 567
 - observed, 514
- Sverdrup balance, 737
 - near the equator, 863
- Sverdrup interior flow, 738
- Symmetric diffusivity tensor, 490
- Tangent plane, 69
- Taylor–Proudman effect, 90
- TEM, 379, 387, 389, 392
- TEM equations, **387**
 - for primitive equations, 581
- Temperature, 16
- Thermal equation of state, 13, 47
 - moist air, 721
- Thermal wind, 87
 - in shallow water equations, 118, 119
- Thermal wind balance, **87–93**
 - pressure coordinates, 91
- Thermobaric parameter, 14
- Thermocline, 761, **774–799**
 - advective scaling, 778
 - boundary-layer analysis, 782
 - diffusive, 805
 - diffusive scaling, 777
 - internal, 780
 - kinematic model, 775
 - main, 774
 - one-dimensional model, 781
 - reduced-gravity, single-layer model, 787
 - scaling for, 776
 - summary and overview, 790
 - ventilated, 786
 - wind-influenced diffusive scaling, 778
- Thermodynamic equation, **21–30**
 - Boussinesq equations, 72
 - for liquids, 25, 30
 - summary table, 26
- Thermodynamic equilibrium, 21
- Thermodynamic potentials, 16–19
- Thermodynamic relations, **14–21**
 - fundamental, 16
 - Maxwell, 16
- Thermodynamics
 - first law, 15
 - fundamental postulate, 14
 - moist, 720
- Thermohaline circulation, 801
- Thickness, 83
- Thickness diffusion, **502**, 504
- Thomas, Dylan, 3
- Tilting and tipping, 149
- Topographic effects
 - atmospheric stationary waves, 609
 - JEBAR, 757
 - oceanic western boundary current, 753
- Toy models, 907
 - El Niño, 898
- Tracer continuity equation, 10
- Tracer homogenization, 487
- Traditional approximation, 65
- Transformed Eulerian Mean, **387**
 - and eddy transport, 506
 - isentropic coordinates, 389
 - primitive equations, 581
 - quasi-geostrophic form, 387
 - spherical coordinates, 581

- Transformed Eulerian mean, 379, 392
- Triad interactions, 415
 - two-layer geostrophic turbulence, 457
- Tropics, 673
- Tropopause, 572–580, **706–708**
 - definition, 573
- Tropopause height
 - in midlatitudes, 578
 - theory of, 706, 707, 725
- Troposphere, 514, 572
 - and potential vorticity transport, 574
 - stratification, 572, 574, 579
 - ventilation and moist convection, 577
- Turbulence, **413**
 - beta-plane, 448
 - closure problem, 413
 - degrees of freedom, 422
 - fundamental problem, 413
 - predictability of, 433
 - three-dimensional, 416
 - two-dimensional, 423
- Turbulent diffusion, **473, 475**, 490
 - and the TEM, 506
 - in the atmosphere and ocean, 493
 - macroscopic perspective, 487
 - potential vorticity, 505
 - requirements for, 485
 - thickness, 502
 - two-dimensional, 483
- Turbulent diffusivity, 478
- Turning line, 589, 607
- Two-box model, 814
- Two-dimensional turbulence, **423–433**
 - beta effect, 445
 - eddy diffusion in, 483
 - energy and enstrophy transfer, 424
 - numerical solutions, 432
- Two-dimensional vorticity equation, 147
- Two-layer instability problem, 356
- Two-layer model
 - of atmospheric mid-latitudes, 554
- Two-layer QG equations, 184
- Two-level QG equations, 194
- Two-particle diffusivity, 480, 482
- Under Milk Wood, 3
- Unit vectors
 - rate of change on sphere, 62
- Vapour pressure
 - saturation, 676
- Vector invariant momentum equation, 66
- Ventilated pool, 794
- Ventilated thermocline, **786–799**
 - reduced-gravity, single-layer model, 787
 - two-layer model, 788
- Vertical coordinates, **79–83**
- Vertical vorticity equation, 155
- Virtual temperature, 674, 676
- Viscosity, 12
 - effect on energy budget, 44
- Viscous scale, 420
- Viscous-advective range, 439
- Vorticity, 96, **143–153**
 - equation for a barotropic fluid, 146
 - equation on beta plane, 156
 - evolution equation, **145**
 - evolution in a rotating frame, 153
 - frozen-in property, 147
 - in two dimensional fluids, 147
 - stretching, 151
 - stretching and tilting, 149
 - vertical component, 155
- Vorticity equation, 96
- Vorticity, relative, 153
- vr vortex, 144
- Walker Cell, 718
- Walker circulation, 328, 887, 888, **892**
- Water, 673
- Water Vapour, 673–676
- Water vapour
 - measures of, 675
- Wave activity, 384
 - and pseudomomentum, 384
 - group velocity property, 384
 - orthogonality of modes, 385
- Wave breaking, 642
- Wave packet, 221
- Wave propagation, 585
- Wave trains, 611
- Wave–mean-flow interaction, **379**, 382
 - quasi-geostrophic, 380
- Wave–turbulence cross-over, 446
- Waveguides
 - for equatorial waves, 314
 - for internal waves, 274
- Wavelength, 216
- Waves, **215**
 - acoustic-gravity, 293
 - barotropic Rossby, 227
 - breaking, 642
 - frequency, 216
 - group velocity property, 240
 - hydrostatic gravity, 261
 - inertial, 126
 - Kelvin, 126, 636
 - kinematics, 215
 - Lamb, 295
 - Poincaré, 124, 125, 244
 - Rossby, 226

- Rossby dispersion relation, 228
- Rossby wave mechanism, 229
- Rossby, continuously stratified, 231
- Rossby, single-layer, 227
- Rossby, two-layer, 230
- rotating shallow water, 124
- shallow water, 123
- sound, 39
- wavevector, 216
- Wavevector, 216
- Weak temperature gradient approximation, 712, 715
 - adjustment to, 716
 - for stratified flow, 717
 - in shallow water equations, 716
- Weather predictability, 437
- Western boundary currents
 - topographic and inviscid, 753
- Western boundary layer, 739
 - frictional, 738
 - inertial, 747
- Western intensification, 731
- Western pool, 793
- Wind-driven gyres, 731
- Wind-driven ocean circulation, **733–770**
 - continuously stratified, 767
 - homogeneous model, 733
 - two-layer model, 761
 - vertical structure, 761
- WKB approximation, **247–249**, 623
 - internal waves, 271
 - Rossby waves, 589, 590, 598
- Zonal boundary layers in ocean gyres, 744
- Zonal flow in turbulence, 446
- Zonal flows in beta-plane turbulence, 448
- Zonally averaged atmospheric circulation, 539